

# 2026 ISBA World Meeting

## Book of Abstracts



This version: June 29, 2026

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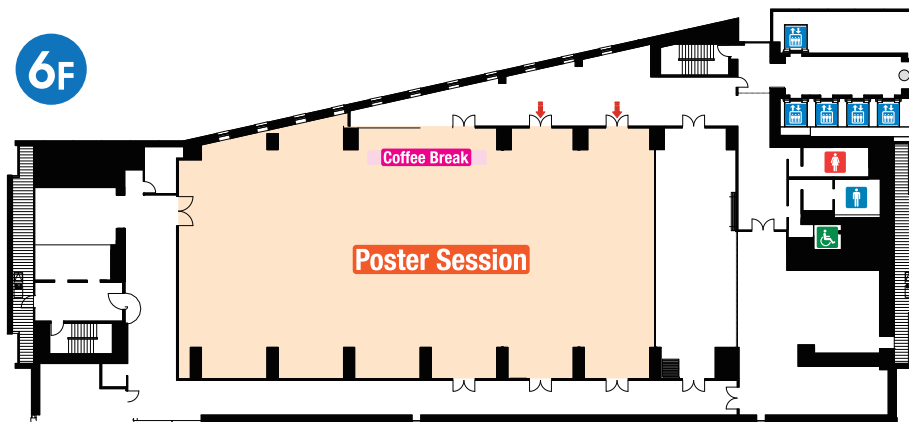
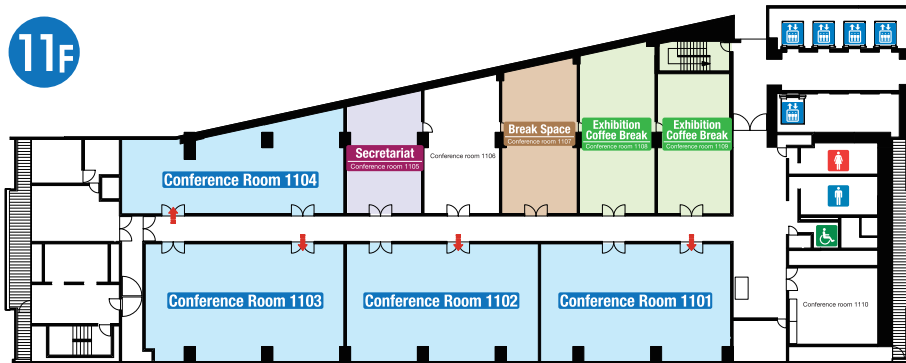
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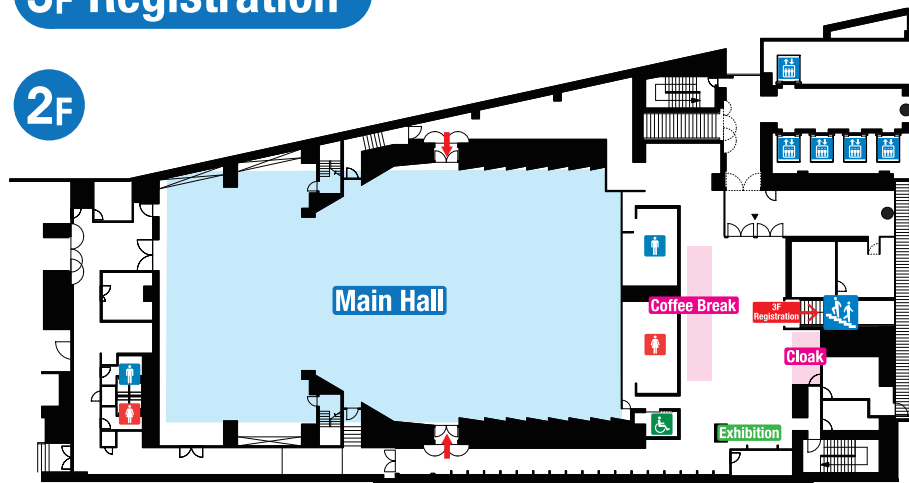
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# Venue



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## Sunday 28 June

### Foundational Lecture 1: 14:30–15:30

#### *Structures and flexibility: a modelling journey in biostatistics*

Presenter: Sylvia Richardson

In this talk, I will journey through some of the modelling tasks in health sciences that I have been confronted with, highlighting in particular different ways of infusing knowledge and borrowing information. Alongside, I will touch on the ever delicate balance between the structural assumptions that we formulate, the degree of flexibility that we try to preserve and the scalability that we aim for.

### Foundational Lecture 2: 15:30–16:30

#### *Bayesian Inference and the Fair Administration of Justice*

Presenter: Hal Stern

Probability modeling and statistical inference have emerged as critical topics in ongoing discussions regarding the use of science to assess forensic evidence. A 2009 U.S. National Academies report on forensic science and a subsequent 2016 report by the U.S. President's Council of Advisers on Science and Technology raised questions about the scientific underpinnings for the analysis of a number of types of forensic evidence. Misapplication of forensic science has been identified as a contributing factor in a number of cases of wrongly-convicted individuals. For these reasons there has been an increased focus on evaluating the ways in which evidence is analyzed, interpreted and reported with an eye towards providing more scientifically justified methods. This talk provides some background on different types of forensic evidence and demonstrates novel approaches to inference for bloodstain pattern evidence. Contributions include a novel approach to representing bloodstain patterns and the application of a Dirichlet Process Mixture Model for assessing the likelihood of observing a given pattern under different causal mechanisms.

### Foundational Lecture 3: 17:00–18:00

#### *Adventures in Bayesian clustering*

Presenter: David Dunson

Clustering is a core statistical problem and there is a vast literature on Bayesian approaches. The canonical approach is to model the likelihood as a discrete mixture of simple distributions (e.g., Gaussians). Observations drawn from the same component of this mixture are assumed to fall in the same cluster. MCMC algorithms can be used to sample from the posterior distribution of the component labels for each observation while allowing the mixture weights and component parameters to be unknown. Even with the abundant focus on refining this approach over the years, there are many fundamental challenges that have motivated recent and ongoing paradigm shifting advances. This talk will briefly review the classical Bayesian model-based clustering approach and then highlight key problems including (1) sensitivity to kernel misspecification; (2) curse of dimensionality; and (3) difficulty of summarizing the clustering posterior in an interpretable manner. Recent solutions to (1)-(3) will be described and illustrated through a variety of applications and ongoing challenges in the field will be highlighted.

### Keynote Lecture 1: 18:00–19:00

#### *Predictive Densities Based on Information Geometry*

Presenter: Fumiyasu Komaki

The construction of predictive densities is an important problem in Bayesian inference. In this talk, Bayesian predictive densities are investigated from the perspective of information geometry. Predictive performance is evaluated by the Kullback–Leibler divergence, and geometric structures induced by the Fisher information are used to study predictive risk across a broad class of statistical models. From this viewpoint, shrinkage priors often yield predictive improvements over Jeffreys priors and other conventional noninformative priors. In several important examples, Bayesian predictive densities based on shrinkage priors asymptotically or exactly dominate standard predictive procedures. These results suggest that prior distributions should not necessarily be selected according to context invariance principles, but rather constructed in relation to predictive objectives and the geometric structure of statistical models. Information geometry provides a framework for understanding predictive

shrinkage, asymptotic risk reduction, and the design of predictive priors. The talk also discusses Bayes extended estimators as possible alternatives to conventional empirical Bayes approaches.

# Monday 30 June

## Multitrack Sessions 1: 9:00–10:30

### Bayesian Regression in Complex Data Environments (Room 1101)

#### *Bayesian Nonparametric Model for Nonseparable Instrumental Variable Regression*

Presenter: Andriy Norets

We propose a flexible Bayesian model for estimation of nonseparable instrumental variable regressions with a univariate endogenous covariate and a univariate unobservable responsible for endogeneity. Our model uses recently developed nonparametric priors for conditional distributions based on covariate dependent mixtures that deliver optimal adaptive posterior contraction rates in settings with discrete and continuous variables under possible smoothness and sparsity. We develop posterior concentration results and a Markov chain Monte Carlo algorithm for the model. The proposed methodology provides a simple parametric baseline model that can be gradually extended to more flexible parametric models and ultimately a nonparametric one by increasing the number of mixture components or setting a prior on it. Thus, robustness to nonlinearities, nonseparability, heteroskedasticity and other deviations from the baseline can be gradually introduced in a computationally tractable way. We illustrate the model performance in simulations and applications.

#### *Compressed Bayesian Tensor Regression*

Presenter: Roberto Casarin

To address the common problem of high dimensionality in tensor regressions, we introduce a generalized tensor random projection method that embeds high-dimensional tensor-valued covariates into low-dimensional subspaces with minimal loss of information about the responses. The method is flexible, allowing for tensor-wise, mode-wise, or combined random projections as special cases. A Bayesian inference framework is provided featuring the use of a hierarchical prior distribution and a low-rank representation of the parameter. Strong theoretical support is provided for the concentration properties of the random projection and posterior consistency of the Bayesian inference. An efficient Gibbs sampler is developed to perform inference on the compressed data. To mitigate the sensitivity introduced by random projections, Bayesian model averaging is employed, with normalizing constants estimated using reverse logistic regression. An extensive simulation study is conducted to examine the effects of different tuning parameters. Simulations indicate, and the real data application confirms, that compressed Bayesian tensor regressions can achieve better out-of-sample predictions while significantly reducing computational costs compared to standard Bayesian tensor regressions.

#### *Local Approximate Weighting-Based Diagnostics for Bayesian Multilevel Regression and Poststratification*

Presenter: Ryan Giordano

Multilevel Regression with Post-stratification (MrP) has become a workhorse method for estimating population quantities using non-probability surveys, and is the primary alternative to traditional survey calibration weights. For simple linear regression models, MrP methods admit “equivalent weights”, allowing for direct comparisons between MrP and traditional survey weights (Gelman 2006). We develop a more general framework for computing and interpreting equivalent weights that can be used for complex, nonlinear MrP models, particularly those whose posterior is estimated with Markov Chain Monte Carlo (MCMC). For linear MrP models, our weighting representation is exact and recovers familiar diagnostics for linear regression. For MrP models based on nonlinear links and / or hierarchical structure, our weights are based on a local approximation, which we show can accurately capture important features of an exact linear weighting, including measures of partial pooling, covariate (im)balance, and frequentist sampling variability. Importantly, our weights are easily computed based on existing MCMC samples and conveniently wrap standard MrP software implementations. We illustrate our approach for several canonical studies that use MrP, including for the binary outcome of vote choice, showing that our weights can interpretably detect and diagnose model misspecification and potential bias, even in complex MrP models, with minimal additional computation.

#### *A fully dynamic Bayesian treatment effects approach for panel outcomes*

Presenter: Liana Jacobi

In the analysis of observational data in economic and many other applications, selection into treatment is commonly present, making the issue of endogeneity into the treatment a pervasive problem for the estimation of causal treatment effects. In the case of panel outcomes, this issue complicates the analysis further and panel dependence across outcomes also needs to be addressed leading to two different sources of unobserved heterogeneity.

Depending on the context, we are likely to face additional heterogeneity in the treatment effects depending on the observed characteristics of a unit. In this paper we propose a flexible Bayesian treatment effects approach for panel outcomes to estimate heterogeneous and dynamic causal effects under endogeneity of the treatment in such settings. We extend the panel treatment effects models with a factor-based covariance structure introduced in Jacobi, Wagner, and Frühwirth-Schnatter (2016) and further expanded augmented treatment effects model in Wagner, Frühwirth-Schnatter, and Jacobi (2023) to allow for time-variant observed heterogeneity within a bi-factor framework. We proceed within a potential outcome framework where all regression effects in the potential outcomes model are modelled as time-varying. We consider random walk priors in the non-centered parameterisation. As appropriate prior specifications will be needed to avoid overfitting, we employ shrinkage priors as in Bitto and Frühwirth-Schnatter (2019) and Cadonna, Frühwirth-Schnatter and Knaus (2020). Given the panel data context, we also investigate a similar prior that allows one to borrow strength across factor loadings. We propose an efficient sampler based MCMC methods and test the methods with simulated data.

In the empirical application we investigate the impacts of a longer labour break after childbirth on female earnings profiles after re-entry into the labour market. Labour market patterns and earnings profiles of women continue to vary significantly from those of men, to a large extent as a result of discontinuities in labour market histories after childbirth and greater variation in the intensity of their labour market involvement. A key implication is the well-documented motherhood wage gap. We exploit a unique administrative data set to analyse female earnings after childbirth and investigate how choices regarding the extent of the time away from the labour market (labour market break) after birth, as well as observed and unobserved heterogeneity, impact longer term earnings trajectories of mothers once they return. The preliminary results show that a fully dynamic model is required to capture the full extent of the time-variation in the effect in the observed and unobserved effects and the dynamics and heterogeneity in the treatment effects. The estimates of the dynamic treatment effects from a long break vary considerably over time but do not disappear. The estimated treatment effects for different demographic groups range from positive to negative and illustrate the considerable heterogeneity in the average treatment effect across sub-groups.

#### ***Efficient Bayesian transfer learning via spike-and-slab priors.***

Presenter: Paul Rognon-Vael

Transfer learning aims at improving performance in a target task on a target dataset by leveraging knowledge acquired from source tasks and datasets. Challenges include integrating multiple sources, computational cost, and avoiding negative transfer, that is, when the transfer is harmful to the performance in the target task. We propose a Bayesian method that solves those challenges in high-dimensional linear regression. While existing Bayesian methods either transfer summary statistics of regression coefficients or perform joint estimation of source and target models, we study a middle ground approach where the complete posterior distribution obtained on the source is transferred to improve estimation in the target. Our method based on spike-and-slab priors then allows for a rich transfer while maintaining high levels of computation and communication efficiency, and of privacy. We show our method outperforms summary-based transfer while remaining competitive against joint estimation methods that assume full access to source data.

### **Optimal Transport Meets Bayesian Statistics (Room 1102)**

#### ***Optimal transport between Markov chains and its application to Bayesian computation***

Presenter: Pierre Jacob

Parallelizing Markov chain Monte Carlo (MCMC) computation is a long-standing problem that can be addressed with unbiased MCMC methods, as in Glynn & Rhee (2014) and Jacob, O’Leary & Atchadé (2020). These methods require the user to implement specific couplings of MCMC algorithms. So far, these couplings have been designed on a case-by-case basis, one MCMC algorithm at a time, with few optimality guarantees. This talk explores the possibility of a unifying framework involving optimal transport problems between Markov chains, and we will discuss the difficulty of finding a convenient parametrization of the set of couplings of MCMC transition kernels.

#### ***Asymptotic transport maps for sampling high-dimensional posteriors***

Presenter: Anya Katsevich

In high-dimensional Bayesian inference, posterior distributions are known to be approximately Gaussian in the regime  $d^2 \ll n$ , where  $d$  is the dimension and  $n$  is the sample size. In this regime, the Laplace approximation provides an explicit Gaussian proxy for the posterior and can be used to simplify inference tasks such as sampling and integration. A natural question is whether analytic, non-Gaussian approximations remain possible in higher-dimensional regimes, beyond the validity of the Gaussian approximation but still below the concentration

threshold  $d = o(n)$ .

We answer this question through a higher-order, transport-based construction. Starting from a Gaussian distribution, we build explicit nonlinear polynomial maps whose push-forwards approximate the true posterior in total variation. These maps are not obtained by solving an optimal transport problem or by training a normalizing flow; instead, they arise analytically from high-dimensional Laplace asymptotics and local derivatives of the log-posterior near its mode. The resulting approximations are easy to sample from, come with quantitative high-dimensional error bounds, and remain valid far beyond the usual Gaussian regime, approaching the natural concentration threshold  $d = o(n)$ .

### ***Distances between random probabilities and random measures in Bayesian Nonparametrics***

Presenter: Hugo Lavenant

Random probabilities and random measures are central concepts in Bayesian nonparametric (BNP) statistics. In this talk, we introduce optimal transport-based distances between the laws of such objects, which come with analytical upper and lower bounds while preserving statistical and computational tractability. We also highlight how these distances can be used to analyze BNP models—both a priori and a posteriori—in tasks such as quantifying the merging of opinions and assessing the quality of finite-dimensional approximations of random measures.

## **Challenges in Bayesian Biostatistics (Room 1103)**

### ***A Bayesian decision theoretic perspective on causal questions: a bridge between randomised controlled trials and real-world evidence***

Presenter: Brieuc Lehmann

Randomised controlled trials (RCTs) and real-world evidence (RWE) each have complementary strengths and limitations for causal inference. RCTs offer internal validity through randomisation but are often costly, slow, and conducted in selective populations. RWE, by contrast, can provide larger and more representative samples at lower cost but raises challenges related to confounding and bias. This work investigates how a Bayesian decision theoretic perspective, through value-of-information (VoI) analysis, can serve as a bridge between these two evidence paradigms. VoI quantifies the expected benefit of reducing uncertainty and has been widely adopted in health economics and health technology assessment to inform whether, and how, additional data collection should occur. The same principles can be extended to causal questions, allowing formal assessment of the information yield and decision value of both randomised and observational study designs. Building on connections between VoI and Bayesian experimental design, the proposed framework compares the expected information gain across study characteristics such as sample size, covariate distributions, and degrees of confounding or effect heterogeneity. Through a wide range of simulation-based examples, we demonstrate how this unified approach enables explicit, quantitative comparisons of RCT and RWE designs. The framework highlights methodological challenges, including computational barriers and the specification of utility functions, priors, and data-generating processes, and outlines opportunities for advancing VoI methods to better integrate Bayesian decision theory, causal inference, and real-world biomedical research.

### ***Data selection and integration in clinical trials***

Presenter: Weining Shen

Gene therapies aim to address the root causes of diseases, particularly those stemming from rare genetic defects that can be life-threatening or severely debilitating. Although an increasing number of gene therapies have received regulatory approvals in recent years, understanding their long-term efficacy in trials with limited follow-up time remains challenging. To address this question, we propose a novel Bayesian framework to selectively integrate relevant external data with internal trial data to improve the inference of the durability of long-term efficacy. We proved that the proposed method can theoretically identify external subsets deemed relevant, where relevance is defined as the similarity, induced by the marginal likelihood, between the generating mechanisms of the internal data and the selected external data. We conducted simulations to evaluate its performance under various scenarios. Furthermore, we apply this method to predict and infer the endogenous factor IX (FIX) levels of patients who receive Etranacogene dezaparvovec long-term. Our estimated long-term FIX levels, validated by recent trial data, indicate that Etranacogene dezaparvovec induces sustained FIX production. Together, the theoretical findings, simulation results, and application of this framework underscore its potential to address long-term effectiveness estimation and inference questions in real-world applications.

### ***A Mixed Hawkes Process to Model Epileptic Seizures***

Presenter: Elizabeth Juarez-Colunga

Epilepsy involves recurrent seizures, often occurring in clusters where a seizure increases the short-term likelihood of another. We propose a mixed Hawkes process model to capture clustering and individual heterogeneity. The Hawkes process increases intensity after each event through background and excitation components. The model is motivated by 3-year seizure diary data from 407 patients with newly diagnosed focal epilepsy in the Human Epilepsy Project (HEP), which show clustering, wide variation in seizure rates between individuals. Challenges include daily aggregation of seizure counts and missing intervals where events may have occurred but were not recorded. A Bayesian approach is implemented, which handles non-informative missingness and aggregated counts. We assess model performance via simulation and apply the model to HEP to study trends, covariates, cluster size, and self-excitation.

## **Beyond the Map: Cutting-Edge Spatial Statistics for Contemporary Applications (Room 1104)**

### ***Unraveling temporal patterns in spatial random partitions driven by spanning tree***

Presenter: Jessica Pavani

Spatially constrained clustering is a significant area of research, especially when addressing changes over time. Dividing a map into partitions is a complex task due to the large number of potential partitions within the search space. This complexity is heightened in spatio-temporal clustering, where we must account for sequences of partitions. To address these challenges, we propose a Bayesian model that combines spanning trees and prior distributions based on product partition models. The use of random spanning trees simplifies the exploration of the partition search space. By integrating spanning trees with product partition models, we reduce the search space by creating a connected subgraph that includes all nodes without cycles. This approach limits partitions to those aligned with the tree structure, facilitating contiguous clustering and the detection of irregular shapes. When pruning trees, the process naturally enforces spatial constraints, maintaining cluster connectivity. Temporal dependence is incorporated through a prior distribution that links the random spatial partitions. The performance of our proposal is evaluated through simulated studies and a real data application.

Joint work: Rosangela H. Loschi and Fernando A. Quintana

### ***The analysis of criminal recidivism: a hierarchical model-based approach for the analysis of zero-inflated, spatially correlated recurring event data***

Presenter: Marcos O. Prates

The life course perspective in criminology has become prominent in recent years, offering valuable insights into various patterns of criminal pathways. Noticeably, the study of criminal trajectories aims to understand crime's beginning, persistence, and desistance. Central to this analysis is the identification of patterns in the frequency of criminal victimization and recidivism, along with the factors that contribute to them. Specifically, this work introduces a new class of models that overcome limitations in traditional methods used to analyse criminal recidivism. The proposed models are designed for recurrent events data characterized by excess of zeros and spatial correlation. In addition to their parametric counterparts, we propose flexible semi-parametric versions approximating the intensity function using Bernstein Polynomials. The performance of these models was evaluated in a simulation study with various scenarios, and we applied them to analyse criminal recidivism data in the Metropolitan Region of Belo Horizonte, Brazil. The results provide a detailed analysis of high-risk areas for recurrent crimes and the behaviour of recidivism rates over time. This research significantly enhances our understanding of criminal trajectories, paving the way for more effective strategies in combating criminal recidivism. The author acknowledged CNPq and FAPEMIG for partial financial support.

Joint work: Alisson C. C. Silva, Fábio N. Demarqui and Bráulio F. Silva.

See details in <https://academic.oup.com/jrsssa/advance-article/doi/10.1093/jrsssa/qnaf061/8152038>

### ***Bayesian modelling for the integration of spatially misaligned health and environmental data***

Presenter: Paula Moraga

Spatially misaligned data is increasingly common primarily due to advancements in data collection and management. In this talk, I will present a flexible and fast Bayesian modelling framework for the combination of data available at different spatial resolutions and from various sources. Inference is performed using INLA and SPDE, which provides a fast approach to fit latent Gaussian models. The approach is flexible and can be applied in preferential sampling and spatio-temporal settings. The Bayesian modelling approach is demonstrated in a range of health and environmental settings. Specifically, a spatial model is developed to combine point and areal malaria prevalence data, and to integrate air pollution data from different sources. The approach presented provides a useful tool in a wide range of situations where information at different spatial scales needs to be combined, and



provides valuable insights for decision-making in health and environmental fields.

## **From Inference to Prediction in Large Alphabets: Nonparametric and Empirical Bayesian Approaches (Main Hall)**

### ***Double trouble: Predicting new variant counts across two heterogeneous populations***

Presenter: Tamara Broderick

Collecting genomics data across multiple heterogeneous populations (e.g., across different cancer types) has the potential to improve our understanding of disease. Despite sequencing advances, though, resources often remain a constraint when gathering data. So it would be useful for experimental design if experimenters with access to a pilot study could predict the number of new variants they might expect to find in a follow-up study: both the number of new variants shared between the populations and the total across the populations. While many authors have developed prediction methods for the single-population case, we show that these predictions can fare poorly across multiple populations that are heterogeneous. We prove that, surprisingly, a natural extension of a state-of-the-art single-population predictor to multiple populations fails for fundamental reasons. Taking a Bayesian nonparametric approach, we provide the first predictor for the number of new shared variants and new total variants that can handle heterogeneity in multiple populations. We show that our proposed method works well empirically using real cancer and population genetics data.

### ***Bayesian nonparametric inference for unobserved events***

Presenter: Alessandro Colombi

Many discovery problems require assessing whether additional sampling is likely to reveal new categories with non-negligible prevalence. We study this question through the lens of the maximum unseen probability, defined as the largest probability among categories not yet observed in the sample, and construct one-sided interval estimates for this quantity. Unlike existing distribution-free approaches, which provide universal but often overly conservative bounds, we adopt a Bayesian nonparametric perspective. By specifying flexible priors, we obtain closed-form expressions for posterior upper bounds that explicitly model structural features of the underlying population, such as power-law behavior or a finite alphabet of unknown size. Our framework encompasses the two most common sampling schemes: Bernoulli product models, widely used for presence-absence (incidence) data, and multinomial sampling models, which arise naturally in abundance settings. In both cases, the resulting bounds are substantially sharper than their frequentist worst-case counterparts. We illustrate the proposed methodology through an application to criminal network data concerning the activities of the 'Ndrangheta in Northern Italy.

### ***Empirical Bayes strategies for the maximum unseen probability***

Presenter: Mario Beraha

Given a sample from a multinomial distribution with possibly infinitely categories, how large can the largest probability associated with unseen categories be? Inference on the maximum unseen probability has recently gained traction as a key problem in large alphabet inference. Recent works proposed either worst case approaches, leading to data-independent bounds, or Bayesian strategies, whereby inference is highly sensitive to the prior distribution. In this talk, I will propose a “hybrid” solution that bridges between the worst case and the Bayesian one, based on the idea of F-localization introduced by Ignatiadis & Wager for empirical Bayesian estimands. A central result is the (asymptotic) validity of such empirical Bayes confidence interval even in the case of fixed parameters under suitable conditions on the empirical process.

## **Multitrack Session 2: 11:00–12:30**

### **Bayesian Uncertainty Quantification and Posterior Sampling for Large-Scale Generative Models (Room 1101)**

#### ***Conformal Tree for AI uncertainty quantification***

Presenter: Max Pachman

This work is concerned with conformal prediction in contemporary applications (including generative AI) where a black-box model has been trained on data that are not accessible to the user. Mirroring split-conformal inference, we design a wrapper around a black-box algorithm which calibrates conformity scores. This calibration is local and proceeds in two stages by first adaptively partitioning the predictor space into groups and then calibrating sectionally group by group. Adaptive partitioning (self-grouping) is achieved by fitting a robust regression tree to the conformity scores on the calibration set. This new tree variant is designed in such a way that adding

a single new observation does not change the tree fit with overwhelmingly large probability. This add-one-in robustness property allows us to conclude a finite sample group-conditional coverage guarantee, a refinement of the marginal guarantee. In addition, unlike traditional split-conformal inference, adaptive splitting and within-group calibration yields adaptive bands which can stretch and shrink locally. We demonstrate benefits of local tightening on several simulated as well as real examples using non-parametric regression. Finally, we consider two contemporary classification applications for obtaining uncertainty quantification around GPT-4o predictions. We conformalize skin disease diagnoses based on self-reported symptoms as well as predicted states of U.S. legislators based on summaries of their ideology. We demonstrate substantial local tightening of the uncertainty sets while attaining similar marginal coverage.

### ***Optimal Transport-Based Generative Models for Bayesian Posterior Sampling***

Presenter: Yuexi Wang

We investigate the problem of sampling from posterior distributions with intractable normalizing constants in Bayesian inference. Our solution is a new generative modeling approach based on optimal transport (OT) that learns a deterministic map from a reference distribution to the target posterior through constrained optimization. The method uses structural constraints from OT theory to ensure uniqueness of the solution and allows efficient generation of many independent, high-quality posterior samples. The framework supports both continuous and mixed discrete-continuous parameter spaces, with specific adaptations for latent variable models and near-Gaussian posteriors. Beyond computational benefits, it also enables new inferential tools based on OT-derived multivariate ranks and quantiles for Bayesian exploratory analysis and visualization. We demonstrate the effectiveness of our approach through multiple simulation studies and a real-world data analysis.

### ***Quantile Martingale Posteriors***

Presenter: Edwin Fong

Quantile estimation and regression within the Bayesian framework is challenging as the choice of likelihood and prior is not obvious. In this talk, we introduce a novel Bayesian nonparametric method for quantile estimation and regression based on the recently introduced martingale posterior (MP) framework. We demonstrate that a recursive estimate of a smooth quantile function, subject to a martingale condition, is entirely sufficient for full nonparametric Bayesian inference. We term the resulting posterior distribution as the quantile martingale posterior (QMP), which arises from an implicit generative predictive distribution. Associated with the QMP is an expedient, MCMC-free and parallelizable posterior computation scheme, which can be further accelerated with an asymptotic approximation based on a Gaussian process. Furthermore, the well-known issue of monotonicity in quantile estimation is naturally alleviated through increasing rearrangement due to the connections to the Bayesian bootstrap.

## **Random Partition Learning: Recent Advances (Room 1102)**

### ***Learning pathways of life events: a sequential allocation Bayesian model with label tracking***

Presenter: Andrea Cremaschi

The collection of socio-economic data in longitudinal studies has become widespread across the social, economic, and health sciences. Such datasets often record event histories showing how key events and personal choices (such as marriage, the birth of a child, or a change in employment status) shape life trajectories. Measuring these mechanisms deepens our understanding of individual behaviour and societal dynamics and, crucially, can inform public policy and community investments. Uncovering representative dynamic life patterns requires clustering techniques that are both interpretable and flexible enough to produce meaningful, low-dimensional summaries of these complex phenomena. Existing model-based methods in the statistical literature remain rather limited in this regard. In this paper, we introduce a flexible Bayesian nonparametric framework for modelling multiple dependent categorical variables observed over time through evolving latent life stages. Our primary goal is to identify the stages individuals pass through during their observed lifetimes and to characterise distinct dynamic behavioural patterns. A key methodological innovation is that each life stage is defined by time-invariant parameters, ensuring a consistent interpretation across all time points. Unlike modern Bayesian dynamic clustering approaches, which re-estimate cluster characteristics at every time step, our strategy greatly improves computational efficiency and enables meaningful longitudinal comparisons. In addition, the model captures temporal dynamics via a time-varying partition of the study population that incorporates both abrupt structural changepoints, where the overall clustering configuration may shift, and individual-level transitions between life stages, where a subset of subjects move between stages, possibly at each time point. This framework provides a powerful tool for understanding life-course dynamics, as it coherently integrates personal decision-

making with macro-level societal influences. We illustrate the methodology using longitudinal data from the Italian Institute of Statistics (ISTAT), demonstrating its ability to reveal both individual behaviours and broader societal shifts. (joint work with Franzolini B., Piccarreta R.)

#### ***Exchangeable Hierarchical Partition Distribution with Direct Prior Specification***

Presenter: David Dahl

We propose a unified framework for constructing exchangeable partition distributions that enables modelers to directly encode prior information on both the number of clusters and the distribution of cluster sizes, encompassing classical models such as the Chinese Restaurant Process (CRP) and the Hierarchical Uniform Partition (HUP) as special cases; the probability mass function is built in three hierarchical stages: first, the modeler specifies a marginal distribution for the number of clusters based on prior knowledge or expert opinion, or alternatively chooses one that matches the distribution implied by a standard model (e.g., CRP); second, given the number of clusters, we define the conditional distribution of cluster sizes — a flexible component that may reflect the behavior of the CRP or the HUP — and incorporate a novel tilt parameter that adjusts the relative probabilities of different cluster-size configurations according to a measure of uniformity in cluster sizes, such that when the tilt parameter is zero, our model reduces to either the CRP or the HUP; negative tilt values favor configurations with lower entropy (i.e., more imbalanced cluster sizes); and positive tilt values favor configurations with higher entropy (i.e., more uniform cluster sizes), with the extreme limits inducing partitions of minimum or maximum entropy, respectively; and third, conditional on both the number of clusters and their sizes, a uniform distribution over the partitions is assumed, ensuring exchangeability. This framework offers enhanced flexibility by enabling direct elicitation of prior information on the number of clusters, the distribution of cluster sizes, and preferences regarding the distribution of relative cluster sizes, thereby providing an intuitive and adaptable tool for model-based clustering in a variety of applications. The distribution has a tractable normalizing constant; therefore, posterior inference on the tilt and other parameters is straightforward. (Joint work with Richard Warr and Talmage Hilton)

#### ***Informed Partition Models for Dependent Random Partitions***

Presenter: Sally Paganin

Model-based clustering is a powerful tool often used to discover hidden structure in data by grouping observational units that exhibit similar response values. Recently, clustering methods have been developed that allow the inclusion of an “initial” partition of the data informed by expert opinions, starting from a probability distribution on the space of partitions. Then, using some similarity criteria, partitions different from the initial one are down-weighted, i.e. they are assigned reduced probabilities. We take a different perspective and model the probability that each unit follows the initial partition via auxiliary variables. Our informed partition model provides flexibility to include varying levels of uncertainty to any subset of the partition (i.e., locally weighted prior information). Additionally, it can accommodate settings with multiple dependent partitions, such as temporal or multi-view data. Theoretical properties of the proposed construction are explored, which can be useful for prior elicitation. We illustrate the gains in prior specification flexibility via simulation studies and an application to a dataset concerning spatio-temporal evolution of PM10 measurements in Germany.

### **Bayesian Biostatistics (Room 1103)**

#### ***Bayesian Nonparametric Intensity Estimation for Modeling Cell Distributions in Spatial Transcriptomics***

Presenter: Huiyan Sang

Spatially resolved molecular imaging technologies have revolutionized tissue biology by enabling spatial visualization of biological processes at single-cell and subcellular resolution. We model cell locations in spatial transcriptomics data as point patterns, where the primary problem of interest is to estimate intensity functions that characterize distributions of cells over the tissue domain. Many existing popular intensity function estimation methods face challenges when domains have irregular boundaries, sharp concavities, or interior holes. We propose a nonparametric spatially aware Bayesian additive model to model the distribution of event occurrences. The method has the advantage of capturing both varying smoothness and sharp changes in intensity functions. We develop an exact likelihood-based and efficient Bayesian inference algorithm to estimate the intensity function with uncertainty measures. We demonstrate the advantages of our model over other methods in simulation studies and spatial transcriptomics real data applications.

#### ***Inference under Dynamical Survival Analysis Models of Epidemics***

Presenter: Jason Xu

Stochastic compartmental models are fundamental tools in epidemic modeling. The Dynamical Survival Analysis (DSA) method replaces the stochastic population-level hazard by its large population limit to approximate the original process. While doing so enables more efficient inference via survival analytic techniques and retains a count-valued stochastic model on the same state space, use of a deterministic hazard underestimates the variance, leading to biased posterior uncertainty and incorrect frequentist coverage. To address this limitation, we augment the deterministic DSA formulation with a latent Gaussian process (GP) layer, which reintroduces missing variability via a hierarchical model. We show that posterior inference is naturally carried out with an easily implementable HMC-within-Gibbs framework. Through simulation studies, we find that the correction via a hierarchical view leads to marked improvements in coverage of the credible intervals.

***Leveraging Bayesian nonparametric mixture models to improve our understanding of dietary exposure patterns over time on cardiovascular disease risk***

Presenter: Briana Stephenson

Many studies have examined the relationship of a univariate dietary exposure (e.g. single food/nutrient group, adherence score) to cardiovascular disease (CVD) outcomes. However, few have addressed the effect of temporal changes in multivariate dietary exposures, such as the collective consumption of various foods and beverages, over a fixed period on CVD risk. Recent advances in Bayesian nonparametric model-based clustering have allowed the analysis and identification of dietary exposure patterns over time. This talk will introduce the development of a new model which extends the Bayesian nonparametric mixture model, under a supervised setting, to link with a proportional odds model. Using dietary consumption and cardiometabolic health data from the Black Women's Health Study, we will demonstrate the model's utility to identify longitudinal dietary exposure patterns that are associated with the onset of major CVD risk factors in US Black women from 1995-2021.

***Posterior Predictive (PoP) Design for Phase I Clinical Trials***

Presenter: Jack Lee

This talk introduces the Posterior Predictive (PoP) design, a novel phase I dose-finding framework that leverages Bayesian predictive learning to optimize both short-term dose transitions and long-run MTD convergence. Our work moves beyond the existing model-assisted interval-based designs by achieving global optimality in dose transition. Theoretically, the global optimality ensures that the proposed design can consistently select the true MTD at an impressive convergence rate of  $n^{-1/2}$ . Through extensive simulation studies, we demonstrate that the PoP design yields substantial improvement in operating characteristics to identify MTD, thereby presenting a valuable upgrade to the popular interval-based designs in practice. <https://doi.org/10.1080/01621459.2025.2484044>.

***Bayesian hierarchical models for predictive stability using non-linear kinetic differential equations***

Presenter: Federico Ferrari

Predictive stability is a crucial component in pharmaceutical and vaccine development, enabling faster and more efficient assessment of product shelf-life and degradation over time. Bayesian hierarchical models offer significant benefits within the predictive stability framework by effectively integrating complex multivariate datasets and providing coherent predictions with credible intervals to quantify uncertainty. These models incorporate multiple levels of information, such as different batches, molecular types, and packaging configurations. We propose a generalized predictive stability modeling approach that accounts for nonlinear degradation kinetics and flexible temperature dependence. Specifically, we develop a hierarchical Šesták-Berggren model capable of capturing a wide range of degradation behaviors over time. To enhance interpretability, we embed a variable selection mechanism that simplifies the model to linear dynamics when appropriate. For temperature effects, we extend the classical Arrhenius formulation by using it as a prior structure while permitting deviations when supported by the data. Our approach can capture more complex degradation pathways, improving predictive power over alternatives for vaccines and biopharmaceuticals exhibiting nonlinear or non-Arrhenius stability behavior.

**Recent BART Extensions (Room 1104)**

***Bayesian additive regression trees for varying-coefficient models with mixed-type outcomes***

Presenter: MingAn Yan

Varying-coefficient models are widely used to study effect heterogeneity, yet most existing methods rely on restrictive linear or low-dimensional smoothness assumptions and struggle to accommodate multivariate effect modifiers without intensive computation. Recent advances such as VCBART overcome some of these challenges through tree-based nonparametric modeling, but current implementations are limited to continuous outcomes. We introduce VCBART-Mix, a flexible Bayesian tree-ensemble framework that extends varying-coefficient modeling to

jointly handle mixed-type outcomes—including continuous, binary, count, and ordinal responses—while allowing coefficient functions to vary over high-dimensional modifier spaces. This unified and computationally efficient approach provides a powerful, generalizable tool for characterizing heterogeneous intervention effects in complex community-based health research. We evaluate the method through simulation studies and illustrate its practical utility using real data from a community-based diabetes intervention.

***Why even Bayesians need to worry about multiple comparisons***

Presenter: George Perrett

Researchers asking causal questions are often interested not only in the average treatment effect but also subgroup specific treatment effects that allow for a more nuanced understanding of who benefits from an intervention. However, this pursuit can lead to issues with multiple comparisons. While previous research has demonstrated that Bayesian methods with regularizing prior distributions are more conservative than their frequentist counterparts and can lead to better outcomes than either ignoring the issue or using corrections (Bonferroni, FDR), the extent to which Bayesian methods eliminate the problem of multiple comparisons has been overstated. Critically, we demonstrate a setting common in social sciences where Bayesian regularizing priors are not sufficient to control false positive claims. Moreover, we show that this is not only limited to false positive claims but extends to sign errors. We characterize this setting as dominated by “shrinkage to the wrong place” and suggest a strategy to improve family-wise error and sign-error rates.

***Learning Conditional Average Treatment Effects in Regression Discontinuity Designs using Bayesian Additive Regression Trees***

Presenter: Rafael Alcântara

This paper develops a performant Bayesian approach to conditional average treatment effect (CATE) estimation in regression discontinuity designs (RDD), an increasingly prevalent form of quasi-experiment that facilitates causal inference. Earlier Bayesian approaches do not easily accommodate CATE estimation while recent frequentist approaches to this problem assume a known basis expansion, a steep model specification requirement that our approach avoids. The new model is a variant of a Bayesian additive regression tree (BART) model with linear leaf-level regressions on the running variable and a treatment dummy (and their interaction). The model adaptively partitions covariate space into regions where the slope on the running variable appreciably differs, providing interpretable Bayesian inference on conditional average treatment effects near the cutoff.

**A Tribute to Herman van Dijk’s “Challenges and Opportunities for Twenty First Century Bayesian Econometricians” (Room Main Hall)**

***Bayesian modelling and analysis for “causal” prediction for decisions***

Presenter: Mike West

I discuss Bayesian analysis for causal prediction, broadly relevant in economic policy and related areas. Problems of assessment following planned or observational interventions in dynamic economic (and other) systems explicitly link decision contexts to counterfactual prediction. Methodological challenges arise in developing counterfactual analyses in sequential, time-series settings. Examples from macroeconomics, economic policy, and commercial or consumer campaign evaluation provide highlights. Recent advances in time-series modeling enable flexible Bayesian analyses in complex sequential intervention studies, and new Bayesian perspectives on causal effect evaluation extend traditional (and not fully Bayesian) practices in counterfactual analysis. I also connect these ideas to themes in Herman’s thought-provoking paper, linking them to Bayesian analysis for policy decisions, economic scenario analysis, and formally recognizing decision goals in settings of model uncertainty.

***Data-Driven Decision Making: Scaling Forecast Combinations for the Big Data Era***

Presenter: Francesco Ravazzolo

While the history of forecast combinations traces back to the foundational linear pooling concepts of Bates and Granger (1969), modern volatile environments demand a shift from fixed averages to dynamic models. We trace this evolution into the “Golden Age of Algorithms,” where we utilize simulation-based Bayesian integration to scale combinations across thousands of predictive densities simultaneously. By leveraging efficient parallel computing, the new methodology enables the real-time updating of self-learning combination weights while explicitly accounting for model set incompleteness. Ultimately, we demonstrate how this historical transition to high-dimensional density combinations transforms massive datasets into robust, risk-adjusted decisions.

***Mapping the Frontier: A Bibliometric Analysis and Text Analysis of Trends in Bayesian Econometrics***



Presenter: Nalan Bastürk

We make a comprehensive bibliometric analysis and text analysis of the scholarly developments in Bayesian econometrics over the past 15 years. We systematically analyze publication trends, thematic shifts, and keyword co-occurrences across leading journals. We apply a new Latent Dirichlet Allocation (LDA) to assess methodological and applied research trajectories in Bayesian econometrics, including key themes. The proposed method, Zoom Enabled Latent Dirichlet Allocation, extends traditional methods by allowing for heterogeneity in the corpora. More specifically, it allows for word-topic distributions to differ between pre-determined groups. In the context of scholarly developments, these pre-determined groups are defined as research fields. In addition, we employ co-citation networks to summarize publication trends and high impact publications. The results highlight the growing relationship between Bayesian methods and computational advances, particularly in simulation-based inference, MCMC improvements, and probabilistic programming. We illustrate thematic relations between a wide range of application areas and relate these to earlier bibliometric research on Bayesian econometrics and Herman K. van Dijk's 2024 views for the scholarly challenges and opportunities for Bayesian econometricians. This research is inspired by Herman K. van Dijk's 2024 paper "Challenges and Opportunities for Twenty First Century Bayesian Econometricians: A Personal View."

## Bayarri Lecture 1: 14:00–15:30

### *Causal conclusions from a cut-off: Bayesian regression discontinuity designs*

Presenter: Stéphanie van der Plas

An opportunity for causal inference presents itself when an intervention is assigned based on a cut-off, as is very common in medical decision-making. Suppose for example that patients aged 65 or younger receive treatment A and patients older than 65 receive treatment B. On average, patients aged 64 will be similar to patients aged 66 in all potentially confounding aspects like BMI or smoking status. So if the outcomes for patients aged 64 are much better than those of patients aged 66, we may reasonably ascribe this difference to the intervention, and claim a causal effect. This is the core concept behind the regression discontinuity design (RDD).

In RDD, the causal is estimated only locally at the cut-off point. Here we focus on the situation where the cut-off is unknown. We introduce a Bayesian approach in which we incorporate prior knowledge about the cut-off location, suitable for the hitherto somewhat neglected fuzzy version of the RDD, where compliance may be imperfect. We compare the new method to the most popular frequentists methods in simulations and on medical data sets.

Joint work with Julia Kowalska and Mark van de Wiel

Discussants: Jennifer Hill and Guido Consonni

## Foundational Lecture 4: 16:00–17:00

### *Model Comparisons at Scale: Using Marginal Likelihoods and Orbits for Discovery*

Presenter: Sid Chib

This talk develops a framework for model discovery that relies on the comparison of a large number of competing models at scale. For example, to find the optimal donor pool in a causal synthetic-control setting, one can consider all possible subsets of those donors and confront those induced subset models with the data, evaluating each model on the basis of model-specific marginal likelihoods.

Problems of this type typically involve model spaces that are both very large and structurally heterogeneous. While marginal likelihoods underpin the evidence calculation, the viability of model comparisons at scale rests on tailored priors that do not require manual tuning yet are automatically calibrated to each model. We construct such priors from a small training portion of the data, reserving the remaining observations for evidence calculation. The other challenge is calculating marginal likelihoods when the model space is exponentially large. We show that the method of Chib (1995) and its M-H variant in Chib and Jeliazkov (2001) scale to large model spaces. The method is based on an identity that decomposes the log marginal likelihood into a fit term and a localization term. Large-sample theory connects the fit term to the Kullback–Leibler closeness of the candidate model to the data-generating process; the localization term induces an automatic form of parsimony. We combine this identity with greedy search to isolate a manageable collection of promising models whose evidence is then calculated exactly.

To summarize evidence in large model spaces, we introduce the orbit, defined as the collection of models whose marginal likelihoods lie within a fixed evidence threshold of the highest-evidence model. The orbit captures the local geometry of the evidence surface and provides a representation of model uncertainty in settings where

many structurally distinct models explain the data nearly equally well. The orbit acts as a credibility region on model space and isolates the stable features shared across high-evidence models. Under standard large-sample conditions, the orbit shrinks toward the models that are KL-closest to the data-generating process.

The ideas are illustrated through five applications: spline regression with knot and variable selection; synthetic-control treatment effects when the donor pool exceeds the pre-treatment sample size; stochastic discount factor discovery in asset pricing; structural breaks in factor models; and moment condition models.

## Poster Session I: 17:15–19:30

### *Bayesian Spatial Gaussian Process Models for Snowfall Prediction in British Columbia*

Presenter: Abhinav Kansal

Snowfall variability in British Columbia poses significant challenges for climate-sensitive sectors such as transportation, agriculture, and infrastructure planning. This project develops a Bayesian spatial modeling framework using Gaussian Processes (GPs) to predict snowfall across the province while quantifying uncertainty in unobserved regions. Using 20 years of data (1998–2017) from 547 weather stations, we preprocess and aggregate climate measurements and apply log transformation and standardization to stabilize variance and improve model performance. Two spatial GP models are implemented in Stan: one using only geographical coordinates and another incorporating temperature as an additional covariate. Leave-one-out cross-validation via PSIS-LOO shows that including temperature improves predictive accuracy modestly and significantly reduces residual spatial variance, indicating that temperature explains broad climatic trends while the GP captures localized effects. Visualizations and spatial prediction maps illustrate how incorporating covariates refines the spatial structure of snowfall. Despite increased computational cost, the Bayesian GP framework provides a flexible and interpretable approach for spatial interpolation of climate variables. This methodology can support climate risk assessment and guide regional planning, with future extensions including more covariates, variational inference for scalability, and model selection strategies for improved predictive performance.

### *Modelling recurrent malari: summing over graphs and identity-by-descent partitions*

Presenter: Aimee Taylor

Bayesian models are increasingly used to classify recurrent malaria infections in clinical trial participants. For malaria caused by the parasite *Plasmodium vivax*, the malaria community relies on a model that builds on the concept of identity-by-descent (genetic identity due to common ancestry). To model identity-by-descent between malaria parasites, we first map *P. vivax* genotypes, which are haploid, onto the nodes of transitive graphs whose edges encode sibling, stranger and clonal relationships. Given a graph, genotypes are then partitioned, locus-by-locus, into cells of identity-by-descent. A probabilistic model relates the partition to the data via population-level allele frequencies. This poster describes a major computational issue that arises when fitting such models. The conditional probabilities of both graphs given recurrent states and of partitions given graphs are uniform distributions over sets of unknown cardinalities. These cardinalities make Metropolis-Hastings type acceptance ratios intractable. Exact summation is feasible when genotype counts are moderate, but exact summation seriously limits the applicability of the model in some relevant settings, e.g. with data on samples recently collected in Ethiopia. We describe advances and on-going questions.

### *Generative multi-fidelity modeling and downscaling via spatial autoregressive transport maps*

Presenter: Alejandro Calle-Salazar

Spatial fields are often available at multiple fidelities or resolutions, where high-fidelity data is typically more costly to obtain than low-fidelity data. Statistical surrogates or emulators can predict high-fidelity fields from cheap low-fidelity output. We propose a highly scalable Bayesian approach that can learn the joint non-Gaussian distribution and nonlinear dependence structure of nonstationary spatial fields at multiple fidelities from a small number of training samples. Our method is based on fidelity-aware autoregressive GPs with suitably chosen regularization-inducing priors. Exploiting conjugacy, the integrated likelihood is available in closed form, enabling efficient hyperparameter optimization via stochastic gradient descent. After training, the method also characterizes in closed form the distribution of higher-fidelity fields given lower-fidelity data. In our numerical comparisons, we show that our approach substantially outperforms existing methods and that it can be used to characterize and simulate high-fidelity fine-scale climate behavior based on output from coarse (low-fidelity) global circulation models.

### *Borrowing from historical control data in a Bayesian time-to-event model with flexible baseline hazard function*

Presenter: Alex Lewin

Leveraging control-group data from historical trials is increasingly important for accelerating drug development. In time-to-event settings, existing Bayesian approaches often rely on restrictive assumptions about the baseline hazard to simplify computation, which can limit borrowing performance. We propose a semi-parametric survival model using smoothing priors to estimate a fully flexible smooth baseline hazard function. Dynamic borrowing is achieved through a lump-and-smear prior on baseline hazard parameters, allowing the model to learn adaptively from historical data. Our work includes a principled strategy for setting values of hyperparameters to control the level of dynamic borrowing, based on the tolerated difference between current and historical baseline hazards. We demonstrate the benefit of the dynamic borrowing prior through a simulation study. We show improved power when historical data are exchangeable with current data, whilst reducing type I error in the presence of prior-data conflict. We further illustrate the method using data splitting on a trial for a treatment used in asthma.

***Bayesian modelling and computation utilising directed cycles in multiple network data***

Presenter: Anastasia Mantziou

Modelling multiple network data is crucial for addressing a wide range of applied research questions. However, there are many challenges, both theoretical and computational, to address. Network cycles are often of particular interest in many applications; for example in ecology a largely unexplored area has been how to incorporate network cycles within the inferential framework in an explicit way. The recently developed Spherical Network Family of models (SNF) offers a flexible formulation for modelling multiple network data that permits any type of metric. This has opened up the possibility to formulate network models that focus on network properties hitherto not possible or practical to consider. In this article we propose a novel network distance metric that measures similarities between networks with respect to their cycles, and incorporates this within the SNF model to allow inferences that explicitly capture information on cycles. These network motifs are of particular interest in ecological studies aimed at understanding competitive and hierarchical interactions. We further propose a novel computational framework to allow posterior inferences from the intractable SNF model for moderate-sized networks. Lastly, we apply the resulting methodology to a set of ecological network data studying aggressive interactions between species of fish. We show our model is able to make cogent inferences concerning the cycle behaviour amongst the species, and beyond those possible from a model that does not consider this network motif.

***Bayesian Distributional Clustering Beyond Identity: A Framework for Clustering Similar Distributions***

Presenter: Andrea Teruzzi

In the analysis of grouped data, the Bayesian setting provides flexible frameworks for clustering both at the observation level and at the group level. The latter problem, i.e., distributional clustering, is appealing in applications ranging from clinical trials to educational assessment, where researchers are interested in detecting latent heterogeneity and borrowing information across related subpopulations. Celebrated models such as the Nested Dirichlet Process and the Common Atom Model suffer from a fundamental misspecification when applied to real data because their distributional clustering is driven by exact identity between distributions. In practice, group-level distributions are rarely identical but are often “similar.” Even minor departures from perfect identity can prevent these models from pooling information across groups, leading to poor cluster recovery. Our first contribution is to theoretically formalize the inadequacy of identity-based clustering assumptions in realistic scenarios. We then propose a new class of models designed to cluster non-identical distributions by allowing for approximate rather than exact correspondence within distributional clusters. Our framework employs a random partition model for the distributional clusters and a joint model for the random probability measures within each cluster that explicitly enforces similarity rather than identity. We specialize our framework by modelling the within-cluster distributions either using the hierarchical Dirichlet process, normalized compound random measures, or hierarchical shot-noise Cox processes. We illustrate the performance of the proposed models through a simulation study and an application to large-scale educational data.

Based on a joint work with Mario Beraha, Federico Camerlenghi, Alessandro Carminati, and Alessandra Guglielmi.

***To select or not to select - predictively consistent priors instead of model selection***

Presenter: Anna Elisabeth Riha

Bayesian modelling workflows often require the consideration of different candidate models, and approaches for model selection in the Bayesian framework aim to support the modeller in navigating potential trade-offs between model complexity and generalisability of the results to yet unobserved data. In this work, we propose a change of perspective towards choosing predictively consistent priors, instead of relying on model selection after the fact.



We revisit the issue of overfitting, and clarify why model selection is not necessarily needed and can even be harmful in some modelling scenarios with finite data. When integrating over the posterior and using predictively consistent priors, even if those priors can be considered weakly informative, we can often safely use flexible models with a large number of parameters. We illustrate the relevance of appropriate prior choices, as well as the limitations and alternatives for model selection in different modelling tasks in extensive numerical experiments. We provide detailed examples for adjusting  $R^2$ -based priors for logistic regression models and models that include a known relevant treatment covariate in the analysis of randomised controlled trials. Additionally, we illustrate that different choices for nonlinear models come with different implications for the predictive consistency of the priors.

#### ***Improved methods to account for complex sampling designs in Bayesian analyses***

Presenter: Anne Cohen

Bayesian statistical methods have become increasingly popular in survey research for their intuitive framework together with their ability to accommodate hierarchical data structures and missing data. However, accounting for complex sample design elements such as weights, stratification, and clustering is not straightforward. We propose a novel extension of the finite population Bayesian bootstrap (FPBB) which is used to generate multiple synthetic populations and to extract FPBB weights. Initial posterior draws are obtained from a systematic sample of a single synthetic population, expanding the original sample proportionally to the sampling weights. The draws are re-weighted using importance sampling theory to approximate the distribution of all FPBB-generated synthetic populations. We evaluate our results in a simulation setting with informative sampling and model misspecification, showing that our method produces results comparable to that of Savitsky and Williams (2021) but is far more efficient to fit and can be easily transferred to any Bayesian model with closed-form posteriors. We also consider a logistic mixed effects modeling application to NHIS data, estimating the average effect of not having a usual place for healthcare on smoking prevalence after accounting for race/ethnicity, rurality, and education. The data application confirms consistent and comparable estimates. This simple and generalizable approach has significant implications by allowing for the implementation of complex Bayesian models while accounting for multistage sampling designs.

#### ***Objectively constrained priors for mixed and hierarchical models***

Presenter: Antoine Van Biesbroeck

Constructing suitable and objective priors for complex statistical models remains a central challenge in Bayesian inference. In this work, we focus on two widely used families of models: mixture models and linear hierarchical models. For those, classical reference priors result in improper posteriors that cannot support meaningful inference. To address this limitation, we propose a principled methodology that extends the reference prior theory by incorporating carefully designed constraints into the classical definition of reference priors. These constraints are sought to ensure their tractability (i.e., they yield proper posteriors), while preserving as much possible their objective characteristics. Using the aforementioned models as representative examples, we demonstrate how our framework can be implemented in practice and integrated in standard Bayesian inference pipelines. We then evaluate the resulting estimators in terms of both efficiency and objectivity, showing that they are robust to the choice of constraint and perform competitively when compared with existing approaches. Our theoretical developments are available as a preprint at: [arXiv.2409.13041](https://arxiv.org/abs/2409.13041).

#### ***A Bayesian Dynamic Latent Space Model for Weighted Networks***

Presenter: Antonio Peruzzi

A new dynamic latent space eigenmodel (LSM) is proposed for weighted temporal networks. It accommodates integer-valued weights, time-varying sparsity and time-varying node latent features. Lagged and contemporaneous dependence across nodes and features are introduced, a characteristic neglected in the LSM literature. A Bayesian approach is used to address two inference challenges: latent feature estimation and choice of latent space dimension. We provide a point-process representation of the network weights, derive the joint posterior distribution of the features, and propose a new efficient sampler for posterior approximation in high dimensional networks. A Laplace approximation of the partial marginal likelihood is used to sample the latent dimension. Two illustrations, that are relevant in international economics and neuroeconomics, show the ability of the proposed framework to uncover dynamic latent structures in trade and neural connectivity. Overall, our procedure is general, as it can be easily adapted to static and dynamic settings, as well as to other discrete or continuous weight distributions.

#### ***Estimating Antarctic subglacial conditions using neural posterior inference***

Presenter: Bao Anh Vu

Ice sheet models are routinely used to quantify and project an ice sheet's contribution to sea level rise. In order for an ice sheet model to generate realistic projections, its parameters must first be calibrated using observational data; this is challenging due to the nonlinearity of the model equations, the high dimensionality of the underlying parameters, and limited data availability for validation. This study leverages the emerging field of neural posterior approximation for efficiently calibrating ice sheet model parameters and boundary conditions. We make use of a one-dimensional (flowline) Shallow-Shelf Approximation model in a state-space framework. A neural network is trained to infer the underlying parameters, namely the bedrock elevation and basal friction coefficient along the flowline, based on observations of ice velocity and ice surface elevation. Samples from the approximate posterior distribution of the parameters are then used within an ensemble Kalman filter to infer latent model states, namely the ice thickness along the flowline. We show through a simulation study that our approach yields more accurate estimates

***Frequency selection in Bayesian spectral modeling of time series data with applications to wearable device measurements***

Presenter: Beniamino Hadj-Amar

This work introduces a Bayesian spike-and-slab framework for spectral analysis of time series data. The proposed method combines frequency selection and dimensionality reduction with a refined grid of candidate frequencies, enabling high-resolution recovery of oscillatory components while promoting sparsity through a structured spike-and-slab prior. A stochastic search algorithm efficiently explores the posterior space, yielding posterior inclusion probabilities that quantify the relevance of each frequency. We extend the framework to multivariate signals via a hierarchical prior on frequency inclusion patterns, allowing the model to capture both shared and component-specific rhythms across multiple time series. Extensive simulation studies demonstrate the method's robustness and superior performance in frequency estimation and spectral power reconstruction compared to existing approaches. Applied to actigraphy data from individuals with partial-onset seizures, the univariate model identifies clinically relevant circadian and ultradian rhythms. In a second application, for the joint analysis of physical activity and skin temperature from a healthy individual, the multivariate model reveals partially overlapping rhythmic components consistent with known physiological coupling. This work establishes a powerful and interpretable approach to spectral analysis, with broad applicability to wearable data, chronobiology, and personalized health monitoring.

***Scalable and robust regression models for continuous proportional data***

Presenter: Benjamin Dahl

Beta regression is used routinely for continuous proportional data, but it often encounters practical issues such as a lack of robustness to misspecification of the beta distribution and sensitivity to outliers. We develop an improved class of generalized linear models starting with the continuous binomial (cobin) distribution and further extending to dispersion mixtures of cobin distributions (micobin). The proposed cobin regression and micobin regression models have attractive robustness, computation, and flexibility properties. A key innovation is the Kolmogorov-Gamma data augmentation scheme, which facilitates Gibbs sampling for Bayesian computation, including in hierarchical cases involving nested, longitudinal, or spatial data. We demonstrate robustness, ability to handle responses exactly at the boundary (0 or 1), and computational efficiency relative to beta regression in simulation experiments and through analysis of the benthic macroinvertebrate multimetric index of US lakes using lake watershed covariates.

***Bayesian Empirical Bayes: Simultaneous Inference from Probabilistic Symmetries***

Presenter: Bohan Wu

Empirical Bayes (EB) improves the accuracy of simultaneous inference "by learning from the experience of others" (Efron, 2012). Classical EB theory focuses on latent variables that are iid draws from a fitted prior (Efron, 2019). Modern applications, however, feature complex structure, like arrays, spatial processes, or covariates. How can we apply EB ideas to these settings? We propose a generalized approach to empirical Bayes based on the notion of probabilistic symmetry. Our method pairs a simultaneous inference problem-with an unknown prior-to a symmetry assumption on the joint distribution of the latent variables. Each symmetry implies an ergodic decomposition, which we use to derive a corresponding empirical Bayes method. We call this method Bayesian empirical Bayes (BEB). We show how BEB recovers the classical methods of empirical Bayes, which implicitly assume exchangeability. We then use it to extend EB to other probabilistic symmetries: (i) EB matrix recovery for arrays and graphs; (ii) covariate-assisted EB for conditional data; (iii) EB spatial regression under shift invariance. We develop scalable algorithms based on variational inference and neural networks. In simulations, BEB outperforms existing approaches to denoising arrays and spatial data. On real data, we demonstrate BEB

by denoising a cancer gene-expression matrix and analyzing spatial air-quality data from New York City.

***Bayesian Deep Gaussian Processes with Neural Encoders for High-Dimensional Climate Applications***

Presenter: Boyun Song

Gaussian Processes (GPs) provide a principled Bayesian framework for uncertainty quantification but become computationally and statistically challenging when applied to high-dimensional input spaces such as climate fields or long time-series. Deep Gaussian Processes (DGPs) offer additional flexibility, yet their performance still depends critically on learning effective low-dimensional representations of complex inputs. In this work, we propose a hybrid Neural Network–Deep Gaussian Process (NN–DGP) architecture in which a neural encoder learns structured representations of high-dimensional climate covariates, and a DGP operates on the resulting latent features. This approach preserves the Bayesian interpretability and calibrated uncertainty of GPs while leveraging the expressive power of neural networks to extract relevant spatio-temporal patterns. We evaluate the method on synthetic benchmarks and real climate datasets, demonstrating substantial improvements in predictive accuracy, robustness to noise, and scalability compared to conventional GPs and DGPs without learned encoders. Our results highlight the potential of encoder-augmented Gaussian process models as flexible Bayesian emulators for environmental and geophysical applications involving high-dimensional data.

***Simulation study to evaluate the effect of aggregation in the APC model with a Bayesian approach***

Presenter: Carlos M. Lopera-Gómez (Universidad Nacional de Colombia, Medellín)

Age-Period-Cohort (APC) Poisson regression models are used in epidemiological studies to estimate the effect of each of these factors on the trends in incidence and mortality rates of different diseases. Generally, information on observed cases or mortality for a disease is summarized in a table with two entries: the age group and the calendar period in which the event of interest occurred. One of the challenges researchers face in various contexts and scenarios requiring APC model applications is data aggregation. In most studies, the available information is aggregated into five-year periods, either by age groups, by periods, or by both. Therefore, it would be pertinent to evaluate the effect of this data aggregation on both the estimation of the model's effects and the predictions obtained from it, since this process can lead to the loss of trends or seasonal patterns. Thus, this work presents a simulation study to analyze the impact of data aggregation on future projections or predictions, using a Bayesian modeling approach. Specifically, the study aims to establish the quality of projections for three time periods: five-year periods, three-year periods, and annual periods.

Co-author: Isabel Ramírez (Universidad Nacional de Colombia, Medellín)

***A Skew-Normal Multinomial Probit Model***

Presenter: Cash Hao Looi

The multinomial probit (MNP) model is widely used to model multinomial outcomes, as it allows for correlated latent utilities between different alternatives. However, the standard MNP specification assumes the latent utilities to be symmetric, with error terms following a multivariate normal (MVN) distribution. This assumption could potentially lead to model misspecification when the true error distribution is skewed. To address this limitation, we develop a skewed MNP model that relaxes the assumption of symmetry by specifying a multivariate skew-normal (MSN) distribution for the error terms. Notably, the proposed skewed MNP model includes the standard MNP model as a special case, offering researchers a more flexible and general modeling approach. A Bayesian data-augmented approach is proposed to enable MCMC sampling and posterior inference. For model identification, we impose a prior that fixes the first diagonal element of the MSN scale matrix to unity. To evaluate the performance of the proposed model, we conduct two simulation studies, in which the data are generated via symmetric and skewed error distributions, respectively. We find that the skewed MNP model accurately recovers the true choice probabilities in both scenarios, whereas the standard MNP model fails to capture choice behaviors when the error terms are skewed.

***Spectral gap of Metropolis-within-Gibbs under log-concavity***

Presenter: Cecilia Secchi

The Metropolis-within-Gibbs (MwG) algorithm is a widely used Markov Chain Monte Carlo method for sampling from high-dimensional distributions when exact conditional sampling is intractable. We study MwG with Random Walk Metropolis (RWM) updates, using proposal variances tuned to match the target's conditional variances. Assuming the target  $\pi$  is a  $d$ -dimensional log-concave distribution with condition number  $\kappa$ , we establish a spectral gap lower bound of order  $O(1/\kappa d)$  for the random-scan version of MwG, improving on the previously available  $O(1/\kappa^2 d)$  bound. This is obtained by developing sharp estimates of the conductance of one-dimensional RWM kernels, which can be of independent interest. The result shows that MwG can mix substantially faster with variance-adaptive proposals and that its mixing performance is just a constant factor worse than that of the

exact Gibbs sampler, thus providing theoretical support to previously observed empirical behavior.

### ***Robust Bayesian decision tree regression***

Presenter: Chaemyeong Kwon

Traditional regression tree algorithms determine node splits based on the reduction in mean squared error. However, the L2-based splitting rule has structural limitations in the presence of outliers, and even a small number of outliers can distort the splitting direction. To reduce the influence of outliers, several attempts have been made using L1-based approaches and related techniques. In this paper, we propose a new robustification strategy that addresses these problems by incorporating a mean-shift concept to adjust leaf-node predictions within a Bayesian framework. The key idea is to prevent the prediction mean from being excessively influenced by extreme observations. To achieve this, we assign the Bayesian lasso prior to the mean-shift parameter, which constrains unnecessary deviations of this parameter from zero. The Bayesian formulation allows the splitting rule to be less sensitive to outliers through Bayes factors that automatically impose penalties for model complexity. The performance of the proposed Bayesian decision tree method is evaluated through simulation studies and real data analysis.

### ***Creating a Visual House Quality Index with Bayesian Neural Networks***

Presenter: Cheuk Yin Ho

The hedonic model of house prices aims to estimate the implicit prices of house qualities that rely heavily on observable attributes, such as square footage and the number of bedrooms. However, the model often misses the unmeasured aesthetic factors that significantly influence property prices. This paper develops a visual house quality index using the Bayesian Neural Network (BNN) with image data. The goal is to build a model that takes house images as inputs and outputs a quality index along with a measure of uncertainty. The index is derived solely from the visual features of houses. The BNN mitigates overfitting and provides a predictive distribution over possible house quality scores. A pre-trained Convolutional Neural Network (CNN), such as ResNet50, is used as a feature extractor and then augmented with Bayesian dense layers on top for the regression. The BNN places priors on weights and uses stochastic variational inference for posteriors. The model is trained using ELBO loss. The results show that the visual house quality index accounts for approximately 10

### ***Optimal Test Planning for Degradation Analysis***

Presenter: Chien-Yu PENG

Before conducting a degradation test, it is essential to determine fundamental issues about decision variables. These include determining the required sample sizes, the termination time of sample testing, and the number of measurements, particularly in initial experiments with limited budgets. Consequently, devising cost-constrained optimal test plans becomes a practical concern in degradation tests for highly reliable products. In this talk, I will provide an overview of stochastic processes and their optimal test plans. Some challenges associated with (accelerated) testing plans will be discussed. The development of these optimal test plans, along with their theoretical properties, serves to bring the engineering and statistical communities closer, enabling them to address real-world problems with a more solid foundation.

### ***From national ecological monitoring to local forecasts and adaptive management***

Presenter: Christian Damgaard

Predictions of plant community dynamics under different management scenarios are made for different heathland habitat types using 20 years of national monitoring data from 40-100 sites. The empirical spatial and temporal modelling assumes a structural equation model of the driving factors in a temperate heathland ecosystem, and the model is fitted using hierarchical Bayesian methods to separate sampling and process errors. The observed change in plant species cover is explained well by the temporal model (0.95

### ***Grid-based Bayesian inference and model selection for complex dynamic systems***

Presenter: Christoph Mark

Time series generated by complex systems like financial markets and the earth's atmosphere often represent superstatistical random walks: on short time scales, the data follow a simple low-level model, but the model parameters are not constant and can fluctuate on longer time scales according to a high-level model. While the low-level model is often dictated by the type of the data, the high-level model, which describes how the parameters change, is unknown in most cases. Here we present a computationally efficient, grid-based Bayesian inference method that reconstructs the time course of the parameter variations from time-series. Importantly, as this method is not based on sampling from the posterior distribution, it can accurately evaluate the model evidence to objectively select between competing high-level models, for example deciding whether parameters

vary gradually over time or exhibit abrupt, regime-switching behavior. We apply this method to detect anomalous price movements in financial markets, characterize cancer cell invasiveness, identify historical policies relevant for working safety in coal mines, and compare different climate change scenarios to forecast global warming.

#### ***Loss-based prior for BART models***

Presenter: Cristiano Villa

We present a novel prior for tree topology within Bayesian additive regression trees (BART) models. This approach quantifies the hypothetical loss in information and the loss due to the complexity associated with choosing the "wrong" tree structure. The methodology is designed to promote sparsity, a crucial feature considering BART models' tendency to overfit. The method incorporates prior knowledge into the distribution via two parameters that govern the tree's depth and balance between its left and right branches. We also propose a default calibration for these parameters, offering a solution that carries an objective flavour into the prior. The method's efficacy is demonstrated on both simulated and real datasets.

#### ***Disentangling the Drivers of Survival Improvement: A Bayesian Decomposition of Systematic Drift versus Covariate Evolution in Cancer Registry Data***

Presenter: Dahhay Lee

Population-based cancer registries often exhibit non-stationary survival trends, yet disentangling general health-care improvement (systematic drift) from the evolving impact of specific covariates is an ill-posed problem in standard frequentist frameworks due to the collinearity between follow-up time and calendar time. Additionally, registry data often lack detailed information on specific treatment regimens, making the disentanglement more statistically challenging. Standard proportional hazard models fail to capture these divergent trends by averaging period effects. In this study, we propose a Bayesian Structured Additive Regression (STAR) survival model with a centered dynamic coefficient parameterization to address this problem. We decompose the hazard function into a static baseline disease trajectory in follow-up time, a flexible global population drift, and covariate-specific deviations over calendar time. Crucially, we enforce a sum-to-zero constraint on these deviations and apply dynamic global-local shrinkage priors (Horseshoe) to their spline coefficients. In contrast to standard Gaussian smoothing, this hierarchical regularization effectively tests the null hypothesis of temporal stability by shrinking subgroup trends towards the global baseline, allowing the model to adapt locally only when the data provides overwhelming evidence of structural breaks. The proposed framework turns limitations of treatment-blind registry data into a feature, providing a robust probabilistic mechanism to isolate latent shocks. We outline its implementation for large-scale data and discuss simulation and case-study evaluations to assess its ability to attribute survival gains to either systemic evolution or specific treatment advances. This study was supported by the National Research Foundation of Korea (grant number: RS-2023-00275999), funded by the Korean Ministry of Science and ICT.

#### ***General Bayesian Inference on Heterogeneous Treatment Effects in Regression Discontinuity Designs with BART***

Presenter: Daisuke Kondo

Regression discontinuity (RD) designs are widely used for causal inference. In many empirical applications, treatment effects vary substantially with covariates, and ignoring such heterogeneity can lead to misleading policy conclusions. Therefore, flexible modeling of heterogeneous treatment effects in RD settings is essential. We propose a Bayesian nonparametric approach to estimating heterogeneous treatment effects based on Bayesian Additive Regression Trees (BART). The key feature of our method lies in adopting a general Bayesian framework using a pseudo-model defined through a loss function for fitting local linear models around the threshold, which gives direct modeling of heterogeneous treatment effects by BART. Optimal selection of the bandwidth parameter for the local model is implemented using Hyvärinen scores. We show that this approach flexibly captures complicated structures of heterogeneous treatment effects as a function of covariates.

#### ***On Forward-KL Gradient Estimators for Variational Inference: SNR Analysis for High-Dimensional Models and Annealed Optimization***

Presenter: Daniel Andrade

Bayesian inference with variational inference using the reverse Kullback-Leibler (KL)-divergence is very popular due to its ease of use and good scalability to high dimensions. However, even with complex variational distributions like normalizing flows, the reverse KL sometimes suffers from mode collapse, and as a consequence fails to provide accurate approximations of multi-modal posteriors. As a remedy, several recent works propose to train normalizing flows with the forward KL. In this work, we analyze two popular types of gradient estimators for the forward KL, and in particular investigate the signal-to-noise ratio (SNR) in the regime of high dimensions. Based on our analysis, we propose a new training method for variational inference with the forward KL that



uses an appropriate annealing scheme for effective variance reduction. Experiments on several synthetic target distributions and a Bayesian latent variable model confirm the usefulness of our proposed method.

### ***Analytic Visualization of Nonresponse Bias via Bayesian Imputation in Survey Data***

Presenter: Darcy Morris

Item nonresponse in sample survey data leads to bias in estimation when nonresponse is nonignorable. Imputation is often used to alleviate nonresponse bias by accounting for missing data patterns through an assumed model specification. Bayesian latent class imputation models allow deep complexity in model specification for the missing data mechanism through modeling items jointly and thus naturally incorporating interactions. Empirical assessments of Bayesian latent class model variants, e.g. sensitivity checks and model evaluation measures, provide insight into the missing data mechanism leading to better estimation. This work presents examples of analytically visualizing sensitivity analysis to aid in synthesizing and comparing the performance of Bayesian latent class imputation models to understand effects of nonresponse bias in the U.S. Census Bureau's Household Pulse Survey.

### ***Efficient adaptive importance sampling in complex geometric and high dimensional settings***

Presenter: Darren Wraith

Population Monte Carlo (PMC) adapts importance sampling proposals through iterative refinement of Gaussian or Student- $t$  mixture distributions (Cappé et al 2004). When the target likelihood is computationally expensive—as in cosmological parameter estimation, systems biology, or climate modeling—the number of likelihood evaluations can be computationally expensive. Standard PMC may generate redundant samples when mixture components overlap, repeatedly evaluating the likelihood in the same posterior regions without improving the estimate.

We present two complementary strategies to reduce likelihood evaluations while maintaining statistical accuracy. First, we introduce a *point-level thinning mechanism* that assigns retention probabilities to drawn samples based on responsibility calculations before likelihood evaluation. Points falling in densely covered regions are stochastically discarded, with importance weights corrected by the inverse retention probability to preserve unbiasedness. Second, for moderate to high-dimensional problems ( $p \sim 30\text{--}100$ ), we propose replacing random sampling with *principal points*—deterministic locations that minimize expected squared distance to the proposal distribution. We develop a practical Monte Carlo algorithm for computing principal points of mixture distributions via iterative Voronoi tessellation, naturally handling component overlap by placing points optimally across the entire mixture rather than per-component. This reduces the number of required samples from  $N$  to  $K \ll N$  while maintaining good posterior coverage.

Both methods preserve the key advantages of PMC—independence of samples, straightforward parallelization, and adaptive refinement—while dramatically reducing computational cost. Preliminary experiments on multimodal test densities and cosmological posteriors indicate reductions of 30–70% in likelihood evaluations for comparable effective sample sizes and perplexity. The principal points approach shows particular promise in higher dimensions, where the deterministic placement avoids the curse of dimensionality more effectively than naive random sampling. We discuss theoretical properties including unbiasedness under thinning, convergence of the mixture adaptation, computational complexity, and practical implementation strategies including variance reduction techniques and dimensionality reduction via PCA. The methods are applicable to any setting where likelihood or posterior evaluations dominate computational cost.

### ***Bayesian predictive semiparametric regression***

Presenter: Davide Agnoletto

The generalized Bayesian predictive framework is a generalization of traditional Bayes where one specifies a sequence of one-step ahead predictive distributions that guarantees asymptotic exchangeability instead of the of the likelihood and the prior. Inference is then carried out through the so-called martingale posterior distribution. However, current nonparametric approaches focus primarily on density estimation. Although recent contributions have investigated parametric martingale posteriors by combining plug-in predictive densities with stochastic gradient descent, they focus mainly on theoretical properties, leaving regression problems unexplored. We introduce a novel method for constructing predictive inference in regression settings that accommodates both parametric and nonparametric components, where the estimate of the quantity of interest is updated using residuals. The approach has important connections with well-established algorithms such as recursive least squares, Kalman filtering and backfitting. We leverage these connections for efficient posterior uncertainty quantification through predictive resampling. Moreover, an insightful link with g-priors is also available for the linear regression case. We demonstrate the effectiveness of the proposed method in simulations and real data examples.

### ***Variational Approximations for Robust Bayesian Inference via Rho-Posteriors***

Presenter: EL Mahdi Khribch

We consider the elegant rho-estimators introduced by Baraud and Birgé (2020), which provide robustness against both model misspecification and outliers. While theoretically appealing, these estimators have remained difficult to implement in practice. We derive PAC-Bayes bounds for rho-estimation under the Hellinger metric and develop variational approximations that make the methodology implementable. Our results not only recover the rates established by Birgé but also provide practical variational approximations that enable real-world application. By bridging the gap between theory and computational feasibility, we hope this work enriches the results of Baraud and Birgé and bridges theory with practice, making these robust methods available to practitioners.

### ***Prediction Powered Posteriors***

Presenter: Edric Tam

In many scientific applications, labeled data is often expensive, whereas unlabeled data is abundant. This imbalance has motivated the development of prediction-powered inference, a framework that leverages accurate machine learning predictions on unlabeled data to support valid frequentist inference in settings with limited labeled data. We propose Prediction Powered Posteriors (PPP), a Bayesian perspective on the prediction powered inference framework that incorporates information from machine learning-predicted labels into estimation of the posterior distribution. Drawing on notions developed in the robust Bayesian statistics literature, our framework combines the classical likelihood, which is applied on the gold standard labeled data, with a coarsened likelihood on the machine learning-predicted labels on the unlabeled data. This framework leads naturally to a tempered posterior formulation. Our framework allows for a theoretical analysis of the convergence properties of PPP as the number of unlabeled samples approaches infinity. We demonstrate our methodology across a range of examples, including linear and logistic regression models with applications in various scientific datasets. Our framework offers a flexible and theoretically grounded tool for Bayesian inference in the low-labeled, high-unlabeled regime.

### ***Bayesian Function on Latent Scalar Regression for Correlated Environmental Mixtures***

Presenter: Eduardo Vasquez

There is an increasing interest in understanding how high-dimensional and strongly correlated chemical exposures relate to time-varying outcomes. Motivated by a recent dataset on early-childhood exposure to phthalates and functional connectivity scores obtained from fMRIs, we propose a Bayesian function-on-(latent) scalar regression model, in which the individual-level exposure is modeled by a latent factor structure that drives age-varying effects on the outcome trajectory via spline coefficients. A key feature is an identified set of induced effects that maps back to the original phthalates, enabling predictor-level interpretation. The model can include covariates and interaction effects, and can handle exposures subject to limits of detection. We fit the model in Stan with shrinkage priors on factor loadings, and assess fit via PSIS-LOO. We find negative associations between early-age somatomotor functional network and specific compounds of phthalates. We also describe an extension of these ideas to a function-on-(latent) function regression setting to handle asynchronous exposure and response observations.

### ***Bayesian Multiresolution Functional Regression and Clustering via Self-Truncating Cumulative Shrinkage Processes***

Presenter: Francesco Denti

We propose a Bayesian multiresolution framework for functional data analysis that captures varying levels of complexity at a local scale. As is commonly done, the model represents each function as a sum of increasingly flexible layers, with each layer defined by a basis expansion (e.g., B-splines) whose resolution increases with depth. Spike-and-slab Lasso priors enable data-driven activation of basis coefficients, allowing each layer to selectively operate on local regions of the domain and producing a principled separation between smooth and highly variable segments of the function. To enforce ordered shrinkage across layers, we introduce the Self-Truncating Cumulative Shrinkage Process (ST-CUSP), a prior that regulates the effective number of active layers. ST-CUSP extends the original Cumulative Shrinkage Process by incorporating a probabilistic self-truncation mechanism that penalizes redundant layers via spike-and-slab distributions applied to the stick-breaking variables. This promotes parsimony and interpretability while retaining the adaptive shrinkage behavior characteristic of CUSP. We further extend the prior to a partially exchangeable framework, enabling nested clustering of functions according to their complexity, interpreted as the number of activated layers. Despite its flexibility, the model preserves a tractable structure that enables the straightforward derivation of full conditionals, efficient Gibbs sampling, and structured variational inference. Simulation studies across a range of complexities and noise levels show that ST-CUSP achieves competitive estimation accuracy while activating substantially fewer layers than the original

CUSP model. We further demonstrate the practical utility and interpretability of the methodology through several applications spanning different domains. For instance, we analyze the local temporal complexity of tidal levels in the Venetian Lagoon, illustrating how ST-CUSP can effectively characterize structure in environmental time-series data.

#### ***Exchangeable random permutations with an application to Bayesian graph matching***

Presenter: Francesco Gaffi

We introduce a general Bayesian framework for graph matching grounded in a new theory of exchangeable random permutations. Linking the cycle representation of permutations to the literature on exchangeable random partitions, we define, characterize, and study the structural and predictive properties of these distributions. A novel sequential metaphor—the position-aware generalized Chinese restaurant process—provides a constructive foundation for this theory and supports practical algorithmic design. Exchangeable random permutations offer flexible priors for a wide range of inferential problems where the parameter of interest is a permutation, including statistical graph matching and unmatched regression. As an application, we develop a Bayesian model for graph matching that integrates a correlated stochastic block model with an edge-discrepancy likelihood. The cycle structure of the matching permutation is linked to latent node partitions that explain connectivity patterns—an assumption consistent with the homogeneity requirement underlying the graph matching task itself. This structural alignment not only grounds the model statistically but also enhances the mixing behavior of the sampling algorithm. Posterior inference is performed through a node-wise blocked Gibbs sampler directly inspired by the proposed sequential construction, allowing coherent updates in the complex permutation space. To summarize posterior uncertainty, we introduce perSALSO, an adaptation of the SALSO algorithm to the permutation domain that provides principled point estimation and interpretable posterior summaries. Together, these contributions establish a unified probabilistic framework for modeling, inference, and uncertainty quantification over permutations.

#### ***A new wavelet-based variational family with copula dependence structures***

Presenter: Giovanni Pastori Piccirilli

Variational Inference (VI) has become a key tool for scalable Bayesian computation, particularly as modern statistical models grow in complexity and datasets continue to expand. However, standard mean-field variational families often fail to capture posterior dependence structures and tend to underestimate uncertainty. To address these limitations, we propose a new variational family that combines wavelet-based density representations with copula dependence structures. The method models each marginal posterior using a wavelet density estimator, leveraging the smoothness and flexibility of Daubechies wavelets, while incorporating dependence through either independence or Gaussian copulas. This construction yields a highly flexible family that remains computationally tractable and fully compatible with automatic differentiation frameworks. Through simulation studies—including logistic regression, hierarchical regression, and sparse linear models with automatic relevance determination—we demonstrate that the proposed approach achieves posterior means comparable to Hamiltonian Monte Carlo while substantially improving variance estimation relative to mean-field VI. A real-data application to hierarchical logistic regression further illustrates the method’s capacity to recover dependence structures and deliver accurate predictive performance.

#### ***Borrowing Information between unaligned time series via Bayesian nonparametrics***

Presenter: Giovanni Poli

We define a Bayesian semi-parametric model to effectively conduct inference with unaligned longitudinal binary data. The proposed strategy is motivated by data from the Human Epilepsy Project (HEP), which collects seizure occurrence data for epilepsy patients, together with relevant covariates. The model is designed to flexibly accommodate the particular challenges that arise with such data. First, epilepsy data require models that can allow for extensive heterogeneity, across both patients and time. With this regard, state space models offer a flexible, yet still analytically amenable class of models. Nevertheless, seizure time-series might share similar behavioral patterns, such as local prolonged periods of elevated seizure presence, which we refer to as *clumping*. Such similarities can be used to share strength across patients and define subgroups. However, due to the lack of alignment, straightforward hierarchical modeling of latent state space parameters is not practicable. To overcome this constraint, we construct a strategy that preserves the flexibility of individual trajectories while also exploiting similarities across individuals to borrow information through a nonparametric prior. On the one hand, heterogeneity is ensured by (*almost*) subject-specific state-space submodels. On the other, borrowing of information is obtained by introducing a Pitman-Yor prior on group-specific probabilities for patterns of clinical interest. We design a posterior sampling strategy that leverages recent developments of binary state space models



using the Unified Skewed Normal family (SUN). The model, which allows the sharing of information across individuals with similar disease traits over time, can more generally be adapted to any setting characterized by unaligned time series.

### ***Separate Exchangeability as Modeling Principle in Bayesian Nonparametrics***

Presenter: Giovanni Rebaudo

We argue for the use of separate exchangeability as a modeling principle in Bayesian nonparametric (BNP) inference. Separate exchangeability is de facto widely applied in the Bayesian parametric case, e.g., it naturally arises in simple mixed models. However, while in some areas, such as random graphs, separate and (closely related) joint exchangeable models are widely used, they are curiously underused for several other applications in BNP. We briefly review the definition of separate exchangeability, focusing on the implications of such a definition in Bayesian modeling. We then discuss two tractable classes of models that implement separate exchangeability, which are the natural counterparts of familiar partially exchangeable BNP models. The first is nested random partitions for a data matrix, defining a partition of columns and nested partitions of rows, nested within column clusters. Many recent models for nested partitions implement partially exchangeable models related to variations of the well-known nested Dirichlet process. We argue that inference under such models in some cases ignores important features of the experimental setup. We obtain the separately exchangeable counterpart of such partially exchangeable partition structures. The second class is about setting up separately exchangeable priors for a nonparametric regression model when multiple sets of experimental units are involved. We highlight how a Dirichlet process mixture of linear models, known as ANOVA DDP, can naturally implement separate exchangeability in such regression problems. Finally, we illustrate how to perform inference under such models in two real data examples.

This is a joint work with Qiaohui Lin and Peter Müller.

### ***General Path Dynamic Model for Degradation Data with Covariates***

Presenter: Guilherme Augusto Veloso

The increasing demand for highly reliable products has been posing big challenges on reliability assessment. A major challenge for life tests of these products is how to quickly and efficiently extract failure information to assess the remaining useful life of the devices. A degradation model which measures the physical degradation path as a function of time can provide a direct connection between the product failure time and the inherent degradation mechanism, and hence improves accuracy and credibility of the predicted reliability. Most existing work in the literature focuses on modeling and analysis of degradation data with a single characteristic. In some degradation tests, multiple characteristics of a degradation process are measured to understand different aspects of the reliability performance. The literature on modeling degradation data with multiple characteristics is scarce. We propose a methodology capable of helping to fill this gap in the literature of degradation models for data with covariates. The proposed methodology is a general path dynamic model for degradation data with covariates that allows the degradation rate to be written as a function of two components. The first component represents the particularities of each unit and has a regression structure that accommodates the covariates. The other component represents the random effects of the common environment and evolve over time. In addition, the inspection times are included in the model from generic functions, allowing for different practical representations to be accommodated. The relation of the model parameters and the failure time is found and methods for estimating the remaining useful life for units under test and a future one are discussed. We applied the methodology to the scar width and train wheel degradation data. Results show that the proposed methodology is competitive in predicting failure times and estimating the remaining useful life. Joint work with Rosângela Loschi (UFMG) and Thiago Santos (UFMG)

### ***Accounting for Underreporting in Zero-Inflated Spatiotemporal Infectious Diseases Counts***

Presenter: Guilherme Lopes de Oliveira

Underreporting of disease cases is a recurring challenge in epidemiology, which introduces bias into the statistical estimation of disease rates. Although some approaches for modeling and correcting underreported count data have been proposed in recent years, there is a lack of methods that consider data quality in a spatiotemporal context, particularly in less aggregated time periods and small geographic areas when an excess of zeros results. Zero inflation can be caused by both the absence of the disease and underregistration. We propose a model that simultaneously captures underreporting and zero inflation. In particular, we assume that the reported counts follow a zero-inflated model, which allows for the absence of the disease (true zeros), and an imperfect counting process that enables the observation of a zero even when the disease is present. This is defined by assuming that, whenever the disease is present, the distribution of the observed count follows a Binomial thinned zero-truncated

negative binomial distribution. We consider a spatiotemporal setting, and inference is performed using a Bayesian approach. We study the necessary conditions for identifiability of the model with special concern for parameters associated with the reporting probabilities. Simulation studies are presented to explore the model's behavior under different levels of presence and underreporting, as well as in distinct data generation processes. The applications focus on epidemiological outcomes, such as cases of dengue and tuberculosis in Brazil. By taking into account underreporting, excess zeros, and spatiotemporal heterogeneity, the proposed modeling strategy aims to provide more realistic estimates for associated disease rates. In this way, decision-makers can make more informed and accurate decisions for disease control and prevention, while also better evaluating the adequacy of the United Nations' 2030 Agenda for Sustainable Development goals related to health and well-being.

#### ***Exploring the effects of prior choices in Gaussian-process-based Bayesian optimization***

Presenter: Guilherme dos Santos

Bayesian Optimization (BO) is a powerful tool for optimizing black-box functions, typically working in a sequential manner, driven by a surrogate model and an acquisition rule that selects the next evaluation point. Gaussian processes (GPs) are the most commonly used surrogates, and their performance depends on choices such as which kernel to use and the priors placed on its hyperparameters. When anisotropic kernels are used, the dimension-specific lengthscales play a key role, as they determine which directions are relevant to exploration. Therefore, prior choices for these lengthscales become especially important in high-dimensional settings.

In this work, we investigate how different priors on GP lengthscales affect BO performance. We experiment with benchmark functions with different properties and compare vague and increasingly informative priors, including priors that scale with the dimensionality of the domain. We also study scenarios with spurious, non-influential, dimensions to evaluate how prior choices impact in the ability to identify active dimensions. Certain dimension-scaled informative priors can behave similarly to a variable-selection mechanism resulting in a few dimensions with small lengthscales (indicating active, influential directions) while pushing many others toward large lengthscales (deemphasizing irrelevant dimensions).

We seek to provide guidance for principled prior selection in GP-based BO. Future directions include applications to real-world problems and extensions to other kernels and domains, such as graph-based optimization relevant to tasks such as model selection and Bayesian network structure optimization.”

#### ***A Bayesian Hyperbolic Signed Latent Space Model for Hierarchical Symptom Structure in Psychopathology***

Presenter: Gwanghee Kim

Psychopathology network analysis views mental disorders as systems of interacting symptoms. However, conventional network approaches relying on centrality measures fail to capture hierarchical relations among symptoms and overlook individual-level response patterns. To address these limitations, we propose a Bayesian hyperbolic signed latent space model that represents activation and inhibition patterns in both individual responses and symptom interactions, allowing hierarchical symptom structures to emerge from the latent geometry. To ensure accurate inference in hyperbolic space, we utilize an MCMC algorithm adapted to its geometric constraints. Applied to mental health survey data, the model reveals hierarchical organization among symptoms, while respondent positions in the latent space elucidate how individuals align with distinct symptom domains through radial distance and angular orientation. This framework further generalizes to other signed networks, such as social networks with positive and negative relations.

#### ***Scalable Gibbs sampler using randomized sketching for massive and high-dimensional Bayesian regression***

Presenter: Gyeongmin Park

Scalable MCMC methods for high-dimensional Bayesian regression have primarily focused on reducing the computational burden of iterative operations with large covariance matrices. Most existing approaches rely on the large-p-small-n assumption to achieve computational efficiency. However, the scenario where both  $n$  and  $p$  are large remains inadequately addressed. We propose a novel methodology that integrates randomized sketching into the Gibbs sampling framework to tackle the large-p-large-n problem. Our approach uses a random sketching matrix to efficiently approximate high-dimensional posterior distributions and enables practical Bayesian inference for massive high-dimensional datasets. The proposed method applies to various shrinkage priors commonly used in high-dimensional regression. Through extensive simulation studies, we demonstrate that our approach achieves substantial reductions in computational cost while maintaining estimation accuracy.

#### ***A Laplace Mixture Method for Multimodal Bayesian Inference***

Presenter: Hanyue Gu

In Bayesian inference, accurately approximating the posterior distribution is essential for deriving meaningful probabilistic insights from data. The Laplace approximation provides a computationally efficient Gaussian approximation of posterior distributions in Bayesian inference. However, its accuracy deteriorates for complex posteriors, such as those exhibiting multimodality or strong non-Gaussianity. To address this, we introduce a method that constructs a proposal distribution as a convex mixture of local Laplace approximations. We prove the theoretical convergence of this Laplace-Mixture towards the target posterior, establishing a bound on the Hellinger distance as observational noise diminishes. Our primary contributions are twofold: first, we provide a theoretical analysis of its performance in Monte Carlo methods, proving the convergence of the Effective Sample Size (ESS) in Importance Sampling and of the acceptance rate and Expected Squared Jumping Distance (ESJD) in Independent Metropolis-Hastings. Second, we demonstrate through numerical examples that our method remains robust and efficient as the number of observations increases and the observational noise decreases, validating our theoretical results.

***Estimating Parametric Income Distributions from Grouped Data using Quantiles and Lorenz Curves***

Presenter: Haruhisa Nishino

In this presentation, we introduce a Bayesian approach to estimating parametric income distributions from grouped data by jointly utilizing information from quantiles and the Lorenz curve. Previous studies typically rely either on quantile-based estimation or on statistics derived from the Lorenz curve. In contrast, our approach exploits the asymptotic covariance matrix of the quantiles and partial income shares.

The main objective is to estimate the parameters of flexible four-parameter distributions, such as the Generalized Beta of the Second Kind (GB2). Although the GB2 distribution encompasses several well-known income distributions, its four-parameter structure makes conventional methods, such as maximum likelihood, difficult to implement. To address this challenge, we employ a Bayesian estimation framework using MCMC methods. Finally, using income data from the Japanese Family Income and Expenditure Survey, we present empirical results for the parameter estimates.

***Bayesian estimation of inter-rater agreement under the existence of grey zones and latent inter-rater correlation***

Presenter: Haydar Demirhan

The agreement level between two raters, inter-rater agreement, is used to assess the validity of measurements in the health and social sciences, evaluating the degree of similarity between diagnoses, scores, or judgments made by the raters. The inter-rater agreement context provides a wide variety of medical and social fields with essential information for critical decision-making and evaluation. The agreement metrics, such as Cohen's kappa, measure the agreement between two raters by adjusting for the possibility of chance agreement. Different sources of bias are encountered in the estimates of agreement metrics in practice. For ordinal agreement tables, grey zones between the raters that correspond to adjacent cells of the agreement table introduce a negative bias in the estimated agreement. Grey zones stem from different assessment behaviours of the raters. Another source of bias, in a more general sense, is the latent correlation between the raters that artificially inflates or deflates the estimated agreement. The difference between a grey zone and correlation between raters is that while a grey zone occurs locally in one or more cells of the agreement table, correlation between two raters results in a pattern of artificial inflation in all cell counts. In this study, we propose a Bayesian model to estimate the agreement level between two raters when a grey zone exists in an ordinal agreement table and a correlation between the raters is present. Our Bayesian model utilises a graded response model and a linear-by-linear association model in conjunction with a recently proposed discrete bivariate distribution. It separately captures the impact of grey zones, the correlation between raters, and chance-corrected agreement between the raters. It is beneficial in eliminating raters-related biases from the agreement estimates, hence providing better precision in estimation and supporting less biased decisions.

***Latent principal stratification for continuous treatment and mediator***

Presenter: Heejun Shin

Principal stratification is a fundamental framework for characterizing treatment effect heterogeneity in the presence of post-treatment variables. Yet existing methods are largely restricted to binary treatments or mediators, making them ill-suited for modern applications involving continuous exposures. We introduce a new principal stratification framework tailored to continuous treatment and continuous mediator settings. Our approach leverages a discrete latent-strata representation that enables data-driven clustering of exposure-mediator trajectories and yields interpretable principal causal estimands. We formally develop identification conditions for these estimands and show how they can be recovered. To flexibly model complex heterogeneity across strata, we integrate

Dirichlet process mixtures with Bayesian Additive Regression Trees (BART), providing a fully nonparametric estimation strategy. We apply the proposed framework to quantify the causal effect of PM2.5 exposure on cardiovascular disease outcomes.

### ***Efficient Non-Stationary Spatial Modeling with BART and Bayesian Stacking***

Presenter: Hwangwan Gwon

Mixed effects models are widely used in the analysis of geospatial data. These models typically decompose the response variable in a mean function, which depends on covariates, and spatial random effects. Various approaches can be used to model the mean structure in a non-linear way, mostly nonparametric methods such as generalized additive models (GAM) or machine learning methods. A common assumption is that the flexible mean function accounts for all the spatial dependence, thus implying that the errors can be treated as independent. Recent work has sought to relax this assumption while still retaining the assumption that the spatially dependent errors are second-order stationary. Retaining the flexibility of a non-linear mean function, we propose a Bayesian Additive Regression Tree (BART) approach that can accommodate both stationary and nonstationary spatial data. In particular, our proposal mitigates the tendency of existing methods to oversmooth data. To improve computational efficiency, we further take advantage of Bayesian stacking instead of exhaustively exploring the full parameter space, offering global and local optimization strategies for estimating the stacking weights.

### ***Cluster-Conditioned IBP for Robust Mass Spectrum–Structure Matching Under Noise***

Presenter: Hyuju Kim

Mass spectrum–structure matching is challenging due to heterogeneous fragmentation patterns and realistic measurement noise. We address this problem by introducing the Cluster-Conditioned Indian Buffet Process (CC-IBP), a Bayesian nonparametric framework designed to capture local spectral structure. We first cluster the original mass spectra and treat each cluster as an independent spectral group. Within each cluster, we learn cluster-specific latent  $m/z$  patterns ( $A$ ) and their usage activations ( $Z$ ) using an IBP-based nonparametric feature model. This separation preserves local structure and prevents latent features from being distorted by cross-cluster variability. For evaluation, we apply synthetic perturbations to the spectra—such as Gaussian noise injection and intensity scaling—to assess robustness under varied noisy conditions. Each noisy spectrum is mapped to its most compatible cluster, and the corresponding cluster-specific IBP representations are combined with multiple similarity signals—such as IDF-weighted  $m/z$  voting and prototype-based cosine similarity—to generate a ranked Top-K list of candidate SMILES. CC-IBP provides a principled Bayesian interpretation of region-specific latent structure and demonstrates robustness to spectral noise by avoiding global sharing of heterogeneous latent patterns. Empirically, the framework maintains consistent Top-K SMILES prediction performance under noise, highlighting the practical advantages of combining clustering with Bayesian nonparametric modeling for mass spectrum–structure matching.

### ***Bayesian nonparametric Clustering of Prostate Cancer Patients with Simultaneous Feature Selection of Metabolites***

Presenter: Inga Huld Armann

Prostate cancer (PrCa) is the most common cancer in men in the UK, with 55,000 cases and 12,000 deaths per year. Metabolomics extracted from biofluids such as blood offer a promising and easily accessible source of data for cancer diagnosis. Building on the success of previous analyses of metabolomics data for PrCa, clustering techniques are employed to identify potential PrCa subtypes. A Bayesian nonparametric clustering method via Dirichlet process mixture models is proposed. The flexibility of the nonparametric approach enables the model to estimate the number of clusters from the data, rather than being pre-specified by the user. The model employs an extension that allows for simultaneous feature selection, aiming to identify the features that drive the clustering. This helps identify which metabolites are linked to the observed PrCa subtypes, providing insight into potential biomarkers. The extension can also improve clustering performance for noisy datasets. The posterior of the proposed model is approximated through variational inference, offering a computationally efficient alternative to traditional Markov chain Monte Carlo methods. The effectiveness of the proposed method is demonstrated on simulated data, followed by an analysis of metabolomics data from two case-control studies on PrCa, namely, PROFILE and IMPACT.

### ***Estimating Inhomogeneous Spatio-Temporal Background Intensity Functions using Graphical Dirichlet Processes***

Presenter: Isaías Bañales

An enhancement in seismic measuring instrumentation has been proven to have implications in the quantity of observed earthquakes, since denser networks usually allow recording more events. However, phenomena such as

strong earthquakes or even aseismic transients, as slow slip earthquakes, may alter the occurrence of earthquakes. In the field of seismology, it is a standard practice to model background seismicity as a Poisson process. Based on this idea, this work proposes a model that can incorporate the evolving spatial intensity of Poisson processes over time (i.e., we include temporal changes in the background seismicity when modeling). In recent years, novel methodologies have been developed for quantifying the uncertainty in the estimation of the background seismicity in homogeneous cases using Bayesian non-parametric techniques. This work proposes a novel methodology based on graphical Dirichlet processes for incorporating spatial and temporal inhomogeneities in background seismicity. The proposed model in this work is applied to study the seismicity in the southern Mexico, using recorded data from 2000 to 2015.

***Efficient Bayesian model selection with improved pseudo-marginal RJMCMC for quasi-copula models***

Presenter: Ivis Kerama

Quasi-copulas arise in applications requiring flexible dependence modelling (but without a specific property that would result in a proper copula). We propose an efficient Bayesian model selection and inference procedure within a specified subclass of quasi-copula models for which likelihoods are intractable and the model space (dimension and family choices) is combinatorially large. We develop a reversible-jump pseudo-marginal MCMC sampler substantially extending the one in Karagiannis et al (2021) targeting the joint posterior (over model and parameter space). The methodology is applied to a toy example and a real world optimal decision problem where dependency modelling over equivalence classes and within those classes is modelled as a large nested (qasi)-copula of arbitrary complexity (fixed for our case). The problem is that of uptake for farms given incentives for exchanging parcels for tree planting while maximising certain goals with a Net-zero target in mind. We showcase the improvement in model and parameter identifiability.

***Positive definiteness in separable matrix priors with applications to Gaussian graphical models***

Presenter: Jack Storrer Carter

A simple and increasingly popular (especially in Gaussian graphical models) prior distribution for a positive definite matrix parameter is to assume independence of the entries of the matrix before truncating the resulting distribution onto the space of positive definite matrices. The pre-truncation marginals then provide a framework for encoding and interpreting prior information. However, it is possible, and common in many default settings, for the truncation to drastically alter the true marginal distributions rendering the pre-truncation interpretation misleading. The post-truncation marginals are generally not available in closed form so rely on expensive sampling methods to approximate. To encode prior information you are therefore left with two choices: inaccurately encode it in the pre-truncation marginals or sample from the post-truncation distribution for different choices of prior parameters until the sampled marginals approximate the desired prior information.

To tackle this problem we introduce some theory to characterise the effect of the truncation on the marginal distributions. This theory is then used to propose strategies for ensuring that the post-truncation marginals remain close to the pre-truncation marginals, both for finite dimensions and asymptotically. In this way one can encode prior information via the pre-truncation marginals with theoretical guarantees that it will remain after truncation and removing the need for sampling to check this. We also demonstrate through simulated examples that this goes beyond prior encoding and interpretation and can also improve posterior inference.

***Addressing Phase Discrepancies In Functional Data: A Bayesian Approach For Accurate Alignment And Smoothing***

Presenter: Jacopo Gardella

In many real-world applications, functional data exhibit considerable variability in both amplitude and phase. This is especially true in biomechanical data such as the knee flexion angle dataset motivating our work, where timing differences across curves can obscure meaningful comparisons. Curves of this study also exhibit substantial variability from one another. These pronounced differences make the dataset particularly challenging to align properly without distorting or losing some of the individual curves characteristics. Our alignment model addresses these challenges by eliminating phase discrepancies while preserving the individual characteristics of each curve and avoiding distortion, thanks to its flexible smoothing component. Additionally, the model accommodates group structures through a dedicated parameter. By leveraging the Bayesian approach, the new prior on the warping parameters ensures that the resulting warping functions automatically satisfy all necessary validity conditions. We applied our model to the knee flexion dataset, demonstrating excellent performance in both smoothing and alignment, particularly in the presence of high inter-curve variability and complex group structures. Moreover, the proposed model naturally allows for the incorporation of prior information on the warping parameters, which is particularly useful in the presence of landmark points. In situations where landmark locations are uncertain or



estimated with error, enforcing traditional landmark constraints may lead to inaccurate alignments. By leveraging the Bayesian framework and the flexibility of our new model, prior information on these uncertain landmarks can be seamlessly integrated, yielding a more robust and hybrid alignment strategy.

#### ***Posterior Contraction of Lévy Adaptive B-Spline Regression in Besov Spaces***

Presenter: Jeunghun Oh

We investigate the asymptotic properties of the Lévy Adaptive B-spline (LABS) regression model, a Bayesian nonparametric method that incorporates B-spline kernels into the Lévy Adaptive Regression Kernel (LARK) prior. LABS allows splines of varying degrees with independently defined knots, yielding a flexible model class capable of local adaptation to heterogeneous smoothness of the true function. Within the nonparametric Gaussian regression framework with univariate random design, we establish that the LABS posterior contracts around the true function in Besov classes at nearly minimax-optimal rates, up to a logarithmic factor, while adapting automatically to unknown smoothness. This study contributes to filling a gap in the literature, where theoretical results on posterior contraction for Bayesian random spline methods in Besov spaces remain scarce. Simulation experiments and real data applications complement the theory and demonstrate the practical utility of LABS.

#### ***Bayesian Nonparametric Models for Cause-of-Death Assignments Using Verbal Autopsy***

Presenter: Jibo Shen

Verbal Autopsy (VA) is a well-developed tool to collect information describing deaths outside of hospitals by conducting surveys to the caregivers of the deceased person. It is routinely-implemented in low and middle income countries, where it often lacks sufficient resources to conduct the autopsy. Several Bayesian models have been developed to assign cause-of-death. One main task is to model the joint distribution of symptoms given cause. Previous work addresses the problem by building latent classes models. Yet, the practice of modeling symptom structure specifically for each cause introduces a substantial number of parameters requiring estimation. In our work, we put the latent class model under the Bayesian Nonparametric framework. Unlike previous work, the symptom distributions are not assumed to be specific to a given cause. Instead, we allow individuals with different causes of death to be allocated to the same latent class in order to quantify shared symptom profiles across causes. By taking advantage of the flexibility from BNP, as well as its feature of clustering, we build a model that is more complex in structure, but potentially has much fewer parameters to estimate in practice, while still achieving a comparable accuracy of cause-of-death assignments. We implement the model on both simulated data and real VA data. This is a joint work with Zehang Richard Li.

#### ***Bayesian Gamma-Poisson-Gamma Tensor Factorization***

Presenter: Jie Jian

We model high-dimensional multi-way data as realizations from a distribution whose parameter admits a low-rank tensor factorization that captures latent dependence. In many applications, observations comprise a point mass at zero and a positive continuous component, requiring joint modeling of occurrence and magnitude. Moreover, dispersion often varies across slices within a mode. To accommodate these features, we propose a Bayesian hierarchical model that imposes a low-rank CP structure on a Poisson latent-rate tensor and couples it with a conditional Gamma model for positive outcomes with mode-specific rate parameters. To enable large-scale inference, we develop a scalable hybrid variational–Monte Carlo algorithm combining fast CAVI updates with a partially collapsed augmented-variable sampler. By applying the method to 60 million international trade flows in dollar indexed by exporter, importer, product, and time, we detect interesting multi-way dependence across these dimensions that conventional bilateral or one-margin analyses typically miss.

#### ***Bayesian Win Ratio Analysis with Power Priors for Placebo Borrowing in Composite Endpoints***

Presenter: Joon Jin Song

Leveraging external control data has been used to enhance the efficiency of clinical trials, especially in rare diseases where recruitment is often challenging. However, directly pooling data from different studies without appropriate adjustments can lead to biased results when populations differ across these studies. In addition to limited sample sizes, trials for rare diseases commonly assess treatment efficacy through multiple clinical endpoints using composite endpoints. The win ratio has gained attention for composite endpoint analysis, as it enables prioritized comparisons that account for the relative clinical importance of each endpoint. Motivated by these two challenges, we proposed a novel two-stage Bayesian win ratio framework that: (1) utilizes a joint model to capture heterogeneity in function progression and correlation between longitudinal and survival outcomes, (2) leveraging historical placebo data via power prior approach thereby reducing sample size requirement and conserve resources, and (3) enables interim decisions using a posterior predictive distribution of win ratios accounting to clinical priorities with better interpretation.

### ***Data Integration for Multiple Imputation of Nonignorable Missing Data: A Copula-based Approach***

Presenter: Joseph Feldman

We present methods for parameter estimation and multiple imputation with Gaussian copula models in the presence of nonignorable missing data. Our approach uses Bayesian data integration to combine (i) a Gaussian copula model for all study variables and missingness indicators, which allows arbitrary marginal distributions, nonignorable missingness, and other dependencies, and (ii) auxiliary information in the form of marginal quantiles for some study variables. We prove that, remarkably, one only needs a small set of accurately-specified quantiles to estimate the copula correlation consistently. The remaining marginal distribution functions are inferred nonparametrically and jointly with the copula parameters using an efficient MCMC algorithm. We also characterize the (additive) nonignorable missingness mechanism implied by the copula model. Simulations confirm the effectiveness of this approach for multivariate imputation with nonignorable missing data. We apply the model to analyze associations between lead exposure and end-of-grade test scores for 170,000 North Carolina students. Lead exposure has nonignorable missingness: children with higher exposure are more likely to be measured. We elicit marginal quantiles for lead exposure using statistics provided by the Centers for Disease Control and Prevention. Multiple imputation inferences under our model support stronger, more adverse associations between lead exposure and educational outcomes relative to complete case and missing-at-random analyses.

### ***Predictive modeling with large language models***

Presenter: Juho Lee

Large language models (LLMs) are often hypothesized to perform implicit Bayesian inference, yet a key coherence condition—the martingale property of predictive beliefs—has been shown to fail in controlled synthetic in-context learning settings. We revisit this question in a more typical usage regime: generic multiple-choice question answering. Exploiting the discrete answer space, we compute exact predictive distributions and study belief dynamics induced by autoregressive answer resampling. We introduce prompted predictive resampling (PPR), where an LLM generates a sequence of answers to the same question. Empirically, PPR reveals early-stage belief drift, indicating martingale violations. However, after sufficient resampling steps, the belief process self-stabilizes and converges to a coherent predictive distribution. Based on this observation, we further propose (i) a seed-answer prompting strategy to accelerate stabilization, and (ii) a self-consistency loss that amortizes early-stage drift into the model via fine-tuning. Experiments on multiple-choice QA benchmarks show that our methods substantially reduce belief drift and improve predictive coherence without sacrificing accuracy.

### ***In-Context Multi-Objective Optimization***

Presenter: Julien Martinelli

Balancing competing objectives is omnipresent across disciplines, from drug design to autonomous systems. Multi-objective Bayesian optimization is a promising solution for such expensive, black-box problems: it fits probabilistic surrogates and selects new designs via an acquisition function that balances exploration and exploitation. In practice, it requires tailored choices of surrogate and acquisition that rarely transfer to the next problem, is myopic when multi-step planning is often required, and adds refitting overhead, particularly in parallel or time-sensitive loops. We present TAMO, a fully amortized, universal policy for multi-objective black-box optimization. TAMO uses a transformer architecture that operates across varying input and objective dimensions, enabling pretraining on diverse corpora and transfer to new problems without retraining: at test time, the pretrained model proposes the next design with a single forward pass. We pretrain the policy with reinforcement learning to maximize cumulative hypervolume improvement over full trajectories, conditioning on the entire query history to approximate the Pareto frontier. Across synthetic benchmarks and real tasks, TAMO produces fast proposals, reducing proposal time by 50-1000x versus alternatives while matching or improving Pareto quality under tight evaluation budgets. These results show that transformers can perform multi-objective optimization entirely in-context, eliminating per-task surrogate fitting and acquisition engineering, and open a path to foundation-style, plug-and-play optimizers for scientific discovery workflows.

### ***Reliable Toxicity Classification under Model Misspecification: Gaussian Process Kernels with Mondrian Conformal Prediction***

Presenter: Junhee Kim

Predictive uncertainty is a central component of Bayesian modeling, especially in scientific domains such as chemical toxicity prediction, where misspecified likelihoods, high-dimensional descriptors, and class imbalance often produce overconfident yet unreliable posterior summaries. In this work, we propose a reliability-enhanced framework that combines a kernel-based Gaussian Process (GP) classifier with Mondrian Conformal Prediction (MCP) to provide distribution-free, finite-sample coverage guarantees for molecular toxicity classification. We



construct a hybrid similarity kernel based on Tanimoto fingerprints and RBF–Matern Gaussian Process components, optimized through marginal likelihood search. Although the resulting GP achieves competitive predictive performance ( $AUC \approx 0.89$ ;  $F1 \approx 0.80$ ), its posterior predictive probabilities exhibit substantial miscalibration when applied to noisy chemical datasets. To address this, we apply class-conditional (Mondrian) conformal calibration to the GP predictive scores. This yields provably valid 90% marginal coverage (empirical coverage  $\approx 0.91$ ) while significantly improving prediction efficiency: the average prediction set size is reduced from 1.75 (standard split-conformal) to 1.23 under MCP. Our results highlight that conformal methods can effectively correct miscalibrated Bayesian predictive distributions without altering the underlying model, and that Mondrian calibration is particularly well-suited for imbalanced toxicity data. This modular, model-agnostic approach provides a practical path toward trustworthy Bayesian decision-making for molecular safety assessment, even under severe model misspecification.

### ***Bayesian Variational Inference for Mixed Data Mixture Models***

Presenter: Junyang Wang

Heterogeneous, mixed type datasets including both continuous and categorical variables are ubiquitous, and enriches data analysis by allowing for more complex relationships and interactions to be modelled. Mixture models offer a flexible framework for capturing the underlying heterogeneity and relationships in mixed type datasets. Most current approaches for modelling mixed data either forgo uncertainty quantification and only conduct point estimation. On the other hand, MCMC approaches suffer from poor mixing and high computational cost in this setting. This paper develops a coordinate ascent variational inference algorithm (CAVI) for mixture models on mixed (continuous and categorical) data, which circumvents the high computational cost of MCMC and avoiding label switching issues. We demonstrate our approach through simulation studies as well as an applied case study of the NHANES risk factor dataset. In addition, we show that the posterior means from CAVI for this model converge to the true parameter value as the sample size tends to infinity, providing theoretical justification for our method.

### ***Presence-only species distribution model that account for false positive and false negative errors***

Presenter: Kabiru Abubakari

Records of species data available on online portals, such as the Global Biodiversity Information Facility (GBIF) and the Botanical Society of Britain and Ireland (BSBI), are collected through citizen scientists. This kind of data is often referred to as presence-only data since only presence records are available. Species distribution models (SDMs) require presence-absence data. In previous studies, SDMs were developed using information from environmental variables (background data). This approach tends to favour nonparametric models intended for prediction. Thus, the effect size of environmental variables on species distribution is ignored. We propose a species distribution model using a presence-pseudo-absence dataset, where pseudo-absence data is used in sites without records. We recognise that our presence-pseudo-absence data is prone to false positive and false negative errors and can bias the effect sizes (i.e., coefficients of environmental covariates). In this work, we follow a Bayesian approach that allows us to account for the effect of false positive and false negative errors on the effect sizes in order to obtain consistent estimates.

### ***Critical Illness Risk Modelling using Bayesian Hierarchical Models for Location, Scale and Zero-inflation Probability***

Presenter: Kaitlyn Louth

Accurate prediction of morbidity risk for critical illnesses such as heart attacks, strokes, multiple sclerosis, cancers, and Parkinson’s disease, is essential for public health planning and actuarial decision-making. We present a hybrid modelling framework that integrates hierarchical Bayesian methods with deep learning architectures to improve morbidity risk prediction in the presence of sparse, zero-inflated, and heterogeneous health data. In this framework, the location, scale and zero-inflation probability components are data-driven by risk factors.

Our approach leverages Bayesian Neural Networks (BNNs), in which all weights and biases are assigned probability distributions, enabling full posterior inference over network parameters. This formulation provides principled uncertainty quantification and helps prevent overfitting, an issue that is particularly acute in morbidity data with high proportions of zero claims. In contrast to conventional black-box neural networks, the Bayesian formulation offers improved interpretability through posterior summaries and credible intervals on both parameters and predictions.

We further extend the BNN framework by incorporating mixed-effects components to model covariate-dependent dispersion and to capture both observed and unobserved heterogeneity across demographic, temporal, and geographic strata. This results in a flexible, fully probabilistic model capable of representing complex hierarchical

structures while remaining data-driven.

The models are trained and validated using large-scale datasets from the Office for National Statistics (UK), NHS Open Data, Public Health Scotland, and the Institute and Faculty of Actuaries. Our findings demonstrate the potential of Bayesian hierarchical models to substantially improve morbidity risk through enhanced predictive performance, robustness, and uncertainty quantification.

### ***Dynamic Adaptive Rerandomization for Efficient Sequential Trials Under Budget Constraints***

Presenter: Kateryna Husar

We introduce the Dynamic Adaptive Rerandomization (DAR) framework to improve the statistical efficiency of the Average Treatment Effect (ATE) estimator in resource-constrained sequential randomized controlled trials (RCTs). DAR combines two linked Bayesian procedures for concurrent adaptive learning and robust estimation. In each batch, a Thompson Sampling-based Multi-Armed Bandit (MAB) policy selects covariates with the highest posterior predictive importance while respecting the measurement budget. Measured covariates are then used in rerandomization to minimize the variance of the batch-specific ATE estimator ( $\hat{\tau}_k$ ). The overall Average Treatment Effect ( $\tau$ ) is estimated sequentially using a Bayesian weighted averaging approach. The  $\hat{\tau}_k$ 's are combined using inverse-variance weighting, where uncertainty in the variance estimates ( $\sigma_k^2$ ) is explicitly modeled. This structure naturally assigns greater weight to batches rerandomized on more predictive covariates, as these estimators exhibit smaller sampling variances. Covariate predictive utility is subsequently reassessed to update the importance prior, closing the adaptive loop. This dual-path mechanism allows for identification of influential covariates, optimizes resource use, and yields a precise, fully quantified posterior for the ATE.

### ***Time-varying expected shortfall with realized stochastic volatilities***

Presenter: Keisuke Sasaki

Expected Shortfall (ES) and Value-at-Risk (VaR) are widely used measures in financial risk management. ES is particularly favored since Basel III adopted it as a more robust measure of tail risk compared to VaR. Although a rich literature exists on estimating and predicting ES, there remains a need for more accurate predictive methods. We introduce a new Bayesian method to predict ES by incorporating the structure of the Realized Stochastic Volatility (RSV) model into the Conditional Autoregressive Expectile (CARE) model. By utilizing realized measures from high-frequency data and a stochastic volatility structure, our method aims to capture dynamic changes in tail risk quickly and effectively. Posterior inference is performed using a Metropolis-Hastings-within-Gibbs algorithm tailored for the expectile likelihood. We apply our model to several datasets and demonstrate its predictive performance in comparison to existing models.

### ***Gradient-Informed Grid Selection for Intractable Likelihoods***

Presenter: Laura Bazahica

Bayesian inference for models with intractable likelihoods, such as Markov random fields, requires a careful tradeoff between accuracy and computational cost. Various MCMC methods have been developed to address this challenge. The exchange algorithm targets the exact posterior, but relies on an expensive perfect sampling step at each iteration, which is often infeasible. In contrast, path sampling approximates the Metropolis acceptance ratio using a precomputed grid of likelihood values, but poorly chosen grids can introduce bias. Our approach builds on the theoretically robust framework of amortized inference for MCMC, providing accuracy comparable with the exchange algorithm, while substantially reducing the computational cost. We propose an efficient, gradient-informed method for selecting the precomputed grid. Hermite interpolation between these grid points results in a surrogate model that is easy to estimate with minimal approximation error. The advantages of our method are demonstrated by applications to a hidden Potts model of satellite data and an autologistic model of Arctic ice floes. A simulation study further quantifies the rate of improvement in accuracy as the number of grid points increases.

### ***High-dimensional Function-on-Scalar Regression with Variable Selection via Multi-Output GPs***

Presenter: Leiv Rønneberg

In biomedical applications, the goal is often to relate a functional response to scalar covariates to understand disease mechanisms or identify biomarkers. Function-on-scalar (FoS) regression is a common tool for this. We propose a novel formulation of FoS regression that arises as a special case of multi-output Gaussian process (MOGP) regression for specific kernel choices. Variable selection is introduced with a horseshoe prior on kernel hyperparameters, and subject-specific random effects are incorporated through additive kernels. The MOGP formulation is basis-free and scales effectively with both the domain of the functional response and the number of covariates. It also enables extensions to more complicated input domains (such as graphs) and to models with interactions through simple kernel modifications. This connection further brings modern GP scalability techniques

directly into the FoS setting. We illustrate the model on simulated data and on a cancer drug-sensitivity dataset, showing that the model can recover known genetic biomarkers of drug efficacy.

#### ***Stochastic Bayes factors: how, when, and why***

Presenter: Leonardo Egidi

The Bayes factor (BF) remains a fundamental tool for Bayesian hypothesis testing and model selection, yet its practical deployment is hindered by strong sensitivity to prior specification, incompatibility with improper priors, and the absence of a principled way to quantify uncertainty in its value. As a result, classical BFs often behave analogously to p-values and primarily measure prior-predictive rather than posterior-predictive performance. In this work we introduce the Stochastic Bayes Factor (SBF), a general framework that redefines the BF as a push-forward measure on replicated data, thereby explicitly incorporating sampling variability. By evaluating Bayes factors on prior- or posterior-predictive replications, the SBF provides a distributional view of model evidence rather than a single, fixed number. We establish key theoretical guarantees: consistency, compatibility, separability, and stochastic dominance, showing that posterior-predictive SBFs overcome important limitations of classical BFs and lead to more reliable discrimination between competing models. We further propose an algorithmic routine that operationalizes SBFs for practical model comparison, offering guidance on when models are compatible, distinguishable, or dominated, and highlighting scenarios in which model selection is inherently inconclusive. Simulation studies and real-data applications—including regression and overdispersed count models—demonstrate that SBFs deliver more robust inference, improved calibration, and a clearer quantification of model uncertainty compared to traditional Bayes factors.

#### ***A Multivariate Pólya Urn scheme for Multiview Clustering***

Presenter: Luigi Malgieri

Multiview data—where multiple heterogeneous measurements are collected on the same subjects—pose unique modeling challenges, requiring methods that capture both view-specific structure and shared dependence across views. We introduce a Multiview Pólya Urn process that extends the classic Pólya urn to this Multiview setting, enabling inference of dependent yet distinct clustering patterns across data modalities. Defined through a sequence of predictive distributions, the process yields an asymptotically exchangeable sequence whose directing random measure has Dirichlet Process marginals, with a parameter  $\lambda$  governing the strength of cross-view dependence. This process can serve as a prior in mixture models, where posterior inference is performed via a collapsed Gibbs sampler that exploits conjugacy in the urn parameters through data augmentation. The proposed framework cleanly separates marginal modeling from cross-view dependence, providing a flexible and computationally efficient approach to clustering complex Multiview data.

#### ***Double BMA for the stochastic discount factor***

Presenter: Luis Gruber

The problem of estimating the stochastic discount factor can be formulated as a multivariate regression model. Well-established sources of uncertainty are parameter uncertainty and the selection of relevant risk factors. Another source of uncertainty, which has been paid much less attention so far, is conceptual uncertainty stemming from defensible yet arbitrary choices in factor construction. Such choices, e.g., portfolio breakpoints or weighting schemes, can lead to thousands of different construction possibilities per factor. We develop a hierarchical Bayesian model averaging (BMA) framework, the double BMA, that systematically integrates trillions of plausible risk factor permutations into the stochastic discount factor. Double BMA, because we not only average over possible factor combinations, but also average over different factor construction possibilities. To that end, we propose a tailor-made Markov Chain Monte Carlo (MCMC) algorithm in which certain steps can be parallelized, resulting in a scalable and efficient sampler. In an empirical application, we demonstrate that accounting for conceptual uncertainty reduces pricing errors and yields out-of-sample Sharpe ratios up to 50% higher relative to established benchmark models. (Joint work with Gregor Kastner, Stefan Voigt, and Patrick Weiss)

#### ***Diffusion piecewise exponential models for survival extrapolation using Piecewise Deterministic Monte Carlo***

Presenter: Luke Hardcastle

The piecewise exponential model is a flexible non-parametric approach for time-to-event data, but extrapolation beyond final observation times typically relies on random walk priors and deterministic knot locations, resulting in unrealistic long-term hazards. We introduce the diffusion piecewise exponential model, a prior framework consisting of a discretised diffusion for the hazard, that can encode a wide variety of information about the long-term behaviour of the hazard, time changed by a Poisson process prior for knot locations. This allows the behaviour of the hazard in the observation period to be combined with prior information to inform extrapolations. Effi-

cient posterior sampling is achieved using Piecewise Deterministic Markov Processes, whereby we extend existing approaches using sticky dynamics from sampling spike-and-slab distributions to more general transdimensional posteriors. We focus on applications in Health Technology Assessment, where the need to compute mean survival requires hazard functions to be extrapolated beyond the observation period, showcasing performance on datasets for Colon cancer and Leukaemia patients.

#### ***Flexible Models for Covariate Effects in Longitudinal Observational Studies***

Presenter: Lynsie Warr

Longitudinal studies typically seek to determine the impact of a covariate of interest on the trajectory of an outcome over time. In many cases, this outcome trajectory is nonlinear. Additionally, the relationship between the time-invariant covariate and the outcome may not be well described by a linear model. We develop a Bayesian hierarchical model which allows for a flexible model for the trajectory of the outcome and a flexible interaction between the covariate and time. The work is motivated by a study of the ways in which early life adversity is associated with health outcomes for children. More specifically, we use our model to explore the association of unpredictability in a child's environment with the development of specific brain regions. An additional interesting aspect of our application is the use of missing data methods to combine data from a modest-sized, well-characterized local cohort and a large, but less well-characterized, national sample.

#### ***Discrete Jumps with Time Varying Intensity in an Inverse Gamma Stochastic Volatility Model***

Presenter: Majoni Blessings

This paper develops a novel Inverse Gamma Stochastic Volatility (IGSV) model that incorporates discrete jumps in volatility with time varying probabilities linked to observable macroeconomic news announcements. The model captures heavy tailed returns and allows for abrupt changes in volatility driven by economic events, a key feature in financial time series. We employ an efficient Bayesian estimation strategy through a Particle Marginal Metropolis-Hastings (PMMH) with an auxiliary particle filter. To address the well known mixing challenges associated with nonlinear, non-Gaussian state-space models, we implement adaptive proposal scalings that are parameter specific, resulting in substantial improvements in sampling efficiency. Monte Carlo simulation studies across multiple sample sizes and computational budgets, show that our method efficiently recovers true parameters. An empirical application on financial time series data then shows the model's capability to capture heavy tails and volatility that is driven by macroeconomic news announcements.

#### ***Convergence Rate of Efficient MCMC with Ancillarity-Sufficiency Interweaving Strategy for Panel Data Models***

Presenter: Makoto Nakakita

Improving Markov chain Monte Carlo algorithm efficiency is essential for enhancing computational speed and inferential accuracy in Bayesian analysis. These improvements can be effectively achieved using the ancillarity-sufficiency interweaving strategy (ASIS), an effective means of achieving such gains. Herein, we provide the first rigorous theoretical justification for applying ASIS in Bayesian hierarchical panel data models. Asymptotic analysis demonstrated that when the product of prior variance of unobserved heterogeneity and cross-sectional sample size  $N$  is sufficiently large, the latent individual effects can be sampled almost independently of their global mean. This near-independence accounts for ASIS's rapid mixing behavior and highlights its suitability for modern "tall" panel datasets. We derived simple inequalities to predict which conventional data augmentation scheme—sufficient augmentation (SA) or ancillary augmentation (AA)—yields faster convergence. By interweaving SA and AA, ASIS achieves optimal geometric rate of convergence and renders the Markov chain for the global mean parameter asymptotically independent and identically distributed. Monte Carlo experiment confirm that this theoretical efficiency ordering holds even for small panels (e.g.,  $N=10$ ). These findings confirm the empirical success of ASIS application across finance, marketing, and sports, laying the groundwork for its extension to models with more complex covariate structures and non-Gaussian specifications.

#### ***Bayesian Markov-Switching Partial Reduced-Rank Regression***

Presenter: Maria Fernanda Pintado

Reduced-rank (RR) regression is a powerful dimensionality reduction technique, but traditional RR models typically overlook any potential group structure among the responses by assuming a low-rank structure on the coefficient matrix. When the observations in the regression model are indexed by time, the relationship between covariates and responses could change over periods. A time-varying grouping structure in the response variables in RR regression is currently understudied. To address this limitation, a Markov-switching Bayesian Partial RR (MSPRR) regression is proposed. First, the response vector is partitioned in two groups to reflect different degrees of complexity of the relationship. A "simple" group assumes a low-rank linear regression, and

a "flexible" group remains agnostic and exploits nonparametric regression via a Gaussian Process. Second, different from traditional approaches that assume known group structure and rank, they are treated as unknown parameters to be estimated. Third, time variation and persistence are accounted for by introducing a Markov-switching process, which examines the changes in the grouping structure and model parameters over time. We perform fully Bayesian inference via a partially collapsed Gibbs sampler, which allows uncertainty quantification. Applications to synthetic and real-world macroeconomic and energy data of the U.S. demonstrate the capability of the proposed method to uncover latent states and hidden structures within the data.

***A clustering-based modeling framework for diffuse astronomical source detection and characterization***

Presenter: Marta Ferrari

The identification of diffuse astronomical sources, such as quasar jets, is a crucial challenge in high-energy astrophysics to understand the formation and evolution of the cosmic environment in the early Universe (e.g., black hole activity and power). Understanding their morphology provides meaningful insights into the surrounding cosmic environment, but these objects often exhibit faint, irregular structures with non-uniform brightness, making their complex spatial emission difficult to analyze. In high-energy astronomy, telescopes register the incoming particles and their position on a detector array, resulting in a two-dimensional matrix where each entry represents the photon counts recorded at a given pixel location.

Existing image analysis algorithms in high-energy astronomy are primarily optimized for the detection and characterization of point sources, that is, sources treated as points, since their spatial extent is negligible compared to the angular resolution of the telescope observing them. They are, however, inadequate for discerning irregular and spatially extended emissions, which are characterized by complex morphologies. As a result, pixel-wise hypothesis testing has often been adopted in the literature: these methods classify each pixel as source or background by evaluating it against a null background distribution. These approaches are able to test for the presence of diffuse emission not captured by the null background model, but they do not precisely characterize the nature of the departures from the null.

Therefore, we introduce a new modeling framework based on Bayesian spatial clustering to analyze this pixel-level data in a single energy band. The methodology consists of a two-component mixture model designed to identify the spatial extent of a source by allocating every pixel of the map into two groups to determine whether it does or does not contain a source. To capture the spatial structure inherent in the image, the model incorporates dependence of spatially neighboring pixels. A key innovation of this approach is that it is capable of investigating the different features of the analyzed source (e.g., compact cores, elongated jets). This is done by defining the source emission component in the mixture as a further mixture of sub-components. Furthermore, we propose an MCMC algorithm in order to perform the challenging posterior inference.

This approach offers significant advantages. Instead of evaluating each pixel in isolation, the model is able to analyze the entire map simultaneously, leveraging the incorporated spatial dependencies. Furthermore, while existing methods often just test for a departure from a known background, this approach provides a specific classification of the source's internal features. We demonstrate the capability of our model to provide a robust and flexible characterization of complex astronomical structures through a series of simulation experiments and case studies using data from the Chandra X-ray telescope.

***Prevalence Incidence Cure models: A Bayesian survival model for Electronic Health Records with incomplete baseline diagnoses***

Presenter: Matilda Pitt

Retrospective cohorts can be extracted from Electronic Health Records, such as from General Practice notes, to study prevalence, time until disease or event occurrence and cure proportion in real world scenarios. However, this data was collected for patient care and not research so has many additional features which if not taken into consideration can give biased results, such as patients with missing results at baseline. Currently Prevalence-Incidence (PI) models provide reliable estimation by using a 2 component mixture model with a prevalent case component and an incident case component. These models may be biased if some individuals will never experience the endpoint (they are "cured"). Within the Bayesian paradigm we combine the PI model framework with the cure model framework and introduce a Prevalence Incidence Cure (PIC) model, a 3 component mixture model. The PIC model enables estimation of, and covariate effects on, prevalence, time to incidence via a survival model, and cure proportion. We focus in particular on the interpretability of the priors to enable expert elicitation. We show that under simulation studies, the PIC model is less biased than PI models at estimating survival probability. We also compare performance metrics across vague, informative, and misspecified priors. To allow comparison between PIC models, an adaptation to the C-statistic is required, which is also discussed. We apply the PIC



model to a dataset of 1964 patients undergoing treatment for Diabetic Macular Oedema. Previous RCTs show the treatment is a success, however treatment adherence differs in real-life clinical settings so needs to be investigated. We show the PIC model is a better fit than a PI model.

### ***Non-asymptotic Guarantees for Preconditioned MCMC under a Wasserstein-2 Contraction***

Presenter: Max Hird

Preconditioning is a common method applied to MCMC algorithms to help them run. In some cases it can vastly increase the efficiency of the underlying sampler. MCMC routines from off-the-shelf software packages such as stan and tensorflow probability learn their preconditioners using information from the chain. We establish non-asymptotic guarantees on the time taken to collect  $N$  approximately independent samples for schemes that learn their preconditioners under the assumption that the underlying Markov chain abides by a contraction in the Wasserstein-2 distance. Specifically we analyse and compare schemes which learn a preconditioner based on the target covariance [Haario et al. 2001] and those which learn a preconditioner based on the so-called ‘Fisher matrix’ as proposed in [Titsias 2023]. We subsequently compare these results with the chains that do not do any preconditioning. An example of a popular MCMC chain that is contractive in the Wasserstein-2 distance is the Unadjusted Langevin Algorithm (ULA) on an appropriately regular target, see e.g. [Dalalyan 2017 Theorem 1]. Therefore we apply our results to establish non-asymptotic guarantees for preconditioned ULA which learns either the covariance or the ‘Fisher’ preconditioner.

Joint work with Jeffrey Negrea and Florian Maire.

### ***Prior distributions for structured semi-orthogonal matrices***

Presenter: Michael Jauch

Statistical models for multivariate data often include a semi-orthogonal matrix parameter. In many applications, there is reason to expect that the semi-orthogonal matrix parameter satisfies a structural assumption such as sparsity or smoothness. From a Bayesian perspective, these structural assumptions should be incorporated into an analysis through the prior distribution. In this work, we introduce a general approach to constructing prior distributions for structured semi-orthogonal matrices that leads to tractable posterior inference via parameter-expanded Markov chain Monte Carlo. We draw on recent results from random matrix theory to establish a theoretical basis for the proposed approach. We then introduce specific prior distributions for incorporating sparsity or smoothness and illustrate their use through applications to biological and oceanographic data.

### ***Robust naive Bayes classifier with many potential non-Gaussian predictors***

Presenter: Mijin Jeong

The naive Bayes classifier, which assumes the conditional independence of predictors, improves classification efficiency and has a great advantage in handling high-dimensional data as well as imbalanced data. However, the success of the naive Bayes classifier hinges on the normality assumption for each continuous predictor and its performance decreases considerably as many irrelevant predictors are included. In this paper, we develop a new way of improving the performance of naive Bayes classifiers when many irrelevant non-Gaussian predictors are considered in model building. To address the problem of non-Gaussian predictors, we propose a robust naive Bayes framework using power transformation. To eliminate the irrelevant predictors, we employ the duality between naive Bayes classifier and quadratic logistic regression via the group penalization approach. We conduct a comparative simulation study to demonstrate the superiority of our proposed classifier over existing classification methods. We also apply our proposed classifier to real data and confirm its effectiveness.

### ***Estimating Hierarchical Tissue Organization via Spatial Point Processes and Latent Factor Models***

Presenter: Morris Greenberg

It is thought that some human tissues and cancers exhibit a hierarchical cellular organization, where the abundance of one cell type may depend on the presence or prevalence of another. A natural way to represent this structure is through a directed acyclic graph (DAG), where nodes correspond to cell types, and edges encode the conditional dependencies. However, learning DAG structures from data is computationally challenging due to the super-exponential growth of the DAG space as the number of nodes increases. Further complications arise because cell-type measurements are typically compositional and modeled via multinomial sampling, which imposes a global sum constraint and complicates standard graphical model inference. To address these challenges, we introduce a novel hierarchical generative model designed specifically for representing the hierarchical organization. The model can be seamlessly incorporated into existing score-based MCMC procedures for structure learning and also recovers interpretable parameters that describe cell-cell relationships. We motivate the model theoretically by considering desiderata for compositional graphical inference, as well as empirically by considering functional forms that capture the realized variation in cell data. We derive and implement an exact MCMC sampler for

the proposed model along with a scalable approximate sampler that enables inference in settings with many cell types. Finally, we apply our framework to several cutting-edge datasets in translational biomedicine, illustrating its ability to recover plausible and biologically interpretable hierarchical tissue organizations.

#### ***Bernstein - von Mises theorem for the scale parameter in inverse problems***

Presenter: Natalia Bochkina

We consider a Bayesian linear inverse problem in sequence space with a Gaussian prior where the variance of the observed noise is small and the prior variance is also estimated to adapt to the unknown smoothness of the function of interest. The posterior contraction rate of the unknown function of interest in this problem, i.e. where the prior scale parameter was estimated to achieve adaptation to the unknown smoothness of the function of interest, was studied for a direct problem over Holder classes by Szabo, van der Vaart and van Zanten (2013) and in inverse problems over Sobolev classes by Bochkina and Rodrigues (2022). Here we present the results about the local concentration of the marginal posterior distribution of the prior scale parameter. As the marginal posterior distribution with a flat hyperprior is not log concave for all true functions of interest in the small noise limit, we discuss two ways to resolve this issue: the first one is to constrain the study to the set of true functions where the log concavity holds, and the second one is use a hyperprior that guarantees the log convexity. We will also discuss concentration of the joint posterior distribution of the unknown function of interest and the prior scale parameter in these cases. The results are illustrated on simulated data.

#### ***Blinded Bayesian Sample Size Re-Estimation in Basket Trials Using Discrepancy-Based Cross-Basket Borrowing***

Presenter: Niamh Fitzgerald

Bayesian basket trials aim to increase efficiency by borrowing information across related subgroups, but the degree of between-basket heterogeneity strongly influences how much borrowing is appropriate. Because this heterogeneity is unknown at the design stage, the required sample size cannot be reliably fixed in advance, motivating the use of blinded Bayesian sample size re-estimation (SSRE). A key feature of the proposed framework is that it adaptively borrows information from both historical data and the interim data of the ongoing trial, enabling substantial sample size reductions when interim evidence suggests that a basket is similar to others.

We develop a fully blinded SSRE methodology that builds on distributional-discrepancy ideas from earlier Bayesian basket-trial work and extends earlier methodological development into a practical, simulation-ready platform. Blinded borrowing operates on two sources of information: (i) historical treatment-effect posteriors, and (ii) interim, basket-specific intercept-based proxies that serve as blinded surrogates for current treatment effects. Similarity between these sources is quantified using Hellinger-distance metrics. These discrepancies determine the slab probability of a spike-and-slab commensurate prior (CPP), and CPP-adjusted moments are then combined across baskets using softmax-weighted marginal predictive priors (MPPs), with the softmax temperature calibrated through blinded historical leave-one-out cross-validation. A robust MPP variant and empirical-Bayes comparators (vague, pooled, hierarchical, and EXNEX-type priors) are also evaluated.

For sample size re-estimation, we construct the predictive distribution of future outcomes under each candidate prior and compute predictive assurance for achieving a posterior Go decision. A binary-search procedure explores feasible total sample sizes, and isotonic (monotone-increasing) smoothing is applied to the predictive-assurance curve to remove Monte Carlo-induced non-monotonicity. Operating-characteristic simulations, scenario sweeps, and sensitivity analyses show that borrowing information jointly from interim and historical sources, under strict blinding, can deliver meaningful sample size reductions in baskets with commensurate evidence while maintaining appropriate type I error behaviour when baskets differ.

#### ***Beating the Odds: a Flexible Bayesian Model for Football Scores***

Presenter: Nick Zhang

Modelling football goals has gained significant scientific interest over recent decades, driven by the sport's cultural significance and the inherent randomness of match outcomes. Among various approaches, the Poisson model remains one of the most popular and celebrated parametric models due to its elegance, interpretability, and empirical performance. However, a key assumption of the Poisson distribution is equi-dispersion in the observations, which does not always hold true in football score data. In this work, we generalize the model using the Conway-Maxwell-Poisson (COM-Poisson) distribution, a flexible two-parameter extension of the Poisson distribution that can directly account for over- and under-dispersion. Our formulation preserves the generalized linear model structure, and inference is carried out under a Bayesian framework with latent team-specific attack, defence, and dispersion effects. Additionally, we include a model selection mechanism by introducing a spike-and-slab prior over the latent dispersion variables. The usefulness of the proposed methodology is studied in data



from the English Premier League and other major European football leagues.

### ***Inverse Probability Weighting in a Post-Bayesian World***

Presenter: Owen Thomas

We develop a Bayesian analogue of inverse probability weighting that enables posterior inference to target an idealised population distribution  $\mathcal{Q}$  rather than the biased data-generating distribution  $\mathcal{G}$ . The method estimates density ratios  $r(x) = q(x)/g(x)$  via a classifier trained to discriminate samples from  $\mathcal{Q}$  and  $\mathcal{G}$  and uses the resulting weights  $w_i = r(X_i)$  to reweight the likelihood in a generalised Bayesian update. We show that this reweighted posterior minimises  $\text{KL}(q \parallel p_\theta)$  instead of the usual  $\text{KL}(g \parallel p_\theta)$ , thereby projecting inference toward the desired target distribution while still leveraging the full sample size of the biased data. We establish some theoretical properties of the resultant posteriors, and practical considerations for classifier-based ratio estimation (cross-fitting, clipping, tempering, ESS diagnostics) are given. Simulated examples demonstrate substantial correction of selection bias in both univariate Gaussian models and multivariable regression with realistic sampling mechanisms. The framework unifies ideas from generalised Bayes, density-ratio estimation, and selection-bias adjustment, offering a principled route for Bayesian inference under heterogeneous data sources.

### ***Mapping Rainfall Events in Complex Domains: A Bayesian Semi-Parametric Approach***

Presenter: Paolo Onorati

Environmental phenomena are influenced by complex interactions among various factors. For instance, the amount of rainfall measured at different stations within a given area is shaped by atmospheric conditions, orography, and physics of water processes. Motivated by the need to analyze rainfall across complex spatial domains, we propose a flexible Bayesian semi-parametric model for spatially distributed data. This method effectively accounts for spatial correlation while incorporating dependencies on geographical characteristics in a highly flexible manner. Indeed, using latent Gaussian processes, indexed by spatial coordinates and topographical features, the model integrates spatial dependencies and environmental characteristics within a nonparametric framework. Posterior inference is conducted using an efficient rejection-free Markov Chain Monte Carlo algorithm, which eliminates the need for tuning parameter calibration, ensuring smoother and more reliable estimation. The model's flexibility is evaluated through a series of simulation studies, involving different rainfall and spatial correlation scenarios, to demonstrate its robustness across various conditions. We then apply the model to a large dataset of rainfall events collected from the Italian North-East, these areas are known for their complex orography and diverse meteorological drivers. By analyzing this data, we generate detailed maps that illustrate the mean and standard deviation of rainfall and rainy days. The method is implemented in a new R package available on GitHub.

### ***Reliable Chemical Property Prediction with BMA-Integrated Sparse Gaussian Processes and Conformal Prediction***

Presenter: Park SeongHyeok

This study proposes a trustworthy prediction framework for chemical property estimation—specifically solubility, melting point, and boiling point—of potential chemical warfare agents. Using SMILES-based molecular descriptors generated via the RDKit library, we constructed feature sets of about 9,900 samples with 216 explanatory variables. To capture nonlinear relationships, we trained Sparse Variational Gaussian Process (SVGP) regression models with multiple kernels (RBF, Matern, Rational Quadratic, etc.) and combined them through Bayesian Model Averaging using BIC-based weights. This approach improved model fit ( $R^2$ ) and reduced error (RMSE) compared to single-kernel baselines. Furthermore, Conformal Prediction was applied to calibrate predictive intervals, achieving empirical coverage close to the target 95% while shortening average interval width. These results demonstrate that integrating kernel-based Bayesian modeling with Conformal Prediction enables both accurate and reliable chemical property predictions, enhancing safety assessment in high-risk chemical contexts.

### ***Uncertainty Analysis for Bayesian Clustering under Noise***

Presenter: Qing Lily Li

Clustering in the presence of noise or inactive observations is central to many modern applications, yet remains difficult to summarise within a Bayesian framework. Although Bayesian clustering represents uncertainty through a posterior over partitions, this uncertainty is hard to interpret in practice: posterior samples can vary in the number of occupied clusters, exhibit label switching in mixture-model representations, and—when noise is present—disagree on which observations should be treated as inactive. Common summaries either collapse inference to a single representative partition or absorb noise into an ordinary cluster, which can obscure multimodality and misrepresent uncertainty. We develop a general framework for describing and summarising uncertainty in clustering under noise, applicable to both classical Bayesian models and generalised Bayesian approaches.

### ***Truncated Inverse-Lévy Representation of the Beta Process***

Presenter: Qiufei Yao

The beta process is a fundamental nonparametric prior in Bayesian modeling and machine learning. Existing inference methods often rely on the stick-breaking representation, which is limited to specific hyperparameter settings. In this work, we propose the truncated inverse-Lévy measure representation (TILE-Rep) of the beta process, which extends the decreasing-atom-weight formulation to general hyperparameters. The proposed TILE-Rep bridges the gap between previous representations and achieves lower truncation error compared to other sequential constructions. This new formulation also supports improved posterior consistency in Bayesian factor models. We demonstrate the usefulness of the TILE-Rep through applications to the beta process factor analysis model and the sparse factor model.

### ***Time-Varying Effects of Climate Risks on Inflation: The Role of Climate Concerns***

Presenter: Rangan Gupta

This paper estimates a Bayesian Time-Varying Parameter Local Projection Model with Stochastic Volatility (TVP-LPM-SV) to analyze the effects of extreme weather shocks on the inflation rate of the United States (US), by controlling for a wide array of macroeconomic and financial factors known to affect the inflation rate. The time-varying impulse responses derived from the TVP-LPM-SV over the monthly period of 2004:01 to 2025:02 is then related to a Google Trends-based Climate Concern Index (CCI) to show that size of the effect of weather shocks on the inflation rate is contingent on climate sentiment. In particular, we show that higher CCI mitigates the inflationary impact of abnormal weather shocks, underscoring the importance of information dissemination in shaping economic behavior. Our findings have significant implications for policymakers as we highlight the need to leverage on communication strategies to address the problem of inflation posed by climate change.

### ***Contraction Rates for Generalized Bayesian Nonparametric Regression under Data Contamination***

Presenter: Riccardo Lazzarini

We study the nonparametric Bayesian regression model under (adversarial) data contamination. Since standard Bayesian methods are highly sensitive to such contamination, we consider a generalized approach based on Gibbs posterior, by either introducing data-dependent weights or clipping the likelihood. We analyze the asymptotic frequentist properties of the resulting generalized posterior. We derive posterior contraction rates under general conditions for contaminated data where the number of corrupted observations can increase with the sample size. The proposed approach is widely applicable and includes, as special cases, Gaussian process and Bayesian Neural Network priors. We demonstrate our theoretical findings in a numerical analysis on synthetic data sets.

This is based on a joint work with Ismael Castillo, Paul Egels and Botond Szabo

### ***Continuous-Time Bayesian Networks with Structured Shrinkage Priors for Modelling Multimorbidity Trajectories in Large-Scale Electronic Health Records***

Presenter: Ridwan Oyebayo Olaniran

The co-occurrence of Multiple Long-Term Conditions (MLTCs) poses substantial methodological and clinical challenges, particularly in understanding how diseases interact and evolve over time. Primary Care data from Electronic Health Records (HER) allow us to follow the development of MLTCs in large numbers of people over several years, offering the ideal framework for studying such irregularly timed, interdependent disease processes. We propose a Continuous-Time Bayesian Network approach to model dependencies amongst a set of MLTCs observed in EHR data. Our model allows for multiway interactions between MLTCs and for the inclusion of exogenous covariates as risk factors. Sparsity of dependence structure amongst MLTCs is controlled via four alternative structured shrinkage priors: Hierarchical Shrinkage (HS), Laplace (LASSO), Horseshoe, and continuous Spike-and-Slab. We include in these priors order-dependent penalisation that shrinks higher-order effects more than main effects, thereby addressing the combinatorial growth of parameters and improving interpretability, sparsity, and robustness. We evaluate the CTBN models using simulation studies and illustrate the approach by applying it to large-scale UK Biobank primary care data, comprising 1.9 million clinical records from 229,725 patients. Focusing on the ten most prevalent conditions among multimorbid individuals, the model uncovers clinically plausible dependency structures.

### ***Scalable Gaussian Process Inference via Deep Generative Models: A New Framework for High-Cadence Time Series***

Presenter: Rodrigo Herrera

Gaussian Processes represent the gold standard for Bayesian time-series modeling, yet their cubic computational complexity remains a barrier for high-cadence datasets. This bottleneck is particularly restrictive when these processes are integrated as latent components within complex likelihood functions, such as Hidden Markov

Models, where iterative inference is computationally prohibitive. We introduce a Generative Surrogate framework to overcome these limitations. By utilizing a Variational Autoencoder to learn a compressed representation of a Gaussian Process prior, we map high-dimensional stochastic dependencies into a low-dimensional manifold. This transition effectively reduces inference complexity from cubic to linear. Our approach allows for the seamless integration of non-parametric priors into hybrid architectures without added cost complexity. We demonstrate this on astrophysical light curves, enabling rigorous, large-scale characterization across massive data archives.

***Bayesian design and analysis methods for decentralized clinical trials***

Presenter: Ruitao Lin

Decentralized clinical trials (DCTs) extend trial activities beyond traditional sites, enhancing access, convenience, efficiency, and result generalizability. They are particularly promising for chronic conditions like diabetes and obesity, which require longer study durations to evaluate drug effects. However, decentralized data collection raises concerns about increased variability and potential biases. This paper presents a novel Bayesian integrated learning procedure to analyze dose-response relationships using longitudinal data from a phase II DCT that combines centralized and decentralized data collection. We generalize a parametric exponential decay model to handle mixed data sources and apply Bayesian spike-and-slab priors to address biases and uncertainties from decentralized measurements. Our model enables data-adaptive integration of information from both centralized and decentralized sources. Through simulations and sensitivity analyses, we show that the proposed approach achieves favorable performance across various scenarios. Notably, the method matches the efficiency of traditional trials when decentralized data collection introduces no additional variability or error. Even when such issues arise, it remains less biased and more efficient than naïve methods that rely solely on centralized data or simply pool data from both sources.

***Bones, Bayes and Better X-rays: Safe AI Emulation for Accessible Osteoporosis Diagnosis***

Presenter: Samuel Jackson

Osteoporosis causes costly yet preventable fractures; however, gold-standard DXA scanning remains expensive, specialist-dependent and unevenly accessible. We therefore investigate whether standard X-rays can support affordable osteoporosis screening by improving simulator-based reconstruction of bone density. The key challenge is that high-fidelity X-ray simulation is too slow for clinical use, so fast surrogate models, or emulators, must be used in its place. These emulators provide predictions of simulator output together with corresponding uncertainty estimates. Our approach compares and combines anatomical image alignment, principal component analysis, Bayesian emulation, and neural-network latent representations to model the residual error between a current state-of-the-art convolution-based emulator and full photon-transport simulations.

Anatomically informed alignment proved central. After vertical and horizontal registration of wrist images, greater information could be captured with fewer principal components or latent variables, improving test-set correlation diagnostics and reducing RMSE. Variational autoencoders were limited by posterior collapse; however, denoising autoencoders produced a stable latent representation with smooth anatomical interpolation and robustness to perturbation. Overall, this work points towards safer medical AI, where black-box methods are embedded within a physics-informed, interpretable and uncertainty-aware framework. By combining AI techniques with effective simulation and statistical emulation, the project establishes a promising pipeline for future low-cost osteoporosis screening using standard X-ray equipment.

***Minimodes in heavy-tailed models and exit times for the Zig-Zag process***

Presenter: Sanket Agrawal

We investigate the existence and severity of local modes in posterior distributions from Bayesian analyses. These are known to occur in posterior tails resulting from heavy-tailed error models such as those used in robust regression. To understand this phenomenon clearly, we consider in detail location models with Student-t errors in dimension  $d$  with sample size  $n$ . For sufficiently heavy-tailed data-generating distributions, extreme observations become increasingly isolated as  $n$  goes to infinity. We show that each such observation induces a unique local posterior mode with probability tending to 1. We refer to such a local mode as a micromode. These micromodes are typically small in height but their domains of attraction are large and grow polynomially with  $n$ . We then connect this posterior geometry to computation. We establish an Arrhenius law for the time taken by one-dimensional piecewise deterministic Monte Carlo algorithms to exit these micromodes. Our analysis identifies a phase transition where a misspecified and overly underdispersed model causes exit times to increase sharply, leading to a pronounced deterioration in sampling performance.

***Bayesian Regression in the Proportional High-Noise Limit: A New Asymptotic Regime***

Presenter: Satoru Tokuda

High-resolution measurement systems often suffer from an intrinsic trade-off: increasing temporal or spatial resolution reduces signal-to-noise ratio. In many practical sensing scenarios, this leads to a situation where the observation noise variance grows proportionally with the sample size. Despite its relevance to modern measurement science, this “high-noise  $\times$  large-data” scenario is not encompassed by existing Bayesian asymptotic theories, which typically assume fixed noise levels or study parameter-dimensional scaling. We introduce the proportional high-noise regime, defined by letting the observation noise variance scale linearly with sample size. In this regime, the log marginal likelihood admits a finite-size scaling form consisting of a deterministic scaling function and a random residual. This scaling structure explains phase-transition-like changes in Bayesian model comparison and clarifies how noise-dominated data yield distinct asymptotic behavior in Bayesian inference.

***Accounting for Selection Bias in Spatial Econometric Data: An Application to Adolescent Safety Perceptions***

Presenter: Seorim Yi

Econometric data—ecological measures derived from individual reports—are increasingly used to study contextual influences on health and behavior. However, these analyses frequently encounter selection bias when the likelihood of observing a location depends on the outcome of interest (e.g., adolescents avoiding areas they perceive as unsafe). We address this challenge using data from the Adolescent Health and Development in Context (AHDC) Study, which records adolescents’ perceptions of safety at routine activity locations in Columbus, OH. We propose a Bayesian hierarchical framework that jointly models the econometric outcome and the observation process to handle informative missingness. We extend this model to account for spatial dependence in both the safety perceptions and the selection mechanism, reflecting the geographic clustering of routine activities. We demonstrate that this strategy reduces bias and improves inference regarding neighborhood safety, highlighting the importance of integrating spatial and selection mechanisms in econometric analysis.

***Moment-relevant sensitivities for Bayesian moment condition models***

Presenter: Shelly Xie

Moment condition models are popular as they offer flexibility in handling potentially complex models and data structures. Based on these models, semi-parametric estimation techniques with desirable large sample properties have been proposed for use even when facing endogeneity or non-linearities. Despite this, in finite samples, statistical analyses may be sensitive to ‘outliers’ in the data which may suggest very different conclusions to that of the original data. This paper investigates the sensitivity of generalized method of moments (GMM) and exponentially tilted empirical likelihood (ETEL) estimators when confronted with data contamination, and illustrates this sensitivity across various scenarios and model specifications. Additionally, the study examines potential moment-related frailties of a Bayesian ETEL posterior given moment conditions and in light of available data. Leveraging off of GMM, we then develop case influence measures suitable to investigate ETEL sensitivities. The approach may help the analyst to better understand both the strengths and weaknesses of (estimated) moment condition models.

***Bayesian Quadratic Factor Regression for High-Dimensional Categorical Data***

Presenter: Shinji Tsukamoto

Categorical covariates frequently arise in various types of data, such as marketing and medical data, and they often exert substantial influence on the outcome variable. We consider settings in which many categorical variables jointly affect a response, and propose a Bayesian quadratic factor regression model tailored to such high-dimensional categorical covariates. Specifically, we assume that each categorical variable is driven by an underlying continuous latent variable, impose a latent factor structure on these latent variables, and then build a quadratic regression model using the latent factors. To obtain a parsimonious yet flexible representation, we employ multiplicative gamma process (MGP) shrinkage priors on both the factor loadings and the coefficient matrices associated with interaction terms. We extend the standard MGP prior for hierarchical Schur complement structures of symmetric coefficient matrices, yielding a natural regularization scheme for the interaction coefficients. Posterior inference is conducted via Markov chain Monte Carlo. Simulation studies demonstrate that the proposed method outperforms well-known multivariate analysis techniques in terms of both predictive accuracy and interpretability, and we further illustrate its practical utility through real-data applications.

***Bayesian Local Influence for Autoregressive Time Series Models***

Presenter: Shuangzhe Liu

Autoregressive (AR) models are fundamental tools in time series analysis. This talk introduces a unified Bayesian diagnostic framework for AR modeling, illustrated with two examples: the skew-normal autoregressive (SNAR) model and the Bayesian vector autoregressive (VAR) model. Using MCMC methods for estimation, we develop

Bayesian local influence diagnostics based on perturbations to priors, variances, and data, evaluated through Bayes factors, phi-divergence, and posterior means. Simulations demonstrate the effectiveness of the approach. Applications to the 2015 Shanghai Composite Index returns (SNAR) and U.S. macroeconomic series (VAR) show how Bayesian diagnostics can identify influential observations and assess model stability in practice.

***Prior Effective Sample Size assessment via predictive sampling***

Presenter: Silvia Calderazzo

Bayesian clinical trials facilitate incorporation of external information via the specification of informative prior distributions. A key challenge in this context is quantifying the amount of information that the inclusion of external data, via the prior distribution, adds to the current trial analysis. This aspect is particularly important when considering that informative priors are often adopted when the number of recruitable patients for the current trial is limited and the prior therefore plays a strong role in the trial's decisions. One intuitive metric for prior informativeness is the Effective Sample Size: such metric equates the information contained in the prior to a certain number of virtual samples added to the current data analysis. Various prior ESS proposals have been developed, e.g., Morita et al. (2007) and the ELIR metric by Neuenschwander et al (2020), while Reimherr et al. (2021) and Wiesenfarth and Calderazzo (2020) focused on combining prior ESS and prior-data conflict assessments. We develop a sampling-based approach to ESS calculation which builds on distributional divergences. Such metric could provide additional insights into the well-established ELIR metric. This work aims to eventually provide an efficient algorithm for ESS computation easily applicable to arbitrary prior and likelihood specifications.

***Posterior Contraction Rates for Subspace Estimation through Dense Factor Models***

Presenter: Soham Ghosh

Principal Component Analysis (PCA) is a heavily used statistical tool for low-rank estimation and dimension reduction. Traditionally, some low-rank structure, like sparsity, is assumed for estimating PCs in high dimensions. However, very recently, consistent estimation of PCs has been established under the low effective rank assumption ( $r_e(\Sigma) = \text{tr}(\Sigma)/\|\Sigma\|$ , where  $\text{tr}$  denotes trace,  $\|\cdot\|$  denotes operator norm). In this work, we propose a Bayesian method for estimating the principal subspace that achieves the frequentist optimal rate. Instead of assuming sparsity, we assume a relatively large signal-to-noise ratio, ensuring that the effective rank remains small. We use a very simple prior that facilitates the formulation of an efficient computational strategy.

***Latent variable estimation with composite Hilbert space Gaussian processes***

Presenter: Soham Mukherjee

We develop a scalable class of models for latent variable estimation using composite Gaussian processes, with a focus on derivative Gaussian processes. We jointly model multiple data sources as outputs to improve the accuracy of latent variable inference under a single probabilistic framework. Similarly specified exact Gaussian processes scale poorly with large datasets. To overcome this, we extend the recently developed Hilbert space approximation methods for Gaussian processes to obtain a reduced-rank representation of the composite covariance function through its spectral decomposition. Specifically, we derive and analyze the spectral decomposition of derivative covariance functions and further study their properties theoretically. Using these spectral decompositions, our methods easily scale up to data scenarios involving thousands of samples. We validate our methods in terms of latent variable estimation accuracy, uncertainty calibration, and inference speed across diverse simulation scenarios. Finally, using a real world case study from single-cell biology, we demonstrate the potential of our models in estimating latent cellular ordering given gene expression levels, thus enhancing our understanding of the underlying biological process.

***A Bayesian Biodiversity Attention Index: Linking Public Attention, Biodiversity Risk, and Capital Assets in South Africa***

Presenter: Sonali Das

Biodiversity loss represents a material, yet under-quantified, risk to financial markets and capital assets. This study develops a Biodiversity Attention Index (BAI) for South Africa using Google Trends data from 2004 to 2025. The index is constructed through two complementary dictionaries: an internationally established biodiversity lexicon (Giglio et al., 2023), and a South Africa-specific biodiversity vocabulary. To address uncertainty in indicator construction and signal extraction, we integrate Bayesian inference, enabling the combination of prior knowledge, namely, that biodiversity loss erodes natural and financial capital, with observed patterns of public and market attention. By updating priors with empirical search data, the Bayesian framework provides posterior probabilities that quantify the likelihood of biodiversity-related risks materialising in capital-intensive sectors in South Africa such as mining, agriculture, and tourism. This approach extends existing biodiversity finance



literature by explicitly incorporating uncertainty into biodiversity attention measures (Rowland et al., 2021; Wilhere, 2021). The findings contribute to ongoing debates on systemic biodiversity-related risks (Mies, 2025; Thompson, 2023) and informs both policymakers and investors seeking to align conservation objectives with financial stability.

***Bayesian Causal Inference Using Identifying Moment Conditions: A Moment-Based Framework with Applications to Clinical Data***

Presenter: Sooha Cho

Outcome-model misspecification remains a fundamental barrier in clinical causal inference, particularly in settings with high-dimensional covariates or weak overlap. We attribute this vulnerability to structural limitations of likelihood-based Bayesian estimators and propose a Bayesian moment condition model that replaces the full outcome likelihood with identifying moment conditions for the causal estimand. The same moment-condition structure naturally extends to instrumental-variable analyses, providing a unified Bayesian approach for both standard causal estimands and settings where identification is inherently moment-based. Simulation studies and clinical applications demonstrate that the proposed framework stabilizes estimation under weak overlap and substantially reduces sensitivity to outcome-model misspecification.

***Frequency-Aware Prior Adaptation for Variational Autoencoders under Class Imbalance***

Presenter: Soomin Kwon

Variational Autoencoders (VAEs), as Bayesian latent variable models optimized via variational inference, often produce distorted latent spaces under class imbalance because the isotropic Gaussian prior excessively penalizes minority classes. We introduce the Frequency-Aware Prior VAE (FAP-VAE), a class-conditional empirical-Bayes framework that jointly learns variational posteriors and class-specific priors. Each class prior is parameterized by its own mean and variance, initialized inversely to class frequency and refined through exponential moving averages (EMAs) of encoder-derived statistics. This adaptive scheme preserves minority-class variance while preventing over-regularization. Training proceeds in a two-stage alternating process that decouples posterior and prior updates, ensuring stable optimization and preventing feedback collapse. FAP-VAE provides a principled interpretation of data-dependent prior adaptation and yields a uniform KL-divergence bound independent of class frequency. Empirically, FAP-VAE achieves consistently higher normalized mutual information (NMI) and adjusted Rand index (ARI) across seven benchmarks, maintaining reconstruction and generation quality. These findings highlight how frequency-aware Bayesian prior modeling enhances the robustness and interpretability of deep generative representations under severe imbalance.

***Bayesian analysis of maxima within time series***

Presenter: Stefano Rizzelli

Bayesian inference for extremes based on the Block Maxima method is widely used, but existing asymptotic theory is largely restricted to independent observations, even though the independence assumption is unrealistic in many practical extreme value applications. For stationary time series, fundamental questions about posterior validity and uncertainty quantification remain unresolved. We develop the first asymptotic framework for Bayesian Block Maxima inference under serial dependence. Working with a misspecified Generalized Extreme Value (GEV) model, we establish posterior consistency (quantifying contraction rates), Bernstein–von Mises theorems, and asymptotic validity of credible intervals for GEV parameters, T-time-horizon return levels, and extreme Value at Risk—popular risk measures in environmental science and finance. We further study inference for the extremal index, whose reciprocal measures the mean size of clusters of consecutive extremes over time, and show that naive posterior constructions can exhibit poor coverage. To address this, we propose an adjusted posterior distribution with substantially improved inferential performance. Simulation studies demonstrate strong finite-sample behavior of the proposed methods across a range of dependent time series models. This is a joint work with David Carl and Simone Padoan (Bocconi University).

***Bayesian Latent Cluster Models for Multivariate Regression with Mixed-Type Responses***

Presenter: Suyeon Kang

We propose a Bayesian latent cluster regression model for multivariate outcomes of mixed types, designed to capture heterogeneous predictor-outcome relationships in county- and state-level public health data. Each latent cluster is characterized by a reduced-rank regression coefficient matrix, enabling the discovery of subpopulation-specific associations while simultaneously performing dimensionality reduction. Pólya–Gamma augmentation provides efficient posterior computation for non-conjugate likelihoods. To achieve adaptive rank selection and cluster-specific shrinkage, we employ Multiplicative Gamma Process priors on the low-rank factors. We apply the method to two real-world datasets: COVID-19 county data and U.S. influenza surveillance. Across both

applications, the model recovers interpretable latent clusters and produces well-calibrated predictive distributions, uncovering meaningful heterogeneity in public health patterns. These results demonstrate the value of cluster-aware low-rank modeling for analyzing heterogeneous multivariate health data with mixed likelihoods.

***A Bayesian Decision-Analytic Framework for Robust Inference in Dynamic and Adversarial Environments***

Presenter: Tahir Ekin

Modern industrial systems increasingly rely on data-driven decision-making under uncertainty, where evolving conditions and adversarial manipulation can compromise model reliability. We develop a Bayesian decision-analytic framework for robust sequential outlier detection in such dynamic environments, integrating adversarial risk analysis (ARA) with dynamic data-driven application systems. The framework models the adversary's decision process through uncertain utilities and beliefs, yielding posterior predictive distributions over attack strategies and their operational impacts. To address scalability challenges in large or continuous decision spaces, we introduce Augmented Probability Simulation (APS) as a computationally efficient method for approximating Bayesian posteriors while preserving coherent uncertainty propagation. Temporal dependencies are captured through embedded time-series components, enabling adaptive defense as new information arrives. Furthermore, Retrieval-Augmented Generation (RAG) modules synthesize contextual industrial text data and expert knowledge into the Bayesian updating process. A healthcare fraud-detection case study illustrates how APS-based inference and RAG-enhanced information fusion improve resilience and reliability. The proposed framework advances Bayesian robustness, decision support, and uncertainty quantification for complex, adversarial industrial data environments.

***Adversarial Risk and Dynamic Defense Decisions in Sequential Bayesian Count Models***

Presenter: Tevfik Aktekin

Count data models are widely used in applications such as cyber-attack detection, financial risk modeling, and online user behavior analysis, where sequential inference is essential. The Poisson-Gamma State-Space (PGSS) model provides a flexible Bayesian framework for modeling non-stationary count processes. However, in adversarial settings, these models are susceptible to data poisoning attacks, where an adversary perturbs observations to manipulate forecasts or disrupt decision-making. We develop an Adversarial Risk Analysis (ARA) framework for PGSS models, formulating an attacker-defender game in which the adversary strategically perturbs count observations to maximize the divergence between predicted and true distributions while minimizing attack costs. Our approach accounts for different attack strategies, varying in the attacker's knowledge of the model. We evaluated the impact of these attacks on Bayesian filtering and demonstrated our framework using both simulated data and real-world data subject to adversarial perturbations. This study extends state-space models to adversarial environments, offering a principled methodology to analyze sequential count data under strategic manipulation. Our findings provide insights into the robustness of Bayesian inference in adversarial settings and contribute to the development of counter-adversarial techniques for stochastic process modeling.

***Bayesian Neuronized Knockoff Supervised Principal Component Analysis***

Presenter: Theodosios Papazoglou

Sparse principal component analysis (SPCA) and sparse supervised principal component analysis (SSPCA) extend traditional PCA and supervised PCA by achieving simultaneous dimensionality reduction and variable selection. While these methods are deterministic and make no distributional assumptions, probabilistic formulations have been proposed that reinterpret the loading matrices as maximum likelihood estimators. Bayesian extensions of SPCA have successfully incorporated suitable priors to promote sparsity and allow for uncertainty quantification; however no analogous Bayesian framework has yet been developed for SSPCA. Moreover, existing approaches—both for SPCA and SSPCA—do not address the false discovery rate (FDR) control, which can result in overly liberal variable selection and potentially unreliable conclusions. To address these limitations, namely, the absence of a principled Bayesian framework for SSPCA and the lack of FDR control, we develop Bayesian Neuronized Knockoff Supervised PCA (BNK-SPCA). Our model integrates the knockoff filter directly into a fully Bayesian supervised probabilistic PCA framework and enforces sparsity via a neuronized prior, providing rigorous FDR-controlled variable selection with respect to response variables. Simulations and real-data analyses demonstrate that BNK-SPCA successfully controls FDR even in high-dimensional and noisy settings where current state-of-the-art methods fail.

***Bayesian Cross-Sectional Analysis of the Development of Future-Time Reference in Second-Language Spanish Learners***

Presenter: Thomas Metzger



Despite the growing prominence of Bayesian methods in social sciences, their adoption within applied linguistics, including second-language acquisition, remains limited. In this study, we apply a Bayesian framework to a cross-sectional investigation of future-time reference (FTR) usage among Spanish second-language learners. In Spanish, a speaker can choose to convey FTR with various verb forms. We model this FTR dependent variable with three levels (the three most frequent verb forms) as a function of linguistic fixed and random effects. Specifically, we consider the presence of words that convey uncertainty and the semester of study as fixed effects, and the individual participants and task items as random effects. By leveraging Bayesian modeling, we capture learner variability and item-level heterogeneity more robustly than frequentist approaches, while still quantifying the strength of linguistic effects. We advocate for broader integration of Bayesian methods in applied linguistic research, emphasizing their capacity to accommodate complex data structures and enhance inferential transparency.

#### ***A Bayesian Panel Data Study on Socioeconomic Factors Influencing Exercise Habits in Japan***

Presenter: Tomoki Toyabe

Exercise habits is a central topic in health research because it is closely linked to dietary behavior, sleep patterns, self-esteem, and a range of related determinants. In this study, we investigate the degree to which individuals' exercise habits are shaped by their socioeconomic characteristics. Using panel data from a nationally representative sample of the Japanese population, we estimate models to capture potential heterogeneity in these associations. Methodologically, we employ a Bayesian panel logit framework, estimated via a Markov chain Monte Carlo algorithm combined with the Ancillarity-Sufficiency Interweaving Strategy to achieve efficient inference.

#### ***Hypergeometric weight priors for density estimation***

Presenter: Tsunehiro Ishihara

Geometric weight prior is a simple alternative of Dirichlet process prior in Bayesian nonparametric density estimation. Recently by De Blasi et al.(2020) extends geometric weight prior with Poisson and general negative binomial prior for component weights. In this poster presentation, first we point out the relationship between geometric weight priors and Bissinger system of distributions. Second, we introduce several extensions of geometric prior including hypergeometric functions and their properties. Finally, we propose an MCMC algorithm for the proposed priors.

#### ***Bayesian Variable Selection on Small Sample Trial Data via Adaptive Posterior-Informed Shrinkage Prior***

Presenter: Vincent Tan

Identifying variables associated with clinical endpoints is a central objective in clinical trials. With the rapid expansion of cell and gene therapies (CGTs) and therapeutics for ultra-rare diseases, there is an urgent demand for statistical methods capable of detecting meaningful associations under severe sample-size constraints. Motivated by data-borrowing strategies for historical controls, we propose the Adaptive Posterior-Informed Shrinkage Prior (APSP), a Bayesian framework that adaptively leverages external information to improve variable-selection efficiency while preserving robustness across diverse scenarios. APSP extends existing Bayesian borrowing methods by incorporating data-driven adaptive information selection, mixture shrinkage informative priors, and empirical-null-based decision rules, thereby enhancing variable-selection performance in small-sample settings. Extensive simulations demonstrated that APSP achieves superior efficiency compared with traditional and widely used Bayesian data-borrowing and variable-selection approaches, while maintaining robustness across a broad range of simulation scenarios. Finally, we applied APSP to the Clinical Islet Transplantation (CIT) Consortium, identifying variables associated with peak C-peptide at Day 75 in study CIT06 by borrowing information from study CIT07.

#### ***Predictive Resampling for High-Dimensional Regression***

Presenter: Wanyue Sun

The martingale posterior framework generalizes traditional Bayesian inference by focusing on a sequence of one-step-ahead predictive densities. In this framework, predictive resampling, which involves the imputation of unseen observables, is conducted to provide a computationally efficient alternative to posterior sampling. Recent work has further combined parametric plug-in predictive with stochastic gradient descent to define a parametric version of the martingale posterior. While this framework is guaranteed to reproduce exact posteriors in low-dimensional regression settings, a fundamental challenge arises in the high-dimensional regime: when the empirical distribution is employed as the design distribution for covariates, it leads to non-identifiability, thus violating Doob's assumptions. This project will propose a generalized predictive resampling framework within the parametric martingale posterior for high-dimensional regression, which induces a design distribution with

full-rank covariance, along with the modified recursive updates to maintain the asymptotic convergence to the correct posterior variance. We will develop a theoretical framework characterizing the role of design distribution in regression, which establishes a fundamental connection to a predictive version of Bayesian ignorability. Our main methodological contribution is the construction of predictive sequences that satisfy a weaker version of the ignorable property and applicable to a broad range of models, including generalized linear models and Student-t regression model.

### ***Bayesian Fusion Learning for Subgroup Analysis***

Presenter: Weitao Hu

Subgroup identification is essential for uncovering heterogeneity among individuals, yet existing Bayesian approaches often rely on mixture modeling assumptions that may perform less effectively when subgroup boundaries are weak or overlapping. This work introduces a structured shrinkage framework that bridges Bayesian and frequentist perspectives for subgroup analysis. By imposing spike-and-slab (lasso) priors with binary latent indicators on pairwise relationships, our approach adaptively shrinks individualized regression coefficients and automatically recovers latent subgroup structures. We further develop an efficient expectation-maximization (EM) algorithm for mode detection and a Gibbs sampling scheme for full posterior inference. Extensive simulations and two real data applications demonstrate that the proposed methods achieve robust subgroup recovery, interpretability, and computational efficiency, even in challenging heterogeneous settings. The framework not only provides a flexible and principled Bayesian foundation for subgroup analysis but also serves as a Bayesian alternative to existing frequentist fusion-based methods, offering uncertainty quantification and reliable inference for complex data.

### ***Bayesian regression framework for models with circular components***

Presenter: Xiang Ye

Advancements in computational power and methodologies have enabled research on massive datasets. However, tools for analyzing data with directional or periodic characteristics, such as wind directions, instrument turning angles and time-of-day effect, remain underdeveloped. While statisticians have proposed circular distributions for such analyses, significant challenges persist in constructing circular statistical models, particularly in the context of regression modeling, since the concept of building a linear predictor from linear combinations of covariates and various random effects, breaks the circular nature of the variable. We provided principled priors for the hyperparameters of circular distributions and developed a well-defined Bayesian circular regression framework for handling concentrated circular data. Our approach extends naturally to joint regression models where we can have several circular and non-circular responses, and allow us to handle a mix of linear covariates, circular covariates and various random effects. Our formulation aligns naturally with the integrated nested Laplace approximation (INLA), which provides fast and accurate Bayesian inference.

### ***Bayesian Forecast Combination with Forward-Looking and Robust Decision Feedback***

Presenter: Xiaorui Luo

We develop a Bayesian forecast combination framework that embeds forward-looking signals—formulated as predictive priors—directly into the latent process governing time-varying combination weights. This allows the weights to adaptively reflect both historical forecast performance and anticipated model behavior. As a concrete implementation, we use model diversity as the forward-looking signal and obtain a diversity-driven time-varying weights (DTVW) scheme. Empirical applications to multi-step-ahead oil price forecasts and to bivariate forecasts of U.S. inflation and GDP growth show that DTVW consistently outperforms classical combination methods, including equal-weighting, Bayesian model averaging, and standard time-varying weights. Beyond accuracy gains, the diversity-based predictive priors offer diagnostic information on model incompleteness and forecast uncertainty, making DTVW more adaptive and more informative than existing Bayesian combination approaches. While these results concern the forecasting stage, many real-world applications ultimately assess forecasts through the decisions they support. Building on this foundation, we then incorporate the decision-making environment, which is often the ultimate driver of value in real-world forecasting. We introduce a prediction–decision feedback loop in which forecasts guide decisions, and the realized consequences of those decisions feed back into the Bayesian posterior of the combination framework. This loop supports dynamic refinement of both the weight structure and the decision rule over time. To enhance robustness and stability, we integrate tools from robust optimization, ensuring that recommended actions remain reliable under model uncertainty and misspecification. The resulting framework delivers a unified Bayesian predictive–decision cycle in which forward-looking signals shape the predictive priors and decision feedback updates the posterior, jointly improving both forecasting accuracy and decision robustness.

### ***A Bayesian Framework for Quantifying Directional Dependence in Gene Regulation***

Presenter: Xiaoying Wei

Understanding directional relationships in gene regulation is essential for unraveling disease mechanisms and identifying therapeutic targets. This work develops a Bayesian framework to quantify directional asymmetry in dependence between variables describing gene expression, extending a frequentist copula-based method originally designed for brain connectivity. The frequentist approach quantified uncertainty only through bootstrap resampling, whereas our proposed Bayesian formulation produces posterior distributions for directional dependence measures, providing information on both the strength and the direction of the dependence. We propose a Bayesian framework that addresses these challenges by replacing bootstrap resampling with Hamiltonian Monte Carlo sampling. This approach produces full posterior distributions for directional dependence measures, enabling robust uncertainty quantification even in small-sample scenarios. Our method quantifies the asymmetry in dependence by comparing the strength of influence in both directions ( $U \rightarrow V$  versus  $V \rightarrow U$ ) and provides probabilistic statements about which direction dominates. Importantly, this probabilistic treatment enhances interpretability and stability compared to bootstrap-based inference, offering researchers direct answers to questions like “what is the probability that gene A drives gene B rather than the reverse?” The Bayesian formulation naturally accommodates hierarchical modeling structures and enables coherent propagation of uncertainty throughout the analysis. Through comprehensive simulations, we demonstrate that the Bayesian approach yields more precise and consistent directional estimates than the frequentist method, with critical improvements emerging at moderate sample sizes. We then apply the framework to multi-omic datasets on chromatin accessibility, histone modifications, and gene expression. The analysis reveals directional patterns consistent with established biological regulatory mechanisms, demonstrating the method’s robustness for small-sample, high-noise genomic analyses. This work provides researchers with a principled, interpretable tool for uncovering directional relationships in genomic data.

### ***Optimal Test Planning for Heterogeneous Wiener Processes***

Presenter: Ya-Shan Cheng

Degradation models based on heterogeneous Wiener processes are commonly used to assess information on the lifetime of highly reliable products. An optimal test plan given limited resources is generally obtained using numerical methods for heterogeneous Wiener processes. However, numerical searches for optimal test plans have the disadvantage of being time-consuming and may provide unclear explanations for the findings. To overcome these difficulties, we derive an explicit expression for decision variables (such as the termination time, number of measurements, and sample size) of D- and V-optimal test plans with cost constraints. The theoretical results not only ensure that the optimal test plan is found, but also provide clear insights into the decision variables affected by model parameters and experimental costs. Some numerical examples are presented to support the efficiency and applicability of the optimal test plans.

### ***Geometric Ergodicity of Gibbs Algorithms for a Normal Model With a Global-Local Shrinkage Prior***

Presenter: Yasuyuki Hamura

We consider Gibbs samplers for a normal linear regression model with a global-local shrinkage prior and show that they produce geometrically ergodic Markov chains. First, under the horseshoe local prior and a three-parameter beta global prior under some assumptions, we prove geometric ergodicity for a Gibbs algorithm in which it is relatively easy to update the global shrinkage parameter. Second, we consider a more general class of global-local shrinkage priors. Under milder conditions, geometric ergodicity is proved for two- and three-stage Gibbs samplers based on rejection sampling. We also construct a practical rejection sampling method in the horseshoe case. Finally, a simulation study is performed to compare proposed and existing methods.

### ***Copula-based Bayesian Reliability Analysis of Multiple Degradation Model***

Presenter: Yi-Fu Wang

To determine how long a product could be used, manufacturers often conduct the reliability analysis to investigate the product’s lifetime. With the widespread adoption of high-tech products, traditional methods that rely on complete failure time data have become less feasible. Alternatively, degradation analysis is now commonly used to predict the product’s lifetime based on the observed quality characteristics (QC) which are strongly associated with lifetime. Sometimes, we may encounter situations that a product can be measured by multiple QC, where the correlation among multiple QC is unavoidable. In this study, we propose a copula-based multivariate degradation process (CMDP) model under a Bayesian framework to capture such correlation. Specifically, the copula function is employed to model the dependence structure among those marginal processes. The Bayesian reliability prediction of CMDP model can be conducted by using Markov chain Monte Carlo (MCMC) method.

Further, the simulation study and real-data analysis are performed to demonstrate the practical utility of the proposed model.

### *The minimax optimal convergence rate of posterior density in the weighted orthogonal polynomials*

Presenter: Yiqi Luo

We study Bayesian density estimation based on orthogonal polynomial expansions under a unified weighted  $L_2$  framework, addressing the challenge of constructing priors that attain minimax optimal convergence rates for mixture models—a crucial problem in non-parametric Bayesian density estimation. The true density is represented via a sequence of orthogonal basis functions on a bounded interval,  $\mathbb{R}^+$  or  $\mathbb{R}$ . While classical results by Ghosal et al. (2000) and Shen & Wasserman (2001) provide general sufficient conditions for achieving the minimax optimal posterior convergence rate  $n^{-p/(2p+1)}$ , constructing priors that satisfy these conditions across diverse polynomial systems remains a challenging open problem. Existing approaches are typically basis-specific, restricted to bounded supports, or yield only suboptimal rates with additional logarithmic factors. To overcome these limitations, we propose a new *Gaussian sieve prior* for expansion coefficients, where the variance structure is linked to the integration constants of the chosen orthogonal polynomials. Our theoretical contributions are threefold:

- The minimax optimal convergence rate  $n^{-p/(2p+1)}$  is achieved by the proposed prior for any integer  $p \geq 1$  under both compact and non-compact parameter spaces;
- The sharper upper bounds on posterior probabilities are obtained by controlling the prior predominantly supported on a set with mass nearly 1;
- A constructive link between weighted  $L_2$ -entropy and Hellinger-entropy is established to address non-uniformly bounded densities.

Numerical experiments using Legendre expansions on bounded intervals, Laguerre expansions on  $\mathbb{R}^+$ , and Hermite expansions on  $\mathbb{R}$  corroborate the theoretical findings. Across all settings, the posterior mean estimator achieves accurate recovery of the true density and exhibits convergence behavior consistent with the derived minimax optimal convergence rate. Overall, this work presents the first general-purpose prior construction for polynomial-based Bayesian density estimation that is simultaneously support-agnostic, basis-flexible, and provably minimax optimal—substantially broadening the applicability of polynomial-expansion Bayesian methods in the statistical modeling.

### *Importance inference of optimal test planning for degradation analysis*

Presenter: Yishian Dong

Determination of the decision variables such as the inspection period, number of measurements, and sample size is crucial for planning an efficient degradation test. For widely used stochastic processes, the necessary and sufficient conditions for the explicit expression of optimal decision variables can be derived by minimizing the approximate variance of an estimator of interest under a limited budget. The importance of the decision variable is proposed to study the rate at which the objective function improves with the decision variable. The necessary and sufficient conditions for determining the importance of the optimal decision variables are theoretically investigated to elucidate the effect of the experimental costs and model parameters. Furthermore, the relative rankings of the importance of the optimal decision variables are illustrated through numerical examples.

### *Divide and conquer sequential Monte Carlo for chained Markov melding*

Presenter: Yixuan Liu

A practical motivation for Bayesian inference is to construct models that integrate information from multiple heterogeneous data sources. To this end, the fully Bayesian analysis requires a joint model that incorporates all parameters and separated datasets. Thus, the joint model often becomes complex and computationally expensive. Alternatively, a more feasible way is to model the sources separately and subsequently combine all submodels. Within this framework, Goudie et al. (2019) propose Markov Melding, a generic method for integrating multiple sources that share a common parameter, and Manderson and Goudie (2023) later extend this approach to more general relationships between submodels. However, this method becomes challenging when the number of submodels is large. Our current work focuses on addressing this problem by adopting a Sequential-Monte-Carlo-based method proposed by Lindsten et al. (2017), which offers efficient sampling for hierarchical structures. Based on the real-world and simulation studies, we demonstrate that our method is competitive with the original Markov Melding approach in settings with three submodels and exhibits comparable performance to full MCMC in cases involving five submodels. Ideally, this new extension will remain robust, enabling applications to scenarios with an arbitrarily large number of submodels.

### *Bayesian bootstrap beyond observation times*

Presenter: Ylenia Francesca Buttiglierio

The bootstrap, with all its extensions, is intrinsically defined on the empirical distribution at observed data points and, therefore, provides inference only for those time points. Our work focuses on extending bootstrap procedures for inference at unobserved time locations or along continuous-time trajectories, in settings with multiple observation times and potentially more samples per time point. We work in a Bayesian framework, which models uncertainty continuously over time by placing a prior on the latent process governing the data. Specifically, we study the Fleming–Viot hidden Markov model and its associated smoothing distribution: the Markov dynamics capture temporal dependence, while the smoothing distribution enables principled interpolation between observations. The nonparametric signal evolves according to Fleming–Viot (FV) dynamics, a stochastic process over the space of probability measures that, in our setting, provides a temporal extension of the Dirichlet process (DP). In analogy with the Bayesian bootstrap, we examine the non-informative prior specification, obtained as the concentration parameter of the FV process tends to zero. We show that the resulting law at intermediate times is a finite mixture of DPs supported on a subset of the observed data, where the retained data points are determined by Kingman’s coalescent, a pure-death process dual to the FV dynamics. Building on this characterization, we propose an extended Bayesian bootstrap that repeatedly samples probability measures from this distribution. This scheme, together with the coding framework we developed, allows computing arbitrary statistics and inferring their values at intermediate times, providing also a measure for the associated uncertainty.

### ***Estimation of the null proportion in large-scale simultaneous hypothesis testing***

Presenter: Yong Wang

We present a new method for estimating the proportion of null effects—an essential yet challenging task in large-scale simultaneous hypothesis testing. Our approach naturally leverages a nonparametric Bayes framework and makes a novel use of the profile likelihood function. We also outline the accompanying computational techniques. Numerical studies show that the resulting estimator exhibits a clear convergence pattern and consistently outperforms existing methods across a wide range of scenarios.

### ***Sampling via Gaussian Mixture Approximation***

Presenter: Yongchao Huang

We present a family of *Gaussian Mixture Approximation* (GMA) samplers for sampling unnormalised target densities, encompassing *weights-only GMA* (W-GMA), *Laplace Mixture Approximation* (LMA), *expectation-maximization GMA* (EM-GMA), and further variants. GMA adopts a simple two-stage paradigm: (i) initialise a finite set of Gaussian components and draw samples from a proposal mixture; (ii) fit the mixture to the target by optimising either only the component weights or also the means and variances, via a sample-based KL divergence objective that requires only evaluations of the unnormalised density, followed by stratified resampling. The method is gradient-free, and computationally efficient: it leverages the ease of sampling from Gaussians, efficient optimisation methods (projected gradient descent, mirror descent, and EM), and the robustness of stratified resampling to produce samples faithful to the target. We show that this optimisation-resampling scheme yields consistent approximations under mild conditions, and we validate this methodology with empirical results demonstrating accuracy and speed across diverse densities.

### ***Applying Bayesian Evidence Synthesis for the evaluation of cognitive models: A case of learning curves***

Presenter: Yui Furukawa

Cognitive modeling promotes understanding by formalizing psychological processes with mathematical models, and examining existing models is a valuable endeavor in this field. Specifically, it is important to examine whether a model is consistently supported across different datasets (generalizability) and to investigate how its behavior differs across datasets obtained under different conditions (context dependency). The framework of secondary data analysis, in which studies are conducted using existing data, allows the use of diverse datasets that differ in experimental conditions and other aspects, and is therefore useful for these investigations. While some studies have examined models using multiple secondary datasets, methods for comparing models in a dataset-integrated manner have not been systematically explored and have largely been left to individual researchers’ discretion. In this study, we aim to investigate the usefulness of Bayesian Evidence Synthesis (BES; Kuiper et al. 2013, *Sociol. Methods Res.*) for examining the generalizability and context dependency of cognitive models. BES synthesizes evidence across datasets by taking the product of Bayes factors and has recently gained attention as a flexible alternative to meta-analysis. As a case, we focus on learning curve models. Following Evans et al. (2018, *Psychol. Rev.*), we compare four models of learning curves using 17 datasets collected by Heathcote et al. (2000, *Psychon. Bull. Rev.*). We compute Bayes factors for each dataset and apply BES to integrate them,

thereby illustrating how this approach contributes to understanding the generalizability and context dependency of cognitive models.

### ***Bayesian variable and hazard structure selection for general hazard model***

Presenter: Yulong Chen

The proportional hazards (PH) and accelerated failure time (AFT) models are the most widely used hazard structures for modelling time-to-event data. When the goal is to identify the variables that explain the times to event, researchers typically begin by choosing their preferred hazard structure and then perform variable selection within that structure. However, by choosing one of these specific hazard structures, restrictions are being imposed on how the covariates could affect the hazard. For example, PH model assumes covariates having effect only on hazard level, while AFT model assumes covariates having the exact same effect on both hazard and time level. To achieve simultaneous selection of both the variables and the hazard structure, we perform Bayesian variable selection within a general class of hazard-based regression models, namely the general hazard (GH) model, that includes the PH and AFT structures, among others, as special cases. The main challenges lie in the specifications of the priors on model coefficients and model space that properly account for the different roles of the variables and result in reasonable posterior distributions of the hazard structures. We consider two types of g-priors for the regression coefficients of such class of models that enable a tractable implementation of the proposed methodology. We show that both priors lead to consistent model selection. We also develop a new type of prior on the model space that accounts for model multiplicity and penalises model size from three levels, allowing inference on sub-models. Correspondingly, an extension of Add-Delete-Swap sampler is also developed to move between these sub-models and obtain samples of both variables and hazard structures. A simulation study demonstrates that the proposed methodology can effectively identify both the hazard structure and the active variables, while also illustrating the interplay between sample size and censoring rates. Finally, we present two real-data applications: one in which the data clearly favour one of the hazard structures under consideration, and another where there is substantial uncertainty about both the active variables and the underlying hazard structure, along with the comparison of selection results using existing methodologies.

### ***Median Bayes factors for robust model selection***

Presenter: Zexian Wang

We propose median Bayes factors for Bayesian model selection under heavy-tailed errors by modifying the definition of the posterior model probabilities to preserve model selection consistency when the working model is misspecified. We illustrate the use of median Bayes factors, leading to pseudo-posterior model probabilities, in the context of variable selection, where existing studies have primarily focused on well-specified settings or the assumption that the errors have sub-Gaussian tails. We demonstrate that a strong form of model selection consistency can still be achieved with less restrictive assumptions under our general framework, which can incorporate a broad class of priors on the model space as well as the regression parameters under a putative model. The superior performance of the pseudo-posterior model probabilities as compared to the conventional posterior model probabilities is illustrated through extensive simulation experiments.



## Tuesday 29 June

### Multitrack Sessions 3: 9:00–10:30

#### Bayesian Deep Learning (Room 1101)

##### *Federated ADMM from Bayesian Duality*

Presenter: Thomas Moellenhoff

ADMM is a popular method for federated deep learning which originated in the 1970s and, even though many new variants of it have been proposed since then, its core algorithmic structure has remained unchanged. In this talk, we present a fundamentally new way to derive and extend federated ADMM. We propose to use a structure called Bayesian Duality which exploits a duality of the posterior distributions obtained by solving a variational-Bayesian reformulation of the original problem. We show that this naturally recovers the original ADMM when isotropic Gaussian posteriors are used, and yields non-trivial extensions for other posterior forms. For instance, full-covariance Gaussians lead to Newton-like variants of ADMM, while diagonal covariances result in a cheap Adam-like variant. This is especially useful to handle heterogeneity in federated deep learning, giving up to 7% accuracy improvements over recent baselines. Our work opens a new Bayesian path to improve primal-dual methods.

##### *Bayesian Unlearning in Generalized Linear(ized) Models*

Presenter: Bálint Mucsányi

In classification tasks, softmax functions are ubiquitously used as output activations to produce predictive probabilities. Such outputs only capture aleatoric uncertainty. To capture epistemic uncertainty, approximate Gaussian inference methods have been proposed. We develop a common formalism to describe such methods, which we view as outputting Gaussian distributions over the logit space. Predictives are then obtained as the expectations of the Gaussian distributions pushed forward through the softmax. However, such softmax Gaussian integrals cannot be solved analytically, and Monte Carlo (MC) approximations can be costly and noisy. We propose to replace the softmax activation by element-wise normCDF or sigmoid, which allows for the accurate sampling-free approximation of predictives. This also enables the approximation of the Gaussian pushforwards by Dirichlet distributions with moment matching. This approach entirely eliminates the runtime and memory overhead associated with MC sampling. We evaluate it combined with several approximate Gaussian inference methods (Laplace, HET, SNGP) on large- and small-scale datasets (ImageNet, CIFAR-100, CIFAR-10), demonstrating improved uncertainty quantification capabilities compared to softmax MC sampling.

##### *Sources of Uncertainty Modeling for Bayesian Structured Sparsity in Neural Networks*

Presenter: Rafael Mouallem Rosa

We introduce Sources of Uncertainty (SoU), a framework that extends Bayesian inference by allowing the researcher to encode confidence in different sources of uncertainty within a model. This provides a principled mechanism for model design and regularization control. From this framework, we derive the SoU Sparse Global–Local mechanism (SoU-SGL), a general approach for inducing sparsity in statistical models. Here we focus on an application of SoU-SGL that extends regularized horseshoe shrinkage in the convolutional neural network setting. The resulting model sharpens the separation between active and inactive filters, providing an effective mechanism for architecture recovery. A key property of our model is stability: across different overparameterized starting widths and under a reasonable range of pruning thresholds, it recovers compact architectures with similar total size and layer-wise allocation. This supports the interpretation that the procedure is identifying a compact structure adapted to the data, rather than merely compressing each initial model separately. Across standard image-classification benchmarks, our model recovers substantially smaller architectures than deterministic structured-sparsity baselines, while maintaining competitive accuracy and consistently improving negative log-likelihood and calibration.

Discussant: Vincent Fortuin

#### Bayesian Nonparametric Methods in Ecology (Room 1102)

##### *Covariate-dependent feature allocation models with imperfect detection*

Presenter: Federica Stolf

Indian Buffet Processes (IBPs) are widely used Bayesian nonparametric models designed for binary latent feature matrices with a potentially infinite number of columns. In biodiversity studies, where features correspond to observed species, this approach is particularly powerful as it allows for the inclusion of an ever-growing number

of species. However, current IBP-based models rely on unrealistic assumptions that limit their applicability to ecological data. In particular, they assume that data are exchangeable, ignoring habitat-specific characteristics. Additionally, existing methods do not account for measurement error, which is a known problem in ecology and can lead to biased estimations of species composition. To address these limitations, we propose a covariate-dependent feature allocation model that accounts for imperfect detection. The proposed approach separates the occupancy and the detection part of the model, while incorporating heterogeneity across sampling sites. We discuss theoretical properties of the proposed modeling paradigm and implement efficient algorithms for posterior computation. Our framework also enables to clearly define the so-called beta-diversity, i.e. the species heterogeneity across different sampling regions, under a coherent and elegant probabilistic framework. Simulation studies and applications to biodiversity data provide compelling support for the new modeling class.

### ***Exact and Efficient Inference for Batch-Mark Data using Bayesian Nonparametric Pólya Trees***

Presenter: Ioannis Rotous

Monitoring open wildlife populations is a critical task in ecology, but for many species of amphibians, birds, and fish, researchers must rely on batch-mark (BM) surveys. In these surveys, individuals are given a shared mark per sampling occasion rather than a unique identifier. While practical in the field, BM data presents significant statistical challenges. Existing models for open populations typically rely on approximate inference, struggle to scale with large numbers of individuals, and do not readily accommodate the joint modelling of diverse observation processes.

In this talk, we will introduce a novel, highly efficient Bayesian nonparametric framework designed specifically for BM data. By defining a bivariate grid to capture latent population entry and exit patterns, we employ a Pólya Tree (PT) prior on the grid cells. This formulation enables exact and highly efficient Bayesian inference on the number of individuals in each cell, naturally yielding estimates for total population size and transition dynamics.

This PT batch-mark (PTBM) approach scales with the number of sampling occasions rather than the number of individuals, offering substantial computational advantages. We will outline the construction of the PTBM model, show how it easily adapts to different observation processes via the likelihood function, and demonstrate its practical utility. Finally, we will share results from extensive simulation studies and two real-world case studies, illustrating the performance of our framework.

### ***Predicting global biodiversity via Hubbell regression***

Presenter: Tomasso Rigon

Understanding global biodiversity patterns and their drivers is a prerequisite for countering the biodiversity crisis. In this paper, we introduce a novel generalized linear model, Hubbell regression, to estimate a key biodiversity descriptor, the fundamental biodiversity number. This can be converted into a set of biodiversity descriptors, including Shannon and Simpson indices, and more. Hence, quantifying the impact of environmental conditions on the fundamental biodiversity number allows us to predict the general properties of local biodiversity in any setting. In addition to having a strong mathematical foundation, Hubbell regression consistently outperformed current state-of-the-art models in predicting global biodiversity. We apply the method to arthropods, which account for the majority of terrestrial biodiversity. By parameterizing the models using samples of 1.78 million arthropods from 2415 samples collected at 135 sites spanning all continents, we pinpoint the drivers of arthropod biodiversity and its features at the global scale. We find that actual evapotranspiration is the single largest predictor of arthropod diversity and explains nearly 30% of the variation in richness. Moreover, we infer that high human activity has led to a 21.3% and 29.2% decrease in potential insect richness in tropical and dry zones, respectively, but increased insect richness in polar regions. These insights bring a new foundation for biodiversity research and action.

## **Decision-theoretic Bayesian Privacy (Room 1103)**

### ***Insufficient Bayesian analysis***

Presenter: Antoine Luciano

We discuss in this presentation tools for reconstructing data characteristics such as those sought in privacy adversarial attacks when conditioning on MCMC finite samples, themselves obtained from posteriors attached with insufficient statistics. The solutions are intended to serve as guarantees to data-providers as well as checks for selecting data-protection statistics, while avoiding randomization steps as in standard differential privacy.

### ***Optimal Scaling of Random Walk Metropolis under Penalty, Contamination and Differential Privacy***

Presenter: Shenggang Hu

We study a differentially private version of the penalty random-walk Metropolis algorithm for Bayesian inference when the observation space is unbounded. Standard privacy analyses for MCMC typically require bounded likelihood ratios, an assumption that fails for many common statistical models. We show that a contamination framework with a heavier-tailed contaminating density removes this obstruction by making the relevant likelihood ratio bounded for fixed parameter values, so that privatising the Metropolis acceptance ratio via the Gaussian mechanism yields an  $(\epsilon, \delta)$ -differentially private penalty algorithm. We then investigate the asymptotic behaviour of this private sampler. We derive a lower bound for the expected squared jump distance (ESJD) that makes the effect of privacy noise explicit and shows that the optimal proposal scale decreases with both dimension and the privacy penalty. In particular, the optimal scaling is of order  $n^{-1/2}$  in the large sample regime and  $d^{-1/2}$  in the high-dimensional regime with the optimal acceptance probability approaching 0.234. These results provide a practical tuning rule and theoretical justification for private MCMC.

#### ***A Bayesian decision-theoretic framework for statistical data privacy***

Presenter: Joshua Bon

The scientific and economic value of data continues to increase as our technology does. New hardware and software technologies allow, but also often require, richer and vaster data to operate. As the value of input data to these systems is recognised, so too is the loss of privacy for data providers. In this landscape, data privacy is a fundamental issue for data science disciplines such as statistics and machine learning. More generally, it is an issue for scientific and industrial pursuits that use sensitive data. We develop a framework for measuring privacy from a Bayesian decision-theoretic point of view. With our framework we can generate new privacy principles that are rigorously justified, assess existing privacy definitions using decision-theory, and create new definitions that are fit for purpose. We take special note of assessing privacy for deterministic algorithms, which are overlooked by current privacy standards, and from Monte Carlo samples from posterior distributions.

#### **Beyond the Data: Bayesian Inference for Tail Risks and Extreme Events (Room 1104)**

##### ***Bayesian inference on extremes in time series: the peaks over threshold method***

Presenter: Simone Padoan

Temporal dependence is common in applications, ranging from financial time series to those in climate science. It affects the behavior of extremes leading to clustering which leaves less information available in time series than in the independent case, leading to greater uncertainty. The joint behavior extremes within a cluster is described by a general theory that accommodate for the short-range dependence. In this work we focus on beta-mixing stationary time series and the Peaks-Over-Threshold (POT) which is unarguably the most popular method for analysing univariate extremes. While Bayesian inference under the BM approach has been widely studied in the independence setting, no asymptotic theory currently exists for time series. To address this gap we first establish a comprehensive likelihood theory for the misspecified Generalized Pareto (GP) model under serial dependence. Our results include uniform convergence of the empirical log-likelihood process, contraction rates for the Maximum Likelihood Estimator, and a local asymptotically Gaussian expansion. Building on this foundation, we develop the asymptotic theory of Bayesian inference for the GP parameters, the extremal index, and extreme quantiles (Value at Risk). Under general conditions on the prior, we prove posterior consistency, contraction rates, Bernstein–von Mises theorems, and asymptotic coverage properties for credible intervals. Simulations show excellent inferential performances for the proposed methodology.

##### ***On the tails of random probability measures***

Presenter: Miguel de Carvalho

While random probability measures have a long tradition in probability and statistics, little is known about their tails. The few available results are derived using subordinators, and therefore only apply to measures that can be represented as normalized subordinators - such as the Dirichlet process - failing to extend to the entire class of Pitman–Yor processes. Our work breaks this barrier. By constructing a new family of transport maps and drawing on recent developments on regular variation and on subordinator theory, we establish that the right tail of a Pitman–Yor process is heavy-tailed if the centring distribution is itself heavy-tailed; the Dirichlet process is the only member of this class that fails to obey this convenient property. Asymptotic envelopes for the tails of the Pitman–Yor processes are also derived. Finally, we discuss some consequences of the main results, including aspects related to the posterior process. (Joint work with M. F. Gil-Leyva; V. Palacios Ramirez)

##### ***Bayesian modelling of sub-hourly precipitation extremes with temperature dependence***

Presenter: Antonio Canale

Extreme sub-hourly precipitation events can trigger flash floods and other natural hazards, making accurate

estimation of their exceedance probabilities increasingly critical under changing climate conditions. A recent study (Marra et al., 2024) introduced a non-asymptotic, non-stationary statistical framework that incorporates physically-based temperature dependence. Specifically, it models all peaks of independent rainfall events using a Weibull distribution, whose parameters are linked to near-surface air temperature, in line with the Clausius–Clapeyron relationship. Building on this approach, we propose a Bayesian extension that enables improved quantification of uncertainty, particularly in modeling air temperature distributions. This not only strengthens inference under current climate conditions but also facilitates predictive assessments under future climate scenarios influenced by global warming.

## **Bayesian Uncertainty Quantification with Machine Learning Tools (Room Main Hall)**

### ***AI-powered Bayesian inference***

Presenter: Veronika Rockova

The advent of Generative Artificial Intelligence (GAI) has heralded an inflection point that changed how society thinks about knowledge acquisition. While GAI cannot be fully trusted for decision-making, it may still provide valuable information that can be integrated into a decision pipeline. Rather than seeing the lack of certitude and inherent randomness of GAI as a problem, we view it as an opportunity. Indeed, variable answers to given prompts can be leveraged to construct a prior distribution which reflects assuredness of AI predictions. This prior distribution may be combined with tailored datasets for a fully Bayesian analysis with an AI-driven prior. In this paper, we explore such a possibility within a non-parametric Bayesian framework. The basic idea consists of assigning a Dirichlet process prior distribution on the data-generating distribution with AI generative model as its baseline. Hyper-parameters of the prior can be tuned out-of-sample to assess the informativeness of the AI prior. Posterior simulation is achieved by computing a suitably randomized functional on an augmented data that consists of observed (labeled) data as well as fake data whose labels have been imputed using AI. This strategy can be parallelized and rapidly produces iid samples from the posterior by optimization as opposed to sampling from conditionals. Our method enables (predictive) inference and uncertainty quantification leveraging AI predictions in a coherent probabilistic manner.

### ***FAB-PPI: Frequentist, assisted by Bayes, prediction-powered inference***

Presenter: Stefano Cortinovis

Prediction-powered inference (PPI) enables valid statistical inference by combining experimental data with machine learning predictions. When a sufficient number of high-quality predictions is available, PPI results in more accurate estimates and tighter confidence intervals than traditional methods. In this paper, we propose to inform the PPI framework with prior knowledge on the quality of the predictions. The resulting method, which we call frequentist, assisted by Bayes, PPI (FAB-PPI), improves over PPI when the observed prediction quality is likely under the prior, while maintaining its frequentist guarantees. Furthermore, when using heavy-tailed priors, FAB-PPI adaptively reverts to standard PPI in low prior probability regions. We demonstrate the benefits of FAB-PPI in real and synthetic examples.

### ***Leveraging free energy in pretraining model selection for improved fine tuning***

Presenter: Susan Wei

Recent advances in artificial intelligence have been fueled by the development of foundation models such as BERT, GPT, T5, and Vision Transformers. These models are first pretrained on vast and diverse datasets and then adapted to specific downstream tasks, often with significantly less data. However, the mechanisms behind the success of this ubiquitous pretrain-then-adapt paradigm remain underexplored, particularly the characteristics of pretraining checkpoints that lend themselves to good downstream adaptation. We introduce a Bayesian model selection criterion, called the downstream free energy, which quantifies a checkpoint’s adaptability by measuring the concentration of nearby favorable parameters for the downstream task. We demonstrate that this free energy criterion can be effectively implemented without access to the downstream data or prior knowledge of the downstream task. Furthermore, we provide empirical evidence that the free energy criterion reliably correlates with improved fine-tuning performance, offering a principled approach to predicting model adaptability.

Discussant: Judith Rousseau

## Multitrack Sessions 4: 11:00–12:30

### Deep Learning and Generalized Bayes (Room 1101)

#### *Scalable and Robust Uncertainty Quantification with Stochastic Gradient MCMC*

Presenter: Jonathan Huggins

Stochastic gradient Markov chain Monte Carlo (SG-MCMC) methods such as stochastic gradient Langevin dynamics (SGLD) promise Bayesian inference at the per-iteration cost of stochastic gradient descent, but in practice they are notoriously hard to tune and their uncertainty quantification (UQ) can be unreliable—especially under large batch sizes, model misspecification, or complex model structure. This poster presents unified approach to leverage rigorous statistical theory to guide trustworthy deployment of SG-MCMC methods. We first develop a joint step-size-sample-size scaling-limit theory that characterizes the large-sample behavior of a SGLD (with and without momentum), including Bernstein–von Mises-type guarantees and mixing time estimates. These results yield practical tuning rules, including for generalized posteriors and misspecified models. We then move beyond continuous-time approximations with discrete-time analysis of SGLD that delivers non-asymptotic error bounds on stationary and iterate-average covariances, enabling accurate UQ in the practically relevant large-batch regime. Finally, we extend the continuous-time framework to latent variable models via a jump–diffusion scaling limit for SGLD–Gibbs, producing principled tuning guidance that outperforms stochastic variational inference. Together, the results show that SG-MCMC can provide a scalable, robust, and statistically calibrated approach to generalized Bayesian inference.

#### *Posterior Covariance Information Criterion for Generalized Bayesian Inference*

Presenter: Keisuke Yano

We propose a computationally efficient method for estimating general predictive measures in generalized Bayesian inference. The method introduces a simple bias-correction term expressed through the posterior covariance. It offers three key advantages: (i) it avoids naive importance sampling, which is sensitive to influential observations, thereby improving numerical stability; (ii) it is theoretically justified through its asymptotic unbiasedness, which connects to the Bayesian local sensitivity formula and the asymptotic equivalence between WAIC and Bayesian LOOCV; and (iii) it enables practical estimation of generalization error by subtracting the posterior covariance from the empirical error, removing the need for unstable computation of information matrices and their inverses. We demonstrate that the proposed method provides accurate and stable generalization error estimates across a wide range of settings, including differentially private learning, hierarchical models, regression with influential observations, strongly regularized models, and high-dimensional scenarios.

#### *Posterior contraction rates for deep BNNs with Gaussian weights*

Presenter: Julyan Arbel

Deep Bayesian neural networks with i.i.d. Gaussian weights induce increasingly heavy-tailed priors as depth grows, due to the multiplicative Gaussian structure of the layers. Using the radial product-of-Gaussians representation (yielding Bessel/Meijer-G tails), we show that this implicit heavy-tailed behavior provides sufficient prior mass to achieve near-minimax posterior contraction rates, without explicitly imposing heavy-tailed priors. Our analysis establishes prior-mass conditions by exploiting the depth-dependent tail amplification at the unit level and connects these results to adaptive approximation properties and benign-overfitting phenomena.

#### *Scalable Bayesian Experimental Design with Diffusions*

Presenter: Florence Forbes

Bayesian Optimal Experimental Design (BOED) is a powerful tool to reduce the cost of running a sequence of experiments. When based on the Expected Information Gain (EIG), design optimization corresponds to the maximization of some intractable expected contrast between prior and posterior distributions. Scaling this maximization to high dimensional and complex settings has been an issue due to BOED inherent computational complexity. In this work, we introduce a pooled posterior distribution with cost-effective sampling properties and provide a tractable access to the EIG contrast maximization via a new EIG gradient expression. Diffusion-based samplers are used to compute the dynamics of the pooled posterior and ideas from bi-level optimization are leveraged to derive an efficient joint sampling-optimization loop. The resulting efficiency gain allows to extend BOED to the well-tested generative capabilities of diffusion models. By incorporating generative models into the BOED framework, we expand its scope and its use in scenarios that were previously impractical. Numerical experiments and comparison with state-of-the-art methods show the potential of the approach. An illustration on a real-world MRI application in medical imaging is also given.

#### *Generalized Bayesian inference for chaotic systems*



Presenter: Antonietta Mira

Chaotic systems arise in many scientific fields, from population dynamics to atmospheric flows, and present major challenges for statistical inference because their likelihood functions are typically intractable or prohibitively expensive to evaluate. Infinitesimal perturbations to model parameters or initial conditions produce divergent trajectories, making parameter recovery an ill-posed inverse problem, an issue further amplified in the presence of stochastic dynamics. We adopt a generalised Bayesian perspective, replacing the likelihood with loss functions that quantify discrepancies between observed and simulated attractors. Trajectories are represented as point clouds in state space and attractors are compared using loss functions based on the Chamfer distance. These losses enter the generalized Bayes update directly, yielding posteriors driven solely by simulation rather than explicit likelihood evaluation. As a by-product, we introduce a simple and computationally efficient data-driven calibration method of the generalized Bayes loss scale parameter based on Rademacher splits of the reference trajectory, providing stable and honest uncertainty quantification. We evaluate the framework on the Lorenz-63 system (with and without additive noise), several one-dimensional chaotic population-dynamics maps, and real ecological data from the Nicholson blowflies experiment, where delayed embeddings reconstruct attractors from scalar observations. We also illustrate how the method applies to spatially extended systems with complex spatio-temporal structure. In all cases, the proposed geometry-aware loss functions lead to accurate parameter recovery, and competitive or improved performance relative to existing simulation-based approaches, while avoiding hand-crafted summaries. Our results demonstrate that the generalized Bayes framework, combined with cloud-based geometric distance metrics, provides a practical, robust, and broadly applicable approach to inference and data assimilation in chaotic and highly nonlinear dynamical systems.

## Applications in Ecology (Room 1102)

### *Incorporating Body Condition into the Analysis of Animal Movement*

Presenter: Vianey Leos Barajas

A long-sought goal in ecology is to connect movement with population dynamics. For many species and especially for ungulates, there is a known link between condition (e.g. fat reserves) and the probability of survival and reproduction. Assuming a particular genetic makeup and physiology, condition reflects the history of behavioral decisions, including movement and habitat use. However, the condition of an animal can also have a direct implication on the types of movements that it performs and the habitats that it visits. Movement data for ungulates are typically collected at a fine temporal scale, e.g. a position recorded by a GPS device every five or ten minutes. However, fat reserves cannot be measured remotely and must be done manually. This in turn creates a mismatch in the temporal scale at which the two data streams are observed, i.e. every five minutes for movement vs approximately once a month for condition. Further, the temporal mismatch leads to various challenges when jointly modeling the two processes.

For the movement model, we use discrete-time, finite-state hidden Markov models (HMMs) with the positional data of the sheep serving as the observation process and the underlying state process serving as a proxy for behaviors of interest. To incorporate condition as a potential covariate affecting the movement, and thus behavioral, process, we make use of the differential equations that describe the evolution of body fat in Merino sheep in order to predict daily values of the condition process. The physiological equations are expressed as a function of the states inferred by HMM, as well as the distance that the sheep travels. Altogether, we present a general modeling framework that describes the feedback between condition and movement, using Merino sheep as a case study.

### *Functional regression modeling of animal movement in response to a sound field*

Presenter: Becky Tang

We aim to develop a functional regression approach to examine three-dimensional movement trajectories and diving behavior of whales in relation to modeled sound fields. We have data on behavioral responses and location of individuals at discrete time points, in addition to (modeled) sound propagation output over the entire region at each time point. The modeling goal is to understand if and how fine scale movement of these individuals responds to sonar, while also accounting for home range and boundary effects in the system. Our work will present some preliminary simulations in two-dimensions. We also aim to provide innovative methods of visualizing a three-dimensional functional coefficient.

### *Wait, who was that? Tracking interacting trajectories with label uncertainty*

Presenter: Henry Scharf

This work is motivated by the goal of understanding the impact that naval sonar has on the behavior of highly



social delphinids. Drones now offer ecologists a new tool for the study of animal behavior by providing extremely high resolution measurements of animals' positions in space without the need to attach physical devices, which is intrusive and expensive. One trade-off for the new, rich source of information is that, for some species, individuals' identities can be difficult to determine from drone imagery, and thus as individuals leave and then re-enter the field of view, the appropriate label for a sequence of location records becomes uncertain. We utilize a traditional state-space model framework in which the "output" or "observation" matrices are random variables capturing this uncertain labeling. For moderate rates of label uncertainty, inference about dolphins' responses to exogenous factors like sonar can be obtained by using a Kalman filter to marginalize over the trajectories and Gibbs updates to sample from the joint posterior distribution of the labeling process and behavior parameters.

### ***Species Distribution Modeling with Expert Elicitation and Bayesian Calibration***

Presenter: Jarno Vanhatalo

Species distribution models (SDM) are key tools in ecology, conservation, and natural resources management. They are traditionally trained with data on direct species observations. However, if collecting species data is difficult or expensive, complementary information sources on species distributions are needed. Expert knowledge has been demonstrated to improve SDM predictions in a number of such applications but there is still no consensus on methods to integrate information from several experts into a single coherent species distribution prediction. Moreover, since expert assessments are inherently subjective and prone to biases, expert driven SDMs should calibrate their assessments. We propose a method to tackle these challenges by extending the hierarchical Bayesian integrated species distribution modeling framework to expert informed species distribution modeling. We treated map-like expert assessments as data and integrated them with calibration data on species recordings. Our integrated SDM has model components to estimate experts' reliability and to adjust for potential biases in their assessments. After integrated inference, we used the model to make predictions over a study area. We tested our approach with an extensive simulation study and a real world case study comprising ten expert assessments and survey data on pikeperch larvae from a coastal area of the Gulf of Finland. Expert assessments significantly improved species distribution predictions compared to predictions conditioned on survey data only. They also improved parameter inference, thus strengthening the ecological interpretation of the results. The skill of the experts, and biases in their assessments, varied considerably in the case study though, emphasizing the importance of formal expert calibration provided by our model. Our results show that expert elicitation can be an efficient tool for improving species distribution model predictions. Our approach is especially useful for applications where any type of species data are expensive to collect but local species experts can easily be reached.

### ***A Bayesian mixture model for clustering fire-driven and seasonal wildlife movement patterns***

Presenter: Martina Amongero

Fire represents a key ecological force in the Miombo woodlands of western Tanzania, where yearly burns, mostly triggered by human activities, play an essential role in shaping vegetation patterns and wildlife dynamics. As climate change intensifies fire regimes by extending dry-season conditions, gaining insight into how fire influences animal behavior has become increasingly important. In this study, we investigate whether, and in what manner, fire events alter wildlife return rates, drawing on a decade of data from motion-activated camera traps deployed across 14 km<sup>2</sup>. Cameras provide long-term records of fires occurrence and species encounter rates. A major challenge of this study is that it is not possible to directly estimate how fire alters animal movement dynamics, because there are no reference data on animal behavior for the same periods and under the same conditions in the absence of fire: the same location cannot be observed both with and without fires, and temporal or spatial comparisons are potentially confounded by climatic variation. We propose an innovative Bayesian mixture model to differentiate post-fire return behavior from typical seasonal movement patterns. A key advantage of the proposed model lies in its ability to simultaneously discriminate between two different behavioral components: the periodic component of animal activity, independent of fire, and the fire-related return dynamics. The first pattern is modeled through a circular distribution that accounts for repetitive seasonal movements, while the fire effect is captured with a more flexible, aperiodic distribution. Additionally, allowing covariates, such as body mass and diet type, to enter the parameters of both distributions further enhances our ability to detect species-specific patterns in both post-fire return behavior and standard seasonal activity. Beyond the application of the proposed Bayesian model to the Miombo dataset, we conduct an extensive simulation study to demonstrate its robustness and estimation performance.

## Bayesian Nonparametric Network Modeling (Room 1103)

### *Rapidly Varying Completely Random Measures for Modeling Extremely Sparse Networks*

Presenter: Valentin Kilian

Completely random measures (CRMs) are fundamental to Bayesian nonparametric models, with applications in clustering, feature allocation, and network analysis. A key quantity of interest is the Laplace exponent, whose asymptotic behavior determines how the random structures scale. When the Laplace exponent grows nearly linearly - known as rapid variation - the induced models exhibit approximately linear growth in the number of clusters, features, or edges with sample size or network nodes. This regime is especially relevant for modeling sparse networks, yet existing CRM constructions lack tractability under rapid variation. We address this by introducing a new class of CRMs with index of variation  $\alpha \in (0, 1]$ , defined as mixtures of stable or generalized gamma processes. These models offer interpretable parameters, include well-known CRMs as limiting cases, and retain analytical tractability through a tractable Laplace exponent and simple size-biased representation. We analyze the asymptotic properties of this CRM class and apply it to the Caron-Fox framework for sparse graphs. The resulting models produce networks with near-linear edge growth, aligning with empirical evidence from large-scale networks. Additionally, we present efficient algorithms for simulation and posterior inference, demonstrating practical advantages through experiments on real-world sparse network datasets.

### *Sparse networks with dynamic communities*

Presenter: Francesca Panero

Many real-world networks of interest are large and sparsely connected, yet most statistical models are designed for small, dense networks. Even fewer methods exist for modeling networks that evolve over time. In this work, we show how Bayesian nonparametric methods can be used to model dynamic, sparse networks with overlapping communities. Nodes can belong to multiple communities with varying affiliation strengths and are allowed to change their affiliations over time. Leveraging a regularly varying completely random measure (CRM), our model also captures power-law degree distributions, a common feature of large-scale networks. We illustrate the utility of our approach on a dynamic word co-occurrence network built from Reuters news articles during the 9/11 attacks. The results highlight how dynamic overlapping communities can reveal the emergence of topics and shifts in word meanings over time.

### *Detecting Spatial Health Disparities Using Disease Maps*

Presenter: Luca Aiello

Epidemiologists commonly use regional aggregates of health outcomes to map mortality or incidence rates and identify geographic disparities. An approach to detect health disparities between regions is to identify “difference boundaries” that separate neighboring regions with significantly different spatial effects. With growing interest in studying disparities across multiple outcomes, cancer rates in California in the current investigation, we turn to accounting for dependence among diseases and across regions. We address detecting multivariate difference boundaries for possibly correlated diseases using a Bayesian nonparametric approach to balance smoothing with disparities across geographical neighbors. Specifically, we estimate posterior probabilities of neighboring spatial effects being different using multivariate Dirichlet process models that introduce spatial and inter-disease dependence. The model endows spatial effects with discrete probability laws, which allows control of false discoveries. We evaluated inferential performance using simulation studies over graphical model-based cancer dependencies. We detect difference boundaries for four cancers using data from the Surveillance, Epidemiology, and End Results Program of the National Cancer Institute.

## Applied Bayesian Models in Sports Analytics (Room 1104)

### *College Football Volatility: A Bayesian state-space model of the transfer portal and NIL impact*

Presenter: Ron Yurko

The landscape of American college football has changed dramatically in recent years with conference realignment, increased usage of the transfer portal, and the seismic legal ruling regarding athletes’ name, image, and likeness (NIL). In this work, we use a Bayesian state-space model to capture the impact of transfers and NIL on the volatility of team strengths over time. Specifically, we extend the classic autoregressive process for team strength by modeling the between-season innovation variance as a function of incoming transfers and NIL rule changes. This approach enables us to predict greater variance for team strengths in an upcoming season based on roster changes. Our results from the playoff era (2014-2024) reveal variation between schools that have embraced the volatility of transfers (e.g., Deion Sanders at Colorado) with those who have been reluctant to change (e.g., Iowa and Clemson). We explore variability in the effects between different positions of incoming transfers, and

compare the predictive performance of our novel approach with simpler state-space models.

### ***Fatigue-Adjusted Plus-Minus***

Presenter: Jared Fisher

Fatigue plays a central role in athletics, where performance often declines as cumulative exertion increases. While Regularized Adjusted Plus-Minus (RAPM) is a standard for estimating player impact, traditional models assume a player's performance remains constant throughout a game. This research proposes a modified RAPM framework to isolate the effect of in-game fatigue in NCAA Division I men's basketball. By allowing regularized adjusted plus-minus to decrease as a function of playing time through a varying-coefficient model, a fatigue effect can be calculated relative to the player's "fresh" baseline. For the post-Covid seasons (2021-2025), our results demonstrate a significant linear decrease in player efficiency as playing time accumulates.

### ***Beyond the Yard Line: Accommodating Rounded Sports Data in Statistical Models***

Presenter: Amanda Glazer

In American football, rushing, passing, and receiving yardage is recorded as an integer using a unique rounding method, despite the underlying yardage being continuous. This discretization induces measurement error that is typically ignored in statistical analyses. Play-level yardage can also be negative and strongly skewed, complicating standard distributional assumptions and amplifying rounding effects. Using play-level data on running backs from the 2023 National Football League regular season, we study the consequences of these issues. We provide a general framework to model yards gained that explicitly accounts for discretization using a chosen base probability distribution. Through model checking and validation, we show that models aligned with the data-generating process provide improved fit. These results highlight the importance of accounting for rounding and asymmetry in sports analytics and other applied settings.

## **Savage Award Session: Theory & Methods (Room Main Hall)**

### ***Hierarchical structures in Bayesian Statistics***

Presenter: Filippo Ascolani

In this talk I will briefly described follows three distinct, but related, directions that have been explored in my thesis: (i) modelling complex dependence structures (e.g. time series, multi-samples data...) via a Bayesian nonparametric approach, (ii) mathematical investigation of the resulting inferential procedures, with methods for measuring and tuning dependence and proving frequentist asymptotic properties, (iii) rigorous analysis of the computational algorithms employed for posterior inference with the aforementioned structures, with a focus on high dimensional problems. A unifying thread shared by all these lines of research is the study of the specific probabilistic structure considered: indeed, the choice of a particular dependence structure (more specifically hierarchical models), which is often selected through modelling considerations (prior information, domain-specific knowledge, etc.), requires the investigation of the associated inferential and computational properties. Indeed, different specifications may have significantly different levels of analytical tractability and the performance of routinely used MCMC algorithms (e.g. gradient-based methods, Gibbs samplers) may greatly vary.

### ***Convergence Analysis of Markov Chain Monte Carlo Methods for Model Selection Problems***

Presenter: Hyunwoong Chang

Understanding the convergence of Markov chain Monte Carlo (MCMC) methods is a fundamental challenge in high-dimensional model selection. In this dissertation, we develop a general framework for analyzing mixing time bounds for various Metropolis–Hastings (MH) algorithms under conditions grounded in high-dimensional statistical theory. This framework is readily applicable to a broad spectrum of Bayesian model selection problems. The dissertation considers sophisticated MH schemes, including the informed MH algorithm and the multiple-try Metropolis (MTM) algorithm, which have gained increasing attention among MCMC practitioners, yet whose theoretical mixing properties have rarely been studied in the literature because of their complex proposal schemes. Guided by this framework, we propose RWGES, a random walk Metropolis–Hastings algorithm on sparse equivalence classes for structure learning in Gaussian directed acyclic graph models. We prove that RWGES is rapidly mixing, demonstrating that its computational complexity scales only polynomially with sample size and dimension.

### ***Bayesian High-dimensional Linear Regression with Sparse Projection-posterior***

Presenter: Samhita Pal

We consider a novel Bayesian approach to estimation, uncertainty quantification, and variable selection for a high-dimensional linear regression model under sparsity. The number of predictors can be nearly exponentially

large relative to the sample size. We put a conjugate normal prior initially disregarding sparsity, but for making an inference, instead of the original multivariate normal posterior, we use the posterior distribution induced by a map transforming the vector of regression coefficients to a sparse vector obtained by minimizing the sum of squares of deviations plus a suitably scaled  $\ell_1$ -penalty on the vector. We show that the resulting sparse projection-posterior distribution contracts around the true value of the parameter at the optimal rate adapted to the sparsity of the vector. We show that the true sparsity structure gets a large sparse projection-posterior probability. We further show that an appropriately recentred credible ball has the correct asymptotic frequentist coverage. Finally, we describe how the computational burden can be distributed to many machines, each dealing with only a small fraction of the whole dataset. We conduct a comprehensive simulation study under a variety of settings and found that the proposed method performs well for finite sample sizes. We also apply the method to several real datasets, including the ADNI data, and compare its performance with the state-of-the-art methods. We implemented the method in the R package called **sparseProj**, and all computations have been carried out using this package.

### ***Bayesian Deep Discrete Latent Structures***

Presenter: Yuren Zhou

Modern scientific datasets are increasingly high-dimensional, structured, and heterogeneous, creating major challenges for statistical modeling and inference. This talk develops Bayesian deep discrete latent structures, a class of hierarchical probabilistic models designed to uncover interpretable latent structure in complex data.

We study this framework in two settings. First, for multiplex network data, we introduce a Bayesian hierarchical model that captures population-level node hierarchies while performing multi-resolution clustering across replicated networks. The model is supported by theoretical guarantees including identifiability and posterior consistency, together with efficient posterior computation algorithms. We demonstrate its performance through simulations and applications to brain connectivity data.

Second, for high-dimensional categorical data with covariates, we develop a deep latent class modeling framework that improves flexibility and robustness relative to classical latent class regression approaches. The model retains key theoretical guarantees and enjoys a Bayes oracle clustering property that is particularly beneficial in high-dimensional settings. Applications to ecological species distribution modeling illustrate its practical utility. Together, these methods provide scalable and theoretically grounded tools for learning discrete latent structure in modern high-dimensional data.

## **Multitrack Sessions 5: 14:00–15:30**

### **Recent Advances in Predictive Bayes (Room 1101)**

#### ***Bayesian Predictive Inference Beyond Martingales***

Presenter: Lorenzo Cappello

There is growing interest in procedures for Bayesian inference that bypass the need to specify a model and a prior, relying instead on a predictive rule that describes how we learn about future observations given the available ones. Which conditions the predictive rule needs to satisfy to produce a valid inference is a key question. In the talk, we will discuss relaxing previous assumptions, building on a generalization of martingales, and opening up the possibility of employing a much wider range of predictive rules that were previously ruled out. We will discuss several examples. We will argue that departing from the martingale framework entails essentially no loss, as many key results for martingales extend naturally to this broader class. Notably, we can construct credible sets both via simulation-based inference and via asymptotic arguments. At the same time, we will discuss potential issues that arise from departing from exchangeability, which is now common for most proposals in this literature. Join work with Marco Battiston (Lancaster University).

#### ***Bayesian uncertainty quantification for predictive algorithms: results, properties, comparisons***

Presenter: Sonia Petrone

We often face a trade-off between statistical inferential procedures, carefully designed but computationally slow, and fast, high-performing algorithms that typically lack formal uncertainty quantification. Understanding and expressing uncertainty for predictive algorithms has been a reason behind the renewed interest in the Bayesian predictive approach - under which one may read predictive algorithms as Bayesian learning rules - possibly not exchangeable but “trained” to become such - and may uncover the implicit model and prior, enabling full Bayesian inference and uncertainty quantification. Computational advantages of these novel quasi-exchangeable rules are evident, but how do they compare with “optimal” (exchangeable) Bayes rules, in terms of efficient use of the information? In the talk, we will explore aspects of this comparison, and provide predictive condi-

tions for frequentist properties. We will also present extensions of predictive Bayes to structured data, beyond exchangeability.

### ***Predictively-oriented posteriors***

Presenter: Yann McLatchie

The Bayesian posterior predictive has long been extolled for its principled and coherent approach to probabilistic forecasting. And its optimality is supported by a string of papers dating back to the middle of the last century. The assumptions underlying these results, however, are known to be violated in virtually all modern machine learning applications. In this paper, we provide a simple strategy for constructing posterior distributions that predictively dominate the Bayes posterior when these assumptions are violated. In fact, our results extend beyond standard Bayes posterior and apply to any generalised posterior distribution constructed with convex scoring rules. We provide two computational avenues for computing these objects, and demonstrate the practical benefits of such a prediction-centric posterior design philosophy across several numerical examples.

Discussant: Lorenzo Cappello

## **Statistical Challenges in Mathematical Cognitive Science (Room 1102)**

### ***Simulation-based Bayesian methods for cognitive models of human decision making***

Presenter: Jennifer S. Trueblood

Computational models have long been central to understanding human decision making, especially given the challenges of directly observing and perturbing the brain. These models allow researchers to formalize and test mechanistic hypotheses about how people accumulate information, evaluate options, and make choices—particularly in complex environments with multiple alternatives and attributes. A prominent class of models in this space is Evidence Accumulation Models (EAMs), which simulate the dynamic process by which decisions unfold over time. EAMs are especially powerful because they jointly predict both the choices individuals make and the time they take to make them—so-called choice-response time (RT) data. However, only the simplest models in this family have sufficiently analytic likelihoods to be tractable, presenting challenges for Bayesian inference. In this talk, I will discuss the strengths and limitations of simulation-based Bayesian methods for estimating parameters in complex EAMs using choice-RT data. One such approach, Probability Density Approximation, estimates the likelihood function by simulating large numbers of stochastic outcomes and constructing a kernel density estimate. While effective, this approach yields a stochastic likelihood, which introduces additional complications for Bayesian inference. I will discuss the implications of these findings for cognitive modeling.

### ***Applications of Bayesian models in complex business domains: opportunities, successes, and challenges***

Presenter: David Landy

In tech/business contexts, black-box ML models tend to dominate Bayesian models in practical applications. This happens for a few clear reasons, including that ML predictive models tend to be more performant and more accurate in almost all practical situations. I will discuss a few classes of cases in which, in my experience as an ML scientist working on supporting making movies and television at Netflix, Bayesian models do nevertheless tend to be very useful. The distinguishing features of these situations include the following: (1) Modeling at scale is designed to support or inform the reasoning process of domain experts with strong theories, which can be translated into sensible models; (2) Related yet distinct business contexts have meaningfully different characteristics and are informed by very different levels of empirical support, such that one would like to model each of them; In such cases, especially when the relationship is indirect or imperfect, model specifications can be used to ‘port’ interpretations and uncertainty from a high-data domain to an important low data domain; (3) Quantitative estimates of counterfactuals are vital, and those counterfactuals can be specified in at best a distributional form, and at worst are ill-specified and vague. All of these conditions arise in the context of screening and producing movies at Netflix, which can be considered as an iterative, generative dialogue between creators and consumers, mediated by statistical models and studio teams. In this talk I will present a conceptual framework for thinking about these and related situations, and describe some of the challenges and successes we have encountered in applying Bayesian approaches.

### ***Challenges to cumulative behavioral science require us to lower our sights***

Presenter: Joachim Vandekerckhove

The behavioral sciences face a well-documented crisis of theory, stemming in part from a reliance on vague, verbally-specified theories that are difficult to rigorously test or falsify. This challenge is compounded by the-



oretical siloing and a methodological attachment to null-hypothesis significance testing, which is poorly suited for quantifying evidence for theories. Bayesian inference offers more than just an alternative set of statistical tools; it provides a principled framework for navigating these challenges and fostering a more cumulative science. Beyond vague theory, progress is further hampered by a paucity of data: Individual studies are often noisy and low-powered, making it difficult to draw strong conclusions from any single experiment. In this talk, I will argue for the use of greatly simplified models that cast psychological theories as logical conjunctions of directional predictions. This process of translation forces researchers to move from abstract verbal statements to the concrete, testable predictions that a theory entails, allowing for a direct, intuitive quantification of evidence that aligns with the actual scientific questions being asked. Moreover, it immediately implies a set of tools for evidence synthesis, which is especially critical in fields characterized by noisy data. Rather than relying on binary conclusions from individual experiments, cumulative aggregation of evidence allows theories to be progressively strengthened or refuted by the full weight of distributed findings. This makes it possible to find a clear signal within the noise that plagues individual studies. My goal in this project is to position Bayesian statistics not merely as a replacement for frequentist methods, but as a fundamental toolkit for building and testing theory in a rigorous and cumulative fashion.

Discussant: Trish Van Zandt

## **Scalable and Interpretable: Advances in Graphical Techniques for Modeling Spatial Dependence (Room 1103)**

### ***When can we adjust for spatial confounding***

Presenter: Abhirup Datta

The problem of estimating the slope parameter in regression between two spatial processes under confounding by an unmeasured spatial process has received widespread attention in the recent statistical literature. Yet, a fundamental question remains unsolved: when is this slope consistently estimable under spatial confounding. In this manuscript, we characterize conditions for consistent estimability of the regression slope between Gaussian random fields (GRFs). Under fixed-domain (infill) asymptotics, we give sufficient conditions for consistent estimability. When estimability holds, we provide consistent estimators of the slope using local differencing (taking discrete differences or Laplacians of the processes of suitable order). Using functional analysis results on Paley-Wiener spaces, we then provide an easy-to-verify necessary condition for consistent estimability of the slope in terms of the relative spectral tail decays of the confounder and exposure. As a by-product, we establish a novel and general spectral condition on the equivalence of measures on the paths of multivariate GRFs with component fields of varying smoothnesses, a result of independent importance. We show that for the Matérn, power-exponential, generalized Cauchy, and coregionalization families, the necessary and sufficient conditions become identical, thereby providing a complete characterization of consistent estimability of the slope under spatial confounding. The results are extended to accommodate measurement error using local-averaging-and-differencing based estimators. Finite sample behavior is assessed via numerical experiments. Practical algorithms are explored using graphical models.

### ***Geospatial Network models via Inside-out Cross-covariance***

Presenter: Michele Peruzzi

Interactions between variables can be represented as edges in a graphical model that encodes conditional independence. Gaussian graphical models are especially tractable in this context, as network sparsity corresponds directly to zeros in the precision matrix. When variables are spatially resolved, traditional spatial smoothing via direct models of covariance are fundamentally limited in their ability to impose or interpret network sparsity, due to the nonlinear and unstable relationship between a covariance matrix and its inverse. We present Geospatial Network models via Inside-out Cross-covariance (GeNIOX) to unify the two perspectives. GeNIOX are a flexible family of multivariate spatial models that offer spatial smoothing while simultaneously enabling interpretable network inference through a structured, sparse precision matrix. The resulting covariance and precision matrices both have a block-Kronecker-product structure that ensures scalability and facilitates inference. Among several possible modeling alternatives, latent models enable tractable and flexible inference via novel factor analytic specifications, also yielding novel ways to parsimoniously model spatially-varying networks. We demonstrate the potential of GeNIOX as a tool for geospatial network modeling through synthetic studies and applications to “omics” data. These results highlight GeNIOX’s ability to simultaneously characterize spatial autocorrelation and direct variable interaction, offering a powerful and interpretable alternative to existing approaches for multivariate spatial analysis.



### ***Bayesian Inference for Single-factor Graphical Models***

Presenter: Adrian Dobra

We introduce efficient MCMC algorithms for Bayesian inference for single-factor models with correlated residuals where the residuals' distribution is a Gaussian graphical model. We call this family of models single-factor graphical models. We extend single-factor graphical models to datasets that also involve binary and ordinal categorical variables and to the modeling of multiple datasets that are spatially or temporally related. Our models are able to capture multivariate associations through latent factors across time and space, as well as residual conditional dependence structures at each spatial location or time point through Gaussian graphical models. We illustrate the application of single-factor graphical models in simulated and real-world examples.

Discussant: Kate Calder

### **Covariance Estimation and Shrinkage (Room 1104)**

#### ***Inference on covariance structures in high-dimensional multi-view data***

Presenter: Lorenzo Mauri

This work focuses on covariance estimation for multi-view data. Popular approaches rely on factor-analytic decompositions that have shared and view-specific latent factors. Posterior computation is conducted via expensive and brittle Markov chain Monte Carlo (MCMC) sampling or variational approximations that underestimate uncertainty and lack theoretical guarantees. Our proposed methodology employs spectral decompositions to estimate and align latent factors that are active in at least one view. Conditionally on these factors, we choose jointly conjugate prior distributions for factor loadings and residual variances. The resulting posterior is a simple product of normal-inverse gamma distributions for each variable, bypassing MCMC and facilitating posterior computation. We prove favorable increasing-dimension asymptotic properties, including posterior contraction, a Bernstein von Mises theorem, and central limit theorems for point estimators. We show excellent performance in simulations, including accurate uncertainty quantification, and apply the methodology to integrate four high-dimensional views from a multi-omics dataset of cancer cell samples. Joint work with David B. Dunson.

#### ***Eigenstructure Inference for High-dimensional Covariance with Generalized Shrinkage Inverse-Wishart Prior***

Presenter: Jaeyong Lee

In multivariate statistics, estimating the covariance matrix is essential for understanding the dependence structure among variables. In high-dimensional settings, where the number of covariates increases with the sample size, it is well known that the sample covariance matrix becomes inconsistent. In particular, the largest sample eigenvalue tends to be substantially larger than the corresponding population eigenvalue, while the smallest sample eigenvalue tends to be substantially smaller than its population counterpart. This phenomenon has been widely recognized in the literature and is often described as the over-dispersion of sample eigenvalues. The inverse-Wishart prior, a standard choice for Bayesian covariance estimation, also suffers from over-dispersion in posterior eigenvalues. To address this issue in high-dimensional settings, the shrinkage inverse-Wishart (SIW) prior has recently been proposed. Despite its conceptual appeal and empirical success, however, the asymptotic justification for the SIW prior remains limited. In this paper, we propose a generalized shrinkage inverse-Wishart (gSIW) prior for high-dimensional covariance modeling. By extending the SIW framework, the gSIW prior accommodates a broader class of prior distributions and enables the derivation of theoretical properties under specific parameter choices. In particular, under the spiked covariance assumption, we establish the asymptotic behavior of the posterior distribution for both eigenvalues and eigenvectors by explicitly evaluating posterior expectations for two parameter settings. This explicit analysis provides insights into the large-sample behavior of the posterior that are difficult to obtain through general posterior asymptotic theory. Finally, simulation studies demonstrate that the proposed prior yields accurate estimation of the eigenstructure, particularly for spiked eigenvalues, while also providing competitive uncertainty quantification across a range of high-dimensional settings. For spiked eigenvectors, its performance is generally comparable to that of competing approaches, including the sample covariance estimator.

#### ***Hierarchical Bayesian Estimation of Covariance Matrices***

Presenter: Daniel Yekutieli

We present a general framework for Normal covariance matrix estimation using two generative models based on eigen-decomposition. The first is an oracle model that takes a sorted eigenvalue vector and samples the eigenvector matrix uniformly from the orthogonal group. The second is a hierarchical Bayesian (hBayes) model that samples the eigenvector matrix and incorporates a finite Pólya tree prior distribution for the eigenvalues. Utilizing

the equivariance properties of common loss functions and shrinkage estimators under the orthogonal group, we establish the oracle model's Bayes rules as theoretical risk-minimizing benchmarks for estimating covariance and precision matrices. The hBayes model generates posterior samples of the eigenvalue vectors and eigenvector matrices, which we use to estimate sorted eigenvalues and approximate the oracle Bayes rules. Simulation results demonstrate that our hBayes approach provides accurate eigenvalue estimates and yields covariance and precision matrix estimates with lower risks than existing methods.

## **Bayesian Approaches to Challenges in Finance, Econometrics, and the Social Sciences (Room Main Hall)**

Panelists: Ray-Bing Chen, Yashushi Omoro, Mike K. P. So, Cathy W. S. Chen, Richard Gerlach, Manabu Asai, Robert Kohn

## **Keynote Lecture 2: 16:00–17:00**

### ***Gaussian Process Approximations***

Presenter: Botond Szabo

Gaussian Processes (GPs) are frequently used in machine learning and various fields of science due to their flexibility, built-in uncertainty quantification, theoretical foundations, and interpretability. However, for large sample sizes, Bayesian methods using GP priors become computationally infeasible. Therefore, various approximations are considered in practice. However, these methods often have limited theoretical underpinning. In this paper, we focus on two standard methods: variational Bayes and Vecchia approximations. We derive theoretical guarantees and limitations for these procedures and apply them to (nonlinear) inverse problems and Deep Gaussian Processes. The theoretical results are accompanied by numerical analyses.

## **Poster Session 2: 17:15–19:30**

### ***Sequential Bayesian Estimation with fixed CPU budget per iteration***

Presenter: Andrea Pandolfi

Sequential Monte Carlo (SMC) samplers are a widely used class of algorithms for approximating sequences of probability distributions using a population of weighted particles that evolve via resampling and propagation steps. Many SMC samplers, such as Iterated Batch Importance Sampling (IBIS) or SMC2, adaptively decide when to propagate the particles based on the effective sample size of the population. This makes the overall cost scale linearly with the number of iterations, but concentrates most computation in a few expensive propagation steps. In this work, we propose a modification of these algorithms that maintains a constant computational cost per iteration, by propagating, at each time, only a decreasing fraction of the particles. We provide theoretical results that show how to properly set the proportions of propagated particles to make the error of the particle estimate and the overall cost equal to the one of adaptive SMC samplers. Numerical illustrations with both real and simulated data confirm the effectiveness of the proposed methodology.

### ***PARTOPT+ for Spatial Clustering: Posterior Approximation and Efficient Search in the Space of Partitions.***

Presenter: Andrew Ricketts

Bayesian nonparametric clustering models are widely used in spatial and spatiotemporal domains. Although these approaches capture the uncertainty in the underlying clustering, exploring the posterior with MCMC is challenging due to the vast discrete space of partitions and the added complexity of spatial dependencies. To balance computational tractability with uncertainty quantification, we reframe the exploration as an approximation of the posterior distribution, using not only the Maximum A Posteriori (MAP) partition, but also several high-probability partitions. We also propose a flexible MCAR-within-cluster GLM model applicable to a broad range of problems. To estimate this approximation we present PARTOPT+, a greedy neighbourhood search algorithm for optimisation over the space of partitions with a corresponding R and Python package. The proposed algorithm combines the search over partitions (based on local and global moves) with an intermediate optimization step to overcome the non-conjugacy of the model. With in-depth simulation studies, we examine the type of moves used and investigate the algorithm's strengths. We show how our heuristics for exploring the space provide a broad enough search to escape local optima but while maintaining computational efficiency. We demonstrate our algorithms effectiveness compared to existing approaches for similar tasks and apply our work to disease mapping.

### ***Bayesian estimation of species richness***

Presenter: Anna Petranzan

Species sampling problems play a central role in ecology, biology, genomics, and other fields where quantifying diversity and predicting the number of unobserved species or taxa are essential. Adopting a nonparametric Bayesian perspective provides a transparent and coherent framework for addressing these questions. Within this framework, Gibbs-type priors occupy a prominent position, as they generalize well-known processes such as the Dirichlet and Pitman–Yor processes. In this work, we focus on a specific Gibbs-type prior introduced by Gnedin (2010), which assumes a finite but random number of species in the population. This feature naturally leads to the Bayesian estimation of the total number of species. Building on recent developments, we derive new distributional results for several key quantities, such as the number of observed species and the number of new species revealed in additional sampling, both a priori and a posteriori. Despite the rich theoretical foundation of Gnedin’s model, empirical studies evaluating its performance remain, to our knowledge, largely unexplored. To address this gap, we develop and illustrate practical empirical procedures for estimating population size and species–discovery curves. Finally, we analyze the influence of hyperparameters and propose strategies for their elicitation.

### ***Collapsed Variational Inference for Dirichlet Process Mixtures with unknown concentration parameter and covariance : Scaling High Dimensional Clustering***

Presenter: Anshel Pal

Dirichlet process mixture models (DPMM) have emerged as a principled framework for density estimation and data clustering. Offering a robust alternative to parametric mixture models that rely on pre-specifying cluster number  $K$ , DPMM provide a flexible way of estimating  $K$  from an infinite choice of partitions induced by a Dirichlet Process  $DP(\alpha, G_0)$  prior on the data. Posterior consistency is guaranteed by an adaptive estimation of the DP concentration parameter  $\alpha$ , and the estimated  $K$  jointly depends on  $\alpha$  as well as the variance hyper-parameter of DP base distribution  $G_0$ . Although MCMC methods have bridged the gap between theory and application for such nonparametric Bayesian models, they scale poorly to high-dimensional data (e.g. with several hundred or even thousands of features) and suffer from slow and difficult convergence. Approximation techniques like Variational Inference (VI) provide an alternative for faster convergence, but lack implementation with increasing model fidelity. For instance, collapsed VI is implemented to integrate out hierarchical parameters, which are otherwise considered independent in mean field VI. However, the resulting complexity imposes a significant barrier to a fully adaptive DPMM inference for high dimensional data. We propose a novel method that performs adaptive clustering with DPMM using collapsed VI, while incorporating weakly-informative priors for  $\alpha$  and  $G_0$ . Our model adopts a stick-breaking representation of the DP and optimises the evidence lower bound (ELBO) leveraging a coordinate ascent algorithm. We also illustrate the importance of model covariance, and the effect of its structure and prior choice by considering different parameterisations of the data covariance matrix. On high-dimensional Gaussian simulated data, our model demonstrates substantially faster convergence than a state-of-the-art MCMC sampler. We further evaluate performances on Negative Binomial simulations and conduct sensitivity analyses to assess robustness on realistic data conditions. Application to a leukemia transcriptomic dataset comprising 2,194 gene expression profiles successfully recovers every known subtype from both complete and reduced gene signatures, all while identifying additional gene expression-based sub-clusters with meaningful biological interpretation.

### ***Weighted Fisher divergence for high-dimensional Gaussian variational inference***

Presenter: Aoxiang Chen

Bayesian inference has many advantages for complex models. However, standard Monte Carlo methods for summarizing the posterior can be computationally demanding, and it is attractive to consider optimization-based variational approximations. Our work considers Gaussian approximations with sparse precision matrices which are tractable to optimize in high-dimensional problems. Although the optimal Gaussian approximation is usually defined as the one closest to the target posterior in Kullback-Leibler divergence, it is useful to consider other divergences when the Gaussian assumption is crude, in order to capture important features of the posterior for a given application. Our work studies the weighted Fisher divergence, which focuses on gradient differences between the target posterior and its approximation, with the Fisher and score-based divergences being special cases. We make three main contributions. First, we compare approximations for weighted Fisher divergences under mean-field assumptions for both Gaussian and non-Gaussian targets with Kullback-Leibler approximations. Second, we go beyond mean-field and consider approximations with sparse precision matrices reflecting posterior conditional independence structure for hierarchical models. Using stochastic gradient descent to enforce sparsity, we develop two approaches to minimize the weighted Fisher divergence, based on the reparametrization trick

and a batch approximation of the objective. Finally, we examine the performance of our methods for examples involving logistic regression, generalized linear mixed models and stochastic volatility models.

#### ***Proximal Hamiltonian Monte Carlo***

Presenter: Apratim Shukla

Bayesian formulation of modern day signal processing problems has called for improved Markov chain Monte Carlo (MCMC) sampling algorithms for inference. The need for efficient sampling techniques has become indispensable for high dimensional distributions that often characterize many core signal processing problems, e.g., image denoising, sparse signal recovery, etc. A major issue in building effective sampling strategies, however, is the non-differentiability of the underlying posterior density. Such posteriors are popular in models designed to recover sparse signals. As a result, the use of efficient gradient-based MCMC sampling techniques becomes difficult. We circumvent this problem by proposing a Proximal Hamiltonian Monte Carlo (p-HMC) algorithm, which leverages elements from convex optimization like proximal mappings and Moreau-Yosida (MY) envelopes within Hamiltonian dynamics. Our method improves upon the current state of the art non-smooth Hamiltonian Monte Carlo as it achieves a relatively sharper approximation of the gradient of log posterior density and a computational burden of at most the current state-of-the-art. A chief contribution of this work is the theoretical analysis of p-HMC. We provide conditions for geometric ergodicity of the underlying HMC chain. On the practical front, we propose guidance on choosing the key p-HMC hyperparameter – the regularization parameter in the MY-envelope. We demonstrate p-HMC's efficiency over other MCMC algorithms on benchmark problems of logistic regression and low-rank matrix estimation.

#### ***Scalable Bayesian inference for high-dimensional mixed-type multivariate spatial data***

Presenter: Arghya Mukherjee

Spatial generalized linear mixed-effects methods are popularly used to model spatially indexed univariate responses. However, with modern technology, it is common to observe vector-valued mixed-type responses, e.g., a combination of binary, count, or continuous types, at each location. Methods that allow joint modeling of such mixed-type multivariate spatial responses are rare. Using latent multivariate Gaussian processes (GPs), we present a class of Bayesian spatial methods that can be employed for any combination of exponential family responses. Since multivariate GP-based methods can suffer from computational bottlenecks when the number of spatial locations is high, we further employ a computationally efficient Vecchia approximation for fast posterior inference and prediction. Key theoretical properties of the proposed model, such as identifiability and the structure of the induced covariance, are established. Our approach employs a Markov chain Monte Carlo-based inference method that utilizes elliptical slice sampling in a blocked Metropolis-within-Gibbs sampling framework. We illustrate the efficacy of the proposed method through simulation studies and a real-data application on joint modeling of wildfire counts and burnt areas across the United States.

#### ***Hierarchical Bayes meets hierarchical forecasting: A global forecasting model for hierarchically structured time series***

Presenter: Arwen Nugteren

Hierarchical time series require forecasts at all cross-sectional levels that capture uncertainty probabilistically in order to inform decision-making. Current hierarchical forecasting methods generate independent forecasts at each level, and reconcile them post-hoc to ensure coherence between upper and lower hierarchical levels. Such post-hoc corrections do not incorporate hierarchical structure or decision goals in the underlying parameter estimation. Instead, we propose a Bayesian hierarchical model in which parameters are grouped according to the natural cross-sectional hierarchy of the system. In this approach, information is shared between hierarchical levels through the Bayesian hierarchy and probabilistic uncertainty quantification is provided in the form of the posterior predictive distribution. Applying a generalised Bayesian update, we penalise incoherence across hierarchical levels, thus generating parameter estimates that maximise log probability subject to a soft coherence constraint. Lastly, we extend this updating paradigm to weight different hierarchical levels relative to their roles in decision-making, focusing the global model on those hierarchical levels most relevant to decision outcomes. We demonstrate improvements in predictive accuracy metrics on real data using our proposed methodology.

#### ***Bayesian Hierarchical Modelling of Multidimensional Migration Flows with Incomplete and Inconsistent Data***

Presenter: Aysha Basheer

Reliable estimation of international migration is hampered by sparse data, inconsistent reporting, and complex demographic structure. We present a hierarchical Bayesian framework for estimating age- and sex-specific bilateral migration flows that explicitly models the multidimensional nature of migration across origin, destination,

age, sex, and time. Reported counts from both sending and receiving countries are linked through a Poisson log-normal model, where latent true flows are modelled with structured demographic effects and socio-economic covariates, including life expectancy, unemployment, and education. The model also has a measurement layer that accounts for systematic reporting biases between data sources through origin–destination–time-specific bias terms, enabling coherent inference from discrepant observations. Hierarchical priors induce partial pooling across countries and demographic groups, stabilizing estimation in sparse cells while preserving meaningful heterogeneity. Covariate effects are allowed to vary by age and sex, with demographic constraints incorporated through structured priors and masking where covariates are not substantively relevant. Variance components are assigned weakly informative hyperpriors, supporting regularization and robust uncertainty propagation. Posterior inference is conducted using Markov chain Monte Carlo implemented in JAGS with non-centered parameterizations to improve computational efficiency. The model yields full posterior distributions of migration flows and derived age–sex proportions, providing principled uncertainty quantification. Counterfactual simulations based on posterior predictive distributions illustrate how changes in socio-economic conditions shift migration patterns across demographic groups. This work demonstrates how hierarchical Bayesian modelling can reconcile multiple noisy data sources while delivering policy-relevant, uncertainty-aware estimates for complex demographic systems.

### ***Flexible Bayesian Variable Selection Under Shape Constraints***

Presenter: Ayumi Mutoh

A Gaussian process (GP) is a popular metamodel for non-parametric Bayesian regression, providing a flexible way to model unknown functions while also quantifying uncertainty. Complex systems often involve a large number of factors and identifying those with the most significant influence on the output is essential for efficient modeling. Variable selection in GPs typically operates through prior distributions on the mean and/or correlation parameters that induce sparsity with the posterior distributions. We introduce a new approach that can impose shape constraints, such as monotonicity, on a hierarchical, additive GP model. Elliptical slice sampling allows for rejection-free sampling of the shape-constrained GPs via transformations of latent, unconstrained GPs. A sparsity-inducing prior is assigned to each factor's GP scale parameter to allow for fast Gibbs sampling. We focus our demonstration on the case of monotonic functions and also propose modifications to detect issues caused by model misspecification.

### ***Bayesian Image Segmentation of Remote Sensing Images***

Presenter: Bao Khanh Nguyen

Accurate and scalable land cover classification is essential for global conservation monitoring and policy-making. While remote sensing images provide a cost-effective alternative to ground surveys, current methods often lack principled uncertainty quantification and require substantial labelled data, limiting their usability and reliability in new regions with distribution shifts. We propose a Bayesian spatial mixture modelling approach for image segmentation, extending the classical Potts model by allowing for a generalised spatial dependence structure and incorporating informative priors estimated from pre-existing labelled data. Our framework, called POTTERS (Potts Model for Enhanced Remote Sensing), offers several advances, including (i) incorporating prior information from pre-existing labelled images, (ii) accounting for class-dependent interactions, (iii) automatically discovering new land-cover types when present and (iv) providing robust uncertainty quantification in the land cover classification setting as well as other applications. To ensure scalability to large remote sensing images, we develop an efficient variational inference algorithm for posterior approximation. We demonstrate the benefits of our approach in numerical simulations and apply it to land cover classification in a case study in Scotland.

### ***Newton's algorithm as a gradient flow: a geometric framework for recursive mixture estimation***

Presenter: Bernardo Flores

Bayesian nonparametric mixture models provide a flexible framework for density estimation but are often hindered by the computational expense of traditional inference methods such as Markov chain Monte Carlo. A fast, recursive algorithm proposed by Newton (2002) offers a practical alternative, yet its formal connection to Bayesian inference and its theoretical properties remain only partially understood. This paper reveals a geometric interpretation of this class of predictive recursions. Newton's recursion is a discrete-time approximation of a gradient flow on the space of probability measures governed by the Fisher-Rao geometry, providing the first rigorous dynamical characterization of this family of estimators. This perspective yields a principled theoretical foundation for studying these recursions, clarifying their convergence behaviour, situating them within the variational Bayes literature, and providing a systematic basis for generalization by modifying the underlying geometry and discretization. In contrast to approaches that construct gradient flows from a prescribed variational objective, this work proceeds in the reverse direction, beginning from an existing recursive estimator and uncovering the



variational problem it implicitly solves.

### ***Bayesian multinomial multilevel logistic regression with fixed- and random-effects selection***

Presenter: Bertil Wegmann

We propose a novel Bayesian multinomial multilevel logistic regression model for any setting where each observational unit belongs to a specific group and a non-ordered category. To allow for automatic selection of both fixed- and random-effects, we use Pólya-Gamma data augmentation and develop an efficient Gibbs sampling algorithm via a hierarchical spike-and-slab prior. Inference is fast and does not rely on any analytical approximations or numerical integration. Guidelines for user-defined prior selection are provided. Simulations show that our approach is accurate and can discriminate well between different configurations of fixed- and random-effects. To demonstrate the general applicability of our approach, we show that the model performs well on three real-world datasets from education, medicine, and political science.

### ***Spatio-temporal modelling of bumblebee populations***

Presenter: Cam Milliken

Bumblebees are critically important pollinators, but many species are in decline. Standardised, long-term monitoring schemes are required to assess the extent of this decline and inform conservation efforts, but monitoring bumblebees at the desired spatiotemporal scale is challenging. The Pollinator Monitoring Scheme (PoMS) has been set up for this purpose. Participants in the Flower-Insect Timed Count (FIT Counts) survey record the number of bumblebees they observe landing on flowers in ten minutes within a 50cm x 50cm quadrat, generating count data of the number of bumblebees observed at a particular space and time. The poster will discuss using the Integrated Laplace Approximation (INLA) with *inlabru* to implement a hierarchical spatiotemporal model that takes into account site-specific and meteorological covariates. The poster will also present related findings, such as predictions of underlying spatiotemporal bumblebee population density and estimates of bumblebee phenology, and possible future directions.

### ***Bayesian Sensitivity Analysis of Linear Regression Models***

Presenter: Carlos Cinelli

We develop a Bayesian framework for sensitivity analysis in linear regression models. We study both the classical Gaussian linear model and more general, misspecified cases. Unobserved confounding is indexed by the residual variation it explains in the treatment and the outcome, and priors on these sensitivity parameters induce a posterior distribution for the treatment effect of interest. Within this setup, we define Bayesian robustness values: critical levels of confounding strength such that, for any prior concentrating most of its mass below this level, the posterior continues to concentrate on a nonzero causal effect. We also show how beliefs about the relative strength of observed and unobserved covariates naturally translate into benchmarking priors that naturally bounds how much unobserved confounding can explain the observed effect. We illustrate the framework in several empirical applications.

### ***Bayesian Adaptive Lasso for Randomized Response Survival Data***

Presenter: Chi-Chung Wen

Surveys on sensitive topics often yield current-status survival data that are prone to response bias due to privacy concerns. The randomized response technique (RRT) provides a principled approach to mitigate such bias by ensuring respondent confidentiality. This study develops a Bayesian adaptive lasso method for simultaneous estimation and variable selection in current-status survival data collected under various RRT designs. The event-time distribution is modeled using a flexible class of transformation models encompassing both proportional hazards and proportional odds structures. Given the complex data structure, Markov chain Monte Carlo algorithms combining Gibbs sampling and Metropolis–Hastings updates are used for posterior computation. Simulation studies are conducted, and an application to extramarital sexual behavior data is presented. The proposed approach provides a sparse variable selection method for privacy-sensitive survival data under a Bayesian framework.

### ***Comprehensive Physical Fitness Testing as a Predictor of Sarcopenia in Community-Dwelling Elderly***

Presenter: Chih-Chun Tsai

This poster presents an analysis of 745 Taiwanese older adults, examining how detailed physical fitness testing predicts sarcopenia risk. Hand grip, agility, gait speed, and endurance were evaluated, revealing that individuals with sarcopenia underperform in nearly all domains compared to controls. Logistic regression identified age, male sex, nutritional assessment score, and 8-foot up-and-go time as independent predictors, with a strong overall model performance (AUC=0.867). This poster is partly based on our previously published study in *BMC Geriatrics* (Lin et al., 2024; doi:10.1186/s12877-024-05528-7), with minor extensions and updates to the analysis. It is



presented here to encourage further discussion on the use of fitness testing for early sarcopenia risk identification and prevention.

***What's in a cubit? Reconstructing the Babylonian Astronomical Diaries with science-informed Bayesian latent variable modeling***

Presenter: Christopher Wolfram

The Astronomical Diaries are a collection of Babylonian clay tablets that are the product of the longest-running data collection project in human history. For several hundred years from the mid-7th century to the 1st century BCE, the Diaries record daily observations of astronomical phenomena, as well as the weather, the level of the Euphrates River, and commodity prices in the market in Babylon. Many of the Diaries have been translated into English and digitized, but owing to the volume and unstructured nature of the data, the bulk of the contained observations has never been studied systematically. Moreover, the data exhibits substantial error and missingness due to damage to the tablets as well as noise in the digitization and translation processes. We develop an LLM-based pipeline to identify and extract observations from the unstructured text. We then apply a novel Bayesian latent variable model which compares the extracted ancient data to predictions from modern astronomical models. In so doing, the Bayesian model can identify errors (both ancient and modern) and impute missing data, while also estimating historically-meaningful latent variables, such as the exact times diarists recorded observations or how many modern degrees are in a Babylonian cubit.

***A Bayesian Nonparametric Approach to the Multi-Armed Bandit Problem in Trait Allocation Models***

Presenter: Claudio Del Sole

In trait allocation models, each observation may display multiple features and may have different levels of belonging for each feature. For example, multiple individuals from distinct species may be observed within a time or spatial window. When data are collected from multiple populations, the same feature may appear in two different observations both within and across populations. We consider the problem of maximizing the cumulative number of distinct observed features by sequentially selecting the population from which the next observation is sampled. This task can be naturally framed as a multi-armed bandit problem, where payoffs correspond to the number of newly discovered features. We propose a Bayesian nonparametric approach relying on hierarchical gamma processes, which promotes borrowing of information among populations by establishing a common set of features, while accounting for their heterogeneity through population-specific parameters. The tractable posterior characterization facilitates the implementation of a Thompson sampling strategy to effectively balance exploration and exploitation. This standard approach is compared with a simpler strategy that selects the population with the highest posterior estimate for the number of new features, conditionally on populations parameters. For further comparison, we also consider a novel frequentist estimator, derived along the lines of the popular Good-Turing estimator for species sampling model. The performances of the proposed algorithms are assessed through simulation studies, and compared on two real datasets: the first contains tree species counts from ecological survey plots at various locations in Japan, the second collects gene expression levels from single-cell sequencing in different subjects. This is a joint work with Federico Camerlenghi and Mario Beraha.

***Variance Deltas: Interactive Visualizations for Explaining Posterior Uncertainty***

Presenter: Collin Cademartori

In theory, the posterior distribution jointly quantifies uncertainty over all unknowns simultaneously. But in practice, common posterior summaries (such as marginal histograms and intervals) leave most of the dependencies between parameters unused. In the context of graphical models (particularly Markov random fields), we introduce an interactive, tree-structured visualization system to enable the discovery of useful posterior dependencies. This system, which we term posterior uncertainty trees, is particularly adapted to complex models where: (I) a small number of quantities of substantive interest are modeled using (potentially many) nuisance parameters, and (II) the quantities of interest are at best weakly identified from the observed data. In this case, posterior uncertainty trees can help the modeler to systematically search for collections of nuisance parameters and/or missing data which can explain their uncertainty about quantities of interest and point to possible remedies. We demonstrate this approach using a model of U.S. presidential election polls. For this model and data, posterior uncertainty trees are used to reveal that a majority of the uncertainty in a state-level estimate can be explained in terms of both the time from the last polls to the election date and a particular bias parameter which is poorly identified from data.

***A Boltzmann Generator Framework for Modeling and Forecasting International Trade Networks***

Presenter: Costanza Bosone

We introduce a Boltzmann Generator framework for modeling and forecasting the global trade network. De-

parting from traditional econometric gravity models, we propose a conditional deep generative architecture that maps macroeconomic and geographic covariates into full probabilistic distributions of trade matrices. The model's energy-based formulation connects statistical physics and international economics, enabling realistic trade simulations that capture both bilateral dependencies and higher-order network effects. Bayesian optimization governs the tuning of the model's hyperparameters, ensuring principled exploration of the high-dimensional loss landscape and posterior consistency. Using trade data for 206 countries between 2001 and 2020, we demonstrate that the Boltzmann model achieves superior in-sample and out-of-sample predictive accuracy compared with established econometric and machine learning benchmarks. Statistical tests confirm the model's robust predictive dominance across time and country pairs. Beyond forecasting, the framework supports counterfactual policy analysis: simulations of scenarios reveal distinct propagation mechanisms through the global trade network, highlighting China's central mediating role in transmitting shocks. The Boltzmann approach thus provides an interpretable, and generative alternative for analyzing trade dynamics, quantifying uncertainty, and evaluating policy in complex economic systems.

#### ***Methods for Informative Priors via Predictives using Predictively Oriented and Generalised Bayes***

Presenter: Craig Dunipace

Direct prior elicitation on the parameter space is often difficult (Mikkola et al, 2024). To address this challenge, recent work has focused on eliciting predictive distributions on the observable space (Bockting et al, 2024), then seeking to translate these distributions directly into informative priors on the parameter space. However, existing approaches are either limited to simple models or are highly computationally expensive. In this work, we highlight the connection between prior specification via elicited predictives and recent 'post-Bayesian' developments: specifically, inferentially-oriented Generalised Bayes (Bissiri et al, 2016) and Predictively Oriented Posteriors (McLatchie et al, 2025). Both approaches aim to condition prior parameter distributions on elicited predictive distributions. The Predictively Oriented approach has been proposed as a way to capture the epistemological uncertainty of a misspecified model, a setting with relevance when predictives, or experts, express partially contradictory information. The Generalised Bayes maintains the interpretable inferential focus of standard Bayes and clearer interpretation of information quantification, via generalisations of ideas such as effective sample size, and greater flexibility in choice of loss function. We compare the behaviour of these approaches in several contexts: varying degrees of model misspecification, combinations of parameter and predictive elicitation, and predictives defined on different levels of marginalisation. Evaluation focuses on faithfulness to elicited information, computational tractability, and suitability of distributions as informative priors. We illustrate with a real world biostatistical regression example with expert elicitations.

#### ***Dynamic Factor Stochastic Volatility in Mean: Evidence from U.S. Macroeconomic Data***

Presenter: Daichi Hiraki

A stochastic volatility in mean (SVM) model is developed within a dynamic factor framework to capture common movements in macroeconomic variables under time-varying uncertainty. Motivated by theoretical considerations in macro-finance, the model allows conditional volatility to directly affect the conditional mean through a volatility-in-mean component. This feature enables the model to account for time-varying risk premiums that are otherwise difficult to capture in standard factor stochastic volatility models. The model is estimated using Bayesian Markov chain Monte Carlo methods and applied to quarterly U.S. macroeconomic data from the FRED-QD dataset. The empirical results illustrate how the SVM structure can be embedded in a latent factor setting to study macroeconomic dynamics under uncertainty, providing a basis for future forecasting and structural analysis.

#### ***Correcting Misspecified Bayesian Causal Models***

Presenter: Daniel Manela

Clinical trial practitioners often rely on simple statistical models to estimate treatment effects, yet these models can be sensitive to subtle forms of misspecification, particularly in how outcomes relate to baseline covariates. We study a class of "frugal" causal models that separate the treatment effect of interest from the complex dependence structure of the data, allowing analysts to model the causal margin directly while treating the covariate-outcome relationship as nuisance parameters. Our aim is to understand when Bayesian inference can remain reliable for key causal quantities even if the nuisance component is misspecified. We explore whether general Bayesian misspecification approaches can offer improved robustness and uncertainty quantification while preserving valid inference for the treatment effect. This work provides a pathway for using flexible Bayesian models in clinical trials without requiring full confidence in high dimensional dependence assumptions.

#### ***Hierarchical Latent-Variable Models for Exchangeable Graphs***

Presenter: Daniele Micheletti

The Aldous–Hoover representation theorem characterizes all exchangeable dense graphs, including classical models such as Erdos-Renyi, Stochastic Blockmodels, and Latent Space Models. Leveraging this framework, we propose a generative latent-variable model for graphs based on a Variational Graph Autoencoder with permutation-equivariant Set Transformer encoder and decoder layers. A hierarchical latent structure combines a global latent variable, modeled via a diffusion process, with node-specific Gaussian mixtures to capture graph-level and node-level dependencies. Beyond full exchangeability, we introduce a partially exchangeable extension in which node metadata define the subgroup of admissible permutations. This generalization allows the model to handle structured graphs where node order or roles matter, violating full exchangeability. Experiments on synthetic benchmarks and real datasets—including community-structured, ego, and protein graphs—demonstrate accurate recovery of key structural statistics, interpretable latent representations, and robustness to deviations from full exchangeability.

#### ***Joint Quantile Shrinkage: A State-Space Approach toward Non-Crossing Bayesian Quantile Models***

Presenter: David Kohns

Crossing of fitted conditional quantiles is a prevalent problem for quantile regression models. We propose a new Bayesian modelling framework that penalises multiple quantile regression functions toward the desired non-crossing space. We achieve this by estimating multiple quantiles jointly with a prior on variation across quantiles, a fused shrinkage prior with quantile adaptivity. The posterior is derived from a decision-theoretic general Bayes perspective, whose form yields a natural state-space interpretation aligned with Time-Varying Parameter (TVP) models. Taken together our approach leads to a Quantile-Varying Parameter (QVP) model, for which we develop efficient sampling algorithms. We demonstrate that our proposed modelling framework provides superior parameter recovery and predictive performance compared to competing Bayesian and frequentist quantile regression estimators in simulated experiments and a real-data application to multivariate quantile estimation in macroeconomics.

#### ***Hierarchical Bayesian accelerated failure time models for clustered survival data***

Presenter: Dongu Han

The proposal is to apply a hierarchical Bayesian spectral analysis regression to accelerate the failure time model for the grouped survival data. The proposed models employ hierarchical Gaussian process priors to lift the restrictive parametric assumption and smooth the estimated functions within and between clusters. This hierarchical approach allows the provision of more accurate estimates than fitting separate models to each cluster when data within clusters are sparse. A Dirichlet process mixture model is investigated with a Weibull base measure for clustered survival data. An efficient Markov chain Monte Carlo algorithm is presented, and it describes how to approximate functional estimands such as hazard and residual life functions that are of interest to researchers as well as practitioners. A theoretical justification is provided for the proposed method through weak posterior consistency. An extensive empirical study based on artificial and real datasets has been conducted to assess the performances of the proposed models.

#### ***Bayesian Nonparametric Multiple Change-Point Detection for Compositional Time Series***

Presenter: Edoardo Marchionni

Compositional time series consist of sequences of real-valued vectors in which each observation is constrained to sum to one. They arise naturally across a broad range of fields, including environmental science, biology, demography, and economics. The most common strategy for analyzing such data relies on applying transformations and then using standard techniques for real-valued time series. However, direct modeling of this type of data, especially from a Bayesian perspective, has received relatively limited attention. In this work, we focus on one of the most widely studied inferential targets in time series analysis: change-point detection. Specifically, we rely on a simplex-valued diffusion process to model such data directly on their natural support. We then combine such a process with a Bayesian nonparametric model for off-line multiple change-point detection, in order to identify the different regimes over which the diffusion process behaves homogeneously. The proposed methodology is illustrated and validated through real-world case studies.

#### ***A Bayesian Approach to Optimal Treatment Combinations while Preserving Patient Safety***

Presenter: Edward L Boone

Physicians are often confronted with treating diseases, using a simultaneous administration of several medicines to obtain the desired outcome. However, many of these medicines have adverse side effects that must be managed by the physician to ensure the safety of the patient, as well as promote treatment compliance to the treatment regime. An example of this is hypertension (affecting more than 30% of the Qatari population) that is often treated with

a combination of an angiotensin converting enzyme inhibitor (ACEI) and a calcium channel blocker (CCB). Currently, physicians have no rigorous systematic data-based approach to determine an optimal combination of medicines; instead, they often use a recursive approach of adjusting the dosages of each of the medicines until a satisfactory combination is achieved. No statistical methodology exists that allows for researchers or physicians to estimate the optimal combination of medicines to treat the disease, while controlling for adverse effects. In this presentation, we will show a new statistical methodology to address this problem in both the linear and non-linear cases. The linear case is where the treatments may only have an additive effect and the non-linear case allows for interaction, antagonistic as well as synergistic effects. This Bayesian methodology will utilize Markov chain Monte Carlo (MCMC) techniques to fit toxicology and treatment models to clinical data, which will result in a large number of samples from the posterior distribution of model parameters. These posterior samples will be used to form instances of mathematical programs to be solved using techniques, such as linear programming and non-linear programming, resulting in a distribution of optimal treatment combinations. Several methodologies will be explored to interrogate the posterior distribution of the optimal treatment combinations to obtain a combination that can be administered to a patient.

### ***Bayesian Inference for Periodic Intraday Volatility***

Presenter: Edward M.H. Lin

We propose an innovative conditional autoregressive range (CARR) model with periodic time-varying parameters, which is referred to as the periodic conditional autoregressive range (PCARR) model. This model is able to display time series in which the volatility distribution changes over time following a seasonal or periodic pattern. The construction of parameter estimation and volatility forecasting involves Bayesian Markov chain Monte Carlo sampling methods. The model's ability to forecast and capture stylized facts is increased by incorporating intraday range data with the day-of-the-week effect, as shown in simulations and empirical analyses.

### ***Non-Asymptotic Error Bounds for Gradient Estimation in Importance-Weighted Variational Inference***

Presenter: El Houssaine Chahboun

This document explores gradient estimators in variational inference with importance weighting, focusing on the Variational Rényi importance weighted autoencoder (VR-IWAE) objective introduced by Daudel and Roueff (2024). Our key observation is that the VR-IWAE gradient estimator can be viewed as a special case of a self-normalized importance sampling (SNIS) estimator with appropriately chosen proposal and target distributions. Leveraging this connection, we adapt the asymptotic analysis of SNIS estimators developed by Deligiannidis et al. (2024) to the tilted VR-IWAE target, thereby obtaining asymptotic expansions that recover and provide an alternative route to the main result of Daudel and Roueff (2024). Additionally, we derive high-probability error bounds for the VR-IWAE gradient estimator by extending the concentration analysis of Chatterjee and Diaconis (2017). Our bounds yield explicit sample-size requirements and reveal that the central quantity governing concentration,  $L_\alpha$ , can be written in terms of Rényi  $\alpha$ -divergences between the model  $p_\theta$  and the variational distribution  $q_\phi$ . Finally, by the same observation, we can directly apply the debiasing technique of Deligiannidis et al. (2024), based on coupled Particle-Independent Metropolis-Hastings (PIMH) chains with common random numbers, to construct an unbiased gradient estimator of  $VR(\alpha)(\theta, \phi; x)$ .

### ***Bayesian Conformal Prediction as a Decision Risk Problem Using Bayesian Quadrature***

Presenter: Fanyi Wu

We introduce BQ-optimised Bayesian Conformal Prediction (BCP), a unified framework that formulates conformal prediction as a decision-risk minimisation problem while integrating Bayesian posterior predictive densities as non-conformity scores. A Bayesian quadrature (BQ) procedure is employed to optimise the predictive threshold by estimating the expected prediction-set size under input variability. At the user-specified miscoverage level  $\alpha = 0.2$ , BCP restores nominal coverage (81.7%) under prior misspecification, where traditional Bayesian credible intervals collapse to 53%, and reduces average interval width by 5.6% compared with split CP on the Diabetes dataset. On the Wisconsin Breast Cancer dataset, BCP achieves 81.2% coverage, with BQ optimisation further shrinking the average prediction-set size by 5.9%. These results show that BCP yields sharper, more efficient, and reliable prediction sets, advancing Bayesian decision-theoretic approaches to uncertainty quantification.

### ***Approximate Bayesian Fusion***

Presenter: Filippo Pagani

Distributed inference plays a pivotal role in modern computational statistics and machine learning. In the frequentist setting, divide-and-conquer methods can often aggregate results by simply averaging estimators computed on data subsets. However, distributed Bayesian inference is more challenging, as it requires reconstructing

the full posterior by combining subposterior distributions calculated across different machines. Bayesian Fusion is a powerful approach that enables distributed Bayesian inference while maintaining exactness, at the price of a sizeable computational cost. In this work, we propose a novel method that incorporates numerical approximations to alleviate the most computationally expensive steps of Bayesian Fusion, thereby achieving substantial reductions in runtime. Additionally, we prove theoretical bounds on the numerical error as a function of the step size of the approximation. Our approach retains the flexibility to approximate the target posterior distribution to an arbitrary degree of accuracy, and is a practical and efficient alternative for large-scale Bayesian inference in distributed environments.

### ***Non-reversible lifts of Metropolis-Hastings type Markov chains with efficiency lower bound***

Presenter: Florian Maire

Lifted samplers are a particular class of non-reversible MCMC algorithms which rely on an artificial momentum variable to avoid diffusive behaviors. While stemming from the statistical physics literature in the 90's, they have been recently, and successfully, applied in Bayesian statistics to a number of inference tasks including Model selection (Gagnon and Doucet, 2021), Logistic regression (Kamatani and Song, 2023) and Mixture models (Ascolani and Zanella, 2025). These papers justify, in some sense, the general belief that “lifting” a reversible Markov chain leads to more efficient MCMC algorithms. However, general results supporting such a claim are lacking. In fact, a recent counter-example (Gagnon and Maire, 2024) shows that lifting a reversible MCMC algorithm sampling from an Ising model in low temperature leads to asymptotic inefficiency, as measured by the asymptotic variance of the MCMC estimator. The question our work addresses can be summarized as follows: “to what degree of asymptotic inefficiency lifting a reversible MCMC can lead to?” We provide an answer to this question under a very general framework, by

1. Introducing lifts based on neighborhood splitting schemes of a “reference” Metropolis-Hastings-type reversible Markov chain defined on any state-space. Several well-known non-reversible MCMC algorithms arising in applications (e.g., the non-reversible jump algorithm of Gagnon and Doucet (2021), the non-reversible simulated tempering of Sakai and Hukushima (2016)) are lifts, in this particular sense, of such a reversible reference and thus fit in our framework.
2. Showing that, under mild assumptions, such lifts cannot be more than twice as asymptotically inefficient as the reversible reference. Moreover, the constant 2 in our main result cannot be improved and is independent both of the observable of interest and of the target distribution. It is thus dimension free and tight.

Our work justifies that, while there is potentially a lot to gain from lifting a reversible Markov chain (as seen empirically in applications), there is not much to lose. Several illustrations (on both discrete and continuous state-spaces) will be presented. We will also discuss extension of our ideas to d-directional liftings, as introduced in Ascolani and Zanella (2025), and the search for optimal lifts.

On the theoretical side, we see our work as complementing pioneering findings in Andrieu and Livingstone (2021) which essentially shows that any lifted MCMC cannot be more inefficient than a specific reversible kernel. However, this specific reversible kernel does not usually correspond to the reference Markov chain used in practice, hence limiting its application. By contrast, our work compares the reversible reference to the lifted counterpart designed by our approach. On the methodological side, our framework is easier to construct from a practical standpoint than the  $\Delta$ -guided walk of Kamatani and Song (2023) relying on the Metropolis-Haar kernel, acknowledging that their approach cannot be more inefficient than the corresponding reversible Metropolis-Haar chain.

This is joint work with Philippe Gagnon.

### ***Bayesian Networks with Beta Regression Nodes for Assessing Driver Gesture Data***

Presenter: Francisco Palmí Perales

Within the framework of the European BERTHA project, which aims to develop a comprehensive Driver Behaviour Model (DBM) for automated driving, this study analyses the probabilistic relationship between specific facial Action Units (AUCs) and the emotional state of rage in drivers. Adopting a fully Bayesian approach, we utilize Bayesian Networks (BNs, Ben-Gal 2008) to structure the complex dependencies between observed facial dynamics and emotional intensity through a Directed Acyclic Graph (DAG). This graphical framework allows for the factorization of the joint probability distribution into local conditional densities, providing a transparent representation of the stochastic associations between variables. Given that the variables of interest—muscular activation and reported emotion—are continuous and strictly bounded to the unit interval (0,1), we model these local conditional probabilities using Beta regression (Cribari-Neto and Zeileis, 2010). To rigorously model the interaction between gestures and emotions under different experimental conditions, we develop and propose three distinct modeling scenarios regarding the experimental phase (basal vs. provocation): treating the phase as a



stochastic node, incorporating it as a fixed covariate, and excluding it to assess structural robustness. Posterior results indicate that while the experimental context is the primary driver of fury levels, the “frowning” gesture serves as a robust predictor of rage, with biological sex playing a relevant but secondary modulating role. These findings exemplify the application of the proposed Bayesian Network, which incorporates Beta Regression in the modelization of some nodes, as an effective tool for predictive modeling.

***Wave model in phylolinguistics: a mathematical modelisation***

Presenter: Grégoire Clarté

The Wave model in linguistics has been a long competitor of the more famous Tree model of phylolinguistics. In the Wave model innovations spread between the languages as waves on a graph. This model accounts for the permanent communication of the populations and can be seen as the orthogonal of the inheritance based Tree model. In this work we propose the first mathematical description of this model along with numerical methods to infer the graph of relationship between the languages and the evolution parameters.

***Duality-based inference for the hidden Wright-Fisher diffusion with selection***

Presenter: Guillaume Kon Kam King

Exact inference for hidden Markov models requires the evaluation of all distributions of interest, filtering, prediction, smoothing and likelihood, with a finite computational effort. We provide sufficient conditions for exact inference for a class of hidden Markov models on general state spaces, given a set of discretely collected indirect observations nonlinearly linked to the signal, and a set of practical inference algorithms. The conditions we obtain are concerned with the existence of a certain type of dual process, which is an auxiliary process embedded in the time reversal of the signal, that in turn allows us to represent the distributions and functions of interest as countable mixtures of elementary densities or products thereof. We explore the applicability of this strategy for exact inference on Wright-Fisher diffusions, with or without selection, and propose a particle algorithm to infer mutation and selection parameters in any dimension.

***Bias-Aware Bayesian Integration of Randomized and Observational Studies***

Presenter: Hajime Ogawa

While experimental data, such as Randomized Controlled Trials (RCTs), enables unbiased estimation of Heterogeneous Treatment Effects (HTE) with minimal assumptions, it is often constrained by limited sample sizes and narrow support, leading to challenges in external validity. In contrast, observational data offers broader population coverage but struggles with internal validity due to potential unobserved confounding. This study addresses the limitations of both data types by integrating experimental and observational datasets for HTE estimation. We propose a method to learn the bias induced by unobserved confounders in observational data within the region covered by the RCT data. By incorporating this learned bias function, our approach enables accurate HTE estimation even when the dataset is dominated by observational data subject to unobserved confounding. Furthermore, explicitly modeling the bias function improves not only prediction accuracy but also the quantification of uncertainty. The resulting Bayesian predictive credible intervals exhibit natural behavior: they remain narrow within the RCT support and widen in regions relying solely on observational data.

***Bayesian Intransitive Modeling in Paired Comparisons***

Presenter: Hisaya Okahara

Traditional models for ranked data analysis, such as the Bradley–Terry model, assume globally transitive orderings among entities. However, real-world pairwise comparison data frequently exhibit intransitivity, particularly in domains involving multiple skills or strategies. This study proposes a Bayesian framework that flexibly models both transitive and intransitive structures simultaneously via the Combinatorial Hodge Decomposition. We develop an efficient Gibbs sampling algorithm by exploiting Pólya–Gamma data augmentation, yielding a fully conjugate posterior inference scheme. Simulation studies under intransitive scenarios demonstrate that the proposed model accurately recovers both transitive and intransitive components, outperforming standard transitive models.

***A Bayesian Model with Tensor Decomposition: Application to Jointly Analyzing Shot Attempts and Shot Successes in Basketball Games***

Presenter: Ho Kwok Wah

We propose a Bayesian Negative Binomial-Bernoulli model to jointly analyze the patterns behind field goal attempts and the factors influencing shot success. We apply nonnegative CANDECOMP/PARAFAC tensor decomposition to study shot patterns and use logistic regression to predict successful shots. The model is applied to shot chart data from the National Basketball Association, focusing on the regular seasons from 2015-16 to



2022-23. We consistently identify three latent features in shot patterns across all seasons and verify a popular claim from recent years about the increasing importance of three-point shots. Additionally, we find that the home court advantage in field goal accuracy disappears in the 2020-21 regular season, which was the only full season under strict COVID-19 crowd control, aside from the short bubble period in 2019-20. This finding contributes to the literature on the influence of crowd effects on home advantage in basketball games.

***Is low response-surface alignment responsible for poor uncertainty reduction in climate models when using observations?***

Presenter: Iain Webb

Whilst observations of the observable response variables of a complex climate model will produce a sizeable reduction in these variables' uncertainty, they don't always significantly reduce the uncertainty in unobservable response variables of the same model. Could the reason for this lie in the alignment of the two variables' response surfaces relative to one another? Can a simple measure offer the means to check the potential for uncertainty reduction prior to collection of data? A measure involving the comparison of partial derivatives for a pair of response surfaces will be presented. An overview of the models used - both linear models and Gaussian processes - will be given, as well as a brief discussion of the methods used for obtaining estimates of the derivatives required for the measure. The computational implications of calculating the measure at thousands of gridboxes across the globe will be covered. The application of the measure to a pair of response variables for a particular climate model will be presented, and the results so far discussed, along with potential issues with the measure that require further exploration.

***Coalescent Inference for Epidemics with Latent Periods***

Presenter: Isaac Goldstein

The effective reproduction number is an important descriptor of the infectiousness of a pathogen during an infectious disease epidemic. Coalescent models are used to study the transmission dynamics of rapidly evolving pathogens from molecular sequence data obtained from infected individuals. However coalescent parameters, such as effective population size, offer limited interpretability for transmission dynamics. In this work, we derive a coalescent model for exposed-infected population dynamics that allows us to infer the number of infected individuals and the effective reproduction number over time from the sample genealogy. The model can be interpreted as a two-deme model in which coalescence is restricted to individuals from different demes (exposed and infected). We propose a new data-augmentation framework with Phase-type distribution for Bayesian inference of epidemiological parameters. We combine this model with a non-parametric Gaussian Markov Random Field prior on the time-varying effective reproduction number to conduct semiparametric inference of the effective reproduction number. We study the performance of our approach on simulations and apply it to re-analyze the 2014 Ebola outbreak in Liberia.

***Multi-Fidelity Gaussian Processes for Translational Modelling of Clinical Outcomes***

Presenter: Isaac Hayden

Bridging the gap between animal and human experiments remains a major challenge in translational medicine, particularly in early drug development. Progress is limited by financial cost, the difficulty of integrating multiple sources of heterogeneous data, and the desire for the reduction of in vivo testing for patient safety and animal welfare. We present a multi-fidelity modelling approach using Gaussian processes, in which animal studies are considered as lower-fidelity but informative approximations to human experiments. This framework learns simultaneously the cross-species similarities and underlying exposure-response relationships, enabling principled extrapolation between species while quantifying uncertainty. By leveraging information from multiple experimental fidelities, our method improves estimation of clinically relevant quantities of interest and supports the replacement, reduction, and refinement of in vivo testing. We illustrate this framework on simulated data for determining the risk of drug-induced QT-interval prolongation, a key cardiac safety assessment for the regulatory approval of a drug. In this context, our approach uniquely permits the integration of in vitro data on cardiac ion channel inhibition alongside all animal and human experimental data, at no added cost to the drug developer. Importantly, it also supports transfer learning across compounds, allowing prediction and uncertainty quantification for new drugs based solely on early-stage experimental data.

***BHMOI: A Bayesian hierarchical model with overlapping indices for dynamic information borrowing across subgroups***

Presenter: Jack Lee

Bayesian hierarchical models (BHMs) are widely used to borrow information across multiple subgroups. However, the common exchangeability assumption becomes questionable when heterogeneity exists among subgroups.

Therefore, it is important to accurately identify heterogeneity and, in turn, determine the appropriate strength of information borrowing. Two main challenges arise: (1) assessing the exchangeability assumption to balance the trade-off between efficiency and robustness in information borrowing, and (2) dynamically determining borrowing strength according to varying levels of homogeneity. To address these challenges, we propose a cluster-based framework called the Bayesian hierarchical model with overlapping indices (BHMOI). BHMOI incorporates two novel indices: the overlapping clustering index (OCI), which facilitates optimal clustering, and the overlapping borrowing index (OBI), which guides the assignment of borrowing strength based on cluster-specific homogeneity. BHMOI is a flexible framework that accommodates different data types and enables efficient, robust information borrowing with desirable properties. Simulation studies and real-data analyses demonstrate the effectiveness and practical utility of BHMOI.

#### ***Federated Variational Inference for Bayesian Mixture Models***

Presenter: Jackie Rao

We present a one-shot, unsupervised federated learning approach for Bayesian model-based clustering of large-scale binary and categorical datasets, motivated by the need to identify patient clusters in privacy-sensitive electronic health record (EHR) data. Our approach combines variational inference with a principled “divide-and-conquer” distributed inference strategy. Local nodes perform clustering with local merge and delete moves within batches of the data in parallel, before a central client employs a ‘global’ merge move across batches to find global clustering structures. We show that these merge moves require only summaries of the data in each batch, enabling scalable, communication-efficient federated clustering without sharing individual-level data. Empirical results on simulated and benchmark datasets demonstrate that our method performs well relative to comparator clustering algorithms, achieving significant computational speedups. We validate the practical utility of the method by applying it to a large-scale British primary care EHR dataset comprising 289,821 individuals to identify clusters of individuals with common patterns of co-occurring conditions (multimorbidity). Finally, we outline ongoing extensions of our distributed variational framework to settings that are both large-scale and high-dimensional, including count-based data such as large single-cell RNA sequencing datasets. We consider a number of approaches for dealing with high dimensions, including variable selection models and variational mixtures of factor analysers.

#### ***Inference for Non-Conjugate Distance-Dependent Chinese Restaurant Process Models using RJMCMC***

Presenter: James Dodsworth

This talk introduces a Bayesian nonparametric (BNP) clustering framework designed to extend the distance-dependent Chinese Restaurant Process (ddCRP) to non-conjugate likelihoods. While the ddCRP effectively incorporates covariate information into the clustering process, its application has often been limited to conjugate models. Our proposed extension overcomes this limitation, enabling the use of more flexible, non-conjugate distributions appropriate for complex data structures. We detail the inference procedure, which relies on a Reversible Jump Markov Chain Monte Carlo (RJMCMC) algorithm to jointly estimate the cluster parameters and the underlying data partition. The method’s performance and robustness are first validated using simulated data. We then demonstrate its practical utility through an application modelling global modern slavery prevalence. By using a flexible Negative Binomial likelihood for country-level counts, the model reveals nuanced global patterns and provides a powerful new tool for policy-relevant statistical analysis.

#### ***A Bayesian High-Dimensional Mediator Selection Model for Microbiome Data***

Presenter: James Michael Papadopoulos

The human microbiome profoundly influences health outcomes, yet identifying which microbial taxa drive the effects of microbiome-directed therapies remains a major challenge. Mediation analysis offers a principled statistical framework for inferring these causal pathways. Microbiome mediation analysis is challenging due to the high-dimensional and compositional nature of the data. Our proposed method combines the centered log-ratio (CLR) transform with Bayesian variable selection to accurately identify individual taxa acting as mediators while accounting for high-dimensional covariate data. A beta-binomial model provides robust uncertainty quantification by fitting taxa count data to the CLR-transformed space. In simulation studies, our method achieved greater sensitivity than published mediator selection models while maintaining comparable positive predictive value. Applying our model to a randomized controlled trial of fecal microbiota transplantation in obese adolescents, we identified two candidate mediating taxa of the observed reduction in waist circumference: *Agathobacter rectalis* and *Bacteroides salyersiae*.

#### ***Optimal dose expansion in early phase oncology trials***

Presenter: James Willard

Early phase clinical trials in oncology investigate first-in-human doses of experimental cancer therapies. Phase 1 dose escalation trials were originally designed for cytotoxic chemotherapies, where it was assumed that larger doses provide greater benefit. Thus, these trials have traditionally focused on identifying the largest safe dose, the so-called maximum tolerated dose (MTD), often used as the recommended Phase 2 dose (RP2D). Recently, the FDA's Project Optimus highlighted how modern targeted therapies may provide benefit at doses lower than the MTD and emphasized the importance of considering efficacy when determining the RP2D. To facilitate this, dose expansion has been proposed. In dose expansion, additional efficacy and toxicity information is collected on a small number of the most promising doses which are selected at the end of dose escalation. A standard approach in practice is to equally randomize the expansion patients to two doses only. However, this approach may be suboptimal from an information theoretic perspective as it prematurely reduces the set of candidate RP2Ds and ignores known characteristics of a desired RP2D, e.g., minimal effectiveness or optimality with respect to some utility function. In this work, we investigate these limitations by comparing the standard approach to several other proposed approaches which utilize concepts from classical and modern Bayesian optimal experimental design. Findings from an extensive simulation study will be discussed and recommendations for performing optimal dose expansion will be provided.

### ***R2 priors for Grouped Variance Decomposition in High-dimensional Regression***

Presenter: Javier Aguilár

We introduce the Group-R2 decomposition prior, a hierarchical shrinkage prior that extends R2-based priors to structured regression settings with known groups of predictors. By decomposing the prior distribution of the coefficient of determination  $R^2$  in two stages, first across groups, then within groups, the prior enables interpretable control over model complexity and sparsity. We derive theoretical properties of the prior, including marginal distributions of coefficients, tail behavior, and connections to effective model complexity. Through simulation studies, we evaluate the conditions under which grouping improves predictive performance and parameter recovery compared to priors that do not account for groups. Our results provide practical guidance for prior specification and highlight both the strengths and limitations of incorporating grouping into R2-based shrinkage priors.

### ***Multi-Resolution Spatial Regression Analysis of Cellular Colocalizations in Cancer Imaging***

Presenter: Jessica Aldous

Hierarchical multiplex imaging studies generate multi-resolution spatial data and cell-type measurements across many spatially organized fields of view (FOVs) within each patient biopsy, enabling investigation of how tumor microenvironment organization varies along biologically meaningful tumor gradients. Existing approaches fail to jointly address this multi-resolution data structure: complex within-FOVs cell patterns, non-ignorable spatial dependence between FOVs, and strong within-/between-patient heterogeneity needed to recover true biological signals. We propose MoSAIC: multi-resolution spatial regression analysis of cell colocalizations, a hierarchical Bayesian spatial regression model designed for multi-resolution spatial data. MoSAIC decomposes variation into (i) global tumor-gradient effects, modeled flexibly using penalized splines with simultaneous band score probabilities used to assess the significance, (ii) patient-specific effects to capture inter-patient variability, and (iii) Gaussian process models to account for spatial dependence between FOVs within each patient biopsy. Simulations demonstrate MoSAIC has improved prediction and model fit compared to existing spatial and non-spatial model alternatives. Our method is motivated by and applied to a renal cell carcinoma multiplex imaging cohort to investigate immune-tumor colocalization patterns across the epithelial-to-mesenchymal transition (EMT) gradient. MoSAIC identifies increased Macrophage-tumor colocalization and decreased Cytotoxic T-tumor colocalization progressing across the EMT, consistent with EMT-associated immune suppression and spatially varying immune engagement. Overall, MoSAIC provides an interpretable, multi-resolution framework for quantifying spatial tumor-gradient effects in cancer imaging studies.

### ***Neural Bayesian Methods for Spatiotemporal Data Fusion of Remote Sensing Observations***

Presenter: Joaquim Teixeira

Hyperspectral and multispectral imaging satellites provide complementary remote sensing data across diverse temporal, spatial, and spectral resolutions. However, these resolutions are often misaligned: satellites with high temporal resolution typically have low spatial resolution, and vice versa. Statistical fusion of multi-sensor data offers the potential to generate high-resolution images across space, time, and spectra at near-global scales. Here, we present a highly scalable Neural Bayesian framework for fusing massive hyperspectral datasets from instruments with varying temporal, spatial, and spectral support. Our approach leverages a Dynamic Linear

Model formulation in which 3-D Convolutional Neural Networks approximate the joint posterior distribution over covariance hyperparameters via simulation-based inference, while sparse matrix techniques enable efficient approximate Bayesian inference over latent fields. By combining deep learning, sparse methods, and high-performance computing, our method delivers well-calibrated uncertainties with orders-of-magnitude speedups over existing approaches, enabling rapid, high-fidelity spatiotemporal data fusion at scales commensurate with modern satellite observation systems.

#### ***Nonnegative Einsum Factorization for Large Tensor Models***

Presenter: John Hood

Non-negative matrix and tensor decomposition provides a powerful framework for building probabilistic and Bayesian models that extract scientifically meaningful latent structure from multiway data in many scientific disciplines, such as neuroscience, genomics, political science, and sociology, among many others. In practice, most models conform to one of a small handful of factorization structures—i.e., the canonical polyadic (CP), Tucker, or tensor train (TT) decompositions—owing partly to the fact that existing inference strategies are highly tailored to the specific factorization structure. However, there is a vast space of generalized tensor factorizations that are often motivated in practical applications but do not conform to the usual decompositions, and the choice of factorization can drastically affect the interpretation of model parameters and the insights practitioners might draw. This paper considers a large family of generalized decompositions defined as any multilinear contraction of a set of non-negative parameter matrices or tensors. To fit any decomposition in this family, we derive a highly scalable and compact majorization-minimization algorithm for maximum likelihood and MAP estimation in the family of Tweedie distributions. This framework provides a direct path to scalable approximate Bayesian inference, yielding a coordinate ascent variational inference algorithm for models incorporating priors on the non-negative parameters. We provide case studies on real data from international relations and urban transport networks, showing that our framework allows one to fit highly non-standard probabilistic tensor factorizations that yield particularly interpretable latent structure and obtain substantially better predictive performance than classical factorizations. The algorithms we derive are simple and easily benefit from GPU-accelerated computation, allowing us to fit such models to tensors with tens of billions of observations in minutes.

#### ***Bayesian Kolmogorov-Arnold Neural Model***

Presenter: Johnny Myung Won Lee

This research presents a novel statistical framework for Kolmogorov–Arnold Neural Networks (KAN) redefining them as a neural extension of generalized additive models. The canonical version of our model employs a doubly-additive approach in a three-layer configuration, while the deep version extends this to a deep additive model. Our approach then motivates to a novel Bayesian group horseshoe-type shrinkage prior, referred to as the picking-and-pruning prior, specifically designed for KAN architectures. This prior facilitates group-wise regularization of the spline coefficients governing the learnable activation functions, applied simultaneously to the inner and outer layers. Thus, the method not only performs variable selection but also streamlines the dense network, leveraging Bayesian inference to enhance interpretability. The proposed method is validated on simulation studies on artificial data and further illustrated to model the risk from chemical exposure.

#### ***Interrogating Prior Beliefs and Uncertainty Quantification in Model-Observation Comparison for Climate Models***

Presenter: Jonathan Owen

Computer models are vital in climate science to study drivers of climate change; quantify uncertainties in future climate predictions; and to guide policy decisions. However, their direct use is inhibited by: their complex structure; high-dimensional input and output, including spatial-temporal fields; and their long evaluation times taking months on HPCs. Earth System Models (ESMs) integrate atmosphere, ocean, land, ice, and biosphere. ESMs are used to investigate natural and anthropogenic aerosol emission interactions with clouds yielding large Aerosol Radiative Forcing (ARF; the temporal change in Earth’s energy balance due to aerosols) induced uncertainty in historical climate change. ARF is unobservable and key to predicting future climate, yet research has resulted in little uncertainty reduction in 30-years of IPCC reports.

We interrogate how prior beliefs and uncertainties propagate through model-observation comparison via Bayesian history matching and the constraint on ARF uncertainty for the UK Met Office UKESM1 model, including exposing implicit prior beliefs in past studies. Bayesian emulators, as fast statistical approximations, are employed for a carefully selected set of observable model outputs enabling an efficient global parameter search. These are embedded within an uncertainty quantification framework incorporating structural model discrepancy to link ESM output and the real-world, as well as representation and observation errors, to obtain a meaningful comparison

of observations with the model.

***From Bandits to Balance: A Bayesian Approach to Evaluating Response-Adaptive Randomization and Optimal Interim Analysis Allocation***

Presenter: Jose Manuel Lema Martinez

Bayesian adaptive designs offer significant advantages in efficiency and ethics over traditional fixed-sample trials. However, their adoption is often hindered by regulatory concerns over frequentist operating characteristics, leading to the use of hybrid designs that may not leverage the full power of a Bayesian approach. This presentation provides a practical framework for converting a hybrid multi-arm, multi-stage (MAMS) trial into a fully Bayesian one, grounded in a case study of an oncology, phase II, non-randomised trial (the WIRE trial).

The methodology is twofold. First, we detail the transformation of the hybrid design into a fully Bayesian framework, where the final analysis is now Bayesian. Through simulation, we demonstrate that this framework can match the original's frequentist power and type I error rate—a key consideration for regulatory acceptance—while offering superior control of false positives via a substantially lower Bayesian type I error. This investigation also addresses a critical design question: the optimal timing of interim analyses. By varying the participant allocation across stages, we show that more balanced allocations (i.e., more evenly spaced interim analyses) consistently enhance trial efficiency, yielding higher assurance, a Bayesian alternative to power which better captures the uncertainty about treatment effect, and a lower expected sample size.

Second, we explore Bayesian response-adaptive randomization as an alternative to the trial's original non-randomized allocation. We compare standard Thompson Sampling against Accelerated Thompson Sampling variants, under different prior (flat, informative and correctly centered, and informative but incorrectly centered) assumptions and under different scenarios of treatment effects, to evaluate the fundamental trade-off between in-trial and future patient benefit. Our findings confirm this choice is goal-dependent: more exploitative methods (e.g., a myopic strategy) are more effective at maximizing benefit for participants within the trial; and more explorative methods (standard TS) are superior for correctly identifying the best treatment for future patients. Crucially, standard TS also provides the most robust performance in the presence of misspecified prior information. This work provides a practical methodology for designing robust, efficient, and fully Bayesian adaptive trials, offering clear guidance on both the optimal scheduling of interim analyses and the strategic selection of RAR methods to align with primary trial goals.

***Bayesian Hierarchical Autoregressive Parametric Estimation with Application to the Analysis of Multilevel Electroencephalogram Signals***

Presenter: Josefina Correa

Estimating the spectral content of electroencephalogram signals is a common approach to characterizing how the brain responds to external stimuli. In clinical studies, electroencephalogram signals are collected over multiple subjects and their spectra are computed using either Fourier-based or parametric approaches. A common analysis entails comparing the spectra of a wide-sense stationary data window after the stimulus onset to the spectra before. However, conventional approaches to analyzing these data do not account for between subject variability and could in turn provide inaccurate inferences. This work develops a Bayesian hierarchical auto-regressive modeling framework to estimate subject-level and population-level spectra. Our formulation provides a principled approach for constructing cohort-level estimates, which can be used to assess the extent to which a new subject is consistent with a cohort-level response. We validate our framework in simulation and apply it to the analysis of electroencephalogram signals from healthy volunteers undergoing propofol-mediated anesthesia.

***Weighted Bayesian Bootstrap for Random Forests: Improving Prediction in Highly Skewed Data***

Presenter: Joseph Kang

Addressing the Random Forest method's limitation in handling highly skewed data necessitates a data weighting scheme to improve prediction of the sparse outcome. We propose naturally integrating this requirement using the Bayesian Bootstrap, which assigns weights (ranging from 0 to 1) to data units. The ordinary Bayesian Bootstrap employs equal Dirichlet priors, leading to an even distribution of weights. Our key innovation is the adjustment of these priors to allow for differential sampling of weights, specifically to favor the sparse outcome and give it greater model attention. We demonstrate a significant improvement in predictive accuracy for sparse outcomes using a simulated dataset, showing enhanced performance over the conventional bootstrap methodology. This provides a robust and theoretically sound Bayesian mechanism to adapt Random Forest for challenging highly skewed data problems.

***Credit Risk Prediction Using Machine Learning and Bayesian Network: A Study on Portfolio of Accredited Clients.***



Presenter: José Francisco Martínez Sánchez

The purpose of this work is to implement statistical methods for classifying and identifying patterns in a portfolio of borrowers to predict the probability of default using machine learning (logistic regression) and Bayesian Network (BN) tools. A BN can be seen as a more sophisticated and more realistic version of a Bayes classifier. Firstly, the dataset was tested and cleaned to improve the quality of the work; then exploration data analysis was conducted to learn more about the data before pre-processing. After that, the methodology was used to differentiate the predicted results from the actual results using the confusion matrix. In this work, the model produces a binary outcome with only two options: good customer or bad customer. To determine the most statistically significant variables, binary logistic regression was performed using Python code. The Bayesian network is constructed using significant variables, and prior probabilities are assigned to each node (random variable). Once we have the BN with a prior probability, the LAZY propagation architecture as a junction tree-based inference will be used to generate a posterior probability. As a result, we have that the probability of a client defaulting on their credit obligations is 30.48%; there is no evidence of any pre-established value for the random variables (accredited clients features) of the BN. If there is evidence of any pre-established value for the random variables, for example: "Credit\_history: no credits taken" and "checking\_account:  $\geq 200$  DM"; the probability of default is 25.87%. We can conclude that the model considers the causal relationship between random variables and allows sensitivity analysis with different input values. The model shows an efficiency of 92%.

### ***Conditional Autoregressive (CAR) Models in a State-Space Framework***

Presenter: João Batista de Moraes Pereira

In models for areal data, it is common to assume that the latent structure responsible for the spatial correlation among observations follows a conditional autoregressive (CAR) model. Such models allow the construction of the joint distribution of random variables from the specification of their conditional distributions, each depending only on a set of neighbors. To perform inference in CAR models, it is almost always necessary to evaluate or sample from the posterior distribution of the latent parameters, which can be costly. Lavine (1998) showed that the inference problem in a CAR model on a finite two-dimensional regular lattice can be treated as an inference problem within a particular class of dynamic linear models (DLMs). One advantage of this connection is that efficient methods commonly used in DLM inference can be employed to conduct inference in CAR models. We propose a generalization of this approach for models with responses in the exponential family, considering dynamic generalized linear models (DGLMs), as well as an adaptation of the conjugate updating method, a sequential inference procedure whose key feature is the use of conjugate prior and posterior distributions for the exponential family. The proposed approach has shown promising performance in terms of efficiency and computational cost. This is joint work with Helio S. Migon and Alexandra M. Schmidt.

### ***Posterior Attraction via Moment Matching: An MMD Minimization Approach***

Presenter: Juan Maroñas

Research in adversarial machine learning (AML) has demonstrated that statistical models are vulnerable to maliciously altered data. While much of this work focuses on frequentist methods, Bayesian inference is not immune. In this work, we focus on posterior attraction, a specific white-box poisoning framework where an attacker steers the posterior toward a target distribution through the strategic deletion and replication of observations. Existing literature on posterior attraction typically formulates the problem by minimizing the Kullback-Leibler (KL) divergence. However, this is often impractical because it requires specifying a complete target density. In contrast, an attacker is usually interested in modifying only a few particular moments of the posterior (such as shifting a mean or inflating a variance) rather than defining the geometry of the entire high-dimensional distribution. To address this, we propose a novel optimization framework based on minimizing the Maximum Mean Discrepancy (MMD). Unlike the KL approach, this formulation allows the adversary to target specific moments by employing a tailored kernel, eliminating the need to define the full target posterior. This flexibility further opens the door to decision-theoretic attacks, where the adversary directly targets the expected utility of actions. To solve the optimization problem, we propose a practical heuristic that first approximately solves a continuous relaxation of the objective and subsequently projects the solution back to the integer domain. Within this framework, we analyze the convexity properties of the relaxed objective and derive a technique to obtain an unbiased stochastic gradient. This estimator enables a projected Stochastic Gradient Descent strategy, making the attack applicable to any model where posterior sampling is feasible, for instance via MCMC. Empirical results demonstrate that with relatively little effort, an attacker can substantively mold Bayesian beliefs to their will.

### ***Energy-based Bayesian phylogenetic inference***



Presenter: Julia Palacios

Bayesian phylogenetic inference is challenging because it involves the exploration of a large discrete combinatorial space, multimodal discrete-continuous posterior and often only weak signal in observed molecular data. In this work, we employ a recently proposed matrix representation of tree topologies and branch lengths to train a variational autoencoder that simultaneously infers a latent lower-dimensional representation of tree-space and infers epidemiological parameters under simple phylodynamic models. Finally, we propose two energy-based models that use the phylogenetic matrix representation to approximate the posterior distribution. We compare the performance of the proposed methods to state-of-the-art Bayesian phylogenetic inference frameworks that use MCMC, on simulations and real data.

#### ***Recent Advances in Non-Gaussian SPDE Modeling of Latent Fields over Complex Domains***

Presenter: Kelvin Jhonson Ribeiro de Sousa Almeida Silva

In this presentation we discuss recent advances in building latent models based on stochastic partial differential equations driven by square-integrable Lévy white noises in what is known as the SPDE approach. The theoretical methodology proposed in our work through the so-called coloration approach allows us to obtain an appropriate parametrization of our model depending on the spatial structure, ensures theoretical guarantees on the convergence of the lumped covariance and the underlying field, and makes it possible to extend methods typically applied to Euclidean domains to metric graphs and surfaces. Finally we present efficient tools available for working with these models and that integrate well with INLA and inlabru, the latter being more focused on the Gaussian latent-field case.

#### ***Bayesian Active Learning with Task-Driven Representations for Messy Pools***

Presenter: Kianoosh Ashouritaklimi

Active learning has the potential to be especially useful for messy, uncured pools where datapoints vary in relevance to the target task. However, state-of-the-art approaches to this problem currently rely on using fixed, unsupervised representations of the pool, focusing on modifying the acquisition function instead. We show that this model setup can undermine their effectiveness at dealing with messy pools, as such representations can fail to capture important information relevant to the task. To address this, we propose using task-driven representations that are periodically updated during the active learning process using the previously collected labels. We introduce two specific strategies for learning these representations, one based on directly learning semi-supervised representations and the other based on supervised fine-tuning of an initial unsupervised representation. We find that both significantly improve empirical performance over using unsupervised or pretrained representations.

#### ***A Bayesian Latent Class Approach for Causal Estimation with Multiple Cognitive Outcomes***

Presenter: Kuan Liu

Heterogeneity in cognitive aging is multidimensional: memory and executive function can decline at different rates, and older adults often follow distinct, unobserved progression patterns. Estimating the causal effects of exposures on multiple cognitive measurements is therefore challenging due to latent heterogeneity, informative attrition, and measurement noise. We develop a Bayesian causal latent class framework that estimates how an exposure affects the probability of following distinct, clinically interpretable multivariate cognitive outcomes phenotypes. Individuals are assigned to  $K$  latent classes where class membership depends on exposure and baseline confounders, and domain-specific cognition is modelled within class. We define the primary causal estimand as the probability contrast between potential class-membership probabilities under different exposure status. Causal estimation proceeds under the g-computation framework via posterior predictive inference under a fully Bayesian joint likelihood for outcomes, treatment, and covariates, enabling coherent uncertainty quantification and straightforward sensitivity analysis, with natural extensions to handle missing data. Simulation studies vary class separation, indicator quality, confounding strength, and sample size to evaluate bias, interval coverage, and class recovery. In an application to the Canadian Longitudinal Study on Aging, we examine how modifiable risk factors such as physical activity shape cognitive health derived from multiple established clinical cognition tests.

#### ***Tuning Sequential Monte Carlo Samplers via Greedy Incremental Divergence Minimization***

Presenter: Kyurae Kim

The performance of sequential Monte Carlo (SMC) samplers heavily depends on the tuning of the Markov kernels used in the path proposal. For SMC samplers with unadjusted Markov kernels, standard tuning objectives, such as the Metropolis-Hastings acceptance rate or the expected-squared jump distance, are no longer applicable. While stochastic gradient-based end-to-end optimization has been explored for tuning SMC samplers, they often incur excessive training costs, even for tuning just the kernel step sizes. In this work, we propose a general

adaptation framework for tuning the Markov kernels in SMC samplers by minimizing the incremental Kullback-Leibler (KL) divergence between the proposal and target paths. For step size tuning, we provide a gradient- and tuning-free algorithm that is generally applicable for kernels such as Langevin Monte Carlo (LMC). We further demonstrate the utility of our approach by providing a tailored scheme for tuning kinetic LMC used in SMC samplers. Our implementations are able to obtain a full schedule of tuned parameters at the cost of a few vanilla SMC runs, which is a fraction of gradient-based approaches.

### ***The Bradley-Terry Stochastic Block Model***

Presenter: Lapo Santi

The Bradley-Terry model is widely used for the analysis of pairwise comparison data and, in essence, produces a ranking of the items under comparison. We embed the Bradley-Terry model within a stochastic block model, allowing items to cluster. The resulting Bradley-Terry SBM (BT-SBM) ranks clusters so that items within a cluster share the same tied rank. We develop a fully Bayesian specification in which all quantities—the number of blocks, their strengths, and item assignments—are jointly learned via a fast Gibbs sampler derived through a Thurstonian data augmentation. Despite its efficiency, the sampler yields coherent and interpretable posterior summaries for all model components. Our motivating application analyzes men’s tennis results from ATP tournaments over the seasons 2000-2022. We find that the top 100 players can be broadly partitioned into three or four tiers in most seasons. Moreover, the size of the strongest tier was small from the mid-2000s to 2018 and has increased since, providing evidence that men’s tennis has become more competitive in recent years.

### ***Warped Stochastic Differential Equations***

Presenter: Lars Keizer

Linear stochastic differential equations (SDEs) have gained increasing popularity in psychological science due to their intuitive interpretation in terms of substantive theories on temporal dynamics and their ability to describe both person-specific and population-level dynamics. However, in practical applications SDE-based models can be limited by stationarity assumptions. Gaussian processes (GPs), a methodology used in Bayesian machine learning, offer high flexibility, but for substantive researchers they are generally less straightforward to interpret and more difficult to link directly to theoretical mechanisms of change. This flexibility and broad applicability arise in part from input warping (to accommodate non-stationarity) and output warping (to allow for non-Gaussian observations). Solutions to linear SDEs can be shown to be GPs, which motivates investigating whether techniques from the GP framework can be extended to a larger class of SDEs. This work examines how warping transformations can be applied directly within the SDE formulation, rather than only to the GP solution. Using both simulations and time-series data from studies of affect dynamics, we evaluate the performance of warped SDE models in capturing non-stationary dynamics while dealing with potentially non-Gaussian data. Our findings suggest that techniques developed in the GP literature can be leveraged to broaden the modeling capacity of SDEs while preserving their parametric interpretability.

### ***Bayesian calculus for extended feature allocation models***

Presenter: Lorenzo Ghilotti

We define the class of extended feature allocation models, which generalizes standard feature allocations by incorporating dependencies – such as repulsion or attraction – among features and their associated weights. This framework enhances modeling flexibility, offering significant benefits for latent feature modeling and enabling novel applications beyond traditional domains. We provide a complete Bayesian analysis of the proposed model, and we further characterize those priors that yield predictive distributions of new features depending either solely on the sample size or on both the sample size and the distinct number of observed features. These predictive characterizations, known as “sufficientness” postulates, have been extensively studied in the literature on species sampling models starting from the seminal contribution of the English philosopher W.E. Johnson for the Dirichlet distribution. Within the feature allocation setting, existing predictive characterizations are limited to very specific examples; in contrast, our results are general, providing practical guidance for prior selection. We further specialize our general theory to noteworthy classes of priors. These include (i) a novel prior based on determinantal point processes, for which we show promising results in a spatial statistics application, and (ii) priors based on shot-noise Cox processes, which we apply to genetic variant embedding tasks and to the modeling of two-level taxonomic libraries in ecological studies.

### ***Bayesian Nonparametric Clustering for the Mallows Rank Model***

Presenter: Lorenzo Zuccato

Preference learning analyzes rank data to infer consensus rankings, individual preferences, and clusters of assessors, and the Mallows model is among the most successful approaches. Previous works developed compu-

tationally tractable methods for Bayesian inference based on a Metropolis Hastings scheme, compatible with any right-invariant distance. Clustering was achieved by fitting finite mixture models, while inference on the number of cluster was conducted by looking at the behavior of the within-cluster sum of distances for different mixture models. This paper presents a Bayesian nonparametric approach in order to perform joint inference on the number of non-empty clusters in the data and the clustering allocation. The method is based on the Dirichlet process mixture model. Posterior inference on cluster-specific parameters is conducted by conditioning on a partition of the data estimated from the co-clustering matrix. The sampling algorithm was implemented in the framework of the R package BayesMallows, enabling it to work with incomplete rankings and pairwise comparisons. We evaluate its convergence on simulated datasets by measuring its ability in recovering the right number of clusters. Real data about movie ratings is used to assess the model's effectiveness in detecting missing pairwise preferences. In both experiments, we further compare the performance of the proposed method with the existing finite mixture model, highlighting the overall better reliability in detecting the number of clusters in the population.

### ***Modeling Spatio-Temporal Dynamics of Obesity in Italian Regions Via Bayesian Beta Regression***

Presenter: Luciano Rota

We explore the spatio-temporal evolution of obesity rates across Italian regions from 2010 to 2022, aiming to detect spatial and temporal patterns and to evaluate potential heterogeneities. We employ a Bayesian hierarchical Beta regression framework to study regional obesity rates, incorporating spatial and temporal random effects together with gender and several external predictors. Stochastic Search Variable Selection is used to identify the predictors most strongly supported by the data. The results highlight regional heterogeneity together with spatial and temporal dependence in obesity rates over the study period, emphasizing the relevance of gender, spatial structure and temporal dynamics in explaining their evolution. The inclusion of structured spatial and temporal random effects effectively captures the complexities of regional differences over time. These random effects, along with gender, emerge as the main drivers of obesity prevalence across regions, whereas the influence of exogenous covariates appears limited at the regional scale. Although socioeconomic and lifestyle factors remain important at the individual level, our findings show that incorporating spatial and temporal structure is essential for understanding macro-level variations in obesity.

### ***Scalable Bayesian methods for cognitive modeling***

Presenter: Luis Baroja

Bayesian hierarchical implementations of cognitive models rely heavily on numerical methods. When working with large data sets and complex models, standard Markov chain Monte Carlo methods quickly become impractical, easily requiring hours or days of computation for a single analysis. In this work, we examine the performance of a family of scalable Bayesian techniques based on divide-and-conquer (DnC) strategies and subposterior recombination, applied across different cognitive models. DnC methods partition the full data set into disjoint subsets and compute independent subposteriors in parallel. These subposteriors are then merged using a recombination strategy (e.g., Consensus Monte Carlo; CMC) to approximate the posterior distribution that would have been obtained from the entire data set at once.

We study the boundary conditions of these methods using three case studies: (1) hierarchical Rasch models in item response theory, (2) hierarchical "EZ" drift diffusion models for decision-making, and (3) softmax Q-learning models in bandit tasks. In these applications, CMC performs well when the target posterior is approximately Gaussian, such as in the Rasch model, recovering the full posterior distribution with high accuracy. In contrast, models with truncated or asymmetric posteriors, such as EZ-DDM or Q-learning, show deficient recombination accuracy, although the recombined distributions still retain important qualitative patterns of the full posterior. Our results highlight both the advantages and limitations of DnC Bayesian inference in cognitive modeling. The methods are relatively easy to implement and can simplify debugging or correcting MCMC issues (e.g., by fixing problematic subposteriors separately rather than rerunning the full model). However, their dependence on approximately symmetric posteriors limits their scope, especially in areas where highly skewed or bounded posteriors are common.

### ***A Modular Hierarchical Bayesian Computational Approach to Migrant Stock Estimation Using Facebook, Surveys and Official Statistics***

Presenter: Maciej Danko

This study is part of the Human Migration Database project developed at the Max Planck Institute for Demographic Research, which aims to build consistent global estimates of both migration flows and stocks. In this paper we focus only on migrant stocks. They are essential for understanding long-term migration patterns and

for describing broader demographic change in a comparable way across countries. Our goal is to produce coherent yearly stock estimates by country of birth and residence from 2001 onward. Many statistical sources exist, but they differ in definitions, quality, and completeness. As a result, noticeable inconsistencies appear over time and across countries. To address this, we construct a hierarchical Bayesian model that links all sources to one latent process  $y(j, t)$ , representing the true log migrant stock for each origin destination corridor  $j$  and year  $t$ .

The model integrates population censuses, annual Eurostat and OECD stock series, Labour Force Survey counts, and Facebook MAU/DAU audiences from the Meta Ads platform. Country of birth is used as the main reference. Citizenship, age 15-64 shares, and Facebook penetration act as auxiliary signals that help explain systematic deviations. To keep the structure transparent, we adopt a two-stage strategy. Stage A estimates CTZ/BTH ratios, age coverage, and platform penetration using partial pooling, ratio and logit models, random-walk components, and heteroscedastic Student-t errors. Posterior means and variances from this stage are then passed to Stage B using a modular cut, so information is shared without double counting.

Stage B combines all sources in a single system. We include left censoring for zeros, interval censoring for rounded MAU values, accuracy-group variance ladders, a shared variance floor, and corridor-specific coverage terms for MAU and DAU. The latent process  $y(j, t)$  evolves through corridor-specific AR(1) dynamics with hierarchical priors on intercepts and slopes.

The framework produces harmonized migrant stock estimates with full uncertainty quantification. It also decomposes discrepancies into interpretable components such as undercounting, duration-of-stay differences, survey noise, Facebook penetration, duplicate and false accounts, and corridor-level coverage. Beyond the statistical contribution, the model supports comparative migration research and helps enable near real-time monitoring of migrant populations by combining official statistics with digital-trace data.

### ***The role of Bayesian uncertainty in assessing the fairness of machine learning models***

Presenter: Marco Scutari

Statistical and machine learning models are increasingly employed to take important societal decision, and verifying whether their outputs are biased against disadvantaged groups or individuals is crucial to ensuring they are fair and applicable. Ideally, a rigorous risk assessment of possible fairness violations should require quantifying the uncertainty associated with the estimation of such models. Yet, this is rarely done in the literature, which focuses instead on the optimisation techniques required to provide point estimates of a model's parameters by trading-off predictive accuracy and fairness. In this talk, we will move beyond the current predominant estimation environment and discuss how Bayesian linear and generalised linear models can enhance the current fair learning landscape. We will illustrate with practical examples on simulated and real data how the Bayesian point of view enriches the present framework and sheds light on the (many!) pitfalls of being too reliant on single point estimates. Finally, we will compare our estimation with other frequentist and likelihood-based methods (developed by us) to assess the interaction of different points of views on uncertainty.

The work is joint with Marco Scutari (IDSIA-SUPSI, Switzerland) and Ernst Wit (USI, Switzerland).

### ***Bayesian Overlap Group Lasso***

Presenter: Marco Stefanucci

In this work, we study the overlap group lasso penalty from a Bayesian perspective. By exploiting a scale-mixture representation of normal distributions, we introduce a new hierarchical formulation for the model and characterize the implicit prior distribution induced on the latent parameters, establishing its properness. Although this prior does not belong to any known family, we derive an efficient MCMC algorithm for posterior computation by obtaining full conditional distributions for all parameters. We give particular attention to the estimation of the tuning parameter  $\lambda$ , which controls the level of sparsity. Estimation can be carried out via an MCEM scheme, through direct inclusion of  $\lambda$  in the MCMC sampler, or by external specification. Because standard MCMC does not yield exact zeros, we further introduce and discuss a spike-and-slab extension of the model. Simulation studies and empirical analyses are presented to illustrate the proposed methodology.

### ***Shot noise cox process random measures in Bayesian nonparametrics***

Presenter: Mario Beraha

Shot-noise Cox processes (SNCPs) are a family of point processes that exhibit a random clustering of atoms. I will show how SNCPs can be used as fundamental building blocks for nonparametric priors in a variety of contexts.

For mixture modeling, by constructing normalized random measures driven by SNCPs and using them as priors on the mixing distribution. The resulting “two-level” clustering – of atoms and of observations – captures complex, nonstandard cluster shapes and eases the classical tension between accurate density estimation and

faithful cluster recovery. For partially exchangeable grouped data, we introduce a nested/hierarchical SNCP mixture that borrows strength across groups without forcing exact atom sharing, enabling Bayesian distributional clustering of related, but not homogeneous, populations.

In feature allocation, by constructing an “enriched” Indian buffet prior that generalizes the enriched Dirichlet process to the feature problem, with applications in ecology, and by proposing a simple text classifier under a bag-of-embeddings assumption that leverages the induced feature structure.

#### ***Multiscale network modeling of migration flows in Austria***

Presenter: Martina Contisciani

Migration plays a central role in socioeconomic development, driven by diverse and increasingly complex factors. In recent years, understanding its causes, mechanisms, and impacts has become a priority for both policymakers and researchers seeking deeper insights into local social dynamics and the effects of policy interventions. For modelling such phenomena, most migration research relies on gravity-based models, which—given their heuristic and effective nature—are not expected to saturate the modeling requirements of mobility data. In this work, we analyze internal migration patterns in Austria over a 20-year period using high-resolution administrative data and a Bayesian inferential clustering approach, which in the context of migration studies remains largely underexplored. Specifically, we employ a class of weighted stochastic block models that fits a nonparametric generative model to the observed data, instead of enforcing a particular functional shape for the migration rates. This model approximates any functional shape by separating the nodes (i.e., Austrian municipalities) into discrete groups, and therefore their corresponding migration rates as piecewise mixtures of elementary distributions. Our procedure includes Bayesian regularization based on the minimum description length principle, yielding a principled and robust framework that avoids overfitting and statistical artifacts while remaining computationally efficient. In addition, our approach infers a hierarchical structure, so that the mixing patterns between geographical regions are simultaneously modeled at multiple scales. Applying our method to Austria’s internal migration data reveals systematic biases in migration flows, with administrative boundaries acting as effective barriers to relocation. These biases lead to inferred clusters in the migration network that closely resemble federal state and district borders with striking accuracy. This results in strong regionalization across multiple spatial scales, accompanied by a pronounced urban–rural divide. These patterns remain consistent across decades of data, and deviate significantly from the relocation patterns predicted by traditional models. By challenging the dominant reliance on gravity-based approaches, our study presents a robust methodological alternative that can be used to improve migration analysis, and can uncover previously unobserved phenomena within migration data.

#### ***Variable Ordering in a Cholesky-MSV model for stock market data***

Presenter: Martina Danielova Zaharieva

Modeling the time-varying structure of the covariance matrix of financial time-series has been widely studied both theoretically and empirically in the finance literature. Recently, the Cholesky-decomposition-based approach to multivariate stochastic volatility has been established as a flexible and interpretable alternative. Despite that, the issue of the order dependence in this type of models has not been the primary focus. In this paper, we propose a prior for variable ordering in a Cholesky-type multivariate stochastic volatility model, where a natural ordering of the time series is not available. We apply the approach to forecasting a large cross-section of stock returns.

#### ***Bayesian Dynamic Asset Allocation under Regime Switching***

Presenter: Masataka Hayashi

Financial markets often experience boom-bust cyclical regimes that make the traditional diversified investment strategies less effective. It is therefore crucial for investors to determine their asset allocation based on the prediction of the future regimes. This paper develops a Bayesian Markov-switching framework for dynamically determining the optimal asset allocation under regime-dependent returns and covariances. We model asset returns using a Markov-switching Normal-Inverse-Wishart distribution and the transition matrix of the regimes using a Dirichlet distribution, and estimate the dynamic model of asset returns with Markov chain Monte Carlo (MCMC) methods. Then we simulate the future paths of regimes and the corresponding asset returns from their predictive distribution and implement a simulated-path approach for multi-period portfolio optimization by Hibiki (2002) to determine the optimal allocation weights. The out-of-sample portfolio performance is compared with various benchmarks. Although the model cannot capture sudden regime changes such as COVID-19, it forecasts the market recovery well, resulting in higher cumulative returns.

#### ***Nonparametric Bayesian intensity estimation for covariate-driven point processes***

Presenter: Matteo Giordano

A central task in the statistical analysis of spatial point patterns is to infer the relationship between the point



distribution and a collection of covariates of interest. This talk will present recent theoretical and methodological advances for covariate-based nonparametric Bayesian intensity estimation, assuming that multiple observations of the covariates and the point pattern are available. We devise a “multi-bandwidth” Gaussian process method, and prove that it achieves optimal and adaptive posterior contraction rates towards (possibly) anisotropic intensity functions. We further show how posterior inference can be implemented in practice via a suitable Metropolis-within-Gibbs sampling algorithm and illustrate the performance of the method via numerical simulations. Lastly, we present an application to a Canadian wildfire dataset. Joint work with Patric Dolmeta.

#### ***Cross-target predictive synthesis for modeling predictive-dependent networks***

Presenter: Matthew Nissen

Ensemble methods generally assume that all predictive distributions target the same outcome. In many forecasting settings, however, predictive information relevant for one outcome may reside in agents that target different outcomes. We develop cross-target predictive synthesis, a generalization of Bayesian predictive synthesis, that synthesized forecasts from distinct target variables to predict each other. This induces a predictive-dependent network, where multivariate series are linked through the structure of their predictive distributions, rather than their observed values alone. The resulting network provides a coherent probabilistic framework for information sharing, cross-learning, and forecasting in high-dimensional panels. To scale, we incorporate cointegration-based sparsity, enabling the synthesis function to identify and exploit low-dimensional predictive relationships while remaining computationally tractable. This yields an interpretable representation of predictive connectivity. A topical application to forecasting monthly inflation across OECD countries illustrates the gains in predictive accuracy and the economic interpretability of the learned predictive network. The results highlight substantial improvements over standard ensemble baselines and reveal meaningful patterns of cross-country predictive dependence.

#### ***Scaling Laws for Uncertainty in Deep Learning***

Presenter: Mattia Rosso

Scaling laws in deep learning describe the predictable relationship between a model’s performance, usually measured by test loss, and some key design choices, such as dataset and model size. Inspired by these findings and fascinating phenomena emerging in the over-parameterized regime, we investigate a parallel direction: do similar scaling laws govern predictive uncertainties in deep learning? In identifiable parametric models, such scaling laws can be derived in a straightforward manner by treating model parameters in a Bayesian way. In this case, for example, we obtain  $O(1/N)$  contraction rates for epistemic uncertainty with respect to dataset size  $N$ . However, in over-parameterized models, these guarantees do not hold, leading to largely unexplored behaviors. In this work, we empirically show the existence of scaling laws associated with various measures of predictive uncertainty with respect to dataset and model size. Through experiments on vision and language tasks, we observe such scaling laws for in- and out-of-distribution predictive uncertainty estimated through popular approximate Bayesian inference and ensemble methods. Besides the elegance of scaling laws and the practical utility of extrapolating uncertainties to larger data or models, this work provides strong evidence to dispel recurring skepticism against Bayesian approaches: “In many applications of deep learning we have so much data available: what do we need Bayes for?”. Our findings show that “so much data” is typically not enough to make epistemic uncertainty negligible.

#### ***Constrain equilibrium climate sensitivity via Bayesian composite likelihood***

Presenter: Mengheng Li

The equilibrium climate sensitivity (ECS) is a key parameter in climate science and other disciplines. This study estimates ECS as a common equilibrium parameter across energy balance models (EBMs). Fitting EBMs to simulated data from 31 climate models under a CO<sub>2</sub> quadrupling experiment, we introduce a Bayesian composite likelihood approach to simultaneously integrate and estimate all the constituent EBMs. Complementing methods of storylines and emergent constraints commonly employed by climate scientists, our econometric alternative provides a data-driven ECS estimator. We find an ECS estimate of 3.65K, characterized by a unimodal and right-skewed posterior distribution that facilitates uncertainty quantification. Our approach also yields a 95% credible interval of (2.5K, 5K), which is narrower and closer to the upper bound of the IPCC’s “very likely” ECS range.

#### ***Matching Prior Pairs that Make Posterior Expectation and MAP Estimation Asymptotically Coincide***

Presenter: Michiko Okudo

While the MAP estimator and the posterior mean are both commonly used in Bayesian analysis, this work seeks



pairs of prior distributions for which these two estimators asymptotically coincide. We derive conditions that such pairs of prior densities must satisfy, and show that for models belonging to the class known as  $\alpha$ -flat in information geometry, these conditions take a particularly simple form. Several analytical and numerical examples are provided, including cases involving exponential families and generalized linear models.

This research is joint work with Keisuke Yano (the Institute of Statistical Mathematics).

### ***Integrated Joint Distribution Modeling to Estimate the Biomass of Commercial Fish in the Baltic Sea from Acoustic and Trawl Data***

Presenter: Mikhail Shubin

Integrated species distribution models (ISDMs) have seen rapid development in the recent years. By integrating multiple data sets under single hierarchical Bayesian model has been demonstrated to improve parameter identifiability and prediction in multiple works. However, applications of ISDMs to multispecies modeling are few. In this work, we developed a Integrated Joint species distribution model (IJSJM) for three commercially important fish in the Baltic Sea. We combine annual acoustic and trawl survey data on herring (*Clupea harengus*), sprat (*Sprattus Sprattus*) and stickleback (*Gasterosteus aculeatus*) within our IJSJM to provide estimates on the spatiotemporal distribution and trends in their biomasses. Herring, sprat and stickleback constitute a major part of the Baltic sea fish biomass so their biomass estimates are of great importance to fisheries management and assessment of the ecological status of the Baltic Sea. We compare our results to the results obtained with the current acoustic stock assessment method StoX, which approximates the fish stock by synthesizing data collected by Baltic International Acoustic Survey (BIAS). Unlike StoX, our IJSJM explicitly models fish abundance and its dependence on environmental covariates and provides uncertainty quantification for the estimates.

### ***Variance Reduction for Bayesian Model Choice***

Presenter: Minh Long Nguyen

Practitioners of Bayesian inference often need to pick from multiple candidate models when modelling complex phenomena, with predictive accuracy often serving as a key criterion. Popular Bayesian model selection approaches such as leave-one-out cross-validation (LOO-CV) and the widely applicable information criterion (WAIC) estimate out-of-sample predictive performance but typically require extensive posterior simulation, which can be computationally demanding. Limited computational resources may lead to unreliable model selection estimates and motivate the use of simpler non-Bayesian criteria that have several limitations. This work aims to improve the statistical and computational efficiency of Bayesian LOO-CV and WAIC by leveraging Stein-based methods such as control variates and black-box importance sampling to obtain reduced-variance estimates. These approaches utilise gradient information, which can be obtained for free from modern gradient-based samplers, enabling practitioners to achieve more accurate model assessment with fewer samples and reduced computation time. Theoretical proofs of optimality of proposed approaches are presented.

### ***Sparse Data and Small-Sample Bias in Matched Case-Control Studies: Implications for Penalization and Bayesian Modeling***

Presenter: Naohiro Yonemoto

Background: Self-controlled designs and matched case-control studies in epidemiological studies are widely used to address confounding. However, when outcomes are rare or exposure variation is limited, the resulting small and sparse matched data often lead to unstable estimates and separation issues. Bayesian approaches may alleviate these problems, but their performance depends heavily on prior specification.

Objective: This study aimed to compare Bayesian methods for analyzing small and sparse matched data, focusing on prior sensitivity, estimation stability, robustness to separation, and computational performance.

Methods: We conducted simulation experiments and applied the methods to real-world matched data. Four Bayesian strategies were evaluated in conditional logistic models : (1) with noninformative priors, (2) with weakly informative priors, (3) hierarchical Bayesian models incorporating pair- or subject-level random effects, and (4) penalized Bayesian models analogous to Firth-type regularization. Performance was assessed using bias, interval coverage, behavior under sparse information, separation tolerance, and convergence.

Results: Noninformative priors produced unstable and highly prior-dependent estimates in sparse settings. Weakly informative priors improved stability and maintained adequate interval coverage. Hierarchical models reduced variance through partial pooling but were sensitive to hyper-prior choices in extremely sparse scenarios. Penalized Bayesian models showed the greatest robustness to separation and the most consistent performance overall.

Conclusion: For small and sparse matched data, Bayesian models with noninformative priors are unreliable. Weakly informative or penalized Bayesian approaches provide more stable and robust inference. These findings

offer practical guidance for selecting Bayesian methods in matched epidemiologic analyses with limited information.

### ***Bayesian Inference for Unnormalized Models Using Classification Likelihood***

Presenter: Naruki Sonobe

Unnormalized models are important in statistics because they can flexibly describe the characteristics of data with complex dependence structures. However, the intractability of the normalizing constant has limited the applicability of conventional Bayesian inference methods for such models. In this study, we recast the inference problem as a binary classification task that distinguishes observed data from noise samples, and we treat the normalizing constant as an additional unknown parameter in the resulting likelihood. We also introduce an efficient sampling algorithm for exponential-families based on the Gibbs sampler. The effectiveness of the proposed method is demonstrated through applications to time-varying density estimation and edge selection in sparse graphical models for multivariate angular data.

### ***Density-Informed Pseudo-Counts for Calibrated Evidential Deep Learning***

Presenter: Nevena Gligic

Evidential Deep Learning (EDL) is a popular framework for uncertainty-aware classification that models predictive uncertainty via Dirichlet distributions parameterized by neural networks. Despite its popularity, its theoretical foundations and behavior under distributional shift remain poorly understood. In this work, we provide a principled statistical interpretation by proving that EDL training corresponds to amortized variational inference in a hierarchical Bayesian model with a tempered pseudo-likelihood. This perspective reveals a major drawback: standard EDL conflates epistemic and aleatoric uncertainty, leading to systematic overconfidence on out-of-distribution inputs. To address this, we introduce Density-Informed Pseudo-count EDL, a new parametrization that decouples class prediction from uncertainty quantification by separately estimating the conditional label distribution and the marginal covariate density. This separation preserves evidence in high-density regions while shrinking predictions toward a uniform prior for out-of-distribution data. Theoretically, we prove that our method achieves asymptotic concentration. Empirically, we show our method enhances interpretability and improves robustness and uncertainty calibration under distributional shift.

### ***Scalable Posterior Uncertainty for Flexible Density-Based Clustering***

Presenter: Nicola Barileto

We introduce a novel framework for uncertainty quantification in clustering that combines martingale posterior distributions with density-based clustering. Unlike classical model-based approaches, which define clusters at the latent level of a mixture model, we treat clusters as explicit functionals of the data-generating density, without assuming any specific parametric form. To characterize density uncertainty, we obtain martingale posterior samples via a predictive resampling scheme driven by model score evaluations. This allows us to leverage state-of-the-art differentiable density estimators, such as normalizing flows, making density resampling efficient in large-scale settings and fully parallelizable on modern GPU hardware. Martingale posterior samples of the clustering structure are then obtained by applying density-based clustering to the density draws, enabling principled inference on any clustering-related quantity. Casting the inference target as a density functional further enables a rigorous theoretical analysis of the procedure's convergence properties. We apply our methodology to image and single-cell RNA sequencing data, demonstrating the computational efficiency afforded by its GPU compatibility as well as its ability to recover meaningful clustering structures, with associated uncertainty, across diverse domains.

### ***A Bayesian logistic actor-attribute latent space model for social influence***

Presenter: Noemi Corsini

A central task in network analysis is to model social influence, that is, how individual behaviours and outcomes are shaped by their social environment. Classical regression models are not suitable for this purpose, as they frequently rely on independence assumptions that are violated in network data, where individuals' behaviours are inherently interdependent. Although several methods have been proposed to address this problem, existing approaches either treat the network as fixed, rely on multi-step estimation procedures, or are limited to continuous outcome variables. We introduce the Bayesian logistic actor-attribute latent space model for social influence, a novel approach that jointly models binary actor-level outcomes and the network structure within a unified model framework. The network is represented through a latent social space that provides an interpretable, low-dimensional characterization of the underlying social structure. Our goal is to model a binary actor-level outcome as a function of both observed covariates and latent positions, where the latent social space captures complex network dependencies not explained by covariates alone, but affecting the outcome of interest. Inference is performed within a fully Bayesian framework via a Gibbs sampling algorithm based on Pólya-Gamma data

augmentation. This scheme enables principled uncertainty quantification, efficient posterior estimation, and scalability to large networks.

### ***Probabilistic Predictions of Option Prices Using Multiple Sources of Data***

Presenter: Ole Maneeesoonthorn

A new modular approximate Bayesian inferential framework is proposed that enables fast calculation of probabilistic predictions of future option prices, exploiting multiple information sources including daily spot returns, high-frequency spot data and option prices. Our approach calibrates Bayesian posteriors via summary statistics, working directly with the theoretical option pricing model in the absence of a statistical model that explicitly prescribes the likelihood for observed option prices. Unlike comparable frequentist methods, such as indirect inference, our approach explicitly allows us to express uncertainty about future option prices through posterior predictive distributions. We show that our approach produces accurate probabilistic predictions of option prices in realistic scenarios and, despite not explicitly modelling pricing errors via an arbitrary statistical model, the method is shown to be robust to their presence. Predictive accuracy based on the Heston stochastic volatility model is illustrated empirically for short-maturity options, with rapid real-time updates of the predictive distributions highlighted.

### ***Bayesian dynamic modelling of realized volatility in financial asset return forecasting***

Presenter: Patrick Woitschig

We develop a new class of Bayesian dynamic models for real-valued, positive time series that are motivated by interests in improved financial time series forecasting. The context is that of financial return forecasting where we have data on both equity prices and additional estimates of realized volatility in such prices. The core applied setting is daily price series with realized volatility data from higher frequency, intra-day price fluctuations. The traditional Bayesian dynamic linear model with discount-based stochastic volatility (SV-DLM) for an asset return series is extended by adding a new model of the related realized volatility data. The latter model represents realized volatilities (variances) as conditionally inverse-gamma observations on the latent variance processes underlying the returns. This is a new theoretical framework that maintains conjugacy in the sequential, forward filtering and forecasting analyses. One-step forecasts of volatilities are scaled F distributions in closed form, and filtering, forecasting, marginal-likelihood evaluation, and retrospective smoothing are all analytic. The resulting bivariate price-volatility "Realized Dynamic Linear Model (RDLM)" provides a coherent Bayesian system for real-time forecasting of both returns and realized volatility, as well as for routine monitoring of forecast performance in trading and risk settings. In empirical studies of a collection of S&P 500 sector ETFs over an extended historical sample, RDLMs produce similar point forecasts but substantially sharper predictive densities for daily prices than traditional DLMs. The analytic structure and negligible additional computational costs make the framework directly scalable to higher-dimensional decouple/recouple portfolio and risk applications.

### ***Robust Amortized Bayesian Inference with Self-Consistency Losses on Unlabeled Data***

Presenter: Paul Bürkner

Amortized Bayesian inference (ABI) with neural networks can solve probabilistic inverse problems orders of magnitude faster than classical methods. However, ABI is not yet sufficiently robust for widespread and safe application. When performing inference on observations outside the scope of the simulated training data, posterior approximations are likely to become highly biased, which cannot be corrected by additional simulations due to the bad pre-asymptotic behavior of current neural posterior estimators. In this paper, we propose a semi-supervised approach that enables training not only on labeled simulated data generated from the model, but also on unlabeled data originating from any source, including real data. To achieve this, we leverage Bayesian self-consistency properties that can be transformed into strictly proper losses that do not require knowledge of ground-truth parameters. We test our approach on several real-world case studies, including applications to high-dimensional time-series and image data. Our results show that semi-supervised learning with unlabeled data drastically improves the robustness of ABI in the out-of-simulation regime. Notably, inference remains accurate even when evaluated on observations far away from the labeled and unlabeled data seen during training.

### ***Exploring functional uniform priors in non-linear (mixed) models***

Presenter: Paul Mostert

The choice of the non-informative prior for the model parameters in a Bayesian analysis of nonlinear (mixed) models has received significant attention in the literature. This presentation considers the use of a functional uniform prior (FUP) within nonlinear (mixed) models, specifically in dose-response and tumor growth inhibition (TGI) model applications. Traditional non-informative priors like uniform and Jeffreys' priors are widely used in the pharmaceutical industry, however, they can be quite informative in nature when mapping them on the

nonlinear functional space. Additionally, Jeffreys' prior is dependent on the full data structure being available when deriving it in the context of clinical trials. Bornkamp (2012) derived the FUP for a few nonlinear regression models, including exponential, power and hyperbolic-Emax models, but did not consider non-linear mixed models. An extensive Bayesian simulation study is conducted to evaluate the operating characteristics of the FUP when compared with these standard traditional priors. The FUP has the theoretical advantages of being transformation invariant and satisfies the likelihood-principle. While the FUP approximates Jeffreys' prior, it also has the advantage of being specified prior to data collection in contrast to Jeffreys' prior. The Bayesian simulation study is extended to mixed-effects models, specifically the exponential one-parameter models and the two-parameter TGI model. Finally, the performance of the FUP is explored when analyzing oncology data on colorectal cancer.

### ***Optimal Transport for Bayesian Computation and Statistical Modelling***

Presenter: Peiwen Jiang

This research is centered on the unifying power of Optimal Transport (OT), where the core idea is to utilize the Wasserstein metric and its associated geometry to develop more powerful models. The initial phase of this long-term agenda focuses on merging OT with two key pillars: Bayesian Computation and Statistical Time Series Modelling.

- OT for Bayesian Computation: The first pillar aims to address a central challenge in Bayesian methods: the trade-off between the accuracy of sampling and the efficiency of Variational Inference. We introduce a novel variational framework that integrates the mass-covering properties of the Importance-Weighted ELBO with a Wasserstein gradient flow on the Bures-Wasserstein space of Gaussian distributions. The objective is to deliver more accurate posterior approximations that faithfully capture the true underlying geometry and variance of the true posterior distribution.
- OT for Distributional Time Series Modelling: The second project focuses on advancing time series analysis for distributional data by introducing a novel autoregressive model that operates natively on the Wasserstein space. The model's dynamics are formulated on the Bures-Wasserstein manifold of Gaussian distributions, using its tangent space to linearize the evolution of a latent process utilizing logarithmic and exponential maps. The observed distribution is then generated from this latent process via a flexible parametric transformation, allowing the model to produce outputs with diverse shapes (e.g., using Student's t or g&k quantile functions). This geometric approach provides a flexible framework for modelling complex temporal dynamics.
- OT for Non-Parametric Distributional Smoothing: The third project introduces Wasserstein Exponential Smoothing, a non-parametric approach for forecasting distributional time series. Unlike traditional scalar-valued exponential smoothing, WES operates directly on the space of quantile functions, utilizing the  $W_2$  metric to update the latent state. By bypassing rigid parametric assumptions, WES effectively captures complex, time-varying features such as changing variance and heavy tails, which are common in real-world data like financial returns and temperature distributions. We establish a Minimum Wasserstein Estimator for parameter recovery and demonstrate that this geometric smoothing framework delivers robust out-of-sample performance across diverse, non-linear data-generating processes.

Together, these foundational projects demonstrate the versatility of the Optimal Transport framework in bridging the gap between rigorous geometry and practical statistical modeling. While the initial pillars establish parametric and non-parametric benchmarks for univariate data, the long-term agenda paves the way for multivariate extensions to tackle higher-dimensional distributional dynamics and the integration of OT-based smoothing into broader Bayesian state-space models.

### ***The Dynamic Triple Gamma Prior as a Shrinkage Process Prior for Time-Varying Parameter Models***

Presenter: Peter Knaus

Many existing shrinkage approaches for time-varying parameter (TVP) models assume constant innovation variances across time points, inducing sparsity by shrinking these variances toward zero. However, this assumption falls short when states exhibit large jumps or structural changes, as often seen in empirical time series analysis. To address this, we propose the dynamic triple gamma prior, a stochastic process that induces time-dependent shrinkage by modeling dependence among innovations while retaining a well-known triple gamma marginal distribution. This framework encompasses various special and limiting cases, including the horseshoe shrinkage prior, making it highly flexible. We derive key properties of the dynamic triple gamma that highlight its dynamic shrinkage behavior and develop an efficient Markov chain Monte Carlo algorithm for posterior sampling. The proposed approach is evaluated through sparse covariance modeling and forecasting of the returns of the EURO STOXX 50 index, demonstrating favorable forecasting performance.

### ***Modeling Spatially Varying Peaks in Skewed Spatio-Temporal Data***

Presenter: Peter Zhu

In many applications such as infectious disease epidemiology, urban planning, and environmental science, spatial and spatio-temporal data display heavy tails. A main focus of spatio-temporal statistical models is to capture these heavy tails along with spatial and spatio-temporal trends and dependencies. In this work, we consider New York City subway ridership data providing hourly counts of the number of entries into each subway station. New York City subway ridership has a general temporal trend with well-defined peaks during morning and evening rush hour periods; yet, the intensity of these known peak periods varies spatially. We propose a spatio-temporal Bayesian hierarchical model that captures spatially varying peak dynamics while allowing for and identifying stations with outlier latent spatial effects. Our model captures general temporal trends through an independent temporal random effect, while allowing for adjustments during well-defined peak periods through spatio-temporal interaction terms. This spatio-temporal interaction is composed of a determined, smooth function of time and a spatial random effect term. This interaction alters the shape of the general temporal trend with the spatial random effect defining the intensity of these peak adjustments. A time-independent spatial random effect is provided with a scaled BYM2 prior [Michal et al. 2025]. We evaluate the flexibility of our model to handle intense variations in peak period behavior and general heavy-tailed anomalies by appealing to both synthetic simulated data and to the hourly New York City subway ridership data. We compare the performance of our model with that of a classical spatio-temporal model containing no adjustments for peak period behavior and a recently developed spatio-temporal dynamic generalized linear model [Yadav et al. 2025].

### ***Deconvolution using Empirical Bayes and Maximum Mean Discrepancy***

Presenter: Ritwik Vashistha

Modern data analyses frequently encounter settings where samples of variables are contaminated by measurement error. Ignoring measurement noise can substantially degrade statistical inference and existing approaches can often be computationally inefficient in high dimensions or make strong assumptions about the form of the prior distribution. Recent advances in kernel methods, particularly those based on Maximum Mean Discrepancy (MMD), have enabled flexible, distribution-free inference, which typically assume precise data and overlook contamination by measurement error. In this work, we introduce a novel framework for deconvolution that uses empirical bayes where the prior is estimated non-parametrically with the help of convolutional MMD (convMMD) which compares distributions after noise convolution and retains metric validity under standard kernel conditions. We assume that the samples corrupted by potentially heteroscedastic noise from a known distribution and we establish finite-sample deviation bounds that are unaffected by measurement error. We also show an equivalence between estimation under noise and kernel smoothing, and provide robustness guarantees for mild misspecification of the noise distribution. Leveraging these insights, we introduce a convMMD-based empirical bayes estimator for inference and deconvolution with noisy, heteroscedastic observations. We establish its consistency and asymptotic normality, and provide an efficient implementation using stochastic gradient descent. We demonstrate the practical effectiveness of our approach through simulations and applications in astronomy.

### ***Bayesian repulsive mixture model for multivariate functional data***

Presenter: Rosangela Helena Loschi

We introduce a repulsive mixture model for multivariate functional data having as one of the main goal to clustering curves with similar shapes. We assume that more than one functional curve is observed for each individual. The clustering process takes into account both individual-specific information, as we assume covariate-dependent mixture weights, and the similarity of the curve shape. To facilitate the identification of well-differentiated clusters while avoiding redundant clusters in relation to their curve shapes, we propose a repulsive prior distribution for the component-specific location vector of the B-spline coefficients. In this proposed prior, the repulsive factor depends on a B-spline curve-tailored distance, extending existent repulsive priors to the context of functional data. To sample from the posterior distribution, we propose a MCMC algorithm with a split-merge step. The split-merge step significantly improves the mixing of the chain. Different features of the proposed model, including the effects of repulsion and covariates in the clustering, are investigated through simulation. The model is used to identify different movement characteristics in individuals with Chronic Ankle Instability (CAI data). We also evaluate the effect of repulsion on clustering of functional NBA data which has a high noise level. Joint work with Ricardo Cunha Pedroso(INSPER-Brazil) and Fernando A. Quintana(PUC-Chile). This work was partially supported by CNPq, CAPES, FAPEMIG (Brazil) and FONDECIT (Chile).

### ***Neural Methods for Multiple Systems Estimation Models***

Presenter: Rowland Seymour



Multiple Systems Estimation models are log-linear models based on capture-recapture methods. They are widely used to estimate the size of hidden populations by modelling the pattern of overlaps across multiple lists of the samples from the population. However, in many social science applications, identifiability concerns often necessitate censoring or omitting parts of the data to protect individual privacy, complicating inference. We develop an approach that uses two complementary neural methods, Neural Bayes Estimators (NBE) and Neural Posterior Estimates (NPE), to perform inference for multiple systems estimation models in the presence of missing and censored observations. These likelihood-free methods enables robust population size estimation while accommodating incomplete data. We carry out various simulation studies and sensitivity analyses to evaluate the performance of our method and evaluate the circumstances it can be applied to real word settings. We then demonstrate our method on two real-world examples estimating the prevalence of populations of modern slavery and drug users.

### ***Rebalancing to Non-Reversible Continuous-Time Jump Samplers***

Presenter: Ruben Seyer

Markov chain Monte Carlo methods are central in computational statistics, and typically rely on detailed balance to ensure invariance with respect to a target distribution. Although straightforward to construct by Metropolization, this can induce diffusion-like exploration of the sample space, requiring careful tuning of parameters such as step size. We introduce a general mechanism for constructing non-reversible continuous-time samplers, without requiring detailed balance. Our approach transforms jump processes satisfying a skew-detailed balance condition for a reference measure into processes sampling a target measure absolutely continuous with respect to it. Unbounded balancing functions allow such samplers to dynamically select favourable transitions. We establish invariance under weak criteria and demonstrate how to verify geometric ergodicity. Numerical experiments demonstrate that the resulting samplers are more robust to parameter tuning and can outperform the Metropolized state-of-the-art.

### ***Stabilizing Thompson Sampling with Point Null Bayesian Response-Adaptive Randomization***

Presenter: Samuel Pawel

Response-adaptive randomization (RAR) methods use accumulated data to adapt randomization probabilities, aiming to increase the probability of allocating patients to effective treatments. A popular RAR method is Thompson sampling, which randomizes patients proportionally to the Bayesian posterior probability that each treatment is the most effective. However, its high variability early in a trial can also increase the risk of assigning patients to inferior treatments. We propose a principled method based on Bayesian hypothesis testing to mitigate this issue. Specifically, we introduce a point null hypothesis that postulates equal effectiveness of treatments. This induces shrinkage toward equal randomization probabilities, with the degree of shrinkage controlled by the prior probability of the null hypothesis. Equal randomization and Thompson sampling arise as special cases when the prior probability is set to one or zero, respectively. Simulated and real-world examples illustrate that the proposed method balances highly variable Thompson sampling with static equal randomization. A simulation study demonstrates that the method can mitigate issues with ordinary Thompson sampling and has comparable statistical properties to Thompson sampling with common ad hoc modifications such as power transformation and probability capping. We implement the method in the open-source R package *brar*, enabling experimenters to easily perform point null Bayesian RAR and support more effective randomization of patients.

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### ***Adaptive Shrinkage with a Nonparametric Bayesian Lasso***

Presenter: Santiago Marin

Modern approaches to perform Bayesian variable selection rely mostly on the use of shrinkage priors. That said, an ideal shrinkage prior should be adaptive to different signal levels, ensuring that small effects are ruled out, while keeping relatively intact the important ones. With this task in mind, we develop the nonparametric Bayesian Lasso, an adaptive and flexible shrinkage prior for Bayesian regression and variable selection, particularly useful when the number of predictors is comparable or larger than the number of available data points. We build on spike-and-slab Lasso ideas and extend them by placing a Dirichlet process prior on the shrinkage parameters. The result is a prior on the regression coefficients that can be seen as an infinite mixture of Laplace distributions, all offering different amounts of regularization, ensuring a more adaptive and flexible shrinkage. We also develop an efficient Markov chain Monte Carlo algorithm for posterior inference. Through extensive simulation studies and real-world data analyses, we illustrate that our proposed method leads to coefficient recovery, variable selection accuracy, and out-of-sample predictions that are comparable to or better than those from state-of-the-art shrinkage priors, highlighting the benefits of the nonparametric Bayesian Lasso over existing methods.



### ***Bayesian Regression and Variable Selection for Joint Ordinal-Continuous Outcomes***

Presenter: Sarah Lee

Mixed responses are common in social science research: survey data are typically ordinal (e.g., Likert-scale items), while related socioeconomic measures are continuous. A common practice is to treat ordinal items as continuous by averaging items within a construct and then fitting a multivariate regression model. While defensible, this approach discards the ordered structure of the response variable and loses information at the item level. We propose a multivariate regression methodology that jointly models the mixed responses, preserving the ordinal information at the item level through an augmented Gaussian latent variable that is modelled as jointly dependent with the continuous response. Bayesian variable selection is performed for both responses by placing spike-and-slab priors on the regression coefficients for each outcome. Analytic marginal likelihoods for the inclusion indicators are obtained by integrating out the regression coefficients under the latent-variable augmentation. This approach i) avoids the information loss caused by averaging and ii) improves inference by modelling the two outcomes jointly rather than independently. Through simulations, we examine the frequentist properties of the estimator and demonstrate superior inference and more reliable identification of influential predictors compared with approaches that treat ordinal responses as continuous. We apply our method to real educational survey data, where self-efficacy represents the ordinal response and academic performance represents the continuous outcome.

### ***Bayesian Joint Modeling for hierarchically structured Medical Data***

Presenter: Seongho Song

Joint modeling has been a useful strategy for incorporating latent associations between different types of outcomes simultaneously, often focusing on a longitudinal continuous outcome characterized by a Linear Mixed Effect (LME) sub-model with a binary process, which is commonly specified by a Generalized linear Mixed Model (GLMM) under hierarchically structured medical data. In this study, we propose a multilevel joint model that encompasses LME and GLMM sub-models through a Bayesian approach. Motivated by the need for timely detection of pulmonary exacerbations and characterization of irregularly observed lung function measurements in people living with cystic fibrosis (CF) receiving care across multiple centers, we apply the model to data arising from the US CF Foundation Patient Registry.

### ***Hierarchical Semi-parametric Framework for Inferring Social Contact Patterns***

Presenter: Shozen Dan

Social contact matrices, which characterize close-range human interaction patterns, are essential tools in infectious disease epidemiology for understanding transmission dynamics, optimizing intervention strategies, and calibrating compartmental models. Existing approaches, however, remain largely confined to age-stratified structures, even though the recent global pandemic demonstrated that socioeconomic and demographic inequalities were major drivers of disease burden. Current methods are fragmented, limited in dimensionality, and unable to model richer forms of stratification while preserving epidemiological constraints.

We develop a hierarchical semiparametric Bayesian framework that unifies existing methodology under a single general formulation and extends it to arbitrary combinations of demographic and socioeconomic variables. The framework provides principled solutions to core mathematical challenges in multidimensional contact modelling: (i) enforcement of reciprocity constraints via a novel latent-space construction based on simplex transformations; (ii) substantial parameter reduction through structured Kronecker representations; and (iii) flexible smoothing of age-age and category-specific surfaces using Gaussian Markov random fields, Gaussian processes, or adaptive splines.

Extensive simulation studies show that this unified approach improves inference accuracy compared with existing methods. Applications to pre-pandemic, COVID-19, and post-pandemic eras reveal mixing patterns previously inaccessible to age-only models. These insights have the potential to meaningfully advance epidemiological modelling by better capturing transmission heterogeneity. All methods are released in an open-source Python package to support broad adoption in research and public-health practice.

### ***Generalized Bayes Compositional Directed Acyclic Graph***

Presenter: Shuangjie Zhang

Bayesian network, represented as Directed Acyclic Graph (DAG), provides a structured way to model probabilistic relationships. However, classical Bayesian inference often relies on likelihood-based inference, which can be sensitive to model misspecification and restrictive assumptions about data distributions. Motivated by complex count data from multi-community microbiome studies, we introduce the generalized Bayes compositional directed acyclic graph (GBC-DAG), a novel framework that extends Bayesian updates by incorporating general-

ized Bayes inference within DAG-based probabilistic models. Our approach uses a Kullback–Leibler divergence based likelihood for compositional data, enabling greater flexibility and robustness in posterior inference. We prove the properness of loss-based likelihood and the identifiability of the DAG. GBC-DAG captures the interaction between microorganisms across various communities and our findings highlight the advantages of generalized Bayesian methods in microbiome compositional data analysis.

#### ***Causal Bandits with Gaussian Process Structural Models***

Presenter: Shunsuke Horii

We explore a nonlinear and nonparametric extension of causal bandits by modeling the structural equations of a causal graph with Gaussian processes. While existing causal bandit algorithms typically rely on linear structural equation models, this assumption limits the expressiveness of intervention effects in many real-world systems. Our approach places GP priors over the unknown causal mechanisms, enabling flexible function learning and uncertainty quantification at each node while preserving the ability to reason about interventions through the causal structure. We sketch how Gaussian-process inference can be combined with exploration strategies such as Thompson sampling or UCB to guide intervention selection, and we discuss conditions under which leveraging the causal graph may yield improved regret guarantees. This framework aims to provide a foundation for causal bandits that operate in rich nonlinear environments, and highlights several challenges and opportunities that arise when moving beyond linear SEMs.

#### ***Parallelizing MCMC Across the Sequence Length: $T$ samples in $O(\log^2 T)$ time***

Presenter: Skyler Wu

Markov chain Monte Carlo (MCMC) algorithms are foundational methods for Bayesian statistics. However, most MCMC algorithms are inherently sequential, and their time complexity scales linearly with the number of samples generated. Previous work on adapting MCMC to modern hardware has therefore focused on running many independent chains in parallel. Here, we take an alternative approach: we propose algorithms to evaluate MCMC samplers in parallel across the chain length. To do this, we build on recent methods for parallel evaluation of nonlinear recursions that formulate the state sequence as a solution to a fixed-point problem, which can be obtained via a parallel form of Newton’s method. We show how this approach can be used to parallelize Gibbs, Metropolis-adjusted Langevin, and Hamiltonian Monte Carlo sampling across the sequence length. Moreover, we prove theoretical results that link the convergence rate of the fixed-point algorithm to the stability of the recursion, yielding sublinear time complexity for certain problems. Finally, to lower memory costs and reduce runtime in practice, we develop quasi-Newton methods that are generally applicable for parallel evaluation of nonlinear recursions. Across several examples, we demonstrate the simulation of up to hundreds of thousands of MCMC samples with only tens of parallel Newton iterations. We find that the proposed parallel algorithms accelerate MCMC, in some cases by more than an order of magnitude compared to sequential evaluation.

#### ***Linkage between agricultural cropping systems and crop yields: Bayesian network meta-analysis***

Presenter: Soyeong Jeong

We propose an arm-based Bayesian network meta-analysis framework to investigate the linkage between agricultural cropping systems and crop yields by synthesizing evidence from multiple independent pairwise meta-analyses of conservation agriculture. We adopt a hierarchical random-effects structure to account for between-study heterogeneity. In addition, our model incorporates study-level covariates directly through a Bayesian meta-regression specification, allowing estimation of system-specific mean yield levels and marginal covariate effects, unlike previous studies that rely primarily on subgroup analyses. We conduct posterior inference via Markov chain Monte Carlo methods, providing full uncertainty quantification for both mean effects and indirect comparisons. The proposed methodology is illustrated by integrating data from existing meta-analytic studies of cropping systems, and provides probabilistic estimates of relative yield performance and system-specific responses to agronomic and environmental covariates.

#### ***A Bayesian Nonparametric Framework for Robust Estimation and Uncertainty Quantification of Toxicological Endpoints Under Sparse Conditions***

Presenter: Spiro Stilianoudakis

Accurate estimation of median lethal concentrations (LC50) from aquatic quantal dose–response data is foundational to chemical risk assessment. Current regulatory practice, as outlined in OECD Test Guideline 203, relies on parametric nonlinear regression models (e.g., the two-parameter log-logistic) or the trimmed Spearman–Kärber (TSK) method. However, both approaches exhibit serious limitations when dose–response data are sparse (i.e. when there are few or no partial mortalities present). In such cases, the TSK method often fails to yield confidence intervals, and parametric models become unstable or non-identifiable.

We propose a Bayesian Gaussian Process Regression (BGPR) framework that replaces restrictive functional forms with a nonparametric prior over latent response curves via a custom kernel. Through extensive simulation studies spanning sparse and non-sparse dose-response scenarios, we demonstrate that BGPR yields unbiased LC50 estimates, well-calibrated credible intervals, and superior predictive performance compared to both the log-logistic and TSK approaches. Additionally, we further apply the model to historical acute fish toxicity studies, where approximately 20% of cases exhibited sparse mortality patterns leading to missing 95% confidence intervals under the TSK method. BGPR successfully produced credible LC50 estimates for all studies, and a paired comparison showed that mean LC50 values from BGPR were not statistically larger than TSK-derived estimates, indicating that the Bayesian approach is no less conservative from a regulatory perspective.

These results illustrate how BGPR can serve as a unified, uncertainty-aware framework for dose-response modeling that is robust to data sparsity. The BGPR can be extended to other likelihoods such as for toxicogenomic data, enabling robust benchmark dose (BMD) estimation. This work highlights the potential for modern Bayesian nonparametric methods to enhance reproducibility, interpretability, and regulatory confidence in toxicological endpoint derivation.

#### ***A full-likelihood framework for Bayesian inference in the Cox model via rank-ordered data modeling***

Presenter: Tomohiro Ohigashi

In Bayesian inference for the Cox proportional hazards model, modeling the baseline hazard function has remained a persistent challenge. Recently, the general Bayesian inference has provided a theoretical justification for performing posterior inference based on the partial likelihood, without specifying a prior model for the baseline hazard function. Meanwhile, several Gibbs sampling algorithms based on approximated partial likelihoods have been proposed, though debate continues over their validity and efficiency. In this study, we propose a novel likelihood extension of the Cox proportional hazards model based on rank-ordered data modeling. Specifically, we develop two Gibbs sampling algorithms that combine the full likelihood formulation under the Plackett-Luce or generalized Plackett-Luce models with Pólya-Gamma data augmentation. We present the results of simulation studies and real-data applications comparing the proposed algorithms with existing methods.

#### ***Bayes-Assisted Imputation for Semi-Supervised Causal Mediation Analysis***

Presenter: Tomoki Okuno

Causal mediation analysis decomposes the average treatment effect into natural direct and indirect effects (NDE and NIE), but requires observing the mediator for every unit. When the mediator is costly to measure, as in genome-wide DNA methylation profiling, it is available only for a budget-limited subset while treatment, covariates, and outcome are recorded for the full cohort.

We develop a semiparametric framework for NDE and NIE estimation in this semi-supervised setting. Starting from the complete-data efficient influence function (EIF), we project the mediation score onto the always-observed variables under missing-at-random labeling, obtaining the observed-data EIF. The framework yields a projection estimator that attains the semiparametric efficiency bound, and an imputation estimator that plugs a predicted mediator into the nonlinear mediation score with labeled-subsample bias correction. The imputation estimator carries a structurally irreducible efficiency loss unique to mediation but enables use of external mediator predictors such as mQTL-derived methylation scores.

The labeled-subsample correction of the imputation estimator, inspired by prediction-powered inference (PPI), targets the gap between true and predicted mediation scores, exactly the PPI rectifier. We place a horseshoe prior on this rectifier following the FAB-PPI framework of Cortinovis and Caron (2025), yielding frequentist confidence intervals with guaranteed asymptotic coverage that contract when the predictor is accurate and revert to standard PPI when it is poor. In simulations, FAB-imputation yields shorter intervals under accurate prediction and reverts to standard imputation under poor prediction, maintaining nominal coverage throughout.

#### ***Efficient Bayesian inference for multinomial probit model ensuring covariance identifiability***

Presenter: Tomotaka Momozaki

The Multinomial Probit (MNP) model has become a cornerstone in consumer behavior analysis owing to its flexible representation of choice mechanisms through latent utilities. However, practitioners face a critical challenge: ensuring an identifiable covariance matrix in utilities through proper model specifications and designing inference algorithms that account for them. While identifiability conditions for the covariance matrix in utility differences and the related inference algorithms are well-established, those addressing identifiability in the original utility specifications remain understudied. The identifiability imposes constraints on the correlation structure of utilities, thereby influencing the interpretability of the model. Therefore, inference that accounts for these constraints is crucial for understanding choice mechanisms in practical applications. In this talk, we present a novel Bayesian

inference framework based on theoretical results about the identifiability of the covariance matrix for the original utility specification in the MNP model. The proposed approach combines sampling from the marginal posterior of the covariance matrix using the Metropolis-Hastings algorithm with sampling from the conditional posterior of the coefficient parameters given the covariance matrix. This can be regarded as a collapsed MCMC algorithm that samples from the joint posterior distribution by marginalizing out the high-dimensional utilities, making it more scalable and efficient than traditional algorithms. We will demonstrate through comprehensive simulation studies that the proposed method outperforms existing approaches in inference accuracy, providing practical insights for decision-making applications with the MNP model.

#### ***Multiview species sampling models for edge-exchangeable networks***

Presenter: Tâm Le Minh

Many interaction datasets can be viewed as networks whose edges connect nodes in different roles or node sets, for example senders and receivers in directed graphs or plants and pollinators in bipartite systems. Each role (or node set) is associated with its own latent labels, and an edge is assigned to a latent edge-type determined by the label pair of its endpoints. Existing edge-exchangeable models (Crane and Dempsey, 2016) typically treat these node-side views independently, limiting their ability to represent structured dependence across roles. I introduce multiview species-sampling models (MVSSMs), which extend classical species-sampling priors to jointly model several latent labelings on the same nodes. MVSSMs flexibly encode dependence between the role-specific labelings. As a concrete instance, I show that the CLIC prior (Dombowsky and Dunson, 2025) can be interpreted as a multiview species-sampling model with a parameter controlling cross-view alignment. Using a representation theorem for edge-exchangeable networks, I then construct a nonparametric model in which the role-specific labels determine each edge-type, and interactions within each type are modeled through species-sampling distributions on product spaces. This separates cross-view dependence from within-type heterogeneity and extends naturally to bipartite networks, such as plant-pollinator systems. Finally, I outline an extension to community detection via a nested construction, in which node-side labels are themselves generated from a higher-level species-sampling prior, inducing clusters of labels and yielding a nonparametric analogue of stochastic block models.

#### ***Summer Temperature Extremes in Japan: a Bayesian Markov-Switching Approach to Heat Wave Analysis***

Presenter: Vincenzo Gioia

Heat wave modelling is challenging for climate models because the coarse resolution of the grid can only approximate the local extremal behaviours of the system. Additionally, the classification of a period as a heat wave depends on the specific definition adopted, which can be tailored to the specific purpose of the analysis. In this work, we use the probabilistic structure of the multi-state Markov-switching regression model to study the local summer temperature dynamics. We model the observable process, accounting for seasonality and a multi-year trend, through a large-scale climate index, while considering different specifications on the transition probability matrices to capture gradual thermal evolution and more abrupt changes in the temperature dynamics. By increasing the number of regimes, the model is able to capture the tail behaviour and isolate periods of high and extremely high temperatures from normal conditions. The Markov-switching model allows for characterising heat waves as prolonged periods during which temperatures remain in the last regime, according to the regime membership probability, and for exploring heat wave trend patterns and regional heterogeneity. Results obtained through the probability-based classification rule are compared with those from the empirical-based definition of heat wave, either in terms of frequency and duration over time. Although a unique characterization of heat wave periods cannot be identified, mainly due to factors related to local climate conditions, results align with extreme or moderately extreme heat wave events. The proposal is illustrated on the maximum daily temperatures over the last three decades, considering six Japanese meteorological stations, which partly reflect the climate heterogeneity of the area.

#### ***Bayesian Multiplicative Errors-in-Variables Modeling for Hyperspectral Unmixing***

Presenter: Vivek Kumar Singh

Estimating endmember spectra and their corresponding abundances from hyperspectral imagery is a fundamental inverse problem in remote sensing. The standard linear mixing model relies on additive, homoscedastic errors, which are often misaligned with the physical properties of light, where noise is approximately proportional to the signal intensity. This can lead to biased estimates and poor uncertainty quantification. To address these limitations, we propose a Bayesian multiplicative Errors-in-Variables model. Motivated by the physics of light propagation, our formulation explicitly captures the multiplicative noise through a multivariate log-normal specification, which naturally accommodates covariance across spectral frequencies. Moreover, our Errors-in-

Variables setup allows for endmember-specific variability, providing a more flexible and realistic representation of the data-generating process. For posterior inference, we develop an MCMC sampler that exploits the geometry of the constrained parameter space. The sampler uses the current state of the Markov chain to approximate local curvature and construct more efficient proposal distributions. When compared to the analogous additive model on simulated data, our multiplicative model demonstrates superior performance, producing tighter credible intervals for abundances while achieving better nominal coverage. We also present theoretical results on the asymptotic behaviour of the multivariate coefficient of variation under our model assumptions, clarifying regimes under which the multiplicative framework is preferable. Finally, we apply our method to the well-known Indian Pines AVIRIS hyperspectral dataset, demonstrating its practical effectiveness in real-world scenarios.

#### ***Learning the distance for ABC and neural posterior estimation***

Presenter: Wang Yuyan

Likelihood-free inference methods can perform Bayesian inference when evaluating the likelihood is impractical but simulating synthetic data from the model is feasible. Approximate Bayesian computation (ABC) is a well-established likelihood-free approach that constructs particle posterior approximations by evaluating the similarity between simulated and observed data using a distance function, which is used in rejection or weighting steps. Here we extend previous work on adaptive distance learning for ABC to misspecified time series, while also exploring applications in neural posterior estimation using prior-data fitted networks (NPE-PFN) with localization. The adaptation of the distance that we consider optimizes out-of-sample predictive performance using a scoring rule. For both ABC algorithms and NPE-PFN methods with localization, adaptive distance learning improves forecasting performance in simulated and real examples.

#### ***A novel class of mixed Poisson distributions***

Presenter: Will Townes

Mixed Poisson families are widely used to model count data with overdispersion, zero inflation, or heavy tails in a variety of applications including finance, biology, and the physical sciences. The Poisson rate is typically assigned a nonnegative-valued mixing distribution (also interpretable as a prior). Surprisingly, it is also possible for the mixing distribution to have negative support. For example, the Hermite distribution is analogous to mixing a Poisson with an unconstrained Gaussian and can be derived using generating functions so long as constraints on the natural parameter are satisfied. I will give a general characterization of this unusual class, provide several concrete examples, and discuss computational and statistical challenges in their practical use.

#### ***Non-linear and Non-Gaussian time-varying parameter models with dynamic shrinkage process priors***

Presenter: Yijie Niu

State-space models with the recently developed dynamic global-local shrinkage process priors make it possible to model time-varying parameters that can remain constant for some time periods and move rapidly or jump in other periods. The available Gibbs posterior sampling algorithms are restricted to conditionally linear Gaussian observation models. In this work, we develop and compare fast approximate posterior sampling algorithms for non-Linear/non-Gaussian state-space models with parameter evolutions following dynamic global-local shrinkage process priors. The samplers are partially based on Gaussian approximations and we propose several strategies to improve the approximation accuracy. The methods are illustrated and compared on a time-varying Poisson and Beta regression, using simulated and real data. The algorithms will be released as a publicly available Julia package.

#### ***Adaptive Bayesian computation for efficient biobank-scale genomic inference***

Presenter: Yiran Li

Motivation: Modern biobanks, with unprecedented sample sizes and phenotypic diversity, have become foundational resources for genomic studies, enabling powerful cross-phenotype and population-scale analyses. As studies grow in complexity, Bayesian hierarchical models offer a principled framework for jointly modeling multiple units such as cells, traits, and experimental conditions, increasing statistical power through information sharing. However, adoption of Bayesian hierarchical models in biobank-scale studies remains limited due to computational inefficiencies, particularly in posterior inference over high-dimensional parameter spaces. Deterministic approximations such as variational inference provide scalable alternatives to Markov Chain Monte Carlo, yet current implementations do not fully exploit the structure of genome-wide multi-unit modeling, especially when biological effects of interest are concentrated in a few units.

Results: We propose an adaptive focus (AF) strategy within a block coordinate ascent variational inference (CAVI) framework that selectively updates subsets of parameters at each iteration, corresponding to units deemed relevant based on current estimates. We illustrate this approach in protein quantitative trait locus



(pQTL) mapping using a joint model of hierarchically linked regressions with shared parameters across traits. In both simulated data and real proteomic data from the UK Biobank, AF-CAVI achieves up to a 50% reduction in runtime while maintaining statistical performance. We also provide a genome-wide pipeline for multi-trait pQTL mapping across thousands of traits, demonstrating AF-CAVI as an efficient scheme for large-scale, multi-unit Bayesian analysis in biobanks.

#### ***Plant-Capture Methods for Estimating Homeless Population Size from Uncertain Plant Captures***

Presenter: Yiran Wang

Plant-capture is a specialized variant of traditional capture-recapture methods used to estimate the size of a population. In epidemiologic literature, a notable application of this method is the estimation of the size of homeless populations through point-in-time street surveys. With this approach, decoys referred to as "plants" are introduced into the population to estimate the capture probability. Previous plant-capture studies have not systematically accounted for uncertainty in the capture status of individual plants. To address this, we propose three complementary approaches within a Bayesian framework to formally incorporate uncertainty into the plant-capture model arising from the capture status of plants and heterogeneity between survey sites: (i) full Bayesian inference via Markov chain Monte Carlo (MCMC), (ii) a Bayesian normal approximation for fast approximate posterior summaries, and (iii) an uncertainty-propagation (UP) method that replaces the equality-constraint step with an approximate normal posterior for the target quantity to enable efficient approximate Bayesian inference. We then apply our methods to estimate the size of the homeless population in large US cities in the context of the "S-Night" study conducted by the US Census Bureau.

#### ***Hyperbolic Latent Space Models for Network Data: Model Specification and Bayesian Inference***

Presenter: Yiwei Gong

Many real-world networks exhibit hierarchical, tree-like structure and have heavy-tailed degree distributions, phenomena that are not readily account for using standard statistical models for network data. As a result, extensions of the continuous latent space modeling framework have been proposed that aim to accommodate networks with these features. In particular, the hyperbolic latent space model – where network nodes are probabilistically embedded in a latent Riemannian manifold with constant positive curvature – has been proposed as a natural framework for capturing such hierarchical structures. However, most statistical implementations simplify the original physics-based model by omitting the temperature parameter, which controls the sharpness of the latent distance-to-probability mapping. We argue this is a critical omission. In this paper, we demonstrate that temperature is the fundamental parameter governing a network's tree-like topology, and that failing to infer the temperature from data weakens the expressiveness of the model. We formalize a Bayesian hyperbolic model with an unknown, learnable temperature parameter. We develop two inferential procedures: a Hamiltonian Monte Carlo approach for rigorous posterior characterization and a scalable Auto-Encoding Variational Bayes algorithm for large-scale networks. Our results show that our model consistently outperforms models with fixed temperature and misspecified Euclidean geometries in graph reconstruction tasks, confirming that temperature is a crucial and inferable feature of complex networks.

#### ***Possibilistic Markov Chains and Their Applications***

Presenter: Yoshiki Hori

Fuzzy is a Pareto optimal solution of a two-dimensional error stochastic differential equation. Hereinto, the simultaneous stochastic differential equation for horizontal and vertical error is solved by using the weighting method. In this instance, the horizontal and vertical fuzzy variables are mapped and transformed onto a single state of nature by using a membership function, and the oceanic current is subject to the possibility principal factor rotation if the mapped fuzzy events is the direct sum (calm state) and the possibility oblique factor rotation if they aren't the direct sum (turbulent state). In this paper, we focus especially on the possibility oblique factor analysis, who refer to the initial and stopping condition for its possibility oblique factor rotation. This rotation is adapted to the modeling of oceanic current. In addition, we propose control of the sea algorithm in a turbulent state on the oceanic current using Vague Sets and Theory until now. Since this algorithm is extremely simple, it is expected to be a mass-producible in research for the development of waterborne drone such as unmanned lifeboat and unmanned fishing vessel. The traditional Markov Chain has the worst problem that we can define the transfer matrix by not subject but object. It is very easy for us to deliver the possibilistic transfer matrix by Possibility Theory. Note that we obtain the subjective membership functions in Possibility Theory by the type 2 Vague Sets and Theory. Therefore, when these membership functions are not direct sum, we introduce the judgement pending for covering the fuzzy-system information. As the object our study is modeling Wave, it is simply to understand Wave Equation by assume existing Other Worlds, This World and An Another World.



Therefore, we introduce Japanese Buddhist concept according to worlds. Of course, no Buddhist can understand this concept for the worlds. Though, we once tried to construct KOTODAMA processing by these worlds, it is fact that we failed it by the biggest fuzziness by using twice fuzzy OR.

#### ***Robust local empirical Bayes correction for Bayesian modeling***

Presenter: Yoshiko Hayashi

Efron (2011) develops a local empirical Bayes correction for the location parameter. This paper investigates a robust modeling for the local empirical Bayes correction. Our method has two advantages. First, the method does not require specifying the distribution of the hyperparameter for the location parameter, which allows a multimodal distribution. Second, we take care of outliers using the Student-t distribution.

#### ***A Flexible Nonparametric Survival Framework for Large-Scale EHR Studies***

Presenter: Yu-Chien Ning

Large-scale electronic health records (EHR) provide an invaluable resource for modern clinical and public health research. However, modeling survival data from EHR often requires more flexible methods than traditional Cox-type approaches. Building upon the Gaussian process survival model introduced by Fernández et al. (2016), we develop a fully nonparametric Bayesian survival framework that accommodates complex functional structures. We establish strong theoretical guarantees for this framework, including optimal posterior contraction and valid uncertainty quantification for the survival function under suitable priors. We further demonstrate that the proposed methodology offers a principled and practical approach for addressing outcome measurement error in EHR datasets.

Joint work with Ge Lin, David Zucker, Molin Wang, and Raymond J. Carroll

#### ***Bayesian Item Response Theory Model with Local Dependence***

Presenter: Yu-Wei Chang

Traditional Item Response Theory (IRT) models assume that the responses to all items by one respondent are conditionally independent, conditional on random-effect parameters, and this is referred to as local independence. However, it is not easy that this assumption is fulfilled in practical applications. Various extensions of IRT models have been proposed in the literature to deal with local dependence (LD) structures arising from situations such as testlets, time limit tests, and different response strategies of individuals. In recent years, a more innovative approach involves directly developing general IRT models that account for local dependence. For example, Chen, Li, Liu, and Ying (2018) proposed the FLaG-IRT model, which incorporates the Ising model into IRT model to capture local dependence. However, their statistical inference relies on proximal gradient-based algorithm for maximum likelihood estimation, which is computationally complex and is difficult to be further generalized. The current study proposes a Bayesian inference for the FLaG-IRT model. The advantage of the Bayesian FLaG-IRT inference is to allow parameters to borrow information through prior distributions and to provide a more concise estimation structure that facilitates future model extensions. The main challenge of Bayesian FLaG-IRT inference is the intractable normalizing function issues, and we adopt variational Bayesian method (Kim, Bhattacharya & Maiti, 2024) with an appropriate pseudo likelihood to overcome this issue. Simulation studies are conducted to evaluate the performance of the proposed method under various conditions with two real data sets illustrating its practical utility.

#### ***Bayesian Nonparametric Inference for the Spatiotemporal Hawkes Process: Application to the ETAS Model***

Presenter: Yuanyuan Niu

Hawkes-type point processes are widely used to model the excitation mechanisms among a series of discrete events that occur in a continuous time domain or a spatiotemporal domain. They have wide applications in fields such as seismology, neuroscience, insurance, finance, and social sciences. In this study, we focus on their classical application in seismology, the Epidemic-Type Aftershock Sequence (ETAS) model, in which earthquake occurrence rates are described as the sum of background seismicity rates and triggering rates. The former represents a spontaneous process, while the latter accounts for triggering effects from historical events. In particular, we focus on nonparametric Bayesian inference of background seismicity rates, which are important parameters for seismic hazard assessment and earthquake risk prediction. Specifically, we model the spatiotemporal background rates using a Gaussian process prior and infer them within a nonparametric Bayesian framework using Markov Chain Monte Carlo (MCMC) sampling techniques. In addition, to improve computational efficiency for large earthquake catalogs, we further introduce several strategies to reduce the computational cost associated with Gaussian process computations. Applications in subduction zone regions demonstrate that the inferred spatiotemporally continuous background seismicity rates provide important insights into the relationship between

seismic activity and tectonic processes. Moreover, such a framework is not only applicable to seismology, but can also be generalized to other Hawkes process applications.

***Bayesian Analysis of Mixed Interval-Censored Data Under the Bernstein Proportional Odds Cure Model***

Presenter: Yuh-Jenn Wu

This study proposes a Bayesian approach based on the proportional odds cure model to analyze mixed interval-censored survival data with a cured subgroup and to estimate the cure rate. Previous studies, such as Li et al. (2019) and Yang et al. (2021), developed generalized non-mixture cure models for mixed interval-censored data using maximum likelihood estimation. Chen (2023) extended this framework by adopting a Bayesian method and modeling the cumulative distribution of latent activation times via Bernstein polynomials instead of step functions. However, his study showed limited accuracy in coverage probabilities and did not establish the consistency of Bayesian estimators. In this work, we focus on a simplified proportional odds cure model and employ the Markov Chain Monte Carlo algorithm to obtain Bayesian estimates. The consistency of the proposed estimators is theoretically proven, and simulation studies demonstrate satisfactory estimation performance and accurate coverage probabilities.

***Loss-Driven Bayesian Active Learning***

Presenter: Zhuoyue Huang

The central goal of active learning is to gather data that maximises downstream predictive performance, but popular approaches have limited flexibility in customising this data acquisition to different downstream problems and losses. We propose a rigorous loss-driven approach to Bayesian active learning that allows data acquisition to directly target the loss associated with a given decision problem. In particular, we show how any loss can be used to derive a unique objective for optimal data acquisition. Critically, we then show that any loss taking the form of a weighted Bregman divergence permits analytic computation of a central component of its corresponding objective, making the approach applicable in practice. In regression and classification experiments with a range of different losses, we find our approach reduces test losses relative to existing techniques.

## Wednesday 1 July

### Multitrack Sessions 6: 9:00–10:30

#### Bayesian Structural Learning (Room 1101)

##### *Bayesian Additive Regression Tree Copula Processes for Scalable Distributional Prediction*

Presenter: Nadja Klein

We show how to construct the implied copula process of response values from a Bayesian additive regression tree (BART) model with prior on the leaf node variances. This copula process, defined on the covariate space, can be paired with any marginal distribution for the dependent variable to construct a flexible distributional BART model. Bayesian inference is performed via Markov chain Monte Carlo on an augmented posterior, where we show that key sampling steps can be realized as those of Chipman et al. (2010), preserving scalability and computational efficiency even though the copula process is high dimensional. The posterior predictive distribution from the copula process model is derived in closed form as the push-forward of the posterior predictive distribution of the underlying BART model with an optimal transport map. Under suitable conditions, we establish posterior consistency for the regression function and posterior means and prove convergence in distribution of the predictive process and conditional expectation. Simulation studies demonstrate improved accuracy of distributional predictions compared to the original BART model and leading benchmarks. Applications to five real datasets with 506 to 515,345 observations and 8 to 90 covariates further highlight the efficacy and scalability of our proposed BART copula process model.

##### *Foundation Models and Bayesian Decision-Making on Structured Spaces*

Presenter: Agustinus Kristiadi

Structured data and search spaces are prevalent in many domains. For instance, in chemistry, one might want to find a molecule—often represented by strings and graphs—that has the most desirable property. Another example is in large language models (LLMs): text generation and reasoning can be seen as exploring huge trees. Bayesian decision-making is an intuitive yet theoretically grounded formalism that enables an AI agent to make decisions efficiently. In this talk, I will discuss how deep learning and foundation models can enable cost-effective decision-making in various structured domains. Furthermore, based on our recent work, I will show that the resulting methods are useful in accelerating scientific discovery and making the notoriously compute-hungry LLMs more efficient.

##### *Feature selection via projected criteria*

Presenter: Maxim Fedotov

Bayesian model selection and L0 criteria possess optimal properties for identifying the subset of features truly associated to an outcome. Although the problem is worst-case NP-hard, recent computational advances showed that it's solvable in computational time with high probability as  $n$  grows, even in  $p \ll n$  settings, in theory and in practice. A key unrealistic assumption for these results is that one must be able to quickly score each subset of features (also referred to as model, in the Bayesian jargon), e.g. obtain its marginal likelihood in constant time. In reality the cost grows with  $p$  and  $n$ , which poses a major bottleneck. We propose a fast way to obtain approximate scores that attains excellent theoretical and computational properties. First, the (frequentist) probability of selecting the data-generating subset of features is at least as good as for standard BMS/L0 criteria, and possibly better. Second, in generalized linear and generalized additive models it defines a binary quadratic optimization problem. This is a canonical problem in the optimization literature which, despite being worst-case NP hard, in practice it can be solved for tens and even hundreds of thousands of features in a matter of seconds.

#### Bayesian Uncertainty Quantification for Complex Computer Simulators: Experimental Design, Emulation, and Calibration (Room 1102)

##### *Generalized Deep Gaussian Process Emulation*

Presenter: Deyu Ming

Gaussian process (GP) emulators have become essential tools for approximating complex simulators, significantly reducing computational demands in optimization, sensitivity analysis, and model calibration. While traditional GP emulators effectively model continuous, Gaussian-distributed simulator outputs with homogeneous variability, they typically struggle with discrete, heteroskedastic Gaussian, or non-Gaussian data, limiting their applicability to the increasingly common class of stochastic simulators. In this talk, we introduce a scalable Generalized Deep Gaussian Process (GDGP) emulation framework designed to accommodate simulators with heteroskedastic

Gaussian outputs and a wide range of non-Gaussian response distributions, including Poisson, negative binomial, and categorical distributions. The GDGP framework leverages the expressiveness of DGPs and extends them to latent GP structures, enabling it to capture the complex, non-stationary behavior inherent in many simulators while also modeling non-Gaussian simulator outputs. We make GDGP scalable by incorporating the Vecchia approximation into Stochastic Imputation (SI) inference for settings with a large number of input locations, while also developing SI methods for handling large numbers of replicates. In particular, we present additional methodological developments that further enhance the computational efficiency of SI for heteroskedastic Gaussian responses. Through a series of synthetic and empirical examples, we demonstrate that these extensions support the practical application of GDGP emulators and provide a unified methodology capable of addressing diverse modeling challenges. The proposed GDGP framework is implemented in our open-source R package **dgpsi**.

#### ***Feature Calibration Methods for Computer Model Tuning***

Presenter: Wenzhe Xu

Computer model calibration involves using partial and imperfect observations of the real world to learn which values of a model's input parameters lead to outputs that are consistent with real-world observations. When calibrating models with high-dimensional output (e.g. a spatial field), it is common to represent the output as a linear combination of a small set of basis vectors. Often, when trying to calibrate to such output, what is important to the credibility of the model is that key emergent physical phenomena are represented, even if not faithfully or in the right place. In these cases, comparison of model output and data in a linear subspace is inappropriate and will usually lead to poor model calibration. To overcome this, we present kernel-based history matching (KHM), generalising the meaning of the technique sufficiently to be able to project model outputs and observations into a higher-dimensional feature space, where patterns can be compared without their location necessarily being fixed. We develop the technical methodology, present an expert-driven kernel selection algorithm, and then apply the techniques to the calibration of boundary layer clouds for the French climate model IPSL-CM.

#### ***Bayesian Experimental Design for the Calibration of Complex Simulators***

Presenter: Victoria Volodina

History matching (HM) is a type of computer model calibration that rules out input parameter settings that are inconsistent between observations and computer model output according to a distance-based implausibility function and uncertainty specifications. The remaining parameter space is termed as Not Ruled Out Yet (NROY). History matching has proven to be more effective when performed in waves. At each wave of HM, a new ensemble is obtained to update an emulator (surrogate model) before finding an NROY space. In this paper, we propose a Bayesian experimental design to optimise the data collection process using a loss function that compares the volume of the NROY space obtained with an updated emulator to the volume of the "true" NROY space obtained using a "perfect" emulator. The intuition behind the proposed loss function corresponds to the aim of history matching. In particular, by performing history matching, we want to ensure that we are not going to incorrectly rule out points that are in fact close to observations, or leave regions of space that give output far from the observations. We compare the proposed methodology to space-filling design approaches, most commonly used in practice, on a number of numerical examples.

### **Advances in Bayesian Nonparametrics I (Room 1103)**

#### ***Nearest neighbor split-merge sampling for non-conjugate Dirichlet process mixtures***

Presenter: Rodney Sparapani

Dirichlet process mixtures (DPM) are one of the most popular Bayesian nonparametric techniques. However, posterior samples can mix poorly if atoms are unevenly distributed within prior space (whether atoms are univariate or multivariate) especially for a non-conjugate prior. Previous work on this problem has proposed solutions based on reversible jump, posterior approximation and so-called "split-merge" (SM for short) methods. But all of these proposed approaches have their own challenges limiting feasibility mainly due to complicated implementations. The latter point has hindered SM so much that no shared software could be found despite a determined search. Therefore, we propose a new SM-like technique that we call "nearest neighbor split-merge" (NNSM). NNSM is far simpler while still computationally efficient. Our new method is based upon Metropolis-Hastings split vs. merge proposals with similarities to SM but important differences for simplicity. We have shown that our approach is valid theoretically and generalizable to a variety of settings including non-smooth posteriors. Further, NNSM has desirable computational properties built-in: simple yet efficient. For the non-conjugate prior setting, we demonstrate NNSM's improved performance beyond that of Neal's algorithm 8 (JCGS 2000) with a

simulation study.

### ***A Bayesian Nonparametric Framework for Dynamic Item - Response Theory***

Presenter: Maria de Iorio

Item-response theory (IRT) is widely used for the statistical analysis of questionnaire data, allowing for the differentiation of respondent profiles and the characterisation of questionnaire items through interpretable parameters. However, conventional IRT models are typically cross-sectional and limited in their ability to capture complex longitudinal and hierarchical data structures. We propose a Bayesian semiparametric extension of IRT that introduces temporal dependence across repeated questionnaire administrations, accommodates repeated measurements, and jointly models responses from related subject groups (e.g., mothers and children) to enable information sharing across hierarchies. The framework further incorporates covariate information, allows for the joint modelling of questionnaire data with other longitudinal markers, and supports clustering of subjects based on their latent response profiles. Our approach is built on Bayesian nonparametric priors, specifically the Dirichlet Process and the Normalized Generalized Gamma Process, facilitating the identification of clinically meaningful subgroups within the population. We demonstrate the utility of the proposed methodology through the analysis of longitudinal psychometric questionnaire data collected from mothers and their children, aiming to investigate how various factors influence growth trajectories, developmental outcomes, and mental health. This application, using data from the Singaporean GUSTO cohort study, highlights the potential of our modelling strategy to provide a nuanced understanding of child development by capturing complex dependencies in questionnaire data as well as the relationships between psychometric measures and other growth markers.

### ***Repulsive mixtures via the sparsity-inducing partition prior***

Presenter: Gregor Kastner

We introduce a novel prior distribution for modelling the weights in mixture models based on a generalisation of the Dirichlet distribution, the Selberg Dirichlet distribution. This distribution contains a repulsive term, which naturally penalises values that lie close to each other on the simplex, thus encouraging few dominating clusters. The repulsive behaviour induces additional sparsity on the number of components. We refer to this construction as sparsity-inducing partition (SIP) prior. By highlighting differences with the conventional Dirichlet distribution, we present relevant properties of the SIP prior and demonstrate their implications across a variety of mixture models, including finite mixtures with a fixed or random number of components, as well as repulsive mixtures. We propose an efficient posterior sampling algorithm and validate our model through an extensive simulation study as well as an application to a biomedical dataset describing children's Body Mass Index and eating behaviour. (This is joint work with Alexander Mozdzen, Timothy Wertz, Maria De Iorio, Andrea Cremaschi, and Johan Eriksson.)

### ***Bayesian inference for partial orders from random linear extensions***

Presenter: Geoff Nicholls

Partial orders are a kind of directed acyclic graph which are used to model rank-order data. They are appealing as they do not impose order relations where none exist, in contrast to classical models for rank data such as the Mallows and Plackett-Luce models. This is important in models used to infer social hierarchies and animal dominance hierarchies. The dimension of a partial order is a well-defined property. A latent space prior for partial orders is given: shrinkage in the latent space is shown to imply shrinkage in the partial order dimension. The latent space prior is extended to parameterise a Hidden Markov Model for an evolving social hierarchy. The evolving partial order can be inferred from rank order data emitted by the process over time. The model framework is illustrated using a data set from the eleventh and twelfth centuries informing power relations between Anglo-Norman bishops and yields insights of interest to historians. A second application to evolving chimpanzee dominance hierarchies will be reported. Striking similarities between human and animal hierarchies are visible. Paper Ann. Appl. Stat. 19(2): 1663-1690 (June 2025). DOI: 10.1214/24-AOAS2002

### ***Graph-event Modeling for Inference on Persistent Homology***

Presenter: Leo Duan

Persistent homology is a cornerstone of topological data analysis, offering a multiscale summary of the topology with robustness to nuisance transformations, such as rotations and small deformations. Persistent homology has seen broad use across domains such as computer vision and neuroscience. Most statistical treatments, however, use homology primarily as a feature extractor and then rely on statistical distance-based tests or simple time-to-event models for inferential tasks. While these approaches can detect global differences, they rarely could localize the source of differences. We address this gap by taking a graph-event modeling approach: we associate each vertex with a population latent position and modeling each bar's key events (birth and death times) by an

exponential distribution, whose rate is a transform of the latent positions according to an event happening on the graph. The low-dimensional bars have simple graph-event representations, such as the formation of a minimum spanning tree and the triangulation of a loop, hence enjoy tractable likelihoods. Taking a Bayesian approach, we infer latent positions and enable model extensions such as mixed-effects structures and filtration selection. Applications to benchmark image datasets and a neuroimaging study of Alzheimer's disease demonstrate that our method localizes sources of difference and provides interpretable, model-based analysis of topological structure in complex data.

## **Bayesian Adversarial Machine Learning in Industry (Room 1104)**

### ***Enhancing Security and Practicality of Large Language Models for Autonomous Vehicles through Bayesian Adversarial Machine Learning***

Presenter: Si Liu

Large Language Models (LLMs) are transforming the fields of machine learning (ML) and artificial intelligence (AI) by enabling advanced decision-making, human-machine interaction, and security applications. However, deploying LLMs in resource-constrained environments such as autonomous vehicles and edge devices presents unique challenges due to their computational demands, security vulnerabilities, and privacy concerns. These challenges are compounded by the growing need for real-time decision-making in safety-critical systems. This presentation will address some of these pressing challenges by advancing the security of LLMs from a Bayesian Adversarial ML perspective, bridging the current focus on vehicles to a future where LLMs underpin a broader ecosystem of intelligent edge devices. This is joint work with Qin Huang (University of Shanghai for Science and Technology, China) .

### ***Bayesian predictive models under adversarial attacks with industrial applications***

Presenter: Pablo G. Arce

Adversarial attacks subtly manipulate data yet can drastically alter predictions from machine learning models. Interest in understanding and countering these vulnerabilities is rapidly growing, but research has primarily focused on classical predictive models. Bayesian predictive models, in contrast, remain largely unexplored regarding their susceptibility to such attacks and potential defenses. This work first introduces a general methodology for crafting optimal evasion attacks specifically against Bayesian predictive models. Additionally, it explores alternative posterior defensive constructions that enhance robustness against these adversarial manipulations. Several industrial cases are used for illustrative purposes. This is joint work with Roi Naveiro (CUNEF Universidad, Spain).

### ***Posterior Attraction with Exponential-Family Likelihoods and their Conjugate Priors with applications in adversarial machine learning***

Presenter: William N. Caballero

Bayesian inference is inherently sensitive to assumptions about the data-generating process, and recent developments in adversarial machine learning reveal an underexplored threat: purposeful data corruption by strategic agents. Therefore, this work examines poisoning attacks on Bayesian models, where an adversary manipulates a fixed-size data set to steer the posterior toward an adversarial (target) distribution. Assuming full knowledge of the Bayesian's prior and likelihood, the attacker modifies the data set within specified perturbation constraints to minimize either the exclusive or inclusive Kullback–Leibler divergence between the induced and adversarial posteriors. Focusing on models with conjugate priors and full exponential-family likelihoods, we derive theoretical properties of these optimization problems and demonstrate how adversarial posteriors can be constructed via entropic tilting. Our analysis reveals that even canonical Bayesian models are vulnerable under oblivious inference, raising important concerns about model robustness. Extensive empirical testing is performed to explore attack performance and elucidate their real-world effects. These findings bridge adversarial machine learning and Bayesian statistics, offering a framework for understanding and mitigating adversarial influence in probabilistic modeling.

Discussant: David Rios Insua

## **Savage Award Session: Applied Methodology (Room Main Hall)**

### ***Bayesian Hierarchical Models for Dependent Ecological Data***

Presenter: Michael R. Schwob

Ecological data are inherently dependent due to spatial and temporal autocorrelation, species interactions, sea-



sonality, and environmental gradients. In this presentation, we will present several statistical methods that account for various forms of dependence that arise in ecological data. In the first project, we propose a dynamic population model for the analysis of abundance data, which often experience a high degree of temporal autocorrelation. The model was fit to mosquito abundance data collected across North America. We accounted for temporal autocorrelation and preferential sampling in a Bayesian hierarchical model, which was specified mechanistically such that inference was made for abundance and phenological quantities of interest. The next project addresses the challenge of inferring spatio-temporal mechanisms that govern gene flow in populations. This is achieved through a Bayesian hierarchical dyadic model that incorporates spatial and temporal dependence using a fully-connected network and normalized powered likelihoods. The model is applied to ancient human DNA data to infer the spatio-temporal mechanism that affected human movement in Bronze Age Europe. The final project focuses on analyzing compositional data without assuming the strict assumptions of existing approaches (i.e., log-ratio transformations or Dirichlet regression). Our method accommodates zero-valued components, positive correlations between components, and is generative for compositional data. Additionally, we propose a spatial hyperspheric regression model that incorporates fixed and random multivariate spatial effects. This methodology is demonstrated on classified bioacoustic signals of the downy woodpecker.

***Bayesian models for the spatiotemporal analysis of extremes: Applications to the analysis of climate change***

Presenter: Jorge Castillo-Mateo

We present a comprehensive Bayesian hierarchical framework for modeling daily temperatures that integrates analyses of the mean, quantiles, and record-breaking events across space and time. We first introduce a spatio-temporal mean model for daily maximum temperatures that incorporates two temporal scales, autoregressive dependence, fixed effects, and multiple random effects to capture spatial variability. The modeling provides model-based tools to estimate probability surfaces and characterize the spatial extent of an event. Building on this framework, we develop a mixed-effects quantile regression model with asymmetric Laplace errors to investigate climate change across the temperature distribution. The approach enables marginal quantile inference from conditional autoregressive structures and reveals substantial spatial heterogeneity in quantile-specific climate signals. We also provide a characterization of the spatial joint quantile autoregressive model that avoids quantile crossing. This formulation allows the model to simultaneously capture temporal persistence and spatial correlation while ensuring that the estimated quantile curves remain monotone across quantile levels. Finally, we introduce a hierarchical modeling framework for record-breaking temperatures. Using a logistic regression formulation with an explicit long-term trend and strong daily spatial random effects, the model analyzes the occurrence of calendar-day records across years and allows inference on the number, spatial distribution, and temporal evolution of record-breaking events under climate change. Most models rely on Gaussian latent representations that enable closed-form Gibbs sampling and spatial prediction at unobserved locations. Applications to long-term temperature data from Spain illustrate trends in daily temperatures, quantile-specific climate change, and the increasing frequency of record-breaking events, providing new statistical tools for the analysis of environmental extremes.

***Mixtures of Directed Graphical Models: A Tractable Alternative to Markov Random Fields for Bayesian Spatial Analysis***

Presenter: Brandon Carter

This dissertation addresses critical challenges in Bayesian spatial modeling for discrete-valued outcomes over areal units, culminating in a novel modeling framework that provides a principled and computationally efficient alternative to traditional Markov random fields (MRFs).

The dissertation develops three contributions. The first is a nonstationary spatial model for collective efficacy that pairs a dimension-expanded latent process with a land-use-based filter, yielding fine-scale ordinal predictions across Columbus, OH using data from the Adolescent Health and Development in Context (AHDC) Study. The second provides a systematic review of Markov random fields (MRFs) for discrete spatial data, comparing alternative codings and parameterizations commonly used in practice. To support this review, we develop prior predictive response functions—conceptually linked to thermodynamic quantities—as a prior analysis tool, and use them to reveal non-standard behaviors of the centered autologistic model under varying spatial dependence. The central contribution, and the focus of this talk, is a mixture of directed graphical models (MDGMs): a new framework for discrete spatial data that sidesteps the computational and theoretical challenges of full posterior inference under MRFs. The mixture components are directed acyclic graphs (DAGs) selected to correspond to an undirected graphical representation of an assumed spatial contiguity structure—the same structure that underlies traditional MRF specifications. Crucially, the MDGM is not proposed as an approximation to an MRF, but as

an alternative that yields valid posterior inference while remaining computationally faster than exact MRF inference and more principled than the pseudo-likelihood approximation commonly used in practice. We introduce a notion of compatibility to show how an undirected graph can serve as a template for constructing collections of DAGs that, taken together, preserve the dependencies represented in the template. We compare highlighted classes of MDGMs to MRFs and to a popular Bayesian MRF approximation through simulation studies and an analysis of econometrics data from the AHDC Study, demonstrating that MDGMs offer a flexible and tractable new framework for Bayesian analysis of discrete-valued, spatially-referenced data.

***Balanced random partition models for entity resolution and flexible community detection in networks with node attributes***

Presenter: Changwoo Lee

In the first part of the talk, based on dissertation chapter 2, we discuss a class of “rich-get-poorer” random partition models that favor balanced cluster sizes. Motivated by entity resolution problem, where records corresponding to the same underlying entity are grouped together, we develop a framework for studying cluster size balance in exchangeable random partition models. We show how this property can be tuned in a principled manner and demonstrate through applications to probabilistic entity resolution tasks.

**de Finetti Lecture: 11:00–12:30**

***A semi-parametric Bayesian GLM and applications to outcome dependent sampling***

Presenter: Peter Mueller

We introduce an instance of a (modified) varying weight dependent Dirichlet process (DDP) model to implement a semi-parametric GLM. The model extends recently developed semi-parametric generalized linear models (SPGLM) by adding a nonparametric Bayesian prior on the centering distribution of the GLM. Building on familiar posterior simulation methods for mixtures with respect to normalized random measures we introduce modification to implement posterior simulation in the resulting semi-parametric GLM model. The motivating application is data analysis for a study with outcome dependent sampling, that is, when participants are sampled into a study based on an outcome variable, as well as some auxiliary covariates. We discuss how the outcome dependent sampling design can be accommodated in the proposed semiparametric GLM model with only minor modification of the inference scheme.

Joint work with Entejar Alam and Paul Rathouz

Discussants: Amy Herring and Steve MacEachern

**Multitrack Sessions 7: 14:00–15:30**

**Recent Advances from Junior Bayesians (Room 1101)**

***Bayesian Model Selection for Conditional Statistical Parity***

Presenter: Deborah Sulem

There is increasing interest in using prediction algorithms to aid decision making, and also in preventing potential discriminatory patterns that could be learnt from historical data. Conditional statistical parity (CDP) is one statistical fairness criterion that can be enforced in the learning process of a prediction algorithm to mitigate both disparate impact on disadvantaged groups and disparate treatment of individuals. CDP is formulated as a conditional independence statement, between the model’s prediction and the sensitive attribute, given a set of legitimate covariates. We propose a new efficient methodology to select the latter set, considering legitimate covariates which are both strong predictors and weakly

*dependent on the sensitive attribute. Specifically, in linear models, we construct a sparse penalized estimate of the parameters, as well as a Bayesian Model Selection method for choosing the set of legitimate covariates. This is joint work with Jack Jewson.*

Presenter: Jack Jewson

The Bayesian approach provides powerful methods for variable selection. The ability to incorporate sparsity through prior beliefs and account for parameter uncertainty allows Bayesian variable selection to consistently identify which of the variables are active and exhibit strong finite-sample performance. However, Bayesian methods require the correct specification of full likelihoods for the data, and there is increasing awareness of the problems that model misspecification causes for variable selection. Current approaches to mitigate misspecification either require complex models, detracting from the interpretability of the variable selection task, or move

outside rigorous Bayesian uncertainty quantification and provide no recognised method for variable selection. This paper establishes the model quasi-posterior as a principled tool for variable selection. We prove that the model quasi-posterior shares desirable properties of Bayesian variable selection without requiring full likelihood specification. Instead, the quasi-posterior combines a prior with a quasi-likelihood and requires only specification of mean and variance functions, and is therefore robust to other aspects of the data. Marginalising the quasi-likelihood is analytically possible for linear regression, and Laplace approximations are used beyond this to ensure computational tractability. Extensive simulation studies illustrate improved variable selection accuracy across diverse data-generating scenarios when compared with likelihood-based Bayesian variable selection and lasso-penalized methods. We further demonstrate practical relevance through applications to real datasets from social science and genomics.

### ***Integrating Real-World Data through Bayesian Nonparametric Common Atoms Synthetic Controls***

Presenter: Noirrit Kieran Chandra

Glioblastoma (GBM) is a highly aggressive tumor for which early-phase studies are often single-arm trials, limiting reliable treatment effect evaluation and causing promising therapies to be discarded. We propose a privacy-preserving Bayesian nonparametric common atoms model for constructing synthetic control arms using external real-world observational data.

Our framework aggregates historical patient data from MD Anderson Cancer Center, Dana-Farber Cancer Institute, Harvard Medical School, and the University of California, San Francisco, creating a substantial historical database that improves statistical precision when evaluating new interventions. To address the lack of randomization in these data, the model clusters patients with similar covariate profiles into homogeneous subpopulations and explicitly adjusts for confounding from non-randomized treatment assignment.

To enable practical use, we are developing an open-access R Shiny application that allows investigators to upload data from new trials and obtain treatment effect estimates. Privacy is ensured through reliance on pre-computed MCMC samples and a common atoms formulation for inference, such that no raw historical patient data are accessed and uploaded trial data are automatically deleted after analysis. Compared with propensity score-based approaches, our method supports secure and privacy-preserving synthetic control inference.

### **Of Trees, Time and Uncertainty: New Frontiers in Bayesian Phylogenetics (Room 1102)**

#### ***Tractable time tree distributions for Bayesian phylogenetic inference***

Presenter: Alexei Drummond

Bayesian phylogenetic analysis with MCMC algorithms generates an estimate of the posterior distribution of phylogenetic trees in the form of a sample of phylogenetic trees and related parameters. The high dimensionality and non-Euclidean nature of tree space complicates summarizing the central tendency and variance of the posterior distribution in tree space. Here we introduce a new tractable tree distribution and associated point estimator that can be constructed from a posterior sample of trees. Through simulation studies we show that this point estimator performs at least as well and often better than standard methods of producing Bayesian posterior summary trees. We also show that the method of summary that performs best depends on the sample size and dimensionality of the problem in non-trivial ways.

#### ***TreePPL: a probabilistic programming language for statistical phylogenetics***

Presenter: Fredrik Ronquist

We present a universal probabilistic programming language (PPL) designed for probabilistic modeling and inference in phylogenetics. The model is expressed as a computer program, which can generate simulations from the model conditioned on some input data. Specialized inference machinery then uses this program to estimate the posterior probability distribution. The aim is to allow the user to focus on describing the model, and provide the inference machinery for free. It currently supports a range of inference strategies—including sequential Monte Carlo, Markov chain Monte Carlo, and combinations thereof—and is based on several recent innovations that are important for efficient PPL inference on phylogenetic models. We illustrate how the PPL approach can be used to address a range of common problem types considered in statistical phylogenetics, from diversification and tree inference to complex trait evolution. We end by discussing how probabilistic programming can facilitate further use of machine learning in addressing important challenges in statistical phylogenetics.

#### ***Hierarchical Bayesian modeling for phylogenetic inference and macroevolution using RevBayes***

Presenter: Tracy Heath

Bayesian methods have become central to phylogenetic inference given advances in algorithms such as Markov chain Monte Carlo (MCMC) and the development of software packages. Existing programs, however, still face

several limitations such as fragmented implementation of models across software tools, difficulty in extending program architectures to accommodate new models, and inflexible interfaces for specifying phylogenetic assumptions. RevBayes is a collaboratively developed Bayesian method for model-based inference of phylogenetic trees and other evolutionary parameters. The key features of RevBayes are its use of a probabilistic graphical modeling framework and an interpreted language (called Rev) for specifying hierarchical models and analyses. Consequently, RevBayes is inherently flexible and modular, providing a framework for Bayesian phylogenetic analysis under complex, biologically realistic evolutionary models. Recent work has greatly extended the range of macroevolutionary models enabling analysis of datasets combining data from living and fossil taxa.

Discussant: Chi Zhang

## **Bayesian Clinical Trial Design (Room 1103)**

### ***A cautious use of auxiliary outcomes for decision-making in randomized clinical trials***

Presenter: Lorenzo Trippa

Clinical trials often collect data on multiple outcomes, such as overall survival (OS), progression-free survival (PFS), and response to treatment (RT). In most cases, however, study designs only use primary outcome data for interim and final decision-making. In several disease settings, clinically relevant outcomes, for example OS, become available years after patient enrollment. Moreover, the effects of experimental treatments on OS might be less pronounced compared to auxiliary outcomes such as RT. We develop a Bayesian decision-theoretic framework that uses both primary and auxiliary outcomes for interim and final decision-making. The framework allows investigators to control standard frequentist operating characteristics, such as the type I error rate, and can be used with auxiliary outcomes from emerging technologies, such as circulating tumor assays. False positive rates and other frequentist operating characteristics are rigorously controlled without any assumption about the concordance between primary and auxiliary outcomes. We discuss algorithms to implement this decision-theoretic approach and show that incorporating auxiliary information into interim and final decision-making can lead to relevant efficiency gains according to established and interpretable metrics.

### ***Bayesian Utility-Based Trial Designs for Treatment Monitoring, Screening, and Selection***

Presenter: Juhee Lee

Bayesian designs are proposed for randomized phase II clinical trials involving two or more clinical outcomes. When multiple outcomes are observed, the risk-benefit trade-off among them should play a central role in principled decision-making. We address the development of decision criteria, including interim rules for dropping unsafe or ineffective treatments and final criteria for selecting optimal treatments at the end of the trial. A joint statistical model for the outcomes is constructed to enable a comprehensive assessment of treatment effects, accounting for patient heterogeneity through prognostic factors. A utility function is elicited to quantify the desirability of outcome combinations, which may also vary with patient characteristics. This utility serves as an explicit criterion for evaluating risk-benefit trade-offs in personalized treatment screening and selection and supports interim monitoring for safety and futility. The methodology is illustrated using a trial of targeted agents for metastatic renal cell carcinoma, which motivated the proposed design. In this context, the designs are evaluated through simulation studies, including comparisons with conventional approaches that consider marginal outcome probabilities separately.

### ***Bayesian Calibration-Free Odds (CFO) Design for Phase I Trials***

Presenter: Guosheng Yin

In clinical trials, the initial step typically involves assessing a new drug's toxicity profile, aiming to identify a tolerable dose level for subsequent studies. In phase I trials, the primary objective is to determine the maximum tolerated dose, defined as the highest dose associated with an acceptable level of toxicity. Numerous methods have been developed to guide dose escalation and de-escalation decisions during trial conduct. Among these approaches, the calibration-free odds (CFO) design has demonstrated superior operating characteristics and has emerged as one of the most effective approaches for dose finding. To facilitate the application of the CFO design in clinical trial practice, an R package and a Shiny app have been released. This study presents CFO decision tables in Excel files to further remove the barrier of applying the CFO design to real trials. Anyone involved in the trial conduct can implement the CFO design with no difficulties. During the trial, dose movement decisions can be made simply by referring to the cumulative data (including numbers of patients treated and observed toxicities) and the pre-generated decision tables, without any additional statistical calculation. This approach significantly enhances the usability of the CFO design and reduces the operational complexity associated with its implementation in clinical trials. The Excel CFO decision tables can be downloaded from CFO Shiny App.

Discussant: Kentaro Sakamaki

## **Bayesian Forecasting, Decisions, and Risk Analysis in Macroeconomics and Finance: Multi-Model Management and Synthesis (Room 1104)**

### ***Bayesian predictive decision synthesis with applications to monetary policy***

Presenter: Tony Chernis

Among the major challenges in forecasting for decision-making are those of coherent Bayesian management and analysis with multiple candidate models defining model-specific predictions and decision recommendations. The issues of how to calibrate, collate and combine for formal Bayesian inference and resulting decisions are broadly important, and in the context of central banking policy advising are fundamental. The macroeconomy is a sophisticated dynamic system involving significant uncertainties that complicate modelling. In response, decision-makers consider multiple models that provide different predictions and policy recommendations which are then synthesized into a policy decision. A main response to the questions and challenges arising has been development of Bayesian predictive decision synthesis (BPDS) to formalize monetary policy decision processes in contexts of model uncertainty. BPDS draws on recent developments in model combination and statistical decision theory that yield new opportunities in combining multiple models, emphasizing the integration of decision goals, expectations and outcomes into the model synthesis process. The talk addresses foundational and modelling questions, and aspects of a core case study in central bank policy decision-making focused on target interest rate setting and implications for multi-step macroeconomic forecasting. This application also motivates new methodological developments in conditional forecasting and BPDS, and raises new questions and challenges for Bayesian modellers, forecasters and decision analysts.

### ***Scenario synthesis and macroeconomic risk***

Presenter: Matteo Luciani

Recent advances in Bayesian analysis address fundamental challenges of reconciling judgmental narrative approaches with statistical forecasting in macroeconomics for policy decisions. The questions of understanding, contrasting, calibrating and synthesizing multiple model-based forecasts and multiple partially specified, subjective opinions underlying posited economic future realities— a.k.a. scenarios— is at the heart of modern policy decision-making across central banks worldwide. This talk explores new methodology to bridge scenario analysis and model-based risk forecasting, leveraging their respective strengths in policy settings. This approach evaluates explicit measures of concordance of scenarios with a reference forecasting model, delivers Bayesian predictive synthesis of the scenarios to best match that reference, and addresses scenario set incompleteness (that is, the core challenge of dealing with “all models are wrong”). Analysis then provides for systematic evaluation and integration of risks from different scenarios, and quantifies relative support for scenarios modulo the defined reference forecasts. The framework offers advances in forecasting for policy institutions that supports clear and rigorous communication of evolving risks. We also discuss broader questions of integrating judgmental information with statistical model-based forecasts in the face of unexpected circumstances, linking to related frontiers of Bayesian decision analysis in macroeconomics and other areas.

### ***Score-Driven Calibration of Predictive Densities***

Presenter: Francesco Ravazzolo

We present and discuss a calibration model that combines and dynamically calibrates predictive densities. While the weights are statically estimated, time-varying calibration is introduced giving observation-driven dynamics to the parameters of the calibrating function which is modelled by the score of the assumed conditional likelihood of the data generating process. The model is very flexible and can handle different shapes, instability and model uncertainty in the data generating process density. We show its effectiveness on various simulated datasets. As examples of motivating macroeconomic and financial applications, two empirical applications are introduced: one on financial index density forecasting, and one on short-term wind speed predictions for economic forecasting and resulting decision making in electricity markets. The simulations and empirical applications document the large instability in the performance of individual models in comparison to that of the combined and calibrated forecasts, favouring the approach in terms of predictive accuracy.

## **Tackling Challenges in Spatial Modeling via Bayesian Approaches (Room Main Hall)**

### ***Markov switching zero-inflated space-time multinomial models for comparing multiple infectious diseases***

Presenter: Alexandra Schmidt

In the second part of the talk, based on dissertation chapter 5, we introduce a graph product partition model for probabilistic clustering of nodes in undirected networks with node attributes. Motivated by applications where the block-connectivity assumption is restrictive, we develop a flexible alternative to supervised Bayesian stochastic block models. The key idea is to formulate a probabilistic model for graphs and partitions in the reverse direction by modeling the partition conditional on the graph. We discuss the associated computational strategies and demonstrate the proposed method through an analysis of gene recombination network.

***Computationally efficient Bayesian inference of latent non-Gaussian models***

Presenter: David Bolin

Latent Gaussian models (LGMs) are perhaps the most commonly used class of models in statistical applications. Nevertheless, in areas ranging from longitudinal studies in biostatistics to geostatistics, it is easy to find datasets that contain inherently non-Gaussian features, such as sudden jumps or spikes, that adversely affect the inferences and predictions made using an LGM. These datasets require more general latent non-Gaussian models (LnGMs) that can handle these non-Gaussian features. However, fast implementation and easy-to-use software are lacking, preventing the broad applicability of LnGMs. In this talk, we present a general class of LnGMs that facilitates non-Gaussian extensions of a wide range of popular models, from simple temporal models to complex spatial-temporal and multivariate models. We introduce approaches for fast and scalable Bayesian inference for the model class and illustrate this in a few applications including novel non-Gaussian spatial and spatio-temporal models.

***Bayesian Transfer Learning for Artificially Intelligent Geospatial Systems: A Predictive Stacking Approach***

Presenter: Luca Presicce

Building artificially intelligent geospatial systems require rapid delivery of spatial data analysis at massive scales with minimal human intervention. Depending upon their intended use, data analysis may also entail model assessment and uncertainty quantification. This article devises transfer learning frameworks for deployment in artificially intelligent systems, where a massive data set is split into smaller data sets that stream into the analytical framework to propagate learning and assimilate inference for the entire data set. Specifically, we introduce Bayesian predictive stacking for multivariate spatial data and demonstrate its effectiveness in rapidly analyzing massive data sets. Furthermore, we make inference feasible in a reasonable amount of time, and without excessively demanding hardware settings. We illustrate the effectiveness of this approach in extensive simulation experiments and subsequently analyze massive data sets in climate science on sea surface temperatures and on vegetation index.

Discussant: Kerrie Mengersen

## **Keynote Lecture 3: 16:00–17:00**

***Adaptive Bayesian Intelligence: the Road to Sustainable AI***

Presenter: Emtiyaz Khan

AI training is so costly that most people cannot afford it, and it is clear that the inflated demand for data and compute is difficult to sustain. In this talk, I will discuss a sustainable alternative: adaptive intelligence systems that can learn quickly and continually through everyday interactions, just like humans and animals. I will present a new variational-Bayesian method called “posterior correction” that not only quantifies the interference between the old and new information, but can also reduce it by simply matching the old and new predictions, gradients, and Hessians. I will show several examples, including continual learning, federated learning, unlearning, and model merging, and some early results boosting LLM training. Overall, I will argue that quick adaptation has the potential to drastically reduce the cost of existing AI systems, and conclude with a proposal to accomplish this in the near future.

## **Poster Sessions 3: 17:15–18:30**

***Bayesian Hierarchical Modeling of RNFL Thinning Using Circular Spatial Smoothers***

Presenter: Abraham Liu

Background: Current research is investigating whether glaucoma progression can be characterized by the progressive thinning of the Retinal Nerve Fiber Layer (RNFL). In clinical practice, RNFL thickness is typically measured across discrete sectors of the optic nerve head. However, standard modeling approaches often treat these sectors as independent or fail to account for the continuous, circular topography of the eye, leading to



”boundary effects” and potential loss of predictive accuracy in longitudinal analysis.

**Proposed Method:** We present a Bayesian hierarchical framework that replaces discrete sector-based random effects with continuous circular spatial smoothers. By utilizing Fourier basis functions (sine and cosine transforms of measurement radians), we model the intercept and slope of RNFL thinning as a periodic functional surface. This approach allows for information pooling across the entire 360-degree range while naturally enforcing continuity across time.

**Implementation:** The model is implemented in a hierarchical structure where individual-specific deviations in both baseline thickness and rate of progression are captured through spatial random effects in a population of advanced glaucoma patients. We utilize MCMC methods (via JAGS/R) to estimate posterior distributions for the functional coefficients, allowing for personalized estimates of thinning rates that are robust to measurement noise and spatial misalignment.

**Preliminary Results:** Initial findings suggest that the circular smoother model provides a more parsimonious fit compared to high-dimensional discrete sector models, reducing the effective number of parameters while maintaining sensitivity to localized thinning. We evaluate model performance using the Watanabe-Akaike Information Criterion (WAIC) and out-of-sample prediction accuracy.

**Conclusion:** Our framework provides a flexible and computationally efficient alternative to traditional spatial models, offering a biologically plausible representation of RNFL that can be easily integrated into clinical decision-support tools for glaucoma monitoring.

### ***Value-Censored Gaussian Process Regression***

Presenter: Adam Gorm Hoffmann

Gaussian process (GP) regression is widely used for flexible and non-parametric Bayesian modeling of data arising from underlying smooth functions. This talk introduces a solution to GP regression when the observations are subject to value-based censoring. We derive exact and closed-form expressions for the conditional posterior distributions of the underlying functions in both the single-curve fitting case and in the case of a hierarchical model where multiple functions are modeled simultaneously. Our method can accommodate left, right, and interval censoring, and is directly applicable as an empirical Bayes method or integrated in a Markov-Chain Monte Carlo sampler for full posterior inference. The method is validated through extensive simulations, where it substantially outperforms naive approaches that either exclude censored observations or treat them as fully observed values. We give an application to a real-world dataset of longitudinal HIV-1 RNA measurements, where the observations are subject to left censoring due to a detection limit.

### ***Improving Random Forest Estimation for Voting Rights Act Language Minority Data Using the Bayesian Bootstrap***

Presenter: Adam Hall

The Voting Rights Act (VRA) requires the U.S. Census Bureau to identify counties with language-minority populations that need language assistance in elections. The Bureau aims to accomplish this task by using data from the American Community Survey to estimate the proportion of people in each county and language minority group who have limited English proficiency and/or are illiterate with a random forest model. These data are often extremely imbalanced and right skewed, because a small number of counties are much more populous than the others. However, the standard bootstrap method often undersamples these populous counties, introducing bias into traditional random forest models. To address this issue, we propose a Bayesian Bootstrap based random forest model that replaces ordinary resampling with Dirichlet-distributed sampling weights, allowing each observation to receive a continuous weight. This continuous weight does not exclude extreme observations and hence yields improved predictions where the distribution is most sparse. Simulation studies designed to mimic the structure and skewness of real VRA data show that the Bayesian Bootstrap based random forest model outperforms the traditional random forest, demonstrating that Bayesian resampling provides an effective enhancement for analyzing highly skewed federal statistical data

### ***Cluster Identification Through Minimum Distance Estimation***

Presenter: Alexander Mozdzen

We recast the problem of finding an optimal clustering into minimizing a suitable distance between the empirical distribution and the density of a mixture. This reformulation leverages the well-established consistency properties of minimum distance estimation even under model misspecification, while directly targeting the clustering structure induced by the mixture. We exploit novel results on sliced measures, specifically the sliced Wasserstein distance, to take advantage of its favourable properties in higher dimensions. Placing the approach within the framework of greedy search algorithms allows the method to be applied seamlessly in both the Bayesian and

frequentist paradigms. The efficiency of the proposed method is evaluated in an extensive simulation study that compares several discrepancy measures (Kolmogorov–Smirnov, Pearson’s chi-squared, and sliced Wasserstein) against established Bayesian and frequentist clustering procedures as well as popular losses in the Bayesian model-based clustering literature, such as the Binder loss and variation of information.

***Compartmentalized proportional hazards regression for latent competing risks with applications to cancer survival***

Presenter: Alexander Wolfgang Jung

Cancer is a heterogeneous disease. Each tumour can be compartmentalized, through sequencing and phylogenetics, into distinct subclones with unique genotypes and drug responses. This heterogeneity constitutes one of the major obstacles in the treatment of cancer but is often neglected or inappropriately modeled in cancer survival studies. Here, we address this problem by equipping each subclone with its own latent survival time,  $T_i$ , and linking it to the patient’s overall survival as  $T_i = \min(T_{i1}, \dots, T_{iC})$ , thereby allowing us to model the clonal substructure of a patient’s cancer along with its subclone-specific drug interactions and overall patient characteristics. Effectively, this can be understood as a competing risk analysis without information on the cause. We show through simulation studies that a Cox model neglecting this compartmental substructure can lead to biased estimates, and we establish the finite sample properties of our approach in terms of unbiasedness and nominal coverage. We also demonstrate improved performance over a Cox model and a machine learning approach in three real-world datasets, including 99 non-small cell lung cancer patients with multi-region whole-exome sequencing, 123 AML patients with targeted single-cell sequencing, and 983 breast cancer patients with whole-exome sequencing. In summary, we developed a new approach to model cancer survival by reflecting the clonal substructure of a patient’s tumour. The proposed model can be generalized to applications where the time-to-event is driven by compartments with latent event times.

***Toward Reliable Graph Analysis: Uncertainty Quantification for fMRI Connectivity***

Presenter: Alice Chevaux

Inferring brain graphs from fMRI data relies on correlation matrices, yet standard estimators are unstable in high-dimensional, low-sample-size settings. We propose a general Bayesian framework that avoids structural assumptions and quantifies uncertainty through credible regions for these matrices. Our method constructs these credible regions to account for the dependencies between all coefficients while maintaining reasonable computational cost. This approach enables applications not feasible with point estimates: (i) diagnosing estimator instability, (ii) robust edge detection with posterior control of the Family-Wise Error Rate (FWER), and (iii) direct comparison of two fMRI scans via the posterior probability of matrix equality. This simple, assumption-light framework improves the reliability and interpretability of downstream connectivity analyses.

***Multi-view dynamic network modeling with node embeddings***

Presenter: Amanda Man Ying Chu

We develop a flexible multi-view dynamic network model that incorporates information from previous time points to model connection probabilities. Inspired by word embeddings used in natural language processing and large language models, which measure the similarity of sentences based on word embeddings, we introduce node embeddings, defined as vectors representing information from the networks at previous time points. Just as large language models utilize word embeddings to assess the similarity between words, we employ node embeddings to evaluate the similarity between nodes. Unlike latent vector embeddings commonly used in network modeling, node embeddings are directly observable, eliminating the need for estimation. Furthermore, it integrates a multi-view feature to represent various relationship types at each time point. The proposed model assumes that the connection probabilities between two nodes are influenced by the distances between node embeddings in different network views at previous time points. Network statistics from earlier time points are also included as covariates. In an empirical study involving intraday financial networks, the proposed model yields high in-sample and out-of-sample AUCs for network prediction. The model offers an intuitive interpretation of the estimates and facilitates a straightforward estimation methodology, thereby making it more scalable for dynamic modeling applications.

***Bayesian Meta-Prediction: An Objective Approach to Combining Heterogeneous Models***

Presenter: Anabel Forte-Deltell

We present a Bayesian meta-prediction framework designed to objectively combine the outputs of heterogeneous predictive models — from mechanistic systems of differential equations to black-box deep learning algorithms. By treating individual model predictions as covariates within generalized linear models, and assigning weights through Bayesian Model Averaging (BMA), this approach provides an interpretable and computationally efficient solution to model combination. Crucially, it avoids the need for prior access to internal model structures, relying

solely on predictive performance. The method ensures rigorous uncertainty quantification and is particularly suited to scenarios where model diversity is high. Applications include real-world data in epidemiology, climate science, agriculture, and finance. A freely available R package and Shiny app are under development to facilitate widespread use.

### ***Probabilistic Local-Geometric Alignment of Data Representations***

Presenter: Andrea Mascaretti

We introduce a probabilistic method to optimise the similarity between two representations of the same data, such as an image and its description. Our approach learns a sparse, low-rank Mahalanobis metric that aligns local geometric structure across the two data spaces using triplet distance comparison constraints. The method relies on two key assumptions: (i) meaningful representations preserve neighbourhood relationships, and (ii) in multimodal representations only a small subset of features contributes to this structural alignment. To capture uncertainty, identify the active feature set, and regularize estimation, we adopt a shrinkage prior within a Bayesian metric-learning framework. This yields a flexible posterior over metrics that reveals both the strength and the sources of agreement between the two representations. We validate the method by conducting experiments on both simulated and real data.

### ***Learning Discrete Causal Representations from Heterogeneous Domains: A Bayesian Approach***

Presenter: Ankur Garg

Causal representation learning (CRL) aims to infer abstract latent causal concepts that give rise to potentially high-dimensional observations. This is particularly relevant when heterogeneous data from multiple related domains, or distributions, is available since domain shifts are often thought to arise from sparse and localized changes to some of the underlying causal mechanisms, whereas most other parts of the causal generative process remain unchanged.

In this project, we explore a Bayesian approach to learning discrete causal concepts from such multi-domain data. We propose a model that encodes common CRL structural assumptions, together with a tailored inference scheme; we demonstrate its effectiveness on political survey datasets, where we seek to recover the causal ordering between partisanship and issue opinions from both real survey data and LLM-generated synthetic data.

### ***BAYESIAN QUANTILE-BASED CORRECTION AND SYNTHESIS OF RIVER FLOW FORECASTS***

Presenter: Antonio Aguirre

Accurate probabilistic forecasts of river flow are crucial for water resource management and flood preparedness, yet operational ensemble systems often underestimate uncertainty, especially during extremes. In this work I develop a Bayesian quantile-based framework that combines observed river flow, retrospective analysis products, and ensemble forecasts within an extended Dynamic Quantile Linear Model (exDQLM). The data likelihood is given by an extended Asymmetric Laplace (exAL) distribution, allowing flexible, quantile-specific skewness and scale across multiple sources (USGS observations, ECMWF/GloFAS, and NWS products). Posterior inference is carried out using a scalable Mean Field Variational Bayes scheme with Laplace-Delta approximations for non-conjugate parameters, avoiding the particle-degeneracy issues of importance-sampling-based VB. Discrepancy states between observations and agency products are modeled dynamically and then propagated into the forecast period to correct ensemble predictions. I separately fit a collection of quantile-specific models and then synthesize them into a single posterior predictive distribution using a monotone quantile-synthesis procedure with isotonic and rearrangement corrections. I illustrate the methodology using daily streamflow for the San Lorenzo River (Santa Cruz, CA), incorporating climate covariates, retrospective analyses, and medium-range ensemble forecasts from GloFAS and NWS. The resulting forecasts show improved calibration and more realistic tail behavior during both drought and flood periods, while remaining computationally tractable and interpretable for operational use.

### ***Laplace asymptotics for high-dimensional posteriors up to the concentration threshold***

Presenter: Anya Katsevich

Concentrated, high-dimensional posterior distributions can be expressed in the form  $\pi_n(x) \propto e^{-nf(x)}$ ,  $x \in \mathbb{R}^d$ , where sample size  $n$  and dimension  $d$  are both large, with  $d = o(n)$ . Standard Gaussian Laplace approximations are rigorously justified only in the regime  $d = o(\sqrt{n})$ , a condition that is too restrictive for many high-dimensional Bayesian problems. I will present a higher-order Laplace theory that remains valid arbitrarily close to the concentration threshold  $d = o(n)$ . Under local smoothness and mild global growth assumptions, we derive explicit asymptotic expansions for normalizing constants and posterior expectations, with error of order  $d^{L+1}/n^L$  at expansion order  $L$ . These results provide a rigorous justification for higher-order corrections beyond the classical

Gaussian approximation and substantially enlarge the range of dimensions for which Laplace-type approximations remain accurate. I will also discuss sampling from concentrated posteriors  $\pi_n$ . We construct explicit polynomial transport maps that push forward a Gaussian reference measure to approximations of the posterior, with total variation error again of order  $d^{L+1}/n^L$ . Thus, by increasing the expansion order, both approximation and sampling remain accurate in regimes arbitrarily close to  $d = o(n)$ . Overall, this work gives a rigorous and complete framework for the approximation and sampling of concentrated posterior distributions in growing dimension to arbitrary accuracy.

### ***Bernstein polynomial models for spatial point processes***

Presenter: Athanasios Kottas

We will present a Bayesian nonparametric modeling approach for spatial point patterns, including the setting where the point process is defined/recorded over an irregular domain. The key building block is a prior model for spatial intensities, based on structured weighted combinations of beta densities, designed to balance inferential flexibility with computational efficiency. The model can be used directly for the Poisson process intensity or for the background intensity in spatial Hawkes processes and space-time Hawkes processes. Combined with the Hawkes process branching structure, the modeling approach allows for full inference for point process functionals, avoiding the need for approximations to the point process likelihood or to the posterior distribution. The methods will be illustrated with crime data from the city of Boston.

### ***Dynamic Policy with Network Interference***

Presenter: Bengusu Nar

We develop a Bayesian framework for dynamic binary treatment allocation in networks with interference. The method leverages kinetic Ising models to represent binary outcomes as evolving latent propensities, capturing both temporal dependence and local spillovers. This enriched latent state enables reinforcement learning agents to discover non-myopic and adaptive treatment policies. In tractable settings, these policies perform comparably to empirical welfare maximization rules. While under persistent carryover effects or complex network structures, they outperform static or myopic heuristics. By combining Bayesian inference with reinforcement learning, our framework provides a principled and practical approach to policy design under dynamic network interference.

### ***Beyond KL: Evaluating $\alpha$ -, Rényi-, and $\chi$ -divergences for variational Bayes in clinical phenotyping models***

Presenter: Brian Buckley

Standard variational Bayes (VB) minimizes the exclusive Kullback–Leibler (KL) divergence,  $\text{KL}(q \rightarrow p)$ ; however, the practical consequences of adopting broader divergence families—such as more general  $\alpha$ -divergence, Rényi divergence, or  $\chi$ -divergence—are not yet well understood, especially in clinical applications. We examine how alternative divergence functions shape posterior estimation, variable and model selection in VB for heterogeneous clinical data with mixed continuous, categorical, and binary selection variables. Our methodology is applied to Bayesian latent class approaches to patient phenotyping using variational Gaussian Mixture Models and variational Logistic Regression models. We extend the VB framework to accommodate these divergences and derive corresponding criteria for variable importance, model adequacy and posterior estimation. Through extensive simulations reflecting realistic clinical data-generating mechanisms — using mixed-type predictors, varying missingness levels, and both missing-at-random (MAR) and missing-not-at-random (MNAR) processes — we characterise how divergence choice influences identifiability, posterior accuracy, and sensitivity to model misspecification. We evaluate these methods on synthetic real-world clinical datasets, comparing predictive performance, stability of variable selection, and robustness of inferred latent structure, across divergence objectives. Our analysis identifies conditions under which alternative divergences yield more reliable or better-calibrated posterior approximations than standard KL-VB, and delineates regimes where VB remains consistent or fails under misspecification. This work provides a principled framework for selecting divergence functions in VB applied to clinical settings, improving interpretability and reliability of model and variable selection in high-dimensional, mixed-type clinical data.

### ***Fast Bayesian Basis Selection for Functional Data Representation with Correlated Errors***

Presenter: Camila de Souza

Functional data analysis finds widespread application across various fields. While functional data are intrinsically infinite-dimensional, in practice, they are observed only at a finite set of points, typically over a dense grid. As a result, smoothing techniques are often used to approximate the observed data as functions. In this work, we propose a novel Bayesian approach for selecting basis functions for smoothing one or multiple curves simultaneously. Our method differentiates from other Bayesian approaches in two key ways: (i) by accounting for correlated errors

and (ii) by developing a variational Expectation-Maximization (VEM) algorithm, which is faster than Markov chain Monte Carlo (MCMC) methods such as Gibbs sampling. Simulation studies demonstrate that our method effectively identifies the true underlying structure of the data across various scenarios, and it is applicable to different types of functional data. Our VEM algorithm not only recovers the basis coefficients and the correct set of basis functions but also estimates the existing within-curve correlation. When applied to the motorcycle, LIDAR (LIght Detection And Ranging) experiment and Canadian weather datasets, our method demonstrates comparable, and in some cases superior, performance in terms of adjusted R-squared compared to regression splines, smoothing splines, least absolute shrinkage and selection operator (LASSO) and Bayesian LASSO.

#### ***Predictive posteriors under hidden confounding***

Presenter: Carlos García Meixide

Predicting outcomes in external domains is challenging due to hidden confounders that potentially influence both predictors and outcomes. Well-established methods frequently rely on stringent assumptions, explicit knowledge about the distribution shift across domains, or bias-inducing regularization schemes to enhance generalization. While recent developments in point prediction under hidden confounding attempt to mitigate these shortcomings, they generally do not provide principled uncertainty quantification. We introduce a Bayesian framework that yields well-calibrated predictive distributions across external domains, supports valid causal inference, and achieves posterior contraction rates that improve as the number of observed datasets increases. Simulations and a medical application highlight the remarkable empirical coverage of our approach, nearly unchanged when transitioning from low- to moderate-dimensional settings.

#### ***Bayesian Methods for comparative judgement modelling with applications in violence against women and girls***

Presenter: Charlotte Norridge

Background: The Bayesian Bradley-Terry model is widely used to estimate the relative strengths of objects in pairwise comparisons. However, standard implementations assume that all judges have equal expertise, which can introduce bias, especially in subjective assessments. This limitation is particularly relevant in cases where direct measurement is challenging, such as estimating the prevalence of violence against women and girls across local authority wards. Failure to account for these variations may result in unreliable conclusions and hinder effective safeguarding. We demonstrate the effectiveness of our model on a simulation study ahead of real data collection on the violence against women and girls.

Methods: In this work, we developed an extension to the Bayesian Bradley-Terry model, where judges are clustered, addressing known variation in judges' experience. We accomplished this using a hierarchical Bayesian framework and clustering models including the Chinese restaurant process. This allows the model to find the number of judge clusters through unsupervised learning, while simultaneously estimating global object risk scores and cluster perceived risk scores. We performed inference via Markov chain Monte Carlo (MCMC) and validated the model using simulation studies. In our framework, cluster perceived risk scores are normally distributed based on the global object risk scores and the cluster variance. Comparisons were made between the hierarchical model and the standard Bayesian Bradley-Terry model to evaluate improvements in estimation accuracy.

Results: Our results demonstrate that cluster specific effects can be incorporated into a Bayesian Bradley-Terry model using a hierarchical modelling framework. Furthermore, we can cluster judges accurately and reliably based on subjective assessments resulting from their differing expertise.

Conclusion: Despite the complex nature of our framework, this model provides a more nuanced approach to the Bradley-Terry model. Our work has a potential to make impactful change by help policy makers target resources to tackle violence against women and girls. Future research involves the addition of temporal measures to the model.

#### ***Forecasting healthcare utilization by integrating wastewater and hospitalization surveillance data***

Presenter: Christian O. Bernal Zelaya

Forecasting healthcare utilization is a critical task for public health crises caused by infectious diseases. Various statistical methods have been used for this task: time series models, machine learning methods, and Bayesian semi-mechanistic models. These methods usually use clinical testing data, which are prone to sampling biases and unaccounted changes in the reporting rates. Wastewater pathogen surveillance offers a more robust assessment of pathogen transmission dynamics, but has not been tested in the context of Bayesian semi-mechanistic forecasting. In this work, we develop a Bayesian semi-mechanistic model that integrates wastewater surveillance and hospitalization data to produce probabilistic forecasts of hospital demand. As a bonus, the model also infers the effective reproduction number ( $R_t$ ). The methods we develop employ both compartmental-based processes



to describe time evolution of latent infections, and other disease states and Bayesian nonparametrics to infer time-varying quantities: the effective reproduction number and hospitalization probability. We demonstrate that our model can jointly nowcast  $R_t$  and forecast hospitalization. We validate the  $R_t$  inference using a simulation study. We apply our model to both incidence and prevalence hospitalization data for the state of California and select California counties and compare performance of our new forecasting method against multiple competing methods. We show that our method based on integration of wastewater and hospitalization data consistently outperforms these competing approaches.

### ***Metropolis Adjusted Diffusion Models***

Presenter: Christopher Williams

Diffusion models are generative models that transform a complicated data distribution into a Gaussian reference distribution through a forward stochastic differential equation (SDE), and learn the score function to parameterise the corresponding reverse-time dynamics. Sampling is performed by drawing from the Gaussian reference and transporting these samples back to the data distribution using the learned reverse process. A common formulation decomposes each reverse-time step into a deterministic ODE predictor and a stochastic corrector step based on unadjusted Langevin dynamics, yielding the well-known predictor–corrector framework. However, the unadjusted Langevin algorithm (ULA) is not an exact Markov chain Monte Carlo (MCMC) method under discretisation and therefore introduces bias. Moreover, in diffusion models only the score function is available, rendering classical bias-corrected schemes such as Metropolis Adjusted Langevin Algorithm (MALA) infeasible. In this talk we show how to perform MALA in the context of diffusion models where only a learned score is available. This creates a MCMC scheme that connects standardised methodology from the sampling literature to the context of generative models.

### ***Bayesian Non-Inferiority Trials for dose finding***

Presenter: Connor Fitchett

Project Optimus is an FDA initiative which encourages early phase dose optimisation: going beyond traditional methods which tend to focus on a single maximum tolerated dose and proceed to later phases of clinical development. However, dose finding can also be done in later phases, notably in non-inferiority trials if we want to change or reduce doses. For example, acute lymphoblastic leukaemia (ALL) is currently treated using a long period of chemotherapy. Recent studies investigating Blinatumomab, a novel drug being used to fight ALL, have suggested that it may be possible to ‘deintensify’ the chemotherapy treatment following the introduction of Blinatumomab. Using Bayesian design and adaptations, we investigate through simulations a late phase trial framework that is able to consider alternative doses efficiently whilst protecting patient benefit metrics, with data being based on the ALL deintensification study. We then compare this design to common early phase dose finding methods to evaluate its strengths and weaknesses.

### ***Bayesian Online Test Time Adaptation: A General Framework***

Presenter: Daniel Corrales

Standard supervised machine learning algorithms typically assume identically independent distributed samples in both training and deployment. Further, these algorithms do not usually leverage learning from unlabelled data, effectively wasting task-relevant information. In the standard test-time adaptation literature, machine learning models are first trained using a labelled source dataset and then are adapted in deployment using only the pre-trained model and the incoming unlabelled data. This work addresses the issue of adaptation in scenarios of test-time distributional shifts by designing a fully probabilistic framework that generalises most common approaches in the literature, and, importantly, provides uncertainty estimates on predictions. We exemplify the applicability of this methodology with a set of use cases, growing in complexity in modelling and data dynamics, thus proving the usefulness and scalability of the framework.

### ***Variational predictive resampling***

Presenter: David Frazier

Bayesian inference provides principled uncertainty quantification, but accurate posterior sampling with MCMC can be computationally prohibitive for modern applications. Variational inference (VI) offers a scalable alternative and often yields accurate predictive distributions, but cheap variational families such as mean-field (MF) can produce over-concentrated approximations that miss posterior dependence. We propose variational predictive resampling (VPR), a scalable posterior sampling method that exploits VI’s predictive strength within a predictive-resampling framework to better approximate the Bayesian posterior. Given a prior–likelihood pair, VPR repeatedly imputes future observations from the current variational predictive, updates the variational approximation after each imputation, and records the parameter value implied by the completed sample. We



establish conditions under which the law of the parameter returned by VPR is well defined and show that its finite-horizon approximation converges to this limit. In a tractable Gaussian location model, we show that VPR with MF variational predictives converges to the exact Bayesian posterior, whereas the optimal MF-VI approximation retains a non-vanishing asymptotic gap. Experiments on linear regression, logistic regression, and hierarchical linear mixed-effects models demonstrate that VPR substantially improves posterior uncertainty quantification and recovers posterior dependence missed by MF-VI, while remaining computationally competitive with, and often more efficient than, MCMC.

#### ***A Bayesian Spatiotemporal Varying-Coefficient Model for Multi-Patient Intracranial Recordings***

Presenter: Dominik Wielath

Recording human brain activity by placing electrodes directly on the brain surface enables capturing fast-paced neural computations with a higher signal-to-noise ratio and greater spatial specificity than commonly used extracranial methods. However, electrode placement guided by surgical needs results in limited and uneven spatial coverage, creating the need for integrating data across patients. We develop a hierarchical spatiotemporal Gaussian process varying-coefficient model to estimate the effects of task-specific covariates on brain activity, accounting for spatial dependence, misaligned electrode locations across patients, and between-patient heterogeneity. We apply the model to high-frequency brain activity recordings at 112 electrodes across 9 neurosurgical patients. Each patient performed up to 200 trials of a simple gambling task, with recordings provided as a 57-point time series over 2,850 ms for each trial. The model estimates spatiotemporal effect surfaces jointly across electrodes and time. Exploiting repeated measurements together with separable space-time kernels yields Woodbury- and Kronecker-based computations that make posterior inference computationally tractable. We use the model to study whether task-related effect surfaces exhibit transferable structure across patients via leave-one-patient-out prediction, comparing a patient-specific intercept-only baseline with a model that incorporates effects learned from the remaining patients. The framework provides a principled approach to inference for multi-patient intracranial data with irregular spatial coverage across patients.

#### ***Hierarchical and Clustering Partial-Order Models for Grouped Ranking Data***

Presenter: Dongqing Li

Ranking data often arise in settings where assessors evaluate overlapping subsets of items, producing heterogeneous, sparse, and noisy preference lists. Classical Bayesian ranking models typically center rankings on a single consensus permutation or mixtures of permutations, which fail to represent genuine incomparabilities and transitive structure. I present a Hierarchical Partial Order (HPO) model that instead places a Bayesian prior on partial orders through latent utility vectors whose coordinatewise dominance induces a poset. A global partial order captures shared structure, while assessor-specific partial orders shrink toward the global hierarchy, allowing principled borrowing of strength across heterogeneous assessors. A Dirichlet/Pitman–Yor process mixture further captures assessor clusters without fixing the number of groups. Posterior computation is carried out via MCMC with reversible-jump moves for the latent poset dimension and approximate linear-extension counting for scalability. Applications to synthetic data, 3D auditory perception experiments, and Ghanaian sweet-potato preference data show improved interpretability, clustering coherence, and predictive accuracy compared to total-order Bayesian baselines. The framework provides a flexible and computationally principled approach for Bayesian analysis of complex ranking data.

#### ***Structural inference over reaction network spaces***

Presenter: Elijah Foo

Dynamical systems in biochemistry are complex, and one often does not have comprehensive knowledge about the interactions involved. Chemical reaction network (CRN) inference aims to identify, from observing time-series of species concentrations, the unknown reactions between the species. Most frequentist approaches to CRN inference focus on identifying a single, most likely CRN, without addressing uncertainty about the network structure. On the other hand, Bayesian treatments of CRN inference typically involve trans-dimensional and multimodal posterior distributions, which are computationally challenging to deal with. In this talk, I will explore how Bayesian CRN inference can be tackled with tempered spike-and-slab distributions, and evaluate the results against exhaustive approaches.

#### ***An Integrated Population Model to Incorporate Spatio-Temporal Heterogeneity in Demographic Rates***

Presenter: Fabian R. Ketwaroo

Demographic processes in populations are inherently heterogeneous across both space and time. Many ecological models explicitly account for temporal heterogeneity in demographic rates while assuming spatial homogeneity. However, ignoring spatial heterogeneity can bias inference, limit predictive performance, and obscure key spatial

structure in demographic rates. Integrated Population Models (IPMs) offer a powerful framework to study spatio-temporal variation in demographic processes by combining multiple ecological data sources sampled at multiple locations. However, to accomplish this, IPMs face significant statistical and computational hurdles, including misalignment between different data sources and the need to efficiently account for residual spatial autocorrelation. We develop a novel Bayesian spatially explicit Integrated Population Model (sIPM) that integrates population count and capture–recapture data from multiple sampling locations to estimate and predict continuous spatio-temporal demographic rates, such as survival, recruitment and population growth rate, across large geographic domains. This framework utilizes the concept of change of support to flexibly accommodate spatial and spatio-temporal data misalignment and incorporates a Nearest Neighbor Gaussian Process to efficiently model residual spatial autocorrelation and generate spatial predictions. We assess the performance of our method through an extensive simulation study. Results show that our approach provides unbiased and precise estimates and predictions of spatio-temporal demographic rates, even in the presence of significant data misalignment and residual spatial autocorrelation. We demonstrate the utility of our method by analyzing data on Gray Catbirds from the North American Breeding Bird Survey and the Monitoring Avian Productivity and Survivorship program across the eastern United States from 2004–2014. This analysis reveals important spatio-temporal variations in demographic rates that would have been obscured by spatially homogeneous IPMs. Our sIPM offers a robust and computationally efficient method of estimating and predicting spatio-temporal demographic rates with high spatial resolution over large geographic domains. This quantification is crucial for developing spatially targeted, effective conservation and management strategies. Therefore, our approach offers ecologists and managers a powerful new analytical tool in a spatio-temporal world.

***Bayesian Robustness for Longitudinal Data: Scale Mixture of Skew-Normal Linear Mixed Models via HMC***

Presenter: Fernanda Schumacher

Longitudinal and clustered data are often analyzed using linear mixed models (LMMs). Standard LMMs assume Gaussian random effects and errors, but these restrictive assumptions can result in a lack of robustness and invalid statistical inferences when data exhibit skewness or heavy tails. To address this, we consider the Scale Mixture of Skew-Normal (SMSN) class of distributions as an interesting extension to relax the model's assumptions by accounting for skewness and heavy tails. However, a significant practical problem arises when modeling distributions derived from the skew-normal: the maximum likelihood estimate (MLE) of the skewness parameter can diverge. We resolve this anomaly by proposing a fully Bayesian estimation of the SMSN-LMM. Our approach utilizes Hamiltonian Monte Carlo (HMC) to perform robust inference and effectively mitigate the divergence issue inherent to the frequentist MLE, ensuring stable parameter estimation when proper prior distributions are considered. This work illustrates the anomaly and demonstrates the improved reliability and superior fit of the Bayesian HMC-based SMSN-LMM using a simulation study and a real-world schizophrenia dataset.

***Improved estimation and uncertainty quantification of virtual experiments through patched local Gaussian random fields***

Presenter: Finn Hughes

Bayesian inference is a statistical approach that applies intuitively to many procedures in measurement science where observations are taken indirectly to perform estimation and uncertainty quantification for a quantity of interest. When repeated observations are scarce and difficult to obtain, the Bayesian approach may be applied robustly and can be used in conjunction with virtual experiments to facilitate data analysis. In many metrological applications that employ a virtual replica of a measurement instrument, the current implementation usually features complex parametric models constituting of a wide range of physically meaningful variables. However, in such a model, some effects, such as small imperfections of a machined specimen, may be challenging to efficiently or adequately model parametrically. Our proposal considers the inclusion of an embedded correction term with a Gaussian random field prior that can account for these higher-frequency effects within the chosen statistical modelling approach. To ensure scalability and computational efficiency, a novel patching approach for local random fields is presented. The procedure enables the modelling of spatial heteroscedasticity and non-stationarity, with the additional benefit of reducing boundary effects such as discontinuities or large gradients which are particularly apparent in the predictive variance. For estimation, an alternating optimisation approach using MAP estimators is introduced to find local optima for the parameters and hyperparameters of the model. Different choices of priors for said parameters are explored, with theoretical concepts such as identifiability and existence of moments also being assessed. Additionally, special cases in which analytical expressions are available are discussed. The overall framework is applied to an application motivated from a virtual experiment in computational optical metrology to investigate if the theoretical deductions hold in a practical sense. This proof-of-principles study

aims to propose a Bayesian procedure which could be used to enhance the modelling of virtual experiments and instruments in metrology, while highlighting key theoretical properties that influence the implementation.

### ***Preference Heterogeneity and Taxation of Sugar-Sweetened Beverages: Estimates from a Finite Mixture of EASI Demand Systems***

Presenter: Ford Ramsey

We allow for preference heterogeneity in the estimation of the linearized Exact Affine Stone Index (EASI) demand system through a finite mixture of such systems. The system is estimated with Bayesian inference while accounting for endogeneity, censoring, and theoretical restrictions. We then apply the model to scanner data on purchases of non-alcoholic beverages and conduct a welfare analysis of the effect of a tax on sugar-sweetened beverages. The linearized EASI, despite being linearized and unlike the linearized Almost Ideal Demand, is fully capable of generating estimates of compensating variation. The welfare effects of the tax are heterogeneous across groups by virtue of the mixture model and measures of uncertainty around these effects are easily established from Markov chain Monte Carlo draws. Attention is paid to the label switching problem, possible solutions, and their effects on inference.

### ***Particle methods for empirical Bayes***

Presenter: Francesca Crucinio

Parameter estimation in the presence of hidden, latent, or unobserved variables is key in modern statistical practice. A pragmatic compromise between frequentist and Bayesian approaches, is the empirical Bayes paradigm in which the MMLE is complemented by uncertainty estimation over the latent variables via the posterior. The standard approach in this context is the expectation-maximisation (EM) algorithm in which expectation w.r.t. the posterior and maximisation w.r.t. the parameters are alternated. The wide use of the EM algorithm is due to the fact that it can be implemented using approximations for both steps: analytic maximisation can be replaced by numerical and the expectation step can be approximated via Monte Carlo sampling. This talk will introduce some novel Monte Carlo algorithms which address the empirical Bayes problem without distinguishing between a maximisation step and an expectation step. This approach is based on minimisation of a functional over the parameter space and the space of probability distributions and can be implemented when the latent variable takes values in discrete spaces. We show that this approach suffers less with issues related to local maximisers.

### ***Scalable inference for mixed membership stochastic block models***

Presenter: Francesco Barile

Standard Markov chain Monte Carlo methods run prohibitively slowly on large-scale problems. An important class of “big data” problems are given by networks, where the goal is often to uncover possibly overlapping communities. Mixed membership stochastic block model provides a flexible framework for this task, but inference requires sampling parameters on the probability simplex. While Markov chain Monte Carlo methods outperform variational approaches on moderate-sized problems, scaling them to massive datasets has motivated stochastic gradient Markov chain Monte Carlo techniques. Existing methods based on Langevin diffusions, however, suffer from severe discretization errors near the simplex boundaries, especially in sparse-data regimes. We introduce a scalable stochastic gradient Markov chain Monte Carlo algorithm based on the Cox–Ingersoll–Ross process, which avoids time-discretization errors by using exact transition densities. A central contribution of our approach is the incorporation of control variates to substantially reduce the variance of the stochastic gradient estimator, thereby enhancing the accuracy and computational efficiency of the algorithm. We provide theoretical results characterizing this variance reduction and demonstrate empirically on large networks that our method achieves superior accuracy and scalability compared with existing approaches.

### ***Zeroth-order parallel sampling***

Presenter: Francesco Pozza

Finding effective ways to exploit parallel computing to speed up MCMC convergence is an important problem in Bayesian computation and related disciplines. Here we consider the zero-order (aka derivative-free) version of the problem, where we assume that (a) the gradient of the target distribution is unavailable (either for theoretical, practical or computational reasons) and (b) we can evaluate the (expensive) target distribution in parallel at  $K$  different locations and use these evaluations to speed up MCMC convergence. We make two main contributions in this respect. First, we show that any method falling within a fairly general “multiple proposal framework” can only speed up convergence by  $\log(K)$  factors in high dimensions. The fundamental limitation of such a framework, which includes multiple-try MCMC as well as many other previously proposed methods, is that it restricts possible moves to the support of the  $K$  evaluation points. We state our results in terms of upper bounds on the spectral gap of the resulting scheme. Second, we discuss how stochastic gradient estimators can be used

to make better use of parallel computing and achieve polynomial speedups in  $K$ . Some of the methods have similarities, but also notable differences, with classical zero-order optimization methods.

#### ***Bayesian motif-aware latent position graphs for higher-order modelling of networks***

Presenter: Francesco Sanna Passino

Real-world networks are often characterised by recurring small substructures involving more than two nodes, called motifs. Because of the edge independence assumption, commonly used in many popular statistical models for networks, the presence of motifs is usually not correctly accounted for, failing to capture the structure of some real-world networks. In particular, a class of models which can be affected by this issue are latent position graphs. In this work, we address this drawback by extending latent position models into a wider class of Bayesian motif-aware latent position models (BMALPM) that incorporate pre-specified motifs via a latent position tensor model (LPTM), collapsed into an adjacency matrix and combined with a standard edge-independent model. Each node is characterised by latent positions encoding its propensity to participate in motifs. Crucially, the choice of suitable prior distributions under a Bayesian paradigm determines how edges and motifs are weighted in the inference process, shaping both the identification of latent positions and the representation of higher-order structures. From this framework, we derive a scalable higher-order binary Poisson matrix factorisation model for triangle motifs and develop an augmentation scheme enabling Gibbs sampling. We demonstrate our method on simulated and real-world network data, showing its superior ability than existing random graph models in modelling triangles while retaining performance in inferring latent positions.

#### ***Comprehensive framework for Bayesian meta-analyses***

Presenter: František Bartoš

Meta-analysis aggregates the ever-growing research record into simple summaries that inform theory, practice, and policy. Yet the dominant classical meta-analytic framework suffers from several limitations. It yields unstable estimates when only a few studies are available, fails to account for model uncertainty, relies on a sampling plan that does not exist at the meta-analytic level, cannot distinguish absence of evidence from evidence of absence, and struggles with publication bias adjustment. In this talk, we present a comprehensive Bayesian meta-analytic framework that addresses these issues. The framework uses empirical prior distributions to regularize parameter estimates, leverages Bayesian model averaging to account for model uncertainty, uses Bayes factors to quantify evidence for and against competing hypotheses in the absence of a sampling plan, and combines several publication bias adjustment models in a single result. It further incorporates meta-regression and multilevel meta-analytic models to address the practical needs of analysts. We illustrate the framework in simulation studies and applications to published meta-analyses and provide implementations in R and JASP.

#### ***Longitudinal Belief Networks: A Bayesian Approach***

Presenter: Gabriel Calvo

Analyzing complex longitudinal data requires robust methods to handle multivariate dependencies, temporal evolution, and subject-specific heterogeneity simultaneously (Weiss, 2005). In this work, we propose a unified modeling framework that integrates longitudinal data analysis with the structural interpretability of Dynamic Bayesian Networks (DBNs). Our methodological approach generalizes classical Bayesian Networks (BNs) that represents variables as nodes in a Directed Acyclic Graph (DAG), where directed arcs capture both probabilistic dependencies and temporal transitions (Diestel, 2017). The proposed modelling structure consists of four key components: observed multivariate response variables, potential latent processes, parameter hierarchies, and the inclusion of random effects and serial correlation terms (Diggle, 2002). This combination allows for a richer representation of temporal structures compared to standard DBNs. We perform full Bayesian inference by using Markov Chain Monte Carlo (MCMC) methods implemented in the NIMBLE software to approximate posterior distributions. To demonstrate the practical applicability and flexibility of this framework, we present a case study using real-world dataset. The application highlights how different BN and DBN specifications can effectively uncover hidden patterns and dynamic relationships in repeated measure data.

#### ***Bayesian Time-Varying Multi-Seasonal ARMA Models: Robust, Stable and Scalable Inference for Nonstationary Time Series***

Presenter: Ganna Fagerberg

We propose a seasonal AR (ARMA) model with time-varying parameter processes in both the regular and seasonal parameters. The model is parameterized to guarantee stability at every time point and can accommodate multiple seasonal periods. The time evolution is modeled by dynamic shrinkage processes to allow for long periods of essentially constant parameters, periods of rapid change, and abrupt jumps. A Gibbs sampler is developed based on a robust, fast and accurate approximate sampler based on the extended Kalman filter. An application to

more than a century of monthly US industrial production data shows interesting clear changes in seasonality over time, particularly during the Great Depression and the recent Covid-19 pandemic.

***Two-sample test for laws of random probabilities via exchangeability and optimal transport***

Presenter: George Kanchaveli

Two-sample testing assesses whether two populations differ by comparing their probability distributions, with the Kolmogorov–Smirnov test as a classical example. While many extensions address multivariate data, modern applications often involve complex objects such as probability distributions. In the frequentist literature, random probability measures are typically observed through sequences of observations drawn from each distribution, which we characterize as exchangeable sequences via de Finetti’s theorem. This provides a link between the frequentist testing problem and Bayesian nonparametric models. We then propose a distance-based two-sample test for distinguishing laws of random probability measures using optimal transport theory, and leverage tools from empirical process theory to establish nonparametric theoretical guarantees. Empirically, we benchmark our method against existing approaches on simulated datasets and apply it to a mortality dataset.

***Multi-View Oriented GPLVM: Expressiveness and Efficiency***

Presenter: Giovani L. Silva

The CINDERELLA project (<https://cinderellaproject.eu>) intends to test and validate the implementation of an AI-based tool for a safer and evidence-based clinical decision for patients proposed for locoregional treatment, making it easily applied in clinical practice anywhere in the world. An App was developed within the Cinderella project, aiming at, among other objectives, to provide women information about the disease, treatment, and the answer to several questions that usually arise. Hence, the analysis of the app usage patterns is relevant for understanding user behavior. We present here survival model extensions to analyze recurrent events data in order to answer better questions arises from this clinical trial: For instance, How does app usage evolve before and after surgery? How long does the app remain useful after surgery? Does usage decrease gradually or abruptly post-surgery? Bayesian generalized additive models can be employed and the results can be applied taking into account different clinical centers after a complete evaluation of the proposed model.

***Objective Model Prior Probabilities in Variable Selection***

Presenter: Gonzalo Garcia-Donato

We review known issues associated with the most commonly used prior model probabilities, namely the uniform and Jeffreys priors (also referred to as Scott-Berger priors). We then introduce several alternatives designed to address these issues. The proposed approaches are evaluated both numerically and theoretically.

***Bayesian Structured Selection with Application to Meta Analysis***

Presenter: Grace Nie

In many high-dimensional problems involving categorical predictors, grouping information is often available in various covariates, such as disease type and cancer type—yet existing variable selection methods rarely leverage this structure for selection of main effects and interactions simultaneously. To address this gap, we extend our recently developed hierarchical Bayesian variable selection framework for categorical predictors—designed to perform sparse selection of categorical-type main effects and interactions under strong hierarchy—by incorporating structured grouping information available in categorical covariates. The original framework uses a hierarchical horseshoe prior to jointly select main and interaction effects while enforcing strong heredity and effectively handling sparsity. Building on this foundation, we introduce a group-level shrinkage parameter and embed it within the local shrinkage structure, enabling two levels of shrinkage: selection of both groups and individual effects within the selected groups. This modification allows grouping information to propagate naturally through the hierarchy to interaction terms, while preserving the interpretability and sparsity-inducing properties of the original model. We evaluate the performance of our method using simulated data and apply our proposed approach to identify important risk factors of treatment-related adverse events in a meta-analysis of cancer therapy.

***Hierarchical models for small area estimation using zero-inflated forest inventory variables: comparison and implementation***

Presenter: Grayson White

National Forest Inventory (NFI) data are typically limited to sparse networks of sample locations due to cost constraints. While design-based estimators provide reliable forest parameter estimates for large areas, there is increasing interest in model-based small area estimation (SAE) methods to improve precision for smaller spatial, temporal, or biophysical domains. SAE methods can be broadly categorized into area- and unit-level models, with unit-level models offering greater flexibility, making them the focus of this study. Ensuring valid inference requires



satisfying model distributional assumptions, which is particularly challenging for NFI variables that exhibit positive support and zero-inflation at the unit level, such as forest biomass, carbon, and volume. Here, we implement a two-stage hierarchical modeling framework for estimation of forest biomass and evaluate candidate estimators. In particular, we compare two-stage unit-level hierarchical Bayesian estimators to their single-stage Bayesian counterparts, and to a two-stage frequentist estimator. We produce estimates of forest biomass at the county level in Nevada and Washington, United States. Estimator performance is assessed using repeated sampling from simulated populations and unit-level cross-validation with NFI data. Results show that estimators incorporating our two-stage approach to account for zero-inflation, county-specific random intercepts and residual variances, and spatial random effects yield the most accurate and well-calibrated county-level estimates, with spatial effects providing the greatest benefits when spatial autocorrelation is present in the underlying population.

***Bayesian Random-Effects Meta-Analysis Integrating Individual Participant Data and Aggregate Data***

Presenter: Hang J. Kim

Meta-analysis using individual participant data (IPD) offers many benefits, including greater analytical flexibility, compared to conventional analyses based on aggregate data (AD). However, it is often hindered by restricted access to IPD. Relying solely on available IPD may introduce "data availability bias," compromising external validity. Integrating IPD with relevant AD addresses this concern, but existing methods are restrictive, requiring precise knowledge of the IPD-to-AD parameter mapping or relying on fixed-effect models that fail to account for study-level heterogeneity. We propose a Bayesian random-effects framework to overcome these limitations. Building on existing methods, we will use estimating equations to derive the conditional distributions of AD parameters, given the corresponding IPD model parameters. We will then apply the multiplier bootstrap method and density ratio models to approximate these conditional distributions based on the observed data, without requiring homogeneity in the covariate distributions. We will introduce how the AD information is interpreted as an informative prior distribution.

***Consistency of variational approximations under bounded Kullback–Leibler divergence***

Presenter: Hien Nguyen

Variational methods are widely used to approximate posterior distributions in Bayesian inference when exact computation is infeasible. We study when such approximations inherit posterior consistency. Our first result shows that, on a general metric space, a uniform bound on the Kullback–Leibler divergence from the approximating measures to a tight sequence of target measures forces the approximating sequence to be tight. It follows that if the target posteriors converge weakly to a Dirac mass at the true parameter, then any variational sequence with bounded Kullback–Leibler divergence to the targets is also consistent. We also give simple logarithmic-moment conditions that verify this boundedness condition, and illustrate them for smooth generalised posterior distributions.

***A Provable Comparison Framework via Scaling Limits for Emerging Monte Carlo Samplers***

Presenter: Hirofumi Shiba

A class of Monte Carlo methods with a qualitatively different dynamical nature from traditional MCMC methods is emerging within the framework of piecewise deterministic Markov processes (PDMPs). Its design space is being actively explored, and our work aims to provide theoretical grounding for comparing their performance. We propose to measure their performance by examining their log-likelihood processes, and prove a theoretically rigorous ordering of asymptotic efficiency between algorithms, through scaling analysis and weak convergence results. Specifically, within this framework, Forward Event-Chain Monte Carlo (FECMC) is asymptotically more efficient than Bouncy Particle Sampler (BPS), regardless of the choice of hyperparameters. Moreover, we show that FECMC attains its best efficiency when this hyperparameter is set to zero, showing its design choices successfully lead to a robust exploration of the target distribution.

***A Bayesian Framework for Parameter Estimation under Partial Order Constraints using Isotone Optimization***

Presenter: Imene Bouafia

In archaeology and geochronology, stratigraphic relationships impose ordering constraints on estimated ages. Current Bayesian models based on uniform order statistics introduce systematic biases, underestimating young ages and overestimating old ones, with effects that intensify as the number of samples grows. We propose a two-step framework: computing an unconstrained posterior, then projecting it onto the constraint space via Wasserstein distance minimization. This projection reduces to isotonic regression, enabling efficient computation while preserving uncertainty. Alternative projection maps based on the Huber loss are introduced to ensure robustness to outlying or contaminated samples. We show the effectiveness of our approach through simulation



studies and applications to real archaeological datasets.

***Bayesian inference of a spatio-temporal nearest neighbor Gaussian process model for pooled genetic data***

Presenter: Imke Botha

Large scale genetic studies often report the total allele counts of distinct genetic markers, but not their co-occurrence. Inferring haplotype frequencies from this pooled data is a challenging problem, particularly as the number of haplotypes increases geometrically with the number of genetic markers. Previous work applied a Gaussian process model in this setting, but only considered 3 genetic markers due to computational limitations. In this work, we propose a nearest neighbor Gaussian process model and develop an efficient sequential Monte Carlo squared algorithm for Bayesian inference of the model parameters. The resulting model is applied to genetic data relating to antimalarial drug resistance in Africa as a test case of the methods.

***The Impact of Heteroskedasticity in Observational Studies of Treatment Effects***

Presenter: Ivan Jeliaskov

There is an extensive causal literature, yet little is known about the impact of heteroskedasticity in popular observational settings. While heteroskedasticity in treatment assignment or response generation can be of intrinsic interest in many contexts, its interaction with nonlinearities in treatment models can bias estimators in ways that standard error adjustments alone cannot correct. To address these challenges, we extend current methodology in regression discontinuity designs, potential outcome regressions, propensity score matching, and inverse probability weighting using a Bayesian framework. This approach incorporates flexible modeling and customized parameter estimation and model comparison algorithms to account for model uncertainty and improve estimation efficiency. Simulation studies assess the consequences of omitted heteroskedasticity and gauge the performance of the proposed modeling and estimation methods, while real-world applications demonstrate their practical implications. Specifically, we examine the effect of academic probation on subsequent academic performance, the influence of Medigap on healthcare expenditures, and the impact of COVID-19 vaccination on mental well-being in the UK. The results provide compelling evidence that the risks of ignored heteroskedasticity in causal analyses should not be overlooked.

***Nonparametric Inference for Nonstationary Spatial Point Processes***

Presenter: Izabel Nola

Point pattern data often exhibit features such as abrupt changes, hotspots, and spatially varying dependence on local intensity. Under a Poisson process framework, these correspond to discontinuities and nonstationarity in the underlying intensity function, features that are difficult to capture with standard modeling approaches. This work proposes a spatial Cox process model in which nonstationarity is induced through a random partition of the spatial domain, with conditionally independent Gaussian process priors specified across the resulting regions. This construction allows for heterogeneous spatial behavior, including sharp transitions in intensity. A discretization-free MCMC algorithm is developed to target the infinite-dimensional posterior distribution without approximation, thus ensuring exact inference. The random partition framework via Voronoi tessellation also reduces the computational burden associated with Gaussian process models. Spatial covariates can be incorporated to account for structured variation in intensity. Recent computational methods are used, enabling scalability to large datasets while preserving exactness. The proposed methodology is evaluated through synthetic examples and real-world applications, demonstrating its ability to flexibly capture complex spatial structures.

***Testing Coherence in Predictive Models***

Presenter: Jamie Reason

Many decision making procedures that make use of probabilistic predictors assume that training on new data acts as Bayesian conditioning. When this assumption breaks, the downstream procedure can perform poorly. E.g. we show that in active learning, incoherent updates cause the procedure to prefer to acquire suboptimal points. To quantify how far a model's predictive updates depart from conditioning, we introduce the incoherence ratio. We empirically find that amortised predictors such as TabPFN can be more coherent than standard parametric approximations to Bayesian inference. In synthetic active learning experiments, more coherent models acquire better data, an effect not attributable to better predictive performance. In many tasks, decisions depend on a predictor only through its induced action. We therefore introduce action coherence, a decision-theoretic relaxation that measures only the incoherence affecting that action. These diagnostics make coherence testable and quantifiable, enabling incoherence to be diagnosed and addressed.

***Efficient estimation of Gaussian process state-space models for learning non-linear dynamics in***

### ***psychology***

Presenter: Jan Failenschmid

Psychological concepts are increasingly viewed as parts of complex dynamic systems. To study how these concepts change and interact over time, researchers in psychology primarily use linear dynamic models (i.e., vector-autoregressive models). However, these linear models cannot capture the more complex features that may be present in psychological time-series and generally require more flexible non-linear dynamics. Specifying the correct functional form for these non-linear relations is difficult in practice, especially given the scarcity of theories about non-linear effects in psychology. One approach to address this challenge is the class of Gaussian process state-spaces models. These models use Bayesian machine learning techniques to learn the functional form of dynamic relations between unobserved variables from noisy indicator data. However, general-purpose MCMC samplers (i.e., HMC-NUTS) and previously proposed specialized samplers for these models are often impractical due to the high computational complexity of the model in conjunction with low sampling efficiency. To overcome these limitations, we adapt a previously proposed approximate Gibbs sampler and substantially improve its efficiency through additional blocking and adaptive Metropolis-within-Gibbs steps. This leads to large gains in efficiency, particularly for estimating the hyperparameters of the Gaussian process. We also show how the most common measurement model in psychology (i.e., the confirmatory factor analysis model) can be seamlessly integrated in the model and efficiently estimated. Our Gibbs sampler is implemented in C++ and R to make it accessible to researchers in the social sciences. Finally, we evaluated the consistency of our sampler using simulation-based calibration.

### ***Approximate Bayesian Inference for Fisheries and Ecological Dynamics***

Presenter: Jia Le Tan

Fisheries and ecological systems are governed by complex biological and environmental processes that are difficult to represent with tractable statistical models. Many commonly used models - ranging from nonlinear population dynamics to stochastic or individual-based simulations - lead to likelihood functions that are analytically unavailable or computationally prohibitive. As a result, exact Bayesian inference is often infeasible in practice, particularly when latent states, strong parameter dependencies, or observation noise structure play a central role. This motivates the use of approximate Bayesian inference techniques that rely on forward simulation, flexible discrepancy measures, or surrogate representations in place of explicit likelihood evaluation. These approaches allow us to work directly with rich ecological models while still quantifying uncertainty in parameters and predictions. In the fisheries context, they enable the analysis of models involving recruitment variability, predator-prey interactions, environmental drivers, and behavioural or agent-based components.

In this contribution, I will present ongoing work applying approximate Bayesian approaches to a range of fisheries modelling scenarios, highlighting the practical considerations that arise when dealing with high-dimensional dynamics, stochastic structure, and partially observed systems. The aim is to provide an overview of how approximate Bayesian inference can be effectively used to study ecological and fisheries processes for which traditional likelihood-based methods are not viable.

### ***Conditionally Conjugate Models of Bounded Support Data with Negative Binomially Randomized Beta Likelihoods***

Presenter: Jimmy Lederman

We study a novel family of distributions on the unit interval  $(0,1)$  that arises by randomizing the parameters of a standard beta distribution with three dependent negative binomial random variables. We call the resulting marginal distribution the *triply-randomized negative binomial beta distribution*, denoted  $TNB(\alpha, p, q)$ , which is parameterized by a median  $p$  and a concentration parameter  $q$ , has a closed-form density function free of complex special functions, and naturally extends both the Libby-Novick and doubly non-central beta distributions. Unlike the beta distribution and its existing generalizations, this family admits many computationally convenient analytic properties, including negative binomial inverse distributions for each of the three latent negative binomials, enabling conditionally conjugate updates for  $p$  and  $q$ . Remarkably, linking  $p$  and  $q$  to Gaussian latent variables with logistic functions yields closed-form updates via Pólya-gamma augmentation, extending this popular technique beyond its traditional application to discrete likelihoods. Together, these properties yield simple auxiliary-variable Gibbs samplers for regression models of bounded-support data, which often outperform standard beta regression approaches in both computational efficiency and predictive fit. In a real-world case study of forest growth dynamics, we demonstrate that this framework can modularly accommodate complex spatial structure. Overall, this work substantially expands the class of Bayesian models for continuous and bounded-support data that can be fit efficiently.

***A scalable method to integrate multiview data for the dynamic prediction of chronic disease complications***

Presenter: Jonathan Hori

Multifactorial chronic diseases like type 2 diabetes mellitus (T2D) often develop through diverse physiological mechanisms, resulting in distinct progression patterns and varying clinical phenotypes. While molecular data from multiple omics technologies may offer complementary views to assess underlying disease types and complication risk, scalable methods to integrate these data for right-censored prediction tasks with valid uncertainty quantification are lacking. We develop a Bayesian supervised factor model for multiview data integration, feature extraction, and outcome prediction by decomposing data variation into interpretable latent factors that are either shared or view-specific. By leveraging stochastic variational inference, our method enables the efficient modeling of large-scale and possibly high-dimensional multiomic and clinical data. We demonstrate the effectiveness of our method through simulations and an application to predicting diabetic kidney disease among a cohort of veterans with T2D. Our findings can guide the development of personalized therapeutic strategies, further advancing precision care for high-risk patients.

***A Unified Spatiotemporal Framework for Modeling Censored and Missing Areal Responses***

Presenter: Jose Alejandro Ordonez

We propose a new Bayesian approach for spatiotemporal areal data with censored and missing observations. The method introduces a flexible random effect that combines the spatial dependence structures of the Simultaneous Autoregressive (SAR) and Directed Acyclic Graph Autoregressive (DAGAR) models with a temporal autoregressive component. We demonstrate that this formulation extends both spatial models into a unified spatiotemporal framework, expressing them as Gaussian Markov random fields in their innovation form. The resulting model captures spatial, temporal, and joint spatiotemporal correlations in an interpretable way. Simulation studies show that the proposed model outperforms common ad hoc imputation strategies, such as replacing censored values with the limit of detection (LOD) or imputing missing data by the sample mean. We further apply the method to carbon monoxide (CO) concentration data from Beijing's air quality network, comparing the proposed DAGAR-AR model with the traditional Conditional Autoregressive (CAR) approach. The results indicate that while the CAR model achieves slightly better predictive performance, the DAGAR-AR specification offers clearer interpretability and a more coherent representation of the spatiotemporal dependence structure.

***Generalized Bayesian Inference for Dynamic Random Dot Product Graphs***

Presenter: Joshua Loyal

The random dot product graph is a popular model for network data with extensions that accommodate dynamic (time-varying) networks. However, two significant deficiencies exist in the dynamic random dot product graph literature: (1) no coherent Bayesian way to update one's prior beliefs about the latent positions in dynamic random dot product graphs due to their complicated constraints, and (2) no approach to forecast future networks with meaningful uncertainty quantification. This work proposes a generalized Bayesian framework that addresses these needs using a Gibbs posterior that represents a coherent updating of Bayesian beliefs based on a least-squares loss function. We establish the consistency and contraction rate of this Gibbs posterior under commonly adopted Gaussian random walk priors. For estimation, we develop a fast Gibbs sampler with a time complexity for sampling the latent positions that is linear in the observed edges in the dynamic network, which is substantially faster than existing exact samplers. Simulations and an application to forecasting international conflicts show that the proposed method's in-sample and forecasting performance outperforms competitors.

***Bayesian Estimation of Latent Network Structures in Spatial Count Data Models: A Link Formation Approach***

Presenter: Jukina Hatakeyama

The arbitrary specification of spatial weight matrices is a persistent issue in spatial econometrics, particularly for count data models. This study proposes a data-driven Bayesian framework to estimate latent network structures within a Spatial Autoregressive Poisson model. We employ a hierarchical Spike-and-Slab prior that incorporates a link formation mechanism, modelling connection probabilities as functions of dyadic covariates. Unlike purely algorithmic variable selection methods, this approach ensures both sparsity and interpretability by explicitly identifying the determinants of network formation. An empirical application to urban mobility count data demonstrates the model's ability to uncover significant positive and negative spillovers missed by conventional geographic weights, confirming its effectiveness for analysing spatial interactions in discrete data.

***Amortized Bayesian Neural Network Inference***

Presenter: Julius Kobiialka

Sampling-based inference is the gold standard for Bayesian deep learning but incurs substantial storage and computational costs when approximating posterior quantities via Monte Carlo. To address these limitations, we introduce a novel amortized inference framework designed to generalize across diverse task spaces. Our approach utilizes conditional generative modeling to map dataset characteristics directly to posterior approximations, enabling efficient sample generation at test time and bypassing traditional iterative optimization routines. By distilling complex distribution behaviors into a flexible, data-conditioned architecture, our method delivers competitive predictive performance and high-quality uncertainty calibration while eliminating the need for costly per-task computation and storage.

***Bayesian infinite interactive fixed effects modeling for causal inference***

Presenter: Junha Seo

Causal inference for single treatment effect estimation is challenging due to the absence of valid control units. The synthetic control method (SCM) offers an innovative way of constructing the so-called data-driven control unit. The generalized synthetic control (GSC) method is proposed as a factor model-based extension of SCM. While GSC improves upon SCM, the performance of GSC heavily depends on the choice of the number of latent factors. To account for the uncertainty associated with the number of factors, we propose to employ a Bayesian infinity factor modeling approach. The key idea of our Bayesian infinity factor modeling is to assign a cumulative shrinkage process prior on the factor loadings. In addition, we apply a Gaussian process approach to infer the non-linear treatment effect. The proposed Bayesian framework enables us to make full Bayesian inference about the time-varying treatment effect. The merits of the proposed Bayesian method are demonstrated through simulation studies and real data analysis.

***Hierarchical Modeling in Bayesian Nonparametric Factor Analysis***

Presenter: Junyi Zhang

Bayesian factor models are a popular tool for factor analysis. Current state-of-the-art Bayesian factor analysis approaches leverage the beta-Bernoulli process prior to characterize the factors and do not require prior knowledge about the factor dimensionality. This prior, however, ignores the potential hierarchical structure within the factor values, a key aspect for a principled interpretation of the analysis. In this presentation, we introduce a new framework based on a new class of nonparametric priors that overcome this limitation. This class of priors allows the development of an innovative hierarchical modelling methodology for Bayesian factor analysis. We present numerical implementations based on simulated and real-world datasets to illustrate the usefulness of our hierarchical modelling methodology.

***Ensemble Doubly Robust Bayesian Inference via Regression Synthesis***

Presenter: Kaoru Babasaki

The doubly robust estimator, which models both the propensity score and outcomes, is a popular approach to estimate the average treatment effect in the potential outcome setting. The primary appeal of this estimator is its theoretical property, wherein the estimator achieves consistency as long as either the propensity score or outcomes is correctly specified. In most applications, however, both are misspecified, leading to considerable bias that cannot be checked. In this paper, we propose a Bayesian ensemble approach that synthesizes multiple models for both the propensity score and outcomes, which we call doubly robust Bayesian regression synthesis. Our approach applies Bayesian updating to the ensemble model weights that adapt at the unit level, incorporating data heterogeneity, to significantly mitigate misspecification bias. Theoretically, we show that our proposed approach is consistent regarding the estimation of both the propensity score and outcomes, ensuring that the doubly robust estimator is consistent, even if no single model is correctly specified. An efficient algorithm for posterior computation facilitates the characterization of uncertainty regarding the treatment effect. Our proposed approach is compared against standard and state-of-the-art methods through two comprehensive simulation studies, where we find that our approach consistently provides reliable estimates and valid uncertainty quantification, often outperforming flexible machine learning baselines by effectively synthesizing information from diverse candidate models. An empirical study on the impact of maternal smoking on birth weight highlights the practical applicability of our proposed method.

***Reconstruct-then-Bootstrap test for the sufficiency of diffusion process***

Presenter: Kathleen Kaye Medriano

Diffusion processes are a go-to model for time series that satisfy stationarity and Markovianity assumptions. However, those two assumptions do not guarantee the sufficiency of diffusion processes for a given time series. In this project, we capitalize on the Kramers–Moyal (KM) expansion: a Taylor expansion of the integral form of the master equation that describes Markov continuous-time processes. In the ideal case, if the observed data

arise from a true diffusion process, the KM expansion should truncate naturally after the second term. In theory, this means that any higher-order ( $\geq 3$ ) KM coefficients should be zero. However, in practice, the discrete nature of measurement introduces artificial higher-order KM coefficients, even when the underlying process is truly diffusive. Nonetheless, for genuinely diffusive systems, it is expected that the sampling distribution of a well-chosen statistic associated with higher-order coefficients will be different than non-diffusive ones. This is a viable avenue for testing the appropriateness of a diffusion model given an observed time series. We take advantage of this and propose a meaningful statistic that could inform whether or not a diffusion model is sufficient to explain temporal variation in a given series. We then build a test that involves reconstructing the diffusion equation to generate surrogate paths, yielding a bootstrap distribution against which the observed statistic could be compared. We report on a Monte Carlo study evaluating the sensitivity and selectivity of the new test.

### ***Rao-Blackwellised Reparameterisation Gradients***

Presenter: Kevin Lam

Latent Gaussian variables have been popularised in probabilistic machine learning. In turn, gradient estimators are the machinery that facilitates gradient-based optimisation for models with latent Gaussian variables. The reparameterisation trick is often used as the default estimator as it is simple to implement and yields low-variance gradients for variational inference. In this work, we propose the R2-G2 estimator as the Rao-Blackwellisation of the reparameterisation gradient estimator. Interestingly, we show that the local reparameterisation gradient estimator for Bayesian MLPs is an instance of the R2-G2 estimator and Rao-Blackwellisation. This lets us extend benefits of Rao-Blackwellised gradients to a suite of probabilistic models. We show that initial training with R2-G2 consistently yields better performance in models with multiple applications of the reparameterisation trick.

### ***On predictive inference for grouped data with copulas***

Presenter: Keyi Jiang

The martingale posterior framework posits that posterior sampling can be fundamentally reinterpreted as predictive imputation of unobserved data. While this predictive paradigm offers a compelling alter-native to traditional inferential Bayesian procedure, its extension beyond conditionally independent and identically distributed (i.i.d.) structures remains largely undeveloped. This paper addresses the systematic generalization of the martingale posterior framework to grouped data settings. Leveraging de Finetti's representation theorems, we delineate the distinct sources of posterior uncertainty governing relevant estimands and demonstrate how structural features inherent to hierarchical Bayesian models directly translate into the design of predictive imputation schemes. We introduce a modularized predictive updating scheme and establish sufficient conditions under which it induces a valid martingale posterior distribution for targeted quantities. The methodology is illustrated through a nonparametric predictive construction based on Gaussian copulas, for which some theoretical properties are established.

### ***Tractable Riemann-Laplace Approximations***

Presenter: Larissa Lemos Afonso

The Laplace approximation yields a computationally efficient approximation to the posterior by Taylor-expanding the (log-)posterior around its mode. However, the resulting Gaussian distribution is unable to capture complex posterior features, such as skewness, heavy tails or multimodality. To address this limitation, recent work has exploited the geometry of the statistical manifold endowed with different Riemannian metric choices, such as the ones defined by Fisher and Monge matrices. Nonetheless, the superior expressiveness of Riemann-Laplace approximation comes at a price — both sampling and posterior evaluation become analytically intractable. The former requires pushing tangent vectors through the exponential map, which implies solving the geodesic ODE. The latter implies computing the logarithmic map, which is defined as the inverse of the exponential map, and therefore is doubly intractable. To achieve flexible yet tractable Riemann-Laplace approximations, we propose analytic approximations to the exponential and logarithmic maps of the Fisher-Rao statistical manifold. Experimental results show that our tractable Laplace approximations are able to reproduce complex posterior shapes despite being orders of magnitude faster than prior Riemann-Laplace methods.

### ***Graphical model-based clustering of categorical data***

Presenter: Laura Ferrini

Clustering categorical data is a pervasive issue in many applied domains, including social and biomedical science. In a multivariate setting, model-based clustering should account for differences in the joint distribution of variables across clusters, thus allowing for both shifts in the marginal distributions and changes in their dependence structure. This is crucial in cases where clusters are not well-separated, and differences are mostly captured by



parameters indexing the joint distribution rather than the marginals. In such contexts, incorporating dependence relations between variables can significantly enhance the clustering performance. To address this issue, we propose a novel Bayesian methodology based on graphical models which explicitly accounts for dependencies in the joint distribution of the variables. Specifically, we consider an infinite mixture model based on a Dirichlet Process prior, where each mixture component corresponds to a decomposable graphical model. We provide full Bayesian inference for the model and develop a Markov Chain Monte Carlo scheme for posterior inference on the clustering and graphs. Our methodology is evaluated through simulation studies and applied to real datasets. Results show that the proposed method outperforms benchmark approaches for clustering multivariate categorical data that do not explicitly account for dependencies in the joint distribution of variables.

#### ***Assessing Bayesian borrowing methods in rare disease basket trials***

Presenter: Laura Morgan

Rare diseases pose unique challenges in clinical trials due to small patient populations making it difficult to recruit enough patients to sufficiently power the trial for traditional frequentist analyses. As a result, the incorporation of external evidence through Bayesian methods has become increasingly desirable. A popular approach is to borrow data across related subgroups in basket trials, but determining the degree of borrowing remains difficult. We are conducting a simulation study to investigate how using expert elicitation to inform on prior distributions for borrowing parameters impacts the performance of Bayesian borrowing models, such as hierarchical and commensurate prior models. Alongside Bayesian metrics like calibration and sharpness, we assess how the posterior mean of each model behaves as a frequentist estimator by considering bias and mean squared error. By placing some attention on frequentist assessments of performance, we aim to demonstrate how these inherently Bayesian methods are applicable, interpretable and acceptable in the frequentist-dominated trial and regulations industry. We anticipate that the use of informative priors, which would be derived from expert elicitation, would improve borrowing behaviour and hence the overall performance of the models.

#### ***A Comparison Hierarchical Bayes and Model-Agnostic Meta-Learning***

Presenter: Leanna House

When relying on training and testing datasets to assess complex analytic models, the potential for data shift presents major challenges. This is particularly true when modeling times series data derived from the physical sciences, where changing environments and instruments can alter the relationship between training and testing data in ways that are not fully understood. Statisticians and deep learning researchers have independently developed methods to address data shift; e.g., statisticians may consider Bayesian hierarchical models (BHM) and deep learners may rely on meta-learning. Although the theoretical link between these approaches is known, there are few empirical demonstrations that compare the approaches on their ability to mitigate data shift explicitly. This work makes such comparisons by assessing similarities and differences in applications of BHM and first-order model-agnostic meta-learning (FOMAML) when analyzing a semi-synthetic seismic time series dataset. We assess interpretability, performance, and uncertainty in the methods under known shift. We find that FOMAML achieves higher accuracy when data are abundant, while BHM exhibits more stable behavior and communicates uncertainty more faithfully, even with a simple feature representation. We also show how BHM's transparency helps reveal the data regimes where information pooling improves performance, a feature with no analog in deep networks.

#### ***Conformalized Model Selection***

Presenter: Leo Murao Watson

Conformal prediction is a frequentist predictive inference method that has attracted significant attention due to its finite sample distribution-free guarantees. In contrast, the validity of Bayesian predictive regions typically hinges on correct specification of the prior and likelihood. Recent work has demonstrated the promise of coupling these ideas together to construct prediction sets maintaining a target frequentist coverage rate while incorporating prior information. However, less is known about the relationship between these conformal prediction sets and the Bayesian posterior predictive sets. We introduce model selection and sensitivity analysis methods based on the discrepancies between conformal and Bayesian predictive sets. Our approach measures the degree of “calibration” the conformal procedure applies to a Bayesian model to maintain frequentist coverage. We apply this approach to a variety of model misspecification examples, demonstrating how the gap between Bayesian and conformal predictive objects reveals model inadequacies that are not visible from conformal or Bayesian analyses alone.

#### ***Model comparison for Cronbach's alpha using the balanced one-way random effects model***

Presenter: Lizanne Raubenheimer

In this paper model comparison for Cronbach's alpha will be considered, where the balanced one-way random



effects model will be used. The Jeffreys independence prior will be used, and thus fractional Bayes factors will also be considered. Cronbach's alpha is a measure used to assess the reliability of a set of test items. The Bayes factor is a ratio of two competing statistical models represented by their evidence, and is used to quantify the support for one model over the other. A large value of  $B_{01}$  (the Bayes factor) means that the data favour model  $M_0$ , while a small value favours  $M_1$ . The first problem in the practical application of the Bayes factor is that the marginal likelihood depends on the normalizing constants of the priors, so an improper prior with arbitrary constant makes the Bayes factor arbitrary. A solution to this is partial Bayes factors, where part of the information from the data is used to create proper priors from improper priors. With fractional Bayes factors, a fraction of the likelihood function is used for this purpose.

### ***Informed Order-Invariant Identifiability for Bayesian Factor Models***

Presenter: Lorenzo Schiavon

We propose a unified Bayesian framework for sparse factor models that reconciles identifiability, order-invariance, and interpretability in the decomposition of high-dimensional covariance structures. Conventional identification schemes rely on arbitrary variable orderings or restrictive triangular constraints, often compromising exchangeability and obscuring substantive interpretation. Our approach defines an order-independent prior on the factor loadings matrix that ensures identifiability through probabilistic structure rather than fixed parameterization. The prior combines rank selection mechanisms with covariate-induced global-local shrinkage, allowing sparsity and interpretability to emerge from the data and, when available, from exogenous information. This formulation flexibly matches identifiability constraints with the structural regularities informed by domain knowledge, without imposing deterministically. The resulting posterior is identifiable up to signed permutations and supports efficient inference through Gibbs or variational approximations. Simulated and financial applications illustrate how the proposed model recovers coherent, economically meaningful latent structures, providing a principled route to identifiable and interpretable Bayesian factor analysis.

### ***Partially exchangeable enriched stochastic block models***

Presenter: Louise Alamichel

Stochastic block models learn group structures between nodes sharing similar connectivity patterns. Recent developments have extended this approach to multiple connected networks, often within multilayer or multiplex architectures. However, most formulations still rely on the strong assumption that all networks share a single node partition. In many applications, this assumption is too restrictive: connected networks may exhibit distinct but hierarchically dependent clustering structures. For example, in criminal networks that track the number of meetings attended by criminals at different levels of the criminal organization, the clustering of nodes at one level may naturally fragment into more detailed communities at another level, reflecting different but related organizational principles. To address this, we introduce partially exchangeable enriched stochastic block models, a new class of Bayesian network models that jointly capture multiple layers of dependency through partially exchangeable priors on node partitions. Building on the partially exchangeable stochastic block model of Durante et al. (2025), we extend its construction to an enriched framework where two partitions are linked by a nested structure. The first partition, which governs a node-colored network, is informed by the covariates of the nodes, while the second is generated by fragmenting this coarser structure into nested subgroups. This joint prior is derived from an enriched Gibbs-type process, ensuring partial exchangeability while flexibly adapting to shared and network-specific clustering behaviors. Inference is performed via a collapsed Gibbs sampler using a cut-posterior perspective to improve mixing. Preliminary results on simulated data and in a study of joint participation in summits within a complex mafia organization highlight the strengths of the proposed formulation and its ability to integrate and learn relevant structures in networks.

### ***Posterior local sensitivity through Otto's calculus***

Presenter: Lucas Schwengber

Assessing sensitivity of a Bayesian posterior to its data and hyperparameters are a key step in the Bayesian workflow. Most existing methods for doing so rely either on re-running Markov Chain Monte Carlo (MCMC) samplers (Han, Gelman, and Vehtari 2025) and/or on importance sampling and differential approximations (Giordano and Broderick 2024; Gelman et al. 2020; Giordano, Broderick, and Jordan 2018; Vehtari, Gelman, and Gabry 2017). These methods can be noisy, computationally expensive, or both. We propose an alternative approach to posterior sensitivity analysis based on Otto's calculus formalism. This formalism allows us to convert infinitesimal log-density posterior perturbations into vector fields to perturb the whole sample, which realize an equivalent perturbation. By iteratively computing and applying such vector fields we are able to obtain approximate local perturbations of a posterior draw which correspond to a given local log-density perturbation.

We illustrate the applicability of the methods in examples from political science.

### ***Hierarchical Bayesian Nonparametrics for Spatial Underreporting***

Presenter: Lucia Gallucci

Underreporting is a crucial and spatially uneven phenomenon that, if ignored, can mask true risk patterns and mislead inference in spatial analyses. Existing approaches for spatially correlated count data often face fundamental identifiability issues between the true relative risk and the reporting process. To address these challenges, we propose a Bayesian nonparametric hierarchical model, namely Spatial Hierarchical Dirichlet Process for Underreporting (SHDP-U), that explicitly disentangles spatial risk patterns from heterogeneous underreporting. In the first stage, areas are clustered by observed relative risk via a nonparametric approach that allows an unspecified number of clusters. Spatial coherence is induced by embedding the cluster labels in a Markov Random Field, which encourages neighboring areas to share the same latent risk structure while preserving the possibility of sharp local changes. Conditional on this spatial cluster membership, we model heterogeneity in reporting rates with a covariate-dependent stick-breaking process. This construction allows latent reporting profiles to be shared across clusters, while allowing the associated weights to adapt to local predictors of underreporting. SHDP-U yields an adaptive and flexible representation of both true risk and reporting behavior, separating spatial patterns in disease incidence from systematic differences in case detection. Thus, our model provides a more accurate characterization of disease burden and a rigorous uncertainty quantification of underreporting. We develop a fast and efficient MCMC algorithm for posterior inference, evaluate performance in extensive simulation studies, and apply the method to Autism Spectrum Disorder (ASD) prevalence across Italian municipalities. We prove practical gains in both estimation accuracy and interpretability.

### ***Optimizing Data Augmentation through Bayesian Model Selection***

Presenter: Madi Matymov

Data Augmentation (DA) has become an essential tool to improve robustness and generalization of modern machine learning. However, when deciding on DA strategies it is critical to choose parameters carefully, and this can be a daunting task which is traditionally left to trial-and-error or expensive optimization based on validation performance. In this paper, we counter these limitations by proposing a novel framework for optimizing DA. In particular, we take a probabilistic view of DA, which leads to the interpretation of augmentation parameters as model (hyper)-parameters, and the optimization of the marginal likelihood with respect to these parameters as a Bayesian model selection problem. Due to its intractability, we derive a tractable Evidence Lower BOund (ELBO), which allows us to optimize augmentation parameters jointly with model parameters. We provide extensive theoretical results on variational approximation quality, generalization guarantees, invariance properties, and connections to empirical Bayes. Through experiments on computer vision tasks, we show that our approach improves calibration and yields robust performance over fixed or no augmentation. Our work provides a rigorous foundation for optimizing DA through Bayesian principles with significant potential for robust machine learning.

### ***Cross-Sectional Dynamics Under Network Structure: Theory and Macroeconomic Applications***

Presenter: Marko Mlikota

Many economic environments involve units linked by a network. I develop an econometric framework that derives the dynamics of cross-sectional variables from the lagged innovation transmission along fixed bilateral links and that can accommodate general patterns of how higher-order network effects accumulate over time. The proposed NVAR rationalizes the SAR model as the limit under an infinitely high frequency of lagged network interactions. The factor-representation of the NVAR suggests that at the cost of restricting factor dynamics, it naturally incorporates sparse factors as locally important nodes in the network. The NVAR can be used to estimate dynamic network effects. When the network is estimated as well, it also offers a dimensionality-reduction technique for modeling high-dimensional processes. In a first application, I show that sectoral output in an RBC economy with lagged input-output conversion follows an NVAR. In turn, I estimate that the dynamic transmission of TFP shocks along supply chains accounts for 61

### ***Bayesian inference for ocean wave spectral density models***

Presenter: Matt Moores

Accurately modelling and predicting ocean wave behaviour is crucial for shipping, coastal engineering, and renewable energy generation. Ocean wave spectra, which describe the distribution of wave energy across different frequencies, play a central role in these applications. Models like Pierson-Moskowitz and JONSWAP have been proposed to represent these spectra. However, these models can fail to capture complex features that are encountered in real-world wave conditions, such as multiple spectral peaks. Our approach employs a mixture model that can flexibly adapt to the presence of multiple peaks, providing well-calibrated uncertainty quantification. To

address the potential challenges of identifiability in mixture models, we leverage informative prior distributions for the model parameters. We validate our modelling approach using simulation-based calibration (SBC) and apply it to ocean buoy data from King George Sound, Western Australia.

This is joint work with Sankalpa Fonseka & David Gunawan (University of Wollongong) and Jeff Hansen (University of Western Australia).

#### ***Bayesian SEM model comparison in R with blavaan***

Presenter: Mauricio Garnier-Villarreal

Model evaluation and comparison are common and necessary steps in many analysis frameworks. Structural Equation Modeling (SEM) has a long history of common methods to do model evaluation and comparison, but from the frequentist inference approach. As well, the Bayesian literature has a history of the recommended ways to performed model evaluation and comparisons. In the last decade, there has been an increase in popularity of Bayesian SEM (BSEM) with user friendly software such a the R package blavaan. This surge in interest also requires an integration between the SEM and Bayesian ways of doing model evaluation and comparisons. There has been developments in this area, but it lacks a comprehensive guide for applied users on what are the different steps and ways to integrate them in BSEM. Here we present a tutorial on how to performed BSEM model evaluation and comparison with the R package blavaan. These steps include prior predictive checks, convergence and posterior stability, overall model fit, local model fit, posterior predictive checks, and model comparison based on the Leave-One-Out and Widely Applicable Information Criteria. This tutorial will present a workflow and user friendly example in R.

#### ***A Bayesian predictive control model for a social autonomous robot for autism treatment***

Presenter: Miguel Santos Pascual

We present a Bayesian predictive control model for a social autonomous robot in autism treatment, based on a simple Dirichlet formulation with model averaging that enables fast decision-making together with continuous online learning. The robot selects actions through a weighted utility function that adapts over time as new interaction data are observed, allowing personalized and responsive behavior under uncertainty. This provides an efficient and interpretable framework for real-time socially assistive intervention.

#### ***Personalized Predictive Modeling with Input-dependent Weighting under Incomplete Information: A Bayesian Treed Regression Approach***

Presenter: Mingya Huang

When multiple candidate models with different perspectives—particularly those with multilevel data structures—are used to explain student achievement, researchers face significant model uncertainty. Current popular ensemble learning methods impose restrictive assumption that candidate models collectively capture all relevant predictive information. This assumption becomes problematic across many research contexts, particularly in educational assessment, where theoretical frameworks often provide incomplete representations of the complex processes underlying student learning. We address this limitation through three key methodological advances: (1) using nonparametric approaches that learn how model weights should vary across different contexts, (2) removing the sum-to-one constraint while adding an intercept term to capture unexplained variation, and (3) delivering personalized predictions with well-calibrated uncertainty quantification within a computationally efficient framework. Our proposed method flexibly identifies the relative importance of different predictors across cluster levels and assigns optimal model weights accordingly. Through simulation studies and empirical analysis of PISA 2018 data, we demonstrate substantial improvements in predictive accuracy, uncertainty calibration, and computational performance over current popular methods. This methodology offers the greatest advantages when applied to heterogeneous student populations across diverse educational settings, where the relevance of different candidate models varies systematically by context.

#### ***Bayesian likelihood-free inference with high-dimensional data***

Presenter: Minhye Park

With the growing availability of high-dimensional data, variable selection has become an essential component of regression analysis. However, existing Bayesian inference methods often depend on correctly specifying the likelihood, which may not always be practical or feasible. Recently, the loss-likelihood bootstrap (LLB) has received considerable attention as a means of performing likelihood-free Bayesian inference. While LLB performs well in low-dimensional settings, it fails in high-dimensional regression contexts. To address this limitation, we propose a likelihood-free Markov Chain Monte Carlo Model Composition (MC3) method via likelihood-free Gibbs sampling. Traditional MC3 requires marginal likelihoods, which are unavailable in a likelihood-free framework. To overcome this obstacle, we introduce a new approach that employs the Laplace approximation to estimate

marginal likelihood ratios without explicit likelihood evaluations. This innovation enables efficient and accurate model comparison within the likelihood-free framework. Our proposed approach is applicable to a wide range of high-dimensional regression problems, such as survival data analysis, longitudinal data analysis, and machine learning applications. The performance of the proposed likelihood-free Bayesian method is demonstrated through simulation studies and real data analysis.

***Unbiased Estimation of Log Normalizing Constants via Coupling and Path Sampling.***

Presenter: Mohamed Mehdi Keteb

Posterior distributions often involve an intractable normalizing constants, referred to as marginal likelihood or evidence. The problem of numerical approximation of the evidence has motivated the development of numerous methods, such as bridge sampling and path sampling, also known as thermodynamic integration. Combining unbiased Markov chain Monte Carlo with path sampling, we propose unbiased estimators for the logarithm of normalizing constants. We provide conditions under which the estimators are consistent and have a finite variance. We demonstrate their appeal relative to existing methods for normalizing constant estimation, including the ease of use on parallel processors and ability to construct confidence intervals. As a by-product, we also consider the approximation of leave-one-out Bayesian cross-validation criteria, and we illustrate the methodology through several examples.

***NeuralSurv: Deep Survival Analysis with Bayesian Uncertainty Quantification***

Presenter: Mélodie Monod

We introduce NeuralSurv, the first deep survival model to incorporate Bayesian uncertainty quantification. Our non-parametric, architecture-agnostic framework captures time-varying covariate-risk relationships in continuous time via a novel two-stage data-augmentation scheme, for which we establish theoretical guarantees. For efficient posterior inference, we introduce a mean-field variational algorithm with coordinate-ascent updates that scale linearly in model size. By locally linearizing the Bayesian neural network, we obtain full conjugacy and derive all coordinate updates in closed form. In experiments, NeuralSurv delivers superior calibration compared to state-of-the-art deep survival models, while matching or exceeding their discriminative performance across both synthetic benchmarks and real-world datasets. Our results demonstrate the value of Bayesian principles in data-scarce regimes by enhancing model calibration and providing robust, well-calibrated uncertainty estimates for the survival function. This paper was accepted at NeurIPS 2025.

***Improving Nonlinear Modeling with Informative and Safe Gaussian Process Priors***

Presenter: Nataliia Zubareva

Many real-world phenomena, especially in the social sciences, are inherently nonlinear. Gaussian processes (GPs), a methodology from the Bayesian machine learning literature, offer a flexible framework for studying such phenomena by specifying not a fixed functional form, but rather a kernel that defines smoothness and general patterns, e.g. periodicity, via a structured covariance matrix of the data.

However, this flexibility achieved by using the default kernels sometimes becomes an issue.

- First, as GPs are extremely flexible, uninformed application of GP methodology may result in unrealistic conclusions; a problem that is much less important in linear modeling.
- Second, when studying nonlinear processes in the case of limited data, which is common in the social sciences, uninformed GPs may yield poor fit and poor predictions.
- Third, natural restrictions are often present and contextual (prior) information is available, such as about the general shape of processes or whether they are naturally bounded, which need to be incorporated in the analysis to ensure the validity of the conclusions.

It is therefore recommendable to apply GPs in a more informed manner, but it is difficult for applied researchers as GPs are constructed via kernels which have a less intuitive interpretation. This work aims to provide general guidelines and a digital modeling toolbox for applied researchers to use the flexible GP methodology in an informed manner to improve the validity and precision of statistical conclusions when studying nonlinear phenomena. Particular focus will be given to selecting appropriate kernels, incorporating boundedness, (partial) monotonicity, controlling anticipated shapes via warping techniques, specifying informative priors of the GP parameters, assessing them by prior predictive analysis, and calibrating their strength.

***Interpretable Latent Distances for Breast Tumors via Geometry-Aware Generative Modeling***

Presenter: Negar Safinianaini

Latent-space geometry provides a rigorous foundation for analyzing the latent variables of deep generative models. Existing geometric approaches, however, typically assume that the decoder follows a known probability distribution, enabling the use of the Fisher–Rao (FR) metric on the ambient space. For i.i.d. data, the FR

metric is diagonal, with each component depending only on its corresponding coordinate. From a Riemannian-geometric perspective, though, one is free to choose a metric intrinsically. While the FR metric is justified by the probabilistic model, it does not reflect geometric structures that may be uniquely tied to a specific problem. To capture such problem-specific structure, we instead propose a custom, problem-adapted metric. Consider a breast cancer setting with five subtypes: Luminal A, Luminal B, HER2-enriched, Basal-like, and Normal-like. Biological evidence indicates that Normal-like tumors form a continuum with the Luminal subtypes, reflecting gradual transitions in hyperplasia, Ductal Carcinoma In Situ (DCIS), and normal-cell proportions. Yet distances generated by existing geometric deep Bayesian models contradict this structure: latent-space distances fail to reflect the continuum between Luminal A/B and Normal-like tumors. In this work, we retain Bayesian generative modeling—for posterior inference and uncertainty quantification—while exploiting a pullback metric tailored to the problem. The metric employs a smooth step-like scaling function on the diagonal entries, resulting in regions of higher and lower curvature. With appropriate parameters, the curvature creates a converging effect that draws the related tumor subtypes—Luminal A, Luminal B, and Normal-like—toward a shared continuum. The resulting geodesic distances, i.e., the locally shortest paths on the manifold, recover the desired biological continuum between these subtypes, overcoming the limitations of existing metrics. The proposed metric has the potential to uncover latent space structure in other cancer subtypes with poorly understood molecular landscapes, providing a latent continuum that enables biologically meaningful interpretations.

### ***Dual-stress accelerated life testing models via approximate Bayesian computation***

Presenter: Neill Smit

Accelerated life testing can be used to estimate the life characteristics of high-reliability products, especially where conventional reliability estimation is not possible due to time and cost constraints. In an accelerated life test, products are exposed to more severe than their normal operating conditions, by applying stressors to induce early failures. A time transformation function can then be used to extrapolate the life characteristics of the products under their normal operating conditions. The resulting models are often complicated and classical parameter estimation is not always possible. Bayesian accelerated life testing models, using the generalized Eyring relationship as the time transformation function, is considered in this paper. The generalized Eyring relationship incorporates one thermal stressor and one non-thermal stressor. A likelihood-free method, using approximate Bayesian computation, is investigated for estimating model parameters. The approximate Bayesian computation method is compared to frequentist methods, such as maximum likelihood estimation, in a simulation study.

### ***Adaptive Generalized Elliptical Slice Sampling***

Presenter: Nicholas Marco

In this work, we introduce an adaptive generalized elliptical slice sampler capable of sampling from a broad class of target distributions. The proposed adaptive scheme can take advantage of the geometry of the target distribution, allowing for an efficient sampler that performs well in relatively high dimensions while maintaining many of the desirable features of slice samplers. Assuming certain regularity conditions, we demonstrate that the proposed adaptive algorithm is ergodic. Moreover, through simulation experiments, we illustrate that the proposed sampling scheme has good sampling performance across a diverse set of target distributions, including those that are non-elliptical, non-differentiable, or high-dimensional.

### ***Scalable and Robust Spatial Prediction via Multi-Resolution Ensembles of Predictive Processes***

Presenter: Nicolas Bianco

Gaussian processes provide a flexible framework for spatial prediction, but their computational cost limits applicability to large-scale data with large sample size  $n$ . Predictive processes (PPs), a popular low-rank approximation, mitigate this burden by projecting the original process onto a reduced set of  $m \ll n$  inducing points. However, existing theory requires  $m$  to grow with  $n$ , creating a trade-off between accuracy and computational efficiency. We address this challenge by introducing an ensemble of PPs based on spatial partitioning, and propose a novel partitioning and patching scheme with desirable properties. By generalizing the convergence results of PPs, it becomes possible to explicitly balance scalability and accuracy: increasing the number of ensemble components slows down the convergence but substantially improves computational efficiency. We further show theoretically that, despite the limited approximation accuracy of PPs with fixed  $m$ , they are asymptotically robust to data contamination. Motivated by this insight, we finally introduce a multi-resolution ensemble that combines PPs with fixed  $m$  with multiple ensembles defined over possibly overlapping coarse to fine partitions. Simulations and large-scale geostatistical applications demonstrate that our approach delivers accurate, robust predictions with computational gains, providing a practical and broadly applicable solution for spatial prediction.

### ***Bayesian Posterior Estimation of Gene Regulatory Networks from Zero-Inflated Single-Cell Data.***



Presenter: Noriaki Sato

Single-cell transcriptomic (SCT) data includes tens of thousands of genes across tens of thousands of cells. Learning a gene regulatory network (GRN) to reveal regulatory pathways commonly relies on Bayesian networks (BNs) to enable causal inference and posterior uncertainty quantification, but faces challenging computational and statistical issues beyond 20-30 genes. However, significant prior information on these pathways is available from annotation databases such as KEGG and Dorothea.

In this contribution, we propose a Bayesian approach that makes the best use of this prior information to improve on our recent work on learning zero-inflated BNs tailored to SCT data and on evaluating them as causal models (doi:10.48550/arXiv.2511.12805). At a structural level, we discuss an energy-based prior that weights candidate GRNs based on their interventional distance from the reference GRNs constructed from annotation databases. At the distribution level, we discuss how to replace likelihood-based parameter estimation with a posterior approach informed by regulation annotations (e.g., up/down regulation, non-functional relationships). In this way, we increase the accuracy of learned GRNs while enabling posterior inference on the GRN and its parameters in a computationally feasible manner.

### ***Continuous Bayesian Latent Space Representations of Networks from Event Data***

Presenter: Owen G. Ward

Network data often consists of underlying continuous time interactions between the nodes in the network. Representing the true underlying dynamics of such interactions is challenging, and often instead utilises an aggregate summary of these events. We propose a continuous time latent space model for the nodes in a network, based directly on these underlying interactions, using modern Bayesian computation methods to scale inference to large scale networks commonly seen in practice. Finally, we interpret the results of this model for real interactions on international relations networks, identifying interpretable latent structure.

### ***Scalable computations for Latent Dirichlet Allocation through informed non-reversible MCMC schemes***

Presenter: Paolo Manildo

The Latent Dirichlet Allocation (LDA) is a probabilistic model which has become very popular in various scientific domains. In the context of natural language processing, for example, it describes each document as a mixture of (latent) topics, i.e. probability distributions over all possible words. Through a suitable hierarchical structure, LDA allows for topics to be shared across a corpus of distinct, but possibly related, documents. Real data applications involve hundreds of documents, for a total of hundreds of thousands of words. Therefore the standard collapsed Gibbs sampler, which updates the allocation of one word conditional on all others, often exhibits slow mixing and becomes infeasible as the total number of words  $n$  grows. Thus variational approximations have been developed, which drastically reduce the computational burden: however they can severely underestimate uncertainty and remain stuck in sub-optimal configurations. Leveraging recent results for non-reversible Markov chains, we introduce a novel sampling scheme for LDA designed to be efficient for large  $n$ . We combine the latter with a locally-balanced proposal distribution, which reweights the proposal probability based on the value of the target distribution in a randomly chosen neighborhood: we show both theoretically and empirically that this significantly speeds up the original algorithm with a modest increase in the cost per iteration. The performances of our new scheme are compared with the available methodologies on real-world applications.

### ***High-Dimensional Bayesian Mediation Analysis with Adaptive Laplace Priors***

Presenter: Qingzhao Yu

The mediation analysis method is used to investigate the effects of mediators that intervene in the pathways between an exposure variable and an outcome variable. Bayesian methods are naturally used in mediation analysis due to the hierarchical structure of Bayesian models. This talk introduces an innovative adaptive Bayesian mediation analysis method that incorporates adaptive Laplace priors into the predictive model to account for high-dimensional mediators. This approach introduces a penalization function on the estimated direct and indirect effects rather than solely on the coefficients of predictive models. Consequently, estimated effects that lack statistical significance may shrink to zero, facilitating a more robust analysis. We demonstrate the efficacy of our adaptive mediation analysis method on simulations and on a Louisiana triple negative breast cancer (TNBC) dataset to examine racial disparity in diagnosed stage among TNBC patients diagnosed between 2010 and 2017. The dataset is linked to the 2017 hazardous air pollutant emissions burden estimation database using patients' residential addresses. We effectively explain a portion of the disparity using currently collected variables. The analysis identifies crucial mediators and confounders, highlighting the significance of variables such as age of diagnosis, insurance status, tumor grades, and the concentration of Naphtha in the air.

### ***Bayesian time series modelling for fault prediction across software versions***



Presenter: Rakesh Ranjan

This paper models the number of software bugs and their detection times across different software versions using a non-homogeneous Poisson process (NHPP). The Goel-Okumoto intensity function is employed as the rate parameter of the Poisson process. To better capture patterns observed in the data, several modelling extensions of the standard Goel-Okumoto NHPP are proposed, in particular modelling the similarity of consecutive versions of the software using a time series model. Due to the high dimensionality and analytical intractability of the resulting posterior distributions, inference and prediction are performed using the Markov Chain Monte Carlo (MCMC) algorithm. A comparative analysis is conducted between the original Goel-Okumoto model and its proposed extensions. Numerical illustrations are provided using both simulated and real-world datasets. In particular, Bugzilla data are used to demonstrate the practical applicability of the models. The results indicate that the proposed methods perform satisfactorily in capturing the bug detection dynamics.

***Bivariate Realized Stochastic Volatility Model with the Correlation between Realized Beta and Returns***

Presenter: Reitaro Yoshida

Stochastic volatility models are widely used to capture time-varying volatility, and their extensions incorporating realized volatility from high-frequency data have become increasingly prominent. In asset pricing, beta measures an asset's sensitivity to market movements, and its realized counterpart has received growing attention. Empirical evidence shows asymmetric effects in which declines in returns tend to be followed by increases in next-day beta. This study proposes a bivariate stochastic volatility model that jointly models realized beta and returns, explicitly reflecting this asymmetry. Using daily data from the TOPIX and Tokyo Stock Exchange sector indices, we estimate the model with Bayesian methods using Markov Chain Monte Carlo techniques and evaluate its performance in a risk-management context. Results based on Value at Risk and Expected Shortfall indicate improved tail-risk forecasting accuracy, with lower losses than the benchmark for several assets.

***Inference from Imperfection: Rapid Gravitational Wave Parameter Estimation with Data Gaps in LISA using conditional Flow Matching***

Presenter: Ruiting Mao

In complex astrophysical time-series analysis, such as data from the Laser Interferometer Space Antenna (LISA), observational records are inevitably corrupted by missing data (gaps) and instrumental noise. These discontinuities induce non-stationarity, rendering standard likelihood-based Bayesian inference—specifically, Whittle likelihood evaluations paired with Markov Chain Monte Carlo (MCMC)—both computationally prohibitive and highly susceptible to bias. While previous approaches have relied on deterministic signal imputation to accelerate subsequent analysis, the presence of realistic noise artifacts demands a fully probabilistic approach that directly models the uncertainty. To achieve robust Bayesian inference under these pathological data conditions, we propose a novel Simulation-Based Inference (SBI) framework. Rather than relying on an intractable analytic likelihood, we directly approximate the posterior distribution using conditional flow-matching. This generative approach enables rapid, likelihood-free parameter estimation directly from the noisy, incomplete observation space. To overcome the curse of dimensionality inherent in high-cadence LISA data, we extend a non-linear dimension reduction technique to extract a compressed, low-dimensional feature representation of the signal. This step effectively acts as a learned summary statistic, ensuring the tractability and efficiency of the downstream flow-matching estimator in the time domain. Furthermore, we leverage multiresolution analysis by adapting the proposed methodology to the wavelet domain, which significantly accelerates the training of the neural density estimator. Validated on simulated Galactic Binary signals, our SBI model demonstrates rapid and accurate performance. The proposed framework yields marginal and joint posterior distributions that are highly concordant with gold-standard MCMC results derived from idealized, complete data, but at a fraction of the computational cost. This work presents a statistically rigorous and computationally efficient solution for Bayesian parameter estimation in the presence of severe data artefacts.

***A Bayesian latent Gaussian conditional autoregressive copula model for analyzing spatially-varying trends in rainfall***

Presenter: Sayan Bhowmik

Understanding spatial and temporal variability in monsoon rainfall is critical for rainfed agriculture and climate impact assessment, particularly in India, where the majority of farming depends on monsoon precipitation. Annual monsoon rainfall totals are positive, right-skewed, and exhibit notable spatial heterogeneity across India's meteorological subdivisions, motivating flexible modeling frameworks that accommodate non-Gaussian marginals and spatial dependence. We propose a Bayesian hierarchical model that integrates spatially varying

gamma regression for marginal distributions with a Gaussian conditional autoregressive (CAR) copula for spatial dependence. The gamma regression component models monsoon totals across 34 subdivisions of mainland India from 1951 to 2014, allowing total rainfall to vary with time through a log-linear trend and permitting regression coefficients to differ spatially to reflect regional climatic differences. Spatial dependence between subdivisions is captured via a Gaussian CAR copula, with adjacency defined by shared boundaries, enabling flexible joint modeling of non-Gaussian areal data. To ensure copula coherence, we appropriately scale the CAR covariance structure and incorporate CAR or intrinsic CAR priors on the spatially varying coefficients. Bayesian inference is conducted using a Metropolis-within-Gibbs sampler, and missing values are imputed under a missing-at-random assumption within the MCMC procedure. Through simulation studies, we demonstrate that models incorporating spatial dependence at the likelihood and prior levels outperform simpler alternatives in terms of fitting accuracy and uncertainty quantification. Applying the proposed approach to the Indian monsoon dataset reveals spatially heterogeneous trends in rainfall totals, providing insight into potential climate change effects across regions. Our modeling framework is broadly applicable to other non-Gaussian areal datasets requiring both flexible marginal modeling and structured spatial dependence, making it relevant for environmental, epidemiological, and climate science applications.

#### ***A Bayesian nonparametric model for heterogeneous treatment effects, under multiple treatments***

Presenter: Sebastiano Bianchi

We propose a Bayesian nonparametric approach for estimating heterogeneous treatment effects (HTE) in causal inference settings with multiple treatments, including a placebo group and  $K \geq 2$  active treatments. After a suitable transformation of the input data, the problem is formulated as a nonparametric regression model in which the regression function represents the HTE. We place a Pitman–Yor process prior on this function, with a spike-and-slab baseline measure composed of a point mass at the zero function and a Gaussian process (GP) slab. The spike component explicitly accounts for placebo subjects, for whom a null HTE is expected. We emphasize the design stage of causal inference, focusing on matching to improve balance in the distribution of confounders across treatment groups and thereby support causal interpretation. To address the potentially high dimensionality of the confounder space, we summarize covariates via generalized propensity scores (GPS) and incorporate matching directly into the GP covariance structure through a newly proposed stationary kernel. Finally, we derive theoretical guarantees on the optimal number of matches per unit, balancing computational tractability with accurate imputation of missing potential outcomes.

#### ***Parallel computations for Metropolis Markov chains with Picard maps***

Presenter: Sebastiano Grazi

As Bayesian methods are increasingly applied to large-scale and complex models, the need for scalable computational tools becomes even critical. In this talk, I will present parallel algorithms for simulating zeroth-order Metropolis Markov chains based on the Picard map. For Random Walk Metropolis Markov chains targeting log-concave distributions  $\pi$  on  $\mathbb{R}^d$ , our algorithm generates samples close to  $\pi$  in  $O(\sqrt{d})$  parallel iterations with  $O(\sqrt{d})$  processors, therefore speeding up the convergence of the corresponding sequential implementation by a factor  $\sqrt{d}$ . I will demonstrate through numerical experiments that these algorithms are straightforward to implement and provide a practical tool for practitioners seeking to reduce the runtime of standard MCMC methods via parallel computing. This is joint work with Giacomo Zanella.

#### ***Optimal Tree Space Exploration via Lookahead SMC and Delayed Sampling***

Presenter: Seong-Hwan Jun

Posterior inference over tree topologies is challenging because the space of possible trees grows super-exponentially with the number of leaves, making sequential construction highly sensitive to early merge decisions. We develop a sequential Monte Carlo framework for improved tree-space exploration using lookahead merge proposals. At each stage, candidate subtree merges are evaluated using a marginalized lookahead criterion that integrates over the new height increment, allowing the algorithm to account for branch-length uncertainty when selecting topological moves. Once a merge is selected, the height increment is sampled, inducing branch lengths that are propagated as part of the particle state and corrected by standard importance weights. For Gaussian latent variables evolving by Brownian diffusion, fixed-branch likelihoods and proposal scores can be computed efficiently using message passing algorithm. Optional Metropolis-Hastings rejuvenation moves, informed by upward and downward messages, can further update node heights and branch lengths while leaving the current SMC target invariant. The resulting framework combines lookahead topology exploration, delayed height instantiation, and continuous-parameter updates for Bayesian inference in tree-structured latent variable models.

#### ***Bayesian Modeling of Issue-Specific Legislative Preferences Using Euclidean Ideal Points***

Presenter: Seungju Lee

Understanding legislative voting behavior requires models that accommodate multidimensional conflict and heterogeneous issue structures, especially in multi-party settings. We present a Bayesian framework that integrates Euclidean ideal points with bill-level covariates to quantify how issue content shapes legislators' voting deviations. The approach begins by modeling roll-call data with a Bayesian Latent Space Item Response Model (LSIRM), which embeds legislators and bills jointly in a Euclidean space and produces posterior distributions of legislator–bill distances. To explain how specific issues contribute to this structure, the model incorporates bill information—continuous topic proportions or categorical indicators—through a Bayesian beta regression applied to LSIRM distances after a transformation to the (0,1) interval. This yields Bayesian posterior distributions for issue-specific coefficients for every legislator, enabling coherent comparison of issue-level polarization, intra-party cohesion, and cross-party alignment. Applied to the 17th Korean Congress, the model reveals substantial variation in how legislators diverge across issues, including domains where major parties overlap and others where smaller parties align unexpectedly with larger blocs. The framework offers a unified Bayesian approach for studying issue-structured legislative behavior in multiparty systems.

### ***Bayesian Clustering and Structured Selection for Correlated High-Dimensional Predictors***

Presenter: Shanta Ghosh

Variable selection has become an essential element of statistical modeling to yield parsimonious models while maintaining high prediction accuracy. However, in high-dimensional data, it can be challenging in the presence of collinearity among covariates. Classical variable selection methods often fail when predictors exhibit complex dependence structures or when the functional form of the predictor–outcome relationship is unknown. To address these challenges, we propose a Bayesian Clustering and Structured Selection (BCSS) framework that jointly learns covariate clusters and performs variable selection under both linear and nonlinear settings. We perform model-based clustering using a Dirichlet Process prior without assuming the correlation structure to be known, grouping covariates that share similar latent structure together. Within this clustering framework, BCSS performs variable selection at both the cluster and predictor levels, improving stability by treating correlated predictors as units rather than individually. Posterior inference is carried out via Markov chain Monte Carlo (MCMC) sampling. We further extend BCSS to a generalized additive model (GAM) setting, by incorporating B-spline basis expansions to capture nonlinear predictor effects. This extension preserves the original framework while enabling flexible estimation of nonlinear functional relationships. A distance correlation-based screening step is used prior to model fitting to reduce the predictor space in high-dimensional settings, and posterior inclusion probabilities are used for final predictor selection in both the linear and GAM settings. Through simulation studies across multiple sample sizes and predictor correlation structures, we demonstrate that BCSS achieves improved selection accuracy and false discovery rate control compared to other standard methods. The proposed methods are applied to cancer biomarker datasets to identify molecular predictors associated with survival and disease progression, as well as to SDOH data to uncover socioeconomic and environmental determinants of health outcomes.

### ***Bayesian Synthetic Control with Spatio-temporal Latent Factor Models***

Presenter: Shigeaki Nishino

Synthetic control methods are widely used for causal inference in observational panel data. However, real-world outcomes often exhibit spatial dependence—such as correlations across neighboring regions or countries—which existing synthetic control methods ignore. This omission can lead to biased counterfactual estimates when spatial structure is present. We propose a Bayesian synthetic control framework that explicitly incorporates spatial dependence through a spatial extension of the multiplicative gamma process shrinkage prior. By placing a spatially structured prior on the factor loadings, our model captures cross-sectional dependence in a latent factor space while automatically shrinking unnecessary factors. This enables flexible modeling of spatio-temporal correlation without requiring the number of factors to be known a priori. Through simulation studies, we show that the proposed method improves treatment effect estimation when spatial correlation exists. Even in the absence of spatial dependence, the method performs comparably to or better than existing Bayesian and frequentist synthetic control approaches. We also demonstrate the effectiveness of the proposed method through real data analysis.

### ***Monitoring Adverse Events Through Bayesian Nonparametric Clustering Across Studies***

Presenter: Shijie Yuan

Ensuring patient safety in clinical trials is essential due to ethical and regulatory considerations. Adverse event (AE) monitoring plays a crucial role in detecting potential safety signals and mitigating risks associated with

investigational treatments. One common approach to AE evaluation involves comparing the incidence rate in an ongoing trial to a predefined background rate to determine whether a specific AE occurs more frequently than expected. However, setting an appropriate background rate is challenging due to differences in study populations (e.g., prognosis, age, sex, race) between external studies and the current trial, as well as inconsistencies in data collection, where external data may be available at the patient level, covariate-specific summaries, or only as aggregate study-level statistics.

To address these challenges, we introduce a model-based Bayesian nonparametric approach to clustering that is designed to define background AE rates based on monitoring multiple studies with heterogeneous patient populations and varying levels of granularity in the available data. This framework effectively incorporates external data while accommodating varying levels of covariate granularity, ranging from individual patient-level details to aggregated study-level summaries. The experimental units in the proposed model are patient cohorts defined by available covariate characteristics. They serve as the fundamental element of analysis. Our approach employs a covariate-dependent product partition model (PPMx) to facilitate data integration across the diverse external studies. The inference model is set up to favor co-clustering of experimental units with similar covariates. Additionally, we introduce a pairwise similarity measure to quantify the relatedness between experimental units based on observed covariates. Our framework is not only capable of handling missing data but also fully incorporating clinical experts' domain knowledge by allowing them to assign appropriate weights to different covariates within the pairwise similarity measure.

Through application examples, we demonstrate the effectiveness of our framework in detecting safety signals under various trial scenarios. The flexibility of the model in handling different levels of data resolution makes it particularly suitable for integrating external data sources, especially when patient-level data from external studies are unavailable.

#### ***Location-Scale Calibration for Generalized Posterior***

Presenter: Shu Tamano

General Bayesian updating replaces the likelihood with a loss scaled by a learning rate, but posterior uncertainty can depend sharply on that scale. We propose a simple post-processing that aligns generalized posterior draws with their asymptotic target, yielding uncertainty quantification that is invariant to the learning rate. We prove total-variation convergence for generalized posteriors with an effective sample size, allowing sample-size dependent priors, non-i.i.d. observations, and convex penalties under model misspecification. Within this framework, we justify and extend the open-faced sandwich adjustment (Shaby, 2014), provide general theoretical guarantees for its use within generalized Bayes, and extend it from covariance rescaling to a location-scale calibration whose draws converge in total variation to the target for any learning rate. In empirical illustration, calibrated draws maintain stable coverage, interval width, and bias over orders of magnitude in the learning rate and closely track frequentist benchmarks, whereas uncalibrated posteriors vary markedly.

#### ***A Weighted Likelihood Approach for Bayesian Causal Inference***

Presenter: Shunichiro Orihara

In causal inference, weighting approaches using propensity scores are commonly employed to adjust for confounding. For instance, weighted likelihood methods for Cox proportional hazards models are used to estimate hazard ratios as causal effects. In the Bayesian context, however, because the propensity scores are included in the outcome likelihood, the posterior for the treatment model may be affected by outcome information. This induces a violation of a fundamental property of propensity scores, known as the feedback problem. In this study, we propose a novel approach using entropic tilting to overcome the feedback problem. Specifically, the weights used in the weighted likelihood for the outcome model and those for treatment effect estimation are sampled separately from the propensity scores, and the information from the propensity scores is incorporated only after the initial sampling steps. This process effectively mitigates the feedback issue. We demonstrate the validity of our method through simulation studies comparing the proposed method with existing approaches.

#### ***Bayes Meets Diffusion and Information Theory***

Presenter: Simon Pedro Galeano Munoz

TBA

#### ***Cascading failure in systems with multiple component types***

Presenter: Simon Wilson

Cascading failure in a system of components refers to the situation where there is a rapid sequence of component failures that cause system failure. Usually this is the result of a component failure causing additional stress on the remaining working components. It can be found in systems such as power supply networks. Such systems

are often characterised by infrequent (but often disastrous) failure events, so predictions of system failure time are both important and rely extensively on prior opinion. In this talk we model this process for a system that consists of more than one type of component as a continuous time Markov chain, and use the theory of survival signatures to derive the system failure time as a mixture of phase-type distributions. We develop an inference implementation for these mixture models through MCMC and apply to data.

### ***Exploring Bayesian Opportunities in Cycling Aerodynamics***

Presenter: Sotirios Pestrivas

Bayesian methods uptake in sports performance analysis is increasing due to their ability to quantify uncertainty and incorporate prior knowledge. However, sports aerodynamics (such as cycling) remains underdeveloped; its costly test data, small performance margins, and high-stakes decisions are ideally suited for Bayesian analysis. This is particularly true for wind tunnel (WT) testing, which is considered the gold standard but typically collapses dynamic time-series data into a single mean, losing valuable information. This work explores novel Bayesian frameworks to leverage this discarded information. Part one involves a framework to model test data by segmenting the rich time series into whole crank cycles, allowing for more robust, uncertainty-quantified estimates of parameter effects. Part two explores a Bayesian non-linear varying-coefficient model where both the non-aerodynamic inertial force and the aerodynamic drag (CdA) are modelled simultaneously as flexible cyclic spline functions of the crank angle. This structure allows for the separation of force components, isolating the true intra-cycle mean CdA( $\theta$ ) from the total measured force and providing insight into within-cycle variation for dynamic tests. Investigating these methods provides opportunities for deeper insight. The hierarchical approach can yield more robust, uncertainty-quantified estimates of parameter effects, while the non-linear model offers a full posterior distribution for mean CdA( $\theta$ ). This allows for the identification of where in the pedal stroke aerodynamic changes occur and provides a clear pathway for integration with complementary test data such as from computational fluid dynamics. This exploration of Bayesian methods offers deeper insight than traditional mean-based analysis, establishing a statistical structure to support decision-making in sports aerodynamics.

### ***Sources of Uncertainty-Variational Autoencoders***

Presenter: Soufiane Atouani

We apply a recently proposed Bayesian paradigm, named Sources of Uncertainty (SoU) Bayesian inference, to Variational Autoencoders (VAEs). The SoU formulation allows assigning confidence parameters to distinct sources of uncertainty, e.g., latent variables, encoder weights, decoder weights. By imposing additional structure on these components, and tuning their associated confidences, we show that the SoU-VAE can induce sparse network weights and achieve more tailored disentanglement of the latent space than existing approaches.

### ***Bernstein-von Mises for Missing Data***

Presenter: Stefan Franssen

Missing data remains a major challenge in statistical modeling, and while Bayesian approaches are widely used, there is a distinct lack of theoretical guarantees. We address this gap by proposing a semiparametric Bayesian framework for regression models with covariates missing at random. Our approach combines nonparametric density estimation for the covariate distribution with a parametric prior on regression parameters. Under this setup, we establish frequentist guarantees by proving a Bernstein-von Mises theorem for the regression parameters. This is a joint work with Judith Rousseau

### ***Flexible and scalable Bayesian modeling of spatiotemporal event data with low-rank Neyman-Scott tensor processes***

Presenter: Sulagna Ghosh

Consider data of the form “news source  $s$  reports that actor  $i$  took action  $a$  toward actor  $j$  at time  $t$  in location  $\ell$ ” where the news source  $s \in [S]$  and actors  $i, j \in [N]$  are categorical, while time  $t$  and location  $\ell$  are continuous. Such data can be modeled as a marked spatiotemporal point process, where each event  $e_n := (t_n, \ell_n, m_n)$  occurs at a time  $t_n$  and location  $\ell_n$  and as a discrete mark which is multivariate  $m_n := (s_n, i_n, j_n, a_n)$ . Such data can also be viewed as forming a four-mode “event tensor”  $Y$ , where the entry  $Y_{s,i,j,a} = \{e_n : m_n = (s, i, j, a)\}$  is a spatiotemporal point process specific to a particular value of the mark. This paper considers a family of models we call Neyman-Scott tensor processes (NSTPs) which assume that each  $Y_{s,i,j,a} \sim \text{NSP}(\lambda_{s,i,j,a})$  comes from a Neyman-Scott process with rate function  $\lambda_{s,i,j,a}(t, \ell)$  over time and space, and that the tensor of rate functions  $\Lambda$  has low-rank structure. To fit such models to large-scale event data, we develop a gradient-based amortized variational inference algorithm that uses another NSTP as the variational family and a recognition network based on self-attention. We show that on real news data of international relations, the inferred latent events that comprise each NSP’s rate function are interpretable as “major stories” which generate related events over long



periods. While tailored to international relations data, our work is broadly applicable to marked spatiotemporal event data and generally builds upon recent advances in neural point processes, probabilistic non-negative tensor decomposition, and scalable approximate Bayesian inference.

#### ***A Forward-Only Scheme for Online Learning of Proposal Distributions in Particle Filters***

Presenter: Sylvain Procope-Mamert

Particle filters are powerful algorithms used to sample from a sequence of distributions. It is useful notably for Bayesian inference with different types of models and real data applications. In particular, when we try to recover a hidden signal from sequentially produced data with state-space models, the canonically defined proposals known as the bootstrap particle filter are rarely well-behaved and need extra work to be turned into useful sampling algorithms. Previous works on iterated methods for the automated construction of sequential Monte Carlo proposals, which were based on a backward scheme, have shown how to gradually improve proposals to reach a global optimality criterion, but they require a good initial proposal and cannot be used online.

We introduce a new online approach for constructing proposal distributions in particle filters using a forward scheme. Our method progressively incorporates future observations to refine proposals. This is in contrast to backward-scheme algorithms that require access to the entire dataset, which leverage all future observations through backward recursion. On simulated and real data applications, this method offers significant computational robustness compared to backward methods.

#### ***Empirical Bayes 1-bit matrix completion***

Presenter: Takeru Matsuda

The problem of predicting unobserved entries of a binary data matrix from observed entries is called the 1-bit matrix completion. We develop an empirical Bayes method for 1-bit matrix completion that utilizes a low-rank structure like the multidimensional item response theory. The proposed method is motivated by an empirical Bayes estimator of a normal mean matrix by Efron and Morris, which is a matrix generalization of the James–Stein estimator and shrinks the singular values towards zero. Simulation results and application to real data are presented

#### ***Smoothing the posterior bootstrap with greedy empirical Bayesian trees***

Presenter: Taole Sha

Conventional Bayesian inference faces challenges regarding model misspecification and computational efficiency, especially in modern settings where datasets are increasingly large and high-dimensional. Within the nonparametric learning framework, the posterior bootstrap has become a viable alternative to posterior sampling due to its robustness and scalability. However, most existing methods are built on the Dirichlet process and its variants, which are known to produce posterior samples of the underlying distribution which are discrete. This discrete nature prohibits the inference of parameters which require the underlying distribution to have a density, and furthermore impede the existence of the posterior predictive density. In this work, we propose the partition posterior bootstrap (PPB): a smooth nonparametric learning method which relies on a data-adaptive partition of the sample space parametrized by a binary tree. Conditioned on this partition, posterior sampling involves a scalable posterior bootstrap which does not require Markov chain Monte Carlo. A key advantage of our method is that the tree partition can be learnt in a greedy empirical Bayesian manner from the marginal likelihood, which is both highly computationally efficient and theoretically tractable. Our proposed method can thus be regarded as a default and efficient nonparametric learning method for posterior inference and prediction when smoothness of the underlying distribution is required, especially for high-dimensional datasets. We demonstrate its excellent performance in both simulated and real-world datasets.

#### ***On the Degenerate Behaviour of Wide Variational Neural Networks***

Presenter: Thibault Randrianarisoa

We study Gaussian variational approximations to the parameter posterior of Bayesian neural networks with Gaussian priors and likelihoods. For sufficiently wide architectures and a broad class of activation functions, we show that the variational objective admits a unique optimum. Exploiting the symmetries of the network and the prior, we prove that the corresponding parameter variational posterior is itself symmetric, constraining its structure. We translate these parameter-space results into function space and establish a fundamental limitation of the resulting variational predictive posterior. For odd activation functions, we prove that the variational predictive posterior has exactly zero mean for sufficiently wide networks, making it incapable of fitting the data. As the width grows further it converges in distribution to the prior predictive distribution, i.e., the neural network Gaussian process (NNGP). For more general activation functions, we show that the mean-field Gaussian variational posterior over parameters induces a predictive posterior whose covariance converges to that



of the NNGP, while its expectation converges (up to a multiplicative factor) to a fixed prior-dependent function. Taken together, these results expose intrinsic shortcomings of Gaussian variational inference in the wide-network regime and demonstrate that the Gaussian variational posterior can fail to meaningfully incorporate statistical information from the data.

### ***Characterization of climate and air pollution patterns in Medellín using Extreme Value Theory***

Presenter: Tomás Rodríguez Taborda

The city of Medellín, Colombia, and its neighbouring municipalities in the Aburrá Valley have experienced rapid economic and demographic growth over the last two decades. This has led to increased emissions, urban expansion into high-risk areas, and growing exposure to critical episodes of air pollution and heavy rainfall. Although the region benefits from a robust weather early-warning and monitoring system (SIATA-AMVA), spatial coverage remains limited, and the current tools employed for weather forecasting are insufficient to characterise the more critical extreme events for climate resilience and public health. Many environmental and societal risks are driven not by average conditions, but by events in the tails of distributions of weather events such as precipitation and air pollution. For events at the tails of the distribution, standard tools such as quantile regression become unstable and unreliable, making it difficult to describe and anticipate truly extreme events. Extreme value theory (EVT) provides a principled framework for modelling the behaviour of rare but high-impact outcomes. In this work, we explore a Bayesian extreme-value modelling framework for key environmental variables, including precipitation and air-quality indicators, in Medellín and its metropolitan area. By combining EVT models with prior expert knowledge, we obtain posterior distributions for quantities of interest that explicitly account for uncertainty and the scarcity of data in the tails. This Bayesian EVT perspective aims to provide a more rigorous and informative basis for understanding how climate change and urban growth reshape the distribution of extremes, and ultimately to support more proactive, evidence-based decision-making in risk management for the Aburrá Valley in Medellín, Colombia.

### ***Symmetrization of Martingale Posterior Distributions***

Presenter: Torey Hilbert

The martingale posterior framework, recently proposed by Fong et al., is based on a sequence of one step ahead predictive distributions. It leads to computationally efficient inference in parametric and nonparametric settings. The predictive distributions implicitly provide a joint model for an infinite sequence of data. The observed data, typically arbitrarily considered to be  $Y_1$  through  $Y_n$  in this infinite sequence, form the beginning of the sequence and the tail of the sequence is missing. Filling in the remainder of the sequence allows one to summarize  $Y_1$  through  $Y_\infty$  (or through  $Y_N$ , for large  $N$  in practice). A typical summary, such as the mean, is regarded as a parameter. Generation of a large number of sequences gives access to the posterior distribution of the parameter. In cases where the martingale posterior model does not match a de Finetti model, the joint distribution over the  $Y$ 's is not exchangeable, and so the indices  $\{1, \dots, n\}$  of the data affect the analysis. We investigate methods of symmetrizing inference in these models. While one can partially resolve this by averaging conclusions over permutations of the data, the inference no longer corresponds to using the posterior distribution of a joint distribution. We find that in some non-exchangeable models, re-indexing the observed data to  $\{a_1 < \dots < a_n\}$  and sending  $a_1$  to infinity is analytically tractable, and we recover classical Bayesian models with known priors. We also investigate using weighted permutations associate the non-exchangeable model with an exchangeable one.

### ***Bayesian Sample Size Determination for Multilevel Models***

Presenter: Ulrich Lösenner

When Bayesian sequential designs are not feasible or economic, Bayesian sample size determination (SSD) is needed to plan an efficient experiment. Unfortunately, software for Bayesian SSD is still limited to simpler models such as ANOVA and t-test, which cannot handle nested observations. This project aims to introduce the R package BayesSSD, an easy-to-use open access software to perform Bayesian SSD for multilevel models in the context of cluster-randomized and longitudinal trials. Attrition can be taken into account and is modeled using various survival functions. The SSD algorithm relies on simulation and employs the Approximate Adjusted Fractional Bayes Factor, which is designed to be accessible to applied researchers without a strong background in Bayesian hypothesis evaluation. Its straightforward computation and applicability to models fitted using maximum-likelihood makes it especially suitable for simulation-based approaches. The user-friendly software package contributes to bridging a critical gap between statistical theory and applied research practice, ultimately contributing to more rigorously and efficiently designed studies.

### ***Empirical PAC-Bayes bounds for Markov chains***

Presenter: Vahe Karagulyan

The core of generalization theory was developed for independent observations. Some PAC and PAC-Bayes bounds are available for data that exhibit a temporal dependence. However, there are constants in these bounds that depend on properties of the data-generating process: mixing coefficients, mixing time, spectral gap... Such constants are unknown in practice. In this paper, we prove a new PAC-Bayes bound for Markov chains. This bound depends on a quantity called the *pseudo-spectral gap*,  $\gamma_{ps}$ . The main novelty is that we can provide an empirical bound on  $\gamma_{ps}$  when the state space is finite. Thus, we obtain the first fully empirical PAC-Bayes bound for Markov chains. This extends beyond the finite case, although this requires additional assumptions. On simulated experiments, the empirical version of the bound is essentially as tight as the one that depends on  $\gamma_{ps}$ .

#### ***hierarchical bayesian analysis for Japanese sake export***

Presenter: Wakuo Saito

This study investigates the determinants of luxury positioning in export markets, using Japanese sake as a case study. By analyzing export data segmented by port and firm size using hierarchical Bayesian estimation, we examine how cultural certification, macroeconomic factors, and global disruptions—specifically the UNESCO designation of Japanese cuisine and the COVID-19 pandemic—have shaped export trends. Findings reveal clear distinctions between large and small-to-medium sake breweries. Although large firms dominate in volume and respond strongly to economic conditions, smaller breweries increasingly drive premiumization in key markets like China and the U.S. These findings suggest that the luxury status of exported sake is best identified by concurrent increases in unit value and volume. Strategic and policy implications drive the need for differentiated export support, digital marketing for smaller firms, and infrastructural investment to sustain quality in premium segments. This study contributes to the literature on retail internationalization and luxury consumption by highlighting the role of firm scale in shaping product perception across global markets.

#### ***Associative Brand Recall and Purchase***

Presenter: Walter Zhang

Consumers recall brands sequentially, and those ordered sets encode associative signals that affect subsequent purchasing behaviors. We propose a Bayesian structural model that links the recall sequence to an associative model that allows for asymmetric pairwise associations across brands. The associative model leads to a network structure with a doubly intractable posterior; we address this with Russian Roulette sampling. Posterior recall probabilities serve as unconditional choice probabilities in a rational-inattention multinomial logit framework, which bridges memory and demand. In our application, we use survey recall data for contact lens solution from 3,749 households and their six months of purchase history. We find that the model reveals substantial heterogeneity in brand associations and improves spending forecasts relative to benchmarks based solely on past behavior and marketing controls. Counterfactual simulations demonstrate how brand-level advertising shifts both recall probabilities and predicted sales.

#### ***Flexible and Scalable Bayesian Modeling Of Spatio-Temporal Hawkes Processes***

Presenter: Wenqing Liu

Existing spatio-temporal Hawkes process models typically rely on either parametric or semiparametric assumptions, limiting the model's ability to capture complex endogenous and exogenous event dynamics. We propose a fully Bayesian nonparametric framework for spatio-temporal Hawkes processes using additive Gaussian processes for the prior distributions on the background rate and the triggering kernel. This nonparametric formulation enhances modeling flexibility, while the additive structure improves the numerical stability of the inference procedure on the whole spatio-temporal domain. To address scalability, we develop a sparse variational inference scheme based on the Gaussian variational family. Simulation studies demonstrate that the proposed method can recover background and triggering structures with competitive or better mean squared error compared to existing alternatives. When applied to real-world datasets, it achieves higher held-out log-likelihoods and reveals interpretable spatio-temporal structures of the self-excitation mechanism. Overall, the framework provides a flexible, scalable, and uncertainty-aware approach for modeling complex excitation patterns in spatio-temporal event data.

#### ***Moment Martingale Posterior for Semiparametric Predictive Bayes***

Presenter: Yiu Yin Yung

The predictive Bayesian view involves eliciting a sequence of one-step-ahead predictive distributions in lieu of specifying a likelihood function and prior distribution. Recent methods have leveraged predictive distributions which are either nonparametric or parametric, but not a combination of the two. This paper introduces a semi-

parametric martingale posterior which utilizes a predictive distribution that is a mixture of a parametric and nonparametric component. The semiparametric nature of the predictive allows for regularization of the nonparametric component when the sample size is small, and robustness to model misspecification of the parametric component when the sample size is large. We call this approach the moment martingale posterior, as the core of our proposed methodology is to utilize the method of moments as the vehicle for tying the nonparametric and parametric components together. In particular, the predictives are constructed so that the moments are martingales, which allows us to verify convergence under predictive resampling. A key contribution of this work is a novel procedure based on the energy score to optimally weigh between the parametric and nonparametric components, which has desirable asymptotic properties. The effectiveness of the proposed approach is demonstrated through simulations and a real world example.

### ***Bayesian nonparametric Poisson modeling reconstructs 3D structure for multi-slice spatially resolved transcriptomic data***

Presenter: Yongyi Luo

Spatially resolved transcriptomics (SRT) technologies provide unprecedented insight into tissue architecture by profiling gene expression while preserving spatial context. When SRT data are collected from parallel tissue slices at different depths within an organ, a natural objective is to reconstruct three-dimensional spatial domains. However, integrating multi-slice, high-dimensional SRT data presents several challenges, including slice-specific batch effects, the high dimensionality of gene expression measurements, and the need to ensure spatial contiguity in the resulting 3D domains. To address these issues, we introduce a unified Bayesian nonparametric Poisson factor analysis framework that simultaneously estimates slice-wise batch effects and projects high-dimensional expression profiles into a low-dimensional latent space. We further impose a Markov random field constraint within a Bayesian nonparametric clustering model to delineate 3D spatial domains while automatically determining the optimal number of clusters. The efficacy of our approach is demonstrated through realistic simulations and applications to 12 SRT datasets spanning three experimental platforms. Our method not only surpasses state-of-the-art approaches in clustering accuracy and model selection for the number of clusters, but also yields novel biological insights into cellular compartmentalization within tissue.

### ***Repulsive g-Priors for Regression Mixtures and Extension to Information Geometric Repulsive Priors***

Presenter: Yuta Hayashida

Mixture regression models are powerful tools for capturing heterogeneous relationships, yet they often suffer from cluster overestimation when separability is weak. While repulsive priors have successfully improved parsimony in density estimation, their direct extension to regression is non-trivial because separation must respect the predictive geometry induced by covariates. To address this, we propose a repulsive g-prior for linear regression mixtures that enforces separation in the Mahalanobis metric, penalizing components that are indistinguishable in the predictive mean space. This construction is advantageous as it preserves conjugacy-like updates, enabling efficient blocked-collapsed Gibbs sampling, and provides theoretical guarantees such as posterior contraction rates. In this presentation, we demonstrate the efficacy of this framework and discuss its extension to general construction of repulsive priors based on information geometry. Specifically, we explore a generalized repulsive priors mechanism based on the Fisher information matrix under generalized linear models. This approach serves as a natural generalization of the linear case, aiming to define geometry-aware separation for broader distributions.

### ***Couplings of stereographic MCMC algorithms***

Presenter: Zhihao Wang

Coupling methods have a long history in the analysis of Markov chains and of Markov chain Monte Carlo (MCMC) algorithms. More recently, they have been employed in designing non-asymptotic diagnostics and unbiased estimators of stationary expectations. Their practical efficiency relies on the speed at which pairs of Markov chains meet (i.e. coincide exactly). We consider novel coupling strategies for stereographic MCMC samplers, which are gradient-free and able to deal with large-dimensional, heavy-tailed target distributions. Leveraging the sphere's geometry and existing results on stereographic MCMC, we study the scaling of the meeting times with respect to the dimension. This work confirms the potential of stereographic MCMC as a scalable baseline for high-dimensional MCMC.

### ***Empirical Bayes Shrinkage of Functional Effects, with Application to Analysis of Dynamic eQTLs***

Presenter: Ziang Zhang

We introduce functional adaptive shrinkage (FASH), an empirical Bayes method for joint analysis of observation units where each unit measures an effect function at several values of a continuous condition variable. FASH is

motivated by dynamic expression quantitative trait locus (eQTL) studies which seek to characterize how genetic effects on gene expressions vary over time or other continuous conditions. FASH integrates a broad family of Gaussian processes defined through linear differential operators into an empirical Bayes framework. This framework provides improved estimation of effect functions and principled measures for large-scale hypothesis testing, including false discovery rates and false sign rates. With the aim of providing conservative inferences, we develop a Bayes-factor-based adjustment to the FASH prior that can be incorporated into any of the empirical Bayes shrinkage procedures. We illustrate FASH by reanalyzing dynamic eQTL data on cardiomyocyte differentiation from induced pluripotent stem cells. FASH identified novel dynamic eQTLs, revealed diverse temporal effect patterns, and provided improved power compared with the original analysis. Beyond dynamic eQTLs, FASH provides a flexible statistical framework for joint analysis of functional effects, with possible applications beyond genomics. The methods are implemented as an R package available at <https://github.com/stephenslab/fashr>.

***Bayesian matrix-variate envelope model for clustering non-linear multivariate longitudinal trajectories.***

Presenter: Zihang Lu

Clustering multivariate longitudinal biomarker trajectories in medical research presents several methodological challenges, including nonlinearity, high dependence among biomarkers, and the need to account for both shared and biomarker-specific variation over time. To this end, we propose a novel Bayesian mixture matrix-variate envelope model for clustering nonlinear multivariate longitudinal trajectories. The proposed Bayesian model incorporates B-spline basis functions to capture trajectory shapes, while decomposing the joint distribution of subject-specific random effects into material and immaterial components. This two-way dimension reduction removes irrelevant variation across biomarkers and time points, and therefore enhances clustering accuracy. We develop a Metropolis-within-Gibbs sampler with efficient reparameterization to improve mixing and computational efficiency. Simulation studies were performed to demonstrate the performance of the proposed method to recover complex cluster structures under varying noise conditions. The newly developed approach is applied to six hormone biomarkers from the Penn Ovarian Aging Study, a longitudinal cohort on menopausal transition, where it identifies three clinically interpretable clusters characterized by distinct patterns of hormonal change and variability. This is joint work with Bei Jiang from the University of Alberta.

***A Longitudinal Multi-Domain Latent Structure Model for Domain-specific Disease Progression***

Presenter: Zijian Ye

Neurodegenerative diseases such as Parkinson's disease (PD) typically progress over many years, with patients experiencing heterogeneous declines in motor, cognitive, and non-motor functions. Conventional progression models reduce all measures into a single disease state, which fail to capture this multidimensionality and thus limits their utility for personalized prognoses or interventions. In this article, we propose a longitudinal multi-domain latent structure model to characterize domain-specific disease progression in PD. Our model partitions multivariate PD-related surrogate markers into distinct, clinically meaningful domains (e.g., motor, cognitive, autonomic), and captures the underlying progression within each domain through the dynamics of latent disease states inferred from longitudinal data. In addition, we treat the (unobserved) time at which each disease domain progresses between visits as an interval-censored event, and associate the event times with risk factors by a Cox proportional hazards model. We infer the domain-specific event times from longitudinal markers that have distinct distributions before and after disease progression. An efficient Markov chain Monte Carlo (MCMC) algorithm is developed, employing a two-stage Poisson data augmentation strategy together with adaptive rejection sampling in Bayesian estimation. We establish theoretical results on recovering the marker-domain partition structure. We apply our methods to the Parkinson's Progression Markers Initiative (PPMI) study and reveal five heterogeneous clinical domains, each reflecting a specific symptom cluster. The model captures individualized, domain-specific progression trajectories and identifies significant associations with multiple risk factors (e.g., gender, education level, putamen area atrophy).

***Improving Frailty in Clinical Trials with Bayesian Kumaraswamy Regression (KR) and Bayesian Model Averaging (BMA)***

Presenter: Ziqian Sun

Frailty is a heterogeneous and multifactorial condition that does not have an agreed definition. This makes it expensive and time consuming to test new interventions in clinical trials. Different existing frailty scores are all Bounded Outcome Scores (BOS) with upper and lower bounds. We proposed to use Kumaraswamy Regression (KR) and Exponentiated Kumaraswamy Regression (EKR) with a Bayesian approach to model them from a set of health-related predictors. We found that the predictability on frailty of the KR and EKR models differs when

using different location-scale parameterisations and they are compared with other modelling approaches including Linear Regression, Beta Regression and Gamma Generalised Linear Model (GLM), regarding the Information Criteria (IC) and predictive errors. Standard statistical approaches choose one single best model, but they ignore model uncertainty. To improve this, we use Bayesian Model Averaging (BMA), combining multiple models together by giving each model a weight based on how well they are, to reduce model uncertainty. Results showed that combining models with BMA provides more accurate and precise frailty predictions than any single models. By knowing the weights assigned to each model, we can help the clinicians to have a better understanding of the models' predictability and the relative importance of the health-related predictors. In this way, we can help them accelerate the process of testing a new intervention for frailty with reduced time and cost.

***Multi-View Oriented GPLVM: Expressiveness and Efficiency***

Presenter: zi yang

The multi-view Gaussian process latent variable model (MV-GPLVM) aims to learn a unified representation from multi-view data but is hindered by challenges such as limited kernel expressiveness and low computational efficiency. To overcome these issues, we first introduce a new duality between the spectral density and the kernel function. By modeling the spectral density with a bivariate Gaussian mixture, we then derive a generic and expressive kernel termed Next-Gen Spectral Mixture (NG-SM) for MV-GPLVMs. To address the inherent computational inefficiency of the NG-SM kernel, we design a new form of random Fourier feature approximation. Combined with a tailored reparameterization trick, this approximation enables scalable variational inference for both the model and the unified latent representations. Numerical evaluations across a diverse range of multi-view datasets demonstrate that our proposed method consistently outperforms state-of-the-art models in learning meaningful latent representations.

## Thursday 2 July

### Multitrack Sessions 8: 9:00–10:30

#### Calibrated Bayes: Model Design and Adaptation Under Limited Resources (Room 1101)

##### *A Bayesian Nonparametric Approach to Adapting Model Size During Training*

Presenter: Guiomar Pescador Barrios

Many machine learning models require setting a parameter that controls their size before training, e.g. number of neurons in DNNs or inducing points in GPs. Setting this parameter is important as a model that is too small will underperform, while one that is too large will waste computational resources. This leads to the question: how big is big enough? In this talk, we will discuss recent work exploring how model capacity can be adjusted during training to achieve near-exact performance without unnecessary computation, and how this connects to broader questions of model selection, learning dynamics, and the role of approximation in inference.

##### *Model Selection for Over-Parameterized Models*

Presenter: Maurizio Filippone

The impressive success of Deep Learning (DL) models in obtaining state-of-the-art predictive performance in several tasks is providing strong signs that these models are here to stay and that they will replace a number of classical alternatives. DL models, which tackle misspecification through overparameterization, pose a number of statistical and computational challenges. In this talk, I will discuss the need for a sound statistical treatment of DL models, and the difficulties associated with this. After discussing some of the recent trends in this literature, including prior specification, approximate inference and scaling laws, I will focus on recent works on model selection for DL and its importance in light of Occam's razor.

##### *Online & Predictively-Oriented Model Fusion*

Presenter: Dan Waxman

We consider the problem of online model fusion; i.e., given an ensemble of  $K$  probabilistic models  $\mathcal{M}_1, \dots, \mathcal{M}_K$ , we must find a fused model  $\mathcal{M}_*$  that combines the available models into a single probabilistic prediction. We further restrict ourselves to the online or streaming setting, whereby the fusion strategy must be learned on-the-fly, as new data arrives. The typical approach in this setting would be the recursive form of Bayesian model averaging (BMA); BMA, however, typically collapses extremely quickly to a single model of the ensemble. We discuss the reasons for this collapse, which is intimately related to the more general problem of Bayesian inference over misspecified models. We then take a predictively-oriented approach, and show that this mathematically reduces to the problem of online portfolio selection, providing robust regret bounds w.r.t. a proper scoring rule. Finally, we discuss the scenario where the models  $\mathcal{M}_k$  are state-space models, where our approach generalizes the classical bank of filters and interacting multiple model filters.

Discussant: Youssef Marzouk

#### Bayesian Machine Learning for Causal Inference (Room 1102)

##### *Causal mediation analysis for longitudinal and survival data in continuous time using Bayesian non-parametric joint models*

Presenter: Saurabh Bhandari

Observational cohort data is an important source of information for understanding the causal effects of treatments on survival and the degree to which these effects are mediated through changes in disease related risk factors. However, these analyses are often complicated by irregular data collection intervals and the presence of longitudinal confounders and mediators. We propose a causal mediation framework that jointly models longitudinal exposures, confounders, mediators, and time-to-event outcomes as continuous functions of age. This framework for longitudinal covariate trajectories enables statistical inference even at ages where the subject's covariate measurements are unavailable. The observed data distribution in our framework is modeled using an enriched Dirichlet process mixture (EDPM) model. Using data from the Atherosclerosis Risk in Communities cohort study, we apply our methods to assess how medication—prescribed to target cardiovascular disease (CVD) risk factors—affects the time-to-CVD death.

##### *Type 2 Tobit Sample Selection Models with Bayesian Additive Regression Trees*

Presenter: Eoghan O'Neill

This paper introduces Type 2 Tobit Bayesian Additive Regression Trees (TOBART-2). BART can produce



accurate individual-specific treatment effect estimates. However, in practice estimates are often biased by sample selection. We extend the Type 2 Tobit sample selection model to account for nonlinearities and model uncertainty by including sums of trees in both the selection and outcome equations. A Dirichlet Process Mixture distribution for the error terms allows for departure from the assumption of bivariate normally distributed errors. Soft trees and a Dirichlet prior on splitting probabilities improve modeling of smooth and sparse data generating processes. We include a simulation study and an application to the RAND Health Insurance Experiment dataset.

### ***Bayesian Machine Learning for Heterogeneous Treatment Effect Estimation and Policy Estimation***

Presenter: Antonio Linero

I will discuss various topics at the intersection of machine learning, Bayesian methods, and the estimation of causal effects, focusing on the estimation of conditional average treatment effects (CATEs). I make the following claims:

1. Judicious, direct, regularization of the treatment effect heterogeneity is essential to get low-RMSE estimates of the CATE, and in high-noise settings this can be more important than specifying the functional form of the model correctly.
2. Bayesian decision tree ensembles with causally-informed priors, that both shrink towards homogeneous treatment effects and incorporate the propensity score, typically perform very well in this context relative to meta-learning approaches; in particular, estimation accuracy for CATEs is high and uncertainty quantification is conservative in the sense of being biased against finding non-existent heterogeneity.
3. Naive applications of Bayesian machine learning approaches typically lead to poor Frequentist performance. We observe that naive approaches tend to also be deficient from a subjective Bayesian perspective, in that they imply tightly-concentrated prior distributions on certain selection bias parameters that we actually wish to express ignorance about.

Our points are illustrated through analysis of both data-informed simulations and analysis of medical expenditure data.

### **Less is More: Shrinkage, Selection, and Prior Structure in the Age of Complex Data (Room 1103)**

#### ***Bayesian computation for high-dimensional Gaussian graphical models with spike-and-slab priors***

Presenter: David Rossell

Bayesian methods are appealing to quantify uncertainty associated with structural learning, i.e., the plausibility of conditional independence statements given the data, and parameter estimates. However, computational bottlenecks have limited their application to graphical models when the number of variables is large, which prompted the use of pseudo-Bayesian approaches. We propose fully Bayesian algorithms for Gaussian graphical models that provably scale to high dimensions when the data-generating precision matrix is sparse, at a similar cost to the best pseudo-Bayesian methods. The main trick is inducing sparsity via spike-and-slab priors with exact zeroes, which speeds up computations relative to shrinkage priors. We propose a Metropolis-Hastings-within-Block-Gibbs algorithm that allows row-wise updates of the precision matrix, using local moves. Second, a global proposal that enables adding or removing multiple edges within a row, which can help explore multi-modal posteriors. We obtain mixing bounds for both samplers relative to an ideal Gibbs sampler that are dimension-free under suitable settings, and prove that ideal Gibbs is geometrically ergodic. Our examples show that the methods extend the applicability of exact Bayesian inference from roughly 100 to 1000 variables (equivalently, from 5,000 edges to 500,000 edges).

#### ***Shrinkage priors for multiple high-dimensional Bayesian graphical models***

Presenter: Francesco Stingo

We develop a novel full-Bayesian approach for multiple correlated precision matrices, called multiple Graphical Horseshoe (mGHS). The proposed approach relies on a novel multivariate shrinkage prior based on the Horseshoe prior that borrows strength and shares sparsity patterns across groups, improving posterior edge selection when the precision matrices are similar. On the other hand, there is no loss of performance when the groups are independent. Moreover, mGHS provides a similarity matrix estimate, useful for understanding network similarities across groups. We implement an efficient Metropolis-within-Gibbs for posterior inference; specifically, local variance parameters are updated via a novel and efficient modified rejection sampling algorithm that samples from a three-parameter Gamma distribution. The method scales well with respect to the number of variables and provides one of the fastest full-Bayesian approaches for the estimation of multiple precision matrices. We

empirically demonstrate that mGHS outperforms competing approaches through both simulation studies and an application to a bike-sharing dataset. Joint work with Claudio Busatto.

***Shrinkage Meets Parametrization: Efficient Bayesian Regression with Categorical Covariates***

Presenter: Dan Kowal

Regression analyses commonly feature categorical covariates such as race, sex, or group, along with interactions to model heterogeneous, group-specific effects. However, the overwhelming default parameterization for linear models—using dummy coding and removing a reference group—leads to asymmetric and alarming (e.g., racial) biases when paired with shrinkage estimators, and otherwise complicates parameter interpretation. To address these limitations, we advocate an alternative parameterization using abundance-based constraints (ABCs) and couple it with a suitable shrinkage prior. ABCs identify parameters using group-weighted averages instead of reference groups. Crucially, ABCs offer a unique invariance property: the joint posterior distribution of all main effects is (nearly) unchanged by the addition of categorical interactions. Thus, ABCs facilitate heterogeneous effect estimation without any notable drawbacks for estimating or interpreting the main effects. Consequently, we anticipate an excess of interaction terms; we design structured shrinkage priors to effectively remove the unimportant ones. We apply these tools to study demographic heterogeneities among the effects of social and environmental factors on STEM educational outcomes for children in North Carolina.

Discussant: Merlise Clyde

**Scalable and Diagnosable Flow-based Models (Room 1104)**

***Asymptotically Exact Variational Inference via Involutive Iterated Random Functions***

Presenter: Trevor Campbell

Most expressive variational families (e.g., normalizing flows) lack practical convergence guarantees, as their theoretical assurances typically hold only at an intractable global optimum. In this talk, I will present a general recipe for constructing tuning-free, asymptotically exact variational flows. The core methodological component is a novel representation of general involutive MCMC kernels as invertible, measure-preserving iterated random function systems, which act as the flow maps of our variational flows. This leads to three new variational families with provable total variation convergence. This framework resolves key practical limitations of existing variational families with similar guarantees (e.g., MixFlows), while requiring substantially weaker theoretical assumptions. The talk will conclude with an empirical investigation of these new flows in comparison with existing state-of-the-art variational families.

***Flexible Selective Inference with Flow-based Transport Maps***

Presenter: Sifan Liu

Selective inference seeks to provide valid statistical guarantees after data-driven model selection. Traditional data-carving methods achieve this by conditioning on the selection event, but they typically rely on explicit characterizations of this event—an assumption that breaks down in modern applications involving adaptive tuning or complex pipelines. In this talk, I will present a new framework for selective inference that uses flow-based transport maps to approximate the post-selection distribution, even when the selection mechanism is analytically intractable. By learning a normalizing flow that pushes forward a simple reference distribution to the conditional distribution given selection, we enable valid hypothesis testing and interval estimation without restrictive assumptions. I'll demonstrate how this approach integrates seamlessly with existing inference procedures and show empirical results highlighting its flexibility and accuracy in both simulated and real data settings.

***Efficient and Exact Posterior Predictive Check in Simulation-based Inference***

Presenter: Yuling Yao

Model check is a necessary part in simulation-based inference as a way to test goodness-of-fit of the model. The widely-used posterior predictive check method often requires costly simulations and lacks proper calibration. We introduce a novel framework for posterior predictive checking that is specifically tailored to simulation-based workflows. Our method is compatible with likelihood-free computation, scales efficiently to expensive simulators, and produces frequentist-calibrated p-values. Additionally, it accommodates complex data structures, making it broadly applicable across modern scientific applications.

**Hierarchical Probabilistic Structures in BNP: Theoretical and Computational Advances (Room Main Hall)**

***Hierarchical models as mixtures: identifiability and posterior contraction***

Presenter: Long Nguyen

In this study we take the viewpoint of hierarchical modeling as a special kind of mixture models for grouped data and leverage our understanding of identifiability and posterior contraction of mixture models to shed some light on that of hierarchical models. As an illustration of this approach, we consider a family of latent variable models, namely topic models, equipped with a hierarchical structure among the topic variables. Such models may be viewed as a finite mixture of the latent Dirichlet allocation (LDA) induced distributions, but the LDA components are constrained by a latent hierarchy, specifically a rooted and directed tree structure, which enables the learning of interpretable and latent topic hierarchies of interest. A mathematical framework is developed in order to establish identifiability of the latent topic hierarchy under suitable regularity conditions, and to derive bounds for posterior contraction rates of the model and its parameters. We demonstrate the usefulness of such models and validate its theoretical properties through a careful simulation study and a real data example using the New York Times articles.

#### ***Hierarchical random measures without tables***

Presenter: Marta Catalano

The hierarchical Dirichlet process is the cornerstone of Bayesian nonparametric multilevel models. Its generative model can be described through a set of latent variables, commonly referred to as tables within the popular restaurant franchise metaphor. The latent tables simplify the expression of the posterior and allow for the implementation of a Gibbs sampling algorithm to approximately draw samples from it. However, managing their assignments can become computationally expensive, especially as the size of the dataset and of the number of levels increase. In this work, we identify a prior for the concentration parameter of the hierarchical Dirichlet process that (i) induces a quasi-conjugate posterior distribution, and (ii) removes the need of tables, bringing to more interpretable expressions for the posterior, with both a faster and an exact algorithm to sample from it. Remarkably, this construction extends beyond the Dirichlet process, leading to a new framework for defining normalized hierarchical random measures and a new class of algorithms to sample from their posteriors. The key analytical tool is the independence of multivariate increments, that is, their representation as completely random vectors.

#### ***Stochastic Block Covariance Matrix Estimation***

Presenter: Surya Tokdar

Motivated by a neuroscience application we study the problem of statistical estimation of a high-dimensional covariance matrix with a block structure. The block model embeds a structural assumption: the population of items (neurons) can be divided into latent sub-populations with shared associative covariation within blocks and shared associative or dis-associative covariation across blocks. Unlike the block diagonal assumption, our block structure incorporates positive or negative pairwise correlation between blocks. In addition to offering reasonable modeling choices in neuroscience and economics, the block covariance matrix assumption is interesting purely from the perspective of statistical estimation theory: (a) it offers in-built dimension reduction and (b) it resembles a regularized factor model without the need of choosing the number of factors. We discuss a hierarchical Bayesian estimation method to simultaneously recover the latent blocks and estimate the overall covariance matrix. We show with numerical experiments that a hierarchical structure and a shrinkage prior are essential to accurate recovery when several blocks are present.

## **Multitrack Sessions 9: 11:00–12:30**

### **Calibrated Bayes: Principled Inference Paradigms for Misspecified Models (Room 1101)**

#### ***Learning with Importance Weighted Variational Inference***

Presenter: Kamélia Daudel

Several popular variational bounds involving importance weighting ideas have been proposed to generalize and improve on the Evidence Lower BOund (ELBO) in the context of maximum likelihood optimization, such as the Importance Weighted Auto-Encoder (IWAE) and the Variational Rényi (VR) bounds. The methodology to learn the parameters of interest using these bounds typically amounts to running gradient-based variational inference algorithms that incorporate the reparameterization trick. However, the way the choice of the variational bound impacts the outcome of variational inference algorithms can be unclear. In this talk, we will present and motivate the VR-IWAE bound, a novel variational bound that unifies the ELBO, IWAE and VR bounds methodologies. In particular, we will provide asymptotic analyses for the VR-IWAE bound and its reparameterized gradient estimator, which enable us to compare the ELBO, IWAE and VR bounds methodologies. Our work advances the understanding of importance weighted variational inference methods and we will illustrate our theoretical

findings empirically.

### ***Predictively Oriented Posteriors***

Presenter: Badr-Eddine Chérif-Abdellatif

In this talk, I will explain how we can construct a family of post-Bayesian methods that predictively dominate Bayes posteriors and Gibbs posteriors alike.

### ***Outlier-Robust Bayesian Inference via Coarsening***

Presenter: Miheer Dewaskar

Given a probabilistic model, Bayesian inference is the fundamental idea of inferring unknown objects by conditioning on available information. When our assumed model truly describes reality, conditioning on all available information is the theoretically optimal approach to inference. Typically, however, our model is only approximately correct. In such cases, standard Bayesian inference can unfortunately be arbitrarily wrong even if the model is only slightly misspecified. Acknowledging minor model misspecification, Miller & Dunson (2019) first described how robust Bayesian inference can proceed by conditioning on less (coarser) information than what is available. In this talk I will describe a version of coarsened inference that leads to a posterior distribution that is robust to the presence of a small fraction of outliers in the data. Unlike most previous outlier-robust Bayesian posteriors proposed in the literature, our posterior arises from a novel application of Bayes's rule. Further, for models with product likelihoods, we show how any standard sampler can be modified to sample from our proposed posterior.

Discussant: Katharine Fisher

## **Advances in Bayesian Causal Inference (Room 1102)**

### ***Design and properties of Bayesian substitute confounders***

Presenter: Yordan P. Raykov

We develop a unified Bayesian framework for designing substitute confounders in high-dimensional causal inference with multiple causes. Our starting point is the observation that many multivariate assignment mechanisms admit a distribution substitute confounder. We formalize this idea by establishing (i) existence results for substitute confounders under broad latent-structure generative models; (ii) posterior concentration rates for Bayesian latent factor estimators around the confounding subspace; and (iii) posterior consistency of causal effect estimators that condition on the posterior distribution of the substitute confounder, rather than a point estimate. Our results encompass a wide class of structured factor models, including Normal-Gamma, multiplicative Gamma process (MGP), and cumulative shrinkage process (CSP) priors, which impose adaptive row- and column-wise shrinkage on loadings and thereby induce identifiable, approximately sparse confounding structure.

Methodologically, we show that conditioning on the distribution of the substitute confounder - rather than a deterministic factor score - produces a coherent Bayesian adjustment mechanism analogous to conditioning on the true latent confounder in the underlying generative model. This resolves several pathologies of the classical “causal deconfounder,” whose dense factor estimates can spuriously align with nuisance directions. In contrast, the structured shrinkage priors we study recover the confounding subspace even under severe off-block nuisance variation, heavy-tailed designs, or misaligned factor-strength scenarios.

Practically, we demonstrate two key benefits. First, we show that Bayesian structured sparse FA via NG, MGP, or CSP priors substantially improves assignment-model fit diagnostics such as residual cross-cause correlation, overlap entropy, and principal-angle recovery in synthetic designs where dense FA fails. Second, when the estimated substitute confounder is used inside regularized causal estimators, such as anisotropic ridge or augmented balancing weights, the Bayesian approach learns directionally adaptive shrinkage tuned to the confounding geometry, yielding large improvements over predictive-risk cross-validation. Across multiple controlled scenarios, we show that naive tuning can systematically overshrink or undershrink confounded directions, whereas the Bayesian approach adapts to the posterior confounding metric and achieves the lowest causal MSE and policy bias.

Taken together, our framework provides both the theoretical guarantees and practical regularization tools needed to make substitute-confounder methods reliable in modern, high-dimensional causal applications.

### ***Horseshoe Forests for High-Dimensional Causal Survival Analysis***

Presenter: Tijn Jacobs

We develop a Bayesian tree ensemble model to estimate heterogeneous treatment effects in censored survival data with high-dimensional covariates. Instead of imposing sparsity through the tree structure, we place a horseshoe prior directly on the step heights to achieve adaptive global-local shrinkage. This strategy allows

flexible regularisation and reduces noise. We develop a reversible jump Gibbs sampler to accommodate the non-conjugate horseshoe prior within the tree ensemble framework. We show through extensive simulations that the method accurately estimates treatment effects in high-dimensional covariate spaces, at various sparsity levels, and under non-linear treatment effect functions. We further illustrate the practical utility of the proposed approach by a re-analysis of pancreatic ductal adenocarcinoma (PDAC) survival data from The Cancer Genome Atlas.

### ***Bayesian Causal Effect Estimation for Categorical Data using Staged Tree Models***

Presenter: Manuele Leonelli

We propose a fully Bayesian approach for causal inference with multivariate categorical data based on staged tree models, a class of probabilistic graphical models capable of representing asymmetric and context-specific dependencies. To account for uncertainty in both structure and parameters, we introduce a flexible family of prior distributions over staged trees. These include product partition models to encourage parsimony, a novel distance-based prior to promote interpretable dependence patterns, and an extension that incorporates continuous covariates into the learning process. Posterior inference is achieved via a tailored Markov Chain Monte Carlo algorithm with split-and-merge moves, yielding posterior samples of staged trees from which average treatment effects and uncertainty measures are derived. Posterior summaries and uncertainty measures are obtained via techniques from the Bayesian nonparametrics literature. Two case studies on electronic fetal monitoring and cesarean delivery and on anthracycline therapy and cardiac dysfunction in breast cancer illustrate the methods.

### ***The cdf of BCF: Posterior contraction rate and Bernstein-von Mises theorem for Bayesian causal forests***

Presenter: Andrej Srakar

Bayesian causal inference has been noted in recent overview contribution on causal inference (Cinelli et al., 2025) as one of important area of future development of causal methods. One of most common Bayesian priors for causal inference remains Bayesian additive regression trees (BART, see Chipman et al., 2010). Theoretical properties of BART have been subject of important recent research by Ročková (Ročková and van der Pas, 2017; Ročková, 2019; Jeong and Ročková, 2022). In her recent contribution on the topic Jeong and Ročková propose a new class of sparse piecewise heterogeneous anisotropic Hölder functions and derive their minimax optimality properties. We extend their work to Bayesian Causal Forests (BCF), proposed by Hahn, Murray and Carvalho (Hahn et al., 2020). BCF is a modular Bayesian approach, based on regression decomposition of Robinson (1988) into prognostic and treatment effect part of the outcome regression specification, which include propensity score parameter. In our article we extend theoretical derivations of Jeong and Ročková to accommodate for the modular nature of BCF. To this end we extend Laplace approximation for cut posteriors used in previous work by Pompe and Jacob (2021) to the sparse piecewise heterogeneous anisotropic Hölder functions framework accounting for the mathematical nature of the BART prior. Posterior contraction rate and an asymptotic Bernstein-von Mises theorem for the BCF approach are derived. It seems feasible to generalize our approach and findings also to other types of both nonparametric BART priors (e.g. generalized BART, spatial BART, RDD BART) as well as BCF (e.g. shrinkage BCF). Short application to causal estimation of the effects of COVID-19 lockdown measures on number of infections and number of deaths in times of the pandemic is provided. Our article delves into a still underexplored and highly important area of asymptotics for Bayesian nonparametric priors, and we discuss also extensions to other BNP approaches outside of the BART framework.

### ***Transporting Principal Causal Effects Across Strata: A Bayesian Causal Inference Approach***

Presenter: Veronica Ballerini

In mediation analysis, natural direct and indirect effects (Robins and Greenland 1992; Pearl 2001) provide a mechanistic decomposition of treatment effects, identified under cross-world independence assumptions involving counterfactuals across treatment conditions (a.k.a. a priori counterfactuals). Within the principal stratification framework (Frangakis and Rubin, 2002), causal effects can be identified within strata defined by joint potential mediator values, avoiding a priori counterfactuals in causal effects definitions. However, whether effects within strata can be meaningfully decomposed into a direct and a “mediated” residual component remains debated. This difficulty is due to the fact that direct effects are identifiable only within those principal strata where, by definition, there is no mediated effect, and the literature still lacks a framework for disentangling the two components in the remaining strata (unless strong assumptions are introduced). In this work, building on recent advances in transportability within the causal inference literature (Chattopadhyay et al., 2024; Jalaldoust et al., 2024), we introduce a formal approach for transporting direct principal effects across principal strata. We specify sets of identifying assumptions that enable full or partial transportability of direct principal effects across these strata, and that allow testing the existence of mediated effects. The key peculiarity of our setting with



respect to the literature on transportability across populations is that principal strata are "latent"; as a result, principal causal effects are weakly identifiable even under strong assumptions. To address this, we develop a Bayesian approach to inference that does not require full identification and propagates uncertainty of the strata memberships throughout the estimation procedure. This aspect ensures that our inference reflects all sources of uncertainty and allows us to analyze the sensitivity of the results to key assumptions. We demonstrate our approach using unique nationwide longitudinal Medicare data for over 30+ million beneficiaries, integrating detailed health claims and high-resolution environmental exposures, including PM2.5.

## Shrinkage, Sparsity, and Discrete Structures (Room 1103)

### *A Class of Non-separable Penalty Functions for Bayesian Lasso-like Regression*

Presenter: Chris Hans

Non-separable penalty functions are often used in regression modeling to enforce group sparsity structure, reduce the influence of unusual features, and improve estimation and prediction by providing a more realistic match between model and data. From a Bayesian perspective, such penalty functions correspond to a lack of (conditional) prior independence among the regression coefficients. We describe a class of prior distributions for regression coefficients, the exchangeability properties of which generate non-separable penalty functions. The priors have connections to L1-norm penalization and the Bayesian lasso (BL) and elastic net (BEN) regression models. The regularization properties of the class of priors can be understood both by studying its tunable parameters directly and via the connections to BL and BEN regression. We discuss full Bayesian inference under these priors, show that posterior computation is incredibly straightforward, and demonstrate that inference and prediction based on these models perform competitively under a range of example data structures.

### *High-dimensional Bayesian transfer learning through skew shrinkage priors: applications to genetic and electronic health records data*

Presenter: Aki Nishimura

The Bayesian paradigm provides a natural framework to encode existing knowledge in the form of a prior. For example, a posterior distribution obtained by training a model on one dataset can be used to construct an informed prior for another. Such a framework is especially relevant to transfer learning across a federated data network, in which data can be shared across sites only at summary-level for privacy protection. Constructing an informed prior is challenging in modern high-dimensional problems, however, since they warrant additional assumptions of sparsity in parameters. To address this need for an informed prior with sparsity structure, we propose a skewed generalization of global-local shrinkage priors. The proposed \*skew shrinkage priors\* continuously interpolate between an informed Gaussian and a density concentrated at zero. To ensure scalability of regression models based on skew shrinkage priors, we adapt the state-of-the-art conjugate gradient and spectral collapsing algorithms to develop a fast Gibbs sampler. The modified zigurat algorithm, additionally developed to enable efficient posterior updates of local scale parameters, is a general sampler of independent interest. We demonstrate statistical gains from the proposed transfer learning method through its applications to 1) an observational study on second-line treatments of type-2 diabetes, using the output from the larger IQVIA Pharmetrics data to inform the model for the smaller Johns Hopkins data; and 2) a polygenic risk prediction problem, using the output from the larger white British population to inform the model for the smaller African population in UK Biobank data.

### *How to do multiple testing with the horseshoe?*

Presenter: Fanny Villers

The horseshoe, and more generally continuous mixture priors, has become commonplace in current high-dimensional statistics practice. Given a model where the unknown parameter is a sparse vector, the idea for this prior is to simultaneously place a lot of mass near zero (to mimic a "spike") and having heavy tails in order to catch large signals. An advantage compared to the spike-and-slab prior is that it enables easier sampling from really high-dimensional posteriors.

Since the horseshoe does not explicitly model exact zeros, the posterior distribution does not come with posterior probabilities of a coordinate being zero. A number of empirical criteria have been proposed to nevertheless perform multiple testing, namely decide which coordinates of the unknown vector are non-zero. Yet we are not aware of a procedure based on the horseshoe (or another continuous mixture) prior that would control the False Discovery Rate (FDR) while keeping the False Negative Rate at the lowest possible level.

We propose a generic method to address this question. In the sparse sequence model we prove that this posterior-based method achieves the optimal information-theoretic boundaries. On the algorithmic side, we consider several samplers; associated simulations confirm that FDR and FNR are close to the levels expected from the theory. We



illustrate this on both sequence model and sparse high-dimensional regression, in particular providing procedures that control the FDR at (close to) any desired target level.

This is joint work with Ismael Castillo (Sorbonne Université) and Sayantan Banerjee (IIM Indore)

### ***Scalable Bayesian Structure Learning for Gaussian Graphical Models Using Marginal Pseudo-Likelihood***

Presenter: Reza Mohammadi

Bayesian methods for learning Gaussian graphical models offer a principled framework for quantifying model uncertainty and incorporating prior knowledge. However, their scalability is constrained by the computational cost of jointly exploring graph structures and precision matrices. To address this challenge, we perform inference directly on the graph by integrating out the precision matrix. We adopt a marginal pseudo-likelihood approach, eliminating the need to compute intractable normalizing constants and perform computationally intensive precision matrix sampling. Building on this framework, we develop continuous-time (birth–death) and discrete-time (reversible jump) Markov chain Monte Carlo (MCMC) algorithms that efficiently explore the posterior over graph space. We establish theoretical guarantees for posterior contraction, convergence, and graph selection consistency. The algorithms scale to large graph spaces, enabling parallel exploration for graphs with over 1,000 nodes, while providing uncertainty quantification and supporting flexible prior specification over the graph space. Extensive simulations show substantial computational gains over state-of-the-art Bayesian approaches without sacrificing graph recovery accuracy. Applications to human and mouse gene expression datasets demonstrate the ability of our approach to recover biologically meaningful structures and quantify uncertainty in complex networks. An implementation is available in the R package **BDgraph**.

### ***Profile Graphical Models***

Presenter: Alejandra Avalos-Pacheco

We introduce a novel class of profile graphical models that represent, within a single graph, how an external factor influences the dependence structure of a multivariate set of variables. This class is quite general and includes multiple graphs and chain graphs as special cases. Profile graphical models capture the conditional distributions of a multivariate random vector given different levels of a risk factor, and learn how the conditional independence structure among variables may vary across profiles. We focus on conditional independence relationships and then on the undirected version of profile graphical models; we formally define this family of models and establish their corresponding Markov properties. Furthermore, we show that the resulting profile undirected graphical models are independence-compatible with two-block LWF chain graph models. We then develop a Bayesian approach for Gaussian undirected profile graphical models based on the use of continuous spike-and-slab priors to learn shared sparsity structures across different levels of the risk factor. We also implement a fast EM algorithm for efficient inference. Inferential properties are explored through simulation studies including the comparison with competing methods. The practical utility of this class of models is demonstrated through the analysis of protein network data from various subtypes of acute myeloid leukemia.

## **Recent Advances in Neural Simulation-Based Inference (Room 1104)**

### ***Amortised and Provably-robust Simulation-based Inference***

Presenter: Ayush Bharti

Complex simulator-based models are now routinely used to perform inference across the sciences and engineering, but existing inference methods are often unable to account for outliers and other extreme values in data which occur due to faulty measurement instruments or human error. In this paper, we introduce a novel approach to simulation-based inference grounded in generalised Bayesian inference and a neural approximation of a weighted score-matching loss. This leads to a method that is both amortised and provably robust to outliers, a combination not achieved by existing approaches. Furthermore, through a carefully chosen conditional density model, we demonstrate that inference can be further simplified and performed without the need for Markov chain Monte Carlo sampling, thereby offering significant computational advantages, with complexity that is only a small fraction of that of current state-of-the-art approaches.

### ***Conservative neural posterior estimation via distributionally robust training***

Presenter: Yuga Hikida

Simulation-based inference with neural posterior estimation (NPE) often yields overconfident and unreliable posteriors under limited simulation budgets. To address this, we propose DRO-NPE, a distributionally robust approach that replaces the standard NPE objective with a worst-case loss over a Wasserstein ambiguity set. We introduce KL-based metrics for miscoverage and miscalibration, and use these to show that the DRO-NPE objective controls overfitting and reduces posterior overconfidence. Our method is tractable, parallelisable, and

readily integrates with standard normalizing flows. Across benchmark SBI tasks, DRO-NPE consistently improves coverage and calibration, while narrowing the gap between empirical and population NPE loss, leading to more reliable inference in low-simulation regimes.

***Diffusion posterior sampling for simulation-based inference in tall data settings***

Presenter: Julia Linhart

Determining which parameters of a non-linear model best describe a set of experimental data is a fundamental problem in science and it has gained much traction lately with the rise of complex large-scale simulators. The likelihood of such models is typically intractable, which is why classical MCMC methods can not be used. Simulation-based inference (SBI) stands out in this context by only requiring a dataset of simulations to train deep generative models capable of approximating the posterior distribution that relates input parameters to a given observation. In this work, we consider a tall data extension in which multiple observations are available to better infer the parameters of the model. The proposed method is built upon recent developments from the flourishing score-based diffusion literature and allows to estimate the tall data posterior distribution, while simply using information from a score network trained for a single context observation. We compare our method to recently proposed competing approaches on various numerical experiments and demonstrate its superiority in terms of numerical stability and computational cost.

**Advances in Bayesian Nonparametric Modeling for Complex Dependent Data (Room Main Hall)**

***Partially Bayes p-values for large-scale inference***

Presenter: Li Ma

We seek to conduct statistical inference for a large collection of primary parameters, each with its own nuisance parameters. Our approach is partially Bayesian, in that we treat the primary parameters as fixed while we model the nuisance parameters as random and drawn from an unknown distribution which we endow with a nonparametric prior. We compute partially Bayes p-values by conditioning on nuisance parameter statistics, that is, statistics that are ancillary for the primary parameters and informative about the nuisance parameters. The proposed p-values have a Bayesian interpretation as tail areas computed with respect to the posterior distribution of the nuisance parameters. Similarly to the conditional predictive p-values of Bayarri and Berger, the partially Bayes p-values avoid double use of the data (unlike posterior predictive p-values). A key ingredient of our approach is that we model nuisance parameters hierarchically across problems; the sharing of information across problems leads to improved calibration. We illustrate the proposed partially Bayes p-values in two applications: the normal means problem with unknown variances and a location-scale model with unknown distribution shape. We model the scales via Dirichlet processes in both examples and the distribution shape via Pólya trees in the second. Our proposed partially Bayes p-values increase power and calibration compared to purely frequentist alternatives.

***Predictive constructions, pre-pended data, and exchangeability***

Presenter: Steve MacEachern

Prequential methods implicitly provide a joint distribution for a sequence of data through one-step ahead predictive distributions. The resulting joint distribution is typically not exchangeable. Developments over the past two decades have provided conditions under which the tail of the sequence approaches exchangeability. In this talk, we argue that the distribution of main interest is often the limiting, exchangeable distribution, and that the device of prepending missing data to our observed sequence enables us to effectively work with this distribution. The resulting procedure provides Bayesians with a novel, “third approach” for the construction of a conditionally independent or (partially) exchangeable model for observable data. We discuss data-analytic considerations and computational methods.

***Probabilistic discovery of new species and homogeneous subpopulations***

Presenter: Augusto Fasano

Bayesian hierarchical models are well-suited for data from multiple groups or studies, as they enable borrowing of information across groups while allowing the subjects in the same group to arise from a common unknown distribution. In the context of species-sampling models under partial exchangeability, a central goal is to predict the number of new species in additional samples. Another challenging task is testing for population homogeneity, which standard species-sampling models cannot handle effectively, as they typically assume population distributions are either identical or entirely distinct almost surely. To overcome these limitations and address both prediction and homogeneity testing, we propose an extension of the Hierarchical Pitman–Yor obtained by adding

a nested structure. We derive an analytically tractable form for the distribution of the induced random partition, offering theoretical insights and enabling the design of an efficient Markov Chain Monte Carlo algorithm for posterior inference and prediction. Finally, we introduce a Bayesian nonparametric test for homogeneity between populations and demonstrate its practical utility through real-data applications.

## Bayarri Lecture 2: 14:00–15:30

### *Skew-symmetric representations of posterior densities*

Presenter: Daniele Durante

The perturbation of symmetric densities via suitably-designed skewing functions yields a broad and similarly-tractable class of asymmetric extensions, known as skew-symmetric family. Although such a family has been mostly overlooked in Bayesian statistics, its potential impact could be substantial. As shown in this presentation, any generic posterior density can, in fact, be re-expressed in a skew-symmetric form by decomposing it as the product of (i) a symmetrized version of such a density and (ii) a tractable closed-form skewing function that only depends on the un-normalized posterior. This representation opens the avenues for the design of broadly-applicable strategies perturbing, at no additional optimization cost, any symmetric posterior approximation (e.g., from the Laplace method, variational Bayes and expectation-propagation) to obtain similarly-tractable skew-symmetric alternatives. This improved class of deterministic approximations provably enhances the finite-sample accuracy of the original symmetric counterparts and, under suitable assumptions, it improves the convergence rate to the exact posterior by at least a  $\sqrt{n}$  factor, in asymptotic regimes. As illustrated in the concluding part of the presentation, additional accuracy gains can be obtained under novel extensions of the skew-symmetric family that leverage more general notions of invariance, beyond the one of central symmetry employed in the original construction of such a family.

Discussants: David Nott and David Rossell

## Keynote Lecture 4: 16:00–17:00

**TBA**

Presenter: Barbara Engelhardt

TBA

## Multitrack Sessions 10: 17:15–18:45

### Latest Developments in Cut Bayes (Room 1101)

#### *A more flexible framework for cutting feedback in Bayesian models*

Presenter: Robert Goudie

Standard Bayesian inference may not be reliable if parts of the model are misspecified. One approach to address this problem is “cut inference” in which models are split into modules and “feedback” from suspect modules is cut to prevent these suspect modules from contaminating other modules. Liu and Goudie (JRSSB, in press) recently proposed a general framework for this approach, but this relies on a specific, restrictive definition of modules in which they are defined in terms of a partition of the observable quantities. In this talk, I will describe a more flexible extension of the Liu and Goudie approach that allows the modeller to specify the flow of information in the model more precisely. This definition incorporates some practical uses of the cutting feedback (e.g. in biostatistics) that are not possible under the Liu and Goudie approach, so I will illustrate how the approach can be applied in these settings. I will also highlight connections between these ideas and existing approaches in the literature.

#### *Optimization-based cutting feedback for semiparametric models*

Presenter: David Nott

Modern statistics and machine learning deal with complex models and datasets, and a complex joint model is often built from submodels or modules joined together. We consider Bayesian inference for semiparametric joint models with both parametric and nonparametric modules, which can be difficult for two reasons. First, there is the risk of likelihood misspecification in parametric modules. Second, prior choices for high or infinite-dimensional parameters in nonparametric modules can be highly influential in inference. Motivated by recent generalized Bayesian extensions of popular cutting feedback methods, which can mitigate the effects of likelihood misspecification in multi-modular models, we consider an “optimization-centric” generalized Bayes approach to

cutting feedback for semiparametric joint models. This optimization-centric framework provides a natural way to addressing both likelihood misspecification and conflicts between prior information and a loss function, with the latter being particularly valuable for managing influential prior choices in Bayesian nonparametric modules. The new generalized Bayesian cut posteriors are defined through a variational optimization problem, and computational methods based on variational inference are used. We demonstrate the methods in a benchmark toy example, as well as real examples involving Gaussian process adjustments for confounding in causal inference and misspecified copula models with nonparametric marginals.

#### ***Neural variational cut Bayes***

Presenter: Jiafang Song

A full Bayesian approach integrates all data sources into a joint model, allowing information to flow across modules for simultaneous parameter inference. However, such integration may introduce bias if one or more modules are misspecified. Cutting feedback methods address this by constructing a cut posterior and restricting information from suspect modules, preventing potentially misspecified components from affecting more reliable parts. In this paper, we extend variational inference for Cut-Bayes models by incorporating flow-based generative models within the variational family. Our method, Neural Network Variational Inference Cut-Bayes (NeVI-Cut), enhances the expressiveness beyond traditional parametric families. Leveraging the approximation capabilities of normalizing flows and neural networks, NeVI-Cut is provably expressive and can estimate complex posterior distributions, including multimodality and skewness. Compared to Monte Carlo (MCMC) cut models, NeVI-Cut achieves significant computational gains and scales efficiently to high-dimensional models. Moreover, it requires only posterior samples from the first module, improving modularity and privacy-protection by avoiding requiring direct access to the first data module. Simulation studies and real-world analysis of verbal autopsy data for cause of death estimation in low-and-middle-income countries data illustrate that NeVI-Cut provides posterior distributions comparable to MCMC while dramatically reducing computational cost.

### **Bayesian Decision Analysis and Counterfactual Predictions: Foundations, Methodology, and Applications (Room 1102)**

#### ***Dynamic causal inference with time series data***

Presenter: Kenichiro McAlinn

We consider causal inference for time series data, where the interest is in estimating the effect of an intervention that occurs at some point in time. Existing methods often fail to adequately address temporal dynamics, leading to considerable bias when the effect varies over time. Foundational issues regarding estimands and assumptions further compound this problem. We propose generalizing the potential outcome causal inference framework to estimate the dynamic treatment effect. Employing stochastic process theory, we explicate unique features of time series, and formally define the potential outcomes, causal estimand, and other necessary concepts. Given our proposed causal estimand, we define the dynamic average treatment effect (DATE), which captures the evolving impact of treatment over time, and develop estimation strategies. For observational data, we propose a dynamic inverse probability weighting estimator and show its unbiasedness and consistency. Under cases with few, or one, units of treatment, we show that the dynamic linear model is the best approximation of the dynamic average treatment effect in terms of quadratic loss. Simulation studies demonstrate that our method consistently reduces bias and improves uncertainty quantification, leading to more accurate causal estimates compared to existing approaches. An analysis of the impact of the COVID-19 lockdown on the unemployment rate in the UK highlights the strength and interpretability of our proposed approach.

#### ***Bayesian doubly robust causal inference via posterior coupling***

Presenter: Shonosuke Sugawara

In observational studies, propensity score methods are central for estimating causal effects while adjusting for confounders. Among them, the doubly robust (DR) estimator has gained considerable attention because it provides consistent estimates when either the propensity score model or the outcome model is correctly specified. Like other propensity score approaches, the DR estimator typically involves two-step estimation: first, estimating the propensity score and outcome models, and then estimating the causal effects using the estimated values. However, this sequential procedure does not naturally align with the Bayesian framework, which centers on updating prior beliefs solely through the likelihood. In this manuscript, we propose novel Bayesian DR estimation via posterior coupling, which incorporates propensity score information via moment conditions directly into the posterior distribution. This design avoids the feedback problem and enables a fully Bayesian interpretation of DR estimation without requiring two-step estimation. We detail the theoretical properties of the proposed method

and demonstrate its advantages over existing Bayesian approaches through comprehensive simulation studies and real data applications.

***A cautious use of auxiliary outcomes for decision-making in randomized clinical trials***

Presenter: Massimiliano Russo

Clinical trials often collect data on multiple outcomes, such as overall survival (OS), progression-free survival (PFS), and response to treatment (RT). In most cases, however, study designs only use primary outcome data for interim and final decision-making. In several disease settings, clinically relevant outcomes, for example OS, become available years after patient enrollment. Moreover, the effects of experimental treatments on OS might be less pronounced compared to auxiliary outcomes such as RT. We develop a Bayesian decision-theoretic framework that uses both primary and auxiliary outcomes for interim and final decision-making. The framework allows investigators to control standard frequentist operating characteristics, such as the type I error rate, and can be used with auxiliary outcomes from emerging technologies, such as circulating tumor assays. False positive rates and other frequentist operating characteristics are rigorously controlled without any assumption about the concordance between primary and auxiliary outcomes. We discuss algorithms to implement this decision-theoretic approach and show that incorporating auxiliary information into interim and final decision-making can lead to relevant efficiency gains according to established and interpretable metrics.

**Efficient Computational Strategies for Bayesian Clinical Trials (Room 1103)**

***Computationally efficient Bayesian design of experiments in presence of nuisance parameters***

Presenter: Shirin Golchi

Design of experiments has traditionally relied on the frequentist hypothesis testing framework where the optimal size of the experiment is specified as the minimum sample size that guarantees a required level of power. Sample size determination may be performed analytically when the sampling distribution of the test statistic is given and therefore the power function is available in analytic form. Bayesian methods have gained popularity in all stages of discovery, namely, design, analysis and decision making. Bayesian decision procedures rely on posterior summaries whose sampling distributions are commonly estimated via Monte Carlo simulations. In the design of scientific studies, the Bayesian approach incorporates uncertainty about the design value instead of conditioning on a single value of the model parameter(s). Accounting for uncertainties in the design is particularly critical when the model includes nuisance parameters. In this manuscript, we propose methodology that utilizes the large-sample properties of the posterior distribution together with Bayesian additive regression trees (BART) to efficiently obtain the optimal sample size and decision criteria in fixed and adaptive designs. We introduce a fully Bayesian design procedure that accommodates the uncertainty associated with the model parameters including the nuisance parameters. The proposed approach significantly reduces the computational burden associated with Bayesian design and enables wide adoption of Bayesian operating characteristics.

***Q-approximation of operating characteristics of clinical trial designs***

Presenter: Daniel Schwartz

Designing clinical trials requires evaluating multiple operating characteristics (OCs), such as the likelihood of an early stopping decision, the probability of detecting a treatment effect, and the Type I error rate. In most cases, these evaluations rely on computationally intensive Monte Carlo simulations. As the complexity of clinical trials and the use of adaptive designs increase, the computational burden can quickly become prohibitive. We introduce a strategy for rapidly approximating OCs, called the Q-approximation. Our approach is based on quadratic approximations of the log-likelihood and asymptotic arguments. The Q-approximation approach can be applied to any trial design that uses data analysis methods coherent with the likelihood principle, including Bayesian designs, multistage designs with early stopping, adaptively randomized designs, and designs that leverage external data. We illustrate the approach with several examples and show that it provides accurate approximation of relevant OCs while reducing the computation time compared to Monte Carlo simulations. In particular, in our experiments, standard Monte Carlo approximation of OCs requires at least 25 times more computing than the Q-approximation to achieve comparable levels of accuracy.

***A fast, flexible simulation framework for Bayesian adaptive designs — the R package BATSS***

Presenter: Thomas Jaki

The use of Bayesian adaptive designs for randomised controlled trials has been hindered by the lack of software readily available to statisticians. We have developed a new software package (Bayesian Adaptive Trials Simulator Software - BATSS for the statistical software R, which provides a flexible structure for the fast simulation of Bayesian adaptive designs for clinical trials. We illustrate how the BATSS package can be used to define and



evaluate the operating characteristics of Bayesian adaptive designs for various different types of primary outcomes (e.g., those that follow a normal, binary, Poisson or negative binomial distribution) and can incorporate the most common types of adaptations: stopping treatments (or the entire trial) for efficacy or futility, and Bayesian response adaptive randomisation - based on user-defined adaptation rules. Other important features of this highly modular package include: the use of (Integrated Nested) Laplace approximations to compute posterior distributions, parallel processing on a computer or a cluster, customisability, adjustment for covariates and a wide range of available conditional distributions for the response.

Discussant: Lara Maleyeff

## **Frontiers in Bayesian Spatial and Environmental Statistics: Methodological Innovation and Real-World Impact (Room 1104)**

### ***Bayesian Statistical Learning for Tuberculosis Drug Resistance under High-Dimensional and Sparse Genomic Data***

Presenter: Clara Grazian

The emergence of drug-resistant tuberculosis (TB) poses major challenges for both public health and statistical inference. Advances in whole-genome sequencing have generated high-dimensional genomic datasets that can potentially support rapid identification of resistance mechanisms and personalised treatment strategies. However, statistical analysis in this setting is complicated by sparsity, strong dependence structures among mutations, heterogeneous phenotypic responses, population stratification, and the limited availability of resistant isolates for recently introduced drugs.

In this talk, I will present Bayesian modelling approaches for inference and prediction in TB drug-resistance studies based on genomic and phenotypic data. The framework combines hierarchical probabilistic modelling, shrinkage priors, latent structure learning, and uncertainty quantification to identify mutations associated with resistance while controlling for the high-dimensional nature of the problem and the rarity of resistant phenotypes. Particular attention will be devoted to inference for emerging resistance to drugs such as bedaquiline and clofazimine, where small sample sizes and weak signal strength make classical approaches unstable.

I will discuss methodological challenges arising from highly correlated genomic features, partial identifiability, and heterogeneous resistance mechanisms across subpopulations, and show how Bayesian regularisation and latent variable models provide a coherent framework for borrowing strength across mutations and resistance profiles. I will further illustrate how posterior uncertainty can be propagated into clinically relevant predictive quantities, enabling probabilistic assessment of resistance and supporting decision-making in precision medicine settings.

The talk will highlight the role of modern Bayesian statistical learning in integrating complex biological data, flexible hierarchical modelling, and principled uncertainty quantification for high-impact biomedical applications.

### ***A New Copula for Circular Variables and Its Extension to Spatial Processes with Arbitrary Marginals***

Presenter: Gianluca Mastrantonio

Modeling dependence for directional variables presents unique challenges due to the periodic nature of the sample space. In this talk, I introduce a novel copula construction specifically designed for circular variables, allowing for flexible and interpretable dependence structures while respecting angular geometry. The proposed copula is then embedded within a spatial framework, yielding a stochastic process for directional data with user-specified marginal distributions. This construction enables the modeling of complex spatial dependence patterns without imposing restrictive assumptions on the marginals, providing a versatile alternative to traditional approaches for directional spatial data. The methodology is illustrated through an application to sea current directions in the Gulf of Naples, where capturing both local dependence and marginal heterogeneity is essential. Results demonstrate the ability of the proposed model to recover meaningful spatial structures and improve descriptive and predictive performance.

### ***Harnessing multiple data sources and model outputs on vegetative stress to detect areas with high risk of wildfire***

Presenter: Veronica Berrocal

With increasing temperature, California is facing more frequent and intense droughts as well as greater risks of devastating wildfires. Fueled by strong winds, recent wildfires in California have destroyed and burned through large areas in part also because of the accumulation of dry wood fuel in forests. In this talk, we present a novel Bayesian spatio-temporal model that identifies areas that can be considered prone to risk of wildfire and provide a probabilistic quantification of the risk. To this goal, we leverage multiple data sources and models measuring or, respectively, estimating soil moisture and evapotranspiration, indicators, respectively, of the amount of water



in the soil pores and the amount of water that moves from the soil to the atmosphere, that we combine using a Bayesian Model Averaging type of approach.

## **BNP for Dependent Random Measures and Structures (Room Main Hall)**

### ***A principled approach to covariate-dependent random partition models***

Presenter: Igor Pruenster

Incorporating continuous covariates into fully nonparametric regression models poses a central challenge within the Bayesian nonparametric paradigm. Existing approaches often involve trade-offs between flexibility in modeling the latent partition structure, analytical tractability, and predictive consistency for new observations. We introduce and thoroughly analyze a novel class of covariate-dependent random probability measures, constructed via the normalization of appropriately defined random measures that incorporate covariates through a kernel structure. Specifically, the jumps of a common discrete random measure are rescaled by a similarity kernel, yielding a random probability measure we term a kernel-weighted normalized random measure. Notable instances arise when the common random measure is a gamma process or a specific transformation of a stable process, leading to kernel-weighted Dirichlet and Pitman–Yor processes, respectively. This construction induces a covariate-dependent random partition model that combines substantial flexibility and predictive consistency with analytical tractability, facilitated by the introduction of suitable latent variables. Conditionally on these latent variables, the resulting partition model closely relates to the popular PPMx framework. We derive the partition probability function and the posterior distribution of the common random measure in closed form, given the latent variables, and we characterize the predictive mechanism for both new observations and their associated latent variables. This framework can be effectively employed as building block for constructing nonparametric regression models, as well as clustering or species sampling models that incorporate additional information conveyed through covariates.

### ***A Hierarchical Random Partition Model for Multi-View Data***

Presenter: Garritt Page

Multi-view data arises when multiple sets of features or "views" describe a common set of entities. These data are often hierarchical in nature and exhibit strong inter-view correlations, posing challenges for model-based clustering approaches that respect such structure. In this work, we detail a random partition model that jointly models a collection of view-specific partitions in a hierarchical fashion. This formulation enables capturing both shared and view-specific patterns, facilitating borrowing strength across views and leading to more coherent global clustering. We further show how prior information associated with each view's partition can be seamlessly incorporated into the model. The benefits of this approach are demonstrated through numerical studies and applications to real-world datasets.

### ***Constructing stationary time series of completely random measures via Bayesian conjugacy***

Presenter: Raffaele Argiento

One flexible approach to building stationary time-dependent processes exploits the mathematical concept of conjugacy in a Bayesian framework. Under this approach, the transition law  $\mathcal{L}(X_t|X_{t-1})$  of the process  $X_t$  is defined as the predictive distribution of an underlying Bayesian model (see, e.g., Pitman and Walker, 2005). When the model is conjugate, the transition kernel can be derived analytically, making this approach particularly attractive. We aim to achieve such mathematical tractability in the context of completely random measures (CRMs), where the temporally dependent variables are CRMs. We first view the CRM as marked point processes, then to exploit conjugacy, we focus on the broad class of exponential family CRMs (see Broderick et al., 2018). This leads to a tractable process with an AR(1)-type structure, providing a convenient framework for further extensions to more complex temporal dynamics. The proposed construction naturally lends itself to extending CRM-based Bayesian nonparametric models, such as feature allocation models, to settings with time-dependent data. A naive approach to introducing time dependence into CRMs often results in models where once a feature disappears, it cannot reappear in the future – an undesirable constraint in many applications. We address this issue by leveraging an attractive point process, namely the Shot-Noise Cox process (see Beraha et al., 2025), which allows us to interpret some features as recurring over time.

Discussant: Michele Guindani

## Friday 3 July

### Multitrack Sessions 11: 9:00–10:30

#### Modern Record Linkage Techniques for a Complex and Ever-changing World (Room 1101)

##### *Scalable Record Linkage for Streaming Data Contexts*

Presenter: Andee Kaplan

With the ubiquity of data, linking data sets has become crucial for myriad applications including healthcare, official statistics, ecology, and fraud detection and national security. Record linkage is the task of resolving duplicates in two or more partially overlapping sets of records, or files, from noisy data sources without a unique identifier. In any field where multiple sources of messy data are available to answer a scientific problem, record linkage is a critical step in the analysis pipeline. In streaming record linkage, files arrive sequentially in time and estimates of links are updated after the arrival of each file. The challenge in streaming record linkage is to efficiently update parameter estimates as new data arrive. In this talk, I present a Bayesian record linkage model for the multi-file case formulated specifically for the streaming data context. This model is fit using recursive updates, incorporating each new batch of data into the model parameters' posterior distribution. A novel Markov chain Monte Carlo algorithm is presented that performs recursive Bayesian updates while avoiding the issue of degradation, common to many recursive algorithms. This sampler achieves near-equivalent posterior inference to non-streaming algorithms at a small fraction of the compute time.

##### *Bayesian Small Area Estimation and Record Linkage Techniques for Official Statistics*

Presenter: Brenda Betancourt

Small area estimation (SAE) is imperative for official data production. In this work, we propose Bayesian approaches for SAE that account for the uncertainty introduced with the use of auxiliary data sources. We present Measurement Error Models (MEM) that naturally incorporate and propagate the uncertainty that arises from record linkage and deduplication processes. We explore different state-of-the-art approaches for record linkage and deduplication for the construction of statistical systems and evaluate their performance in terms of precision and practicality in the context of MEM for SAE. The aim is to provide a unified approach for the construction of statistical systems and the estimation of official statistics at subnational and subpopulation levels.

##### *A Comprehensive Bayesian Approach to Entity Resolution for Data with Multiple Truths*

Presenter: Matthew Koslovsky

In many applications, from government to ecology, integrating data from diverse and noisy sources is critical for downstream inference. However, a unique identifier to link records from the same entity may not exist. Record linkage (also referred to as de-duplication or entity resolution) merges such databases to find duplicates within and across them. A popular method represents the truth of each entity as a latent variable, linking records by clustering observations to the truth, allowing for potential data distortions. It assumes the truth is a single fixed value, which may not match reality. For example, survey participants may not recall the exact value of their net income and provide an approximation. Any attempts to link this to official data necessarily encodes the response as random distortion rather than approximate truth. We present a novel generalization of the latent variable record linkage model, allowing values to be considered a "multiple truth" instead of random distortions and handling discrete and continuous fields. We provide options to fit the model: Markov chain Monte Carlo and variational inference for massive data, and demonstrate its value via simulation and linking a longitudinal survey of Italian household wealth.

#### Advances in Nonparametric Empirical Bayes (Room 1102)

##### *BNP Empirical Bayes. Consistency and efficiency in Robbins' framework*

Presenter: Claudia Contardi

We propose a Bayesian nonparametric (BNP) estimator for the Poisson compound decision problem, analysing its performance from a regret-based perspective. Motivated by recent efficiency studies on various g-modelling empirical Bayes (hence frequentist) procedures, we propose a fully BNP counterpart through the adoption of a "nonparametric Bayes Empirical Bayes" (BNP-EB) framework, and introduce a minimum-distance estimator based on a Dirichlet process prior. While it is known that such procedure is consistent in the Bayesian sense, to the best of our knowledge no results are available on posterior contraction rates (PCRs). We fill this gap by showing that it attains the minimax regret rate over a broad class of mixing distributions. Further, we show

that the BNP-EB estimator is competitive in terms of efficiency, achieving the optimal regret rate. Finally, we validate our results empirically, on both synthetic and real datasets, comparing against widely used empirical Bayes competitors.

This talk is based on joint, partly ongoing work with E. Dolera and S. Favaro.

#### ***Quasi-Bayes empirical Bayes: a sequential approach to the Poisson compound decision problem***

Presenter: Sandra Fortini

The Poisson compound decision problem is a long-standing problem in statistics, for which empirical Bayes methods are commonly used to estimate Poisson means in static or batch settings. We consider this problem in a streaming, or online, framework. Building on a quasi-Bayesian approach based on Newton's algorithm, we develop a sequential estimate that is easy to evaluate, computationally efficient, and has constant per-observation cost as the data accrue. We establish frequentist guarantees for the proposed estimate, including consistency and asymptotic optimality, with optimality understood as vanishing excess Bayes risk, or regret. Empirical performance is assessed through simulation studies and comparisons with benchmark procedures.

#### ***Compound decisions and empirical Bayes via Bayesian nonparametrics***

Presenter: Sid Kankanala

We study the Gaussian sequence compound decision problem and analyze a Bayesian nonparametric estimator from an empirical Bayes, regret-based perspective. Motivated by sharp results for the classical nonparametric maximum likelihood estimator (NPMLE), we ask whether an analogous guarantee can be obtained using a standard Bayesian nonparametric prior. We show that a Dirichlet-process-based Bayesian procedure achieves near-optimal regret bounds. Our main results are stated in the compound decision framework, where the mean vector is treated as fixed, while we also provide parallel guarantees under a hierarchical model in which the means are drawn from a true unknown prior distribution. The posterior mean Bayes rule is, a fortiori, admissible, whereas we show that the NPMLE plug-in rule is inadmissible.

### **Principled Tuning of Markov Chain Monte Carlo (Room 1103)**

#### ***Scaling limits for adaptive MCMC***

Presenter: Sam Livingstone

We study the random walk Metropolis with a proposal scale allowed to adapt according to a separate process, as is conventional in adaptive MCMC. We consider various different stochastic approximation schemes for the proposal scale (Robbins–Monro, Polyak-type approaches) and in each case derive a high-dimensional limiting process consisting of a Langevin diffusion whose speed is driven by an ordinary differential equation. This limiting form allows a careful analysis of optimal parameter choices within each stochastic approximation scheme and a comparison of different schemes. This is joint work with Krzysztof Zakrzewski & Błażej Miasojedow.

#### ***Optimised Annealed Sequential Monte Carlo Samplers***

Presenter: Saif Syed

Sampling from complex, unnormalised probability distributions is a fundamental challenge in statistical inference. When direct sampling is intractable, a natural strategy is to introduce an annealing path that gradually interpolates between a tractable reference distribution (e.g. a prior or Gaussian) and the intractable target (e.g., a posterior). Annealed Sequential Monte Carlo (ASMC) samplers traverse this path by generating weighted particle approximations, and can achieve state-of-the-art performance, but existing methods require careful on-line schedule tuning, are sensitive to hyperparameters, and have no guarantee of terminating within a specified compute budget.

We address both challenges. By studying the variance of the normalising constant estimator, we identify a phase transition: below a critical threshold of particles and annealing density, cost scales exponentially in dimension and problem complexity, but above it, cost scales sublinearly, explaining why ASMC can efficiently tackle modern high-dimensional sampling problems. These insights directly motivate Optimised Annealed SMC (OASMC): a GPU-friendly, black-box algorithm with predictable runtime, achieving state-of-the-art performance with no hyperparameter tuning. We close with practical implementation guidelines and a comparison with parallel tempering, illuminating when each approach is preferable.

#### ***Tuning MCMC on GPUs***

Presenter: Charles Margossian

In principle, GPUs allow us to run hundreds or even thousands of Markov chains in parallel. Running algorithms on GPUs presents several challenges but also some opportunities. The main challenge is that operations on a

GPU need to be synchronized, a condition which is not met by many locally adaptive algorithms. The main advantage is that the problem of variance can be entirely solved by running a sufficiently large number of chains. It then remains to determine for how long each chain should be warmed up in order to achieve an acceptable bias. For the latter, I propose to monitor a nonstationary variance, which measures how well the Markov chains forget their starting point. I will show that, under broad conditions, this nonstationary variance decays at the same rate as the squared bias.

## Bayesian Social Sciences (Room 1104)

### *Correcting historical mortality rate bias in big crowd-sourced online genealogies*

Presenter: Michael Chong

In recent years, online communities have created genealogical records that span multiple continents over several centuries and contain demographic information for millions of people. However, these data are unrepresentative, and extracting accurate population information presents a major methodological challenge. We construct a Bayesian model that combines structured mortality estimation and smoothing techniques to correct mortality rates derived from FamiLinx, a large crowd-sourced genealogical dataset. Our model estimates and extrapolates a set of adjustment factors that capture the discrepancy between genealogy-derived rates and more reliable data from the Human Mortality Database. We apply our method to estimate 19th-century mortality rates for countries and time periods that are not covered by the high-quality data to demonstrate out-of-sample performance. Our results illustrate a wide range of underreporting patterns across age, time, and between countries.

### *Posterior Sampling for Causal Experimental Design Under Unknown Structure*

Presenter: Alex Volfovsky

Modern experiments are often designed as if the relevant structure is known in advance: which covariates matter, which units interfere, and which measurements are worth collecting. In many scientific and policy settings, however, these features are uncertain and must be learned while the experiment unfolds. This talk discusses how posterior sampling and sequential decision methods can be used to learn design-relevant structure during experimentation. I will use two recent examples: adaptive covariate acquisition for randomized trials, where Thompson sampling learns which covariates are worth measuring before balancing and adjustment; and adaptive experimentation under network interference, where posterior learning over an unknown graph guides treatment allocation. Together, these examples suggest a broader role for Bayesian ideas in causal experimentation: not only estimating effects after data are collected, but learning the experimental substrate needed for valid and efficient design.

### *Saddlepoint Monte Carlo and its Application to Exact Ecological Inference of Voting Patterns*

Presenter: Robin Ryder

Assuming  $X$  is a random vector and  $A$  a non-invertible matrix, one sometimes need to perform inference while only having access to samples of  $Y = AX$ . The corresponding likelihood is typically intractable. One may still be able to perform exact Bayesian inference using a pseudo-marginal sampler, but this requires an unbiased estimator of the intractable likelihood. We propose saddlepoint Monte Carlo, a method for obtaining an unbiased estimate of the density of  $Y$  with very low variance, for any model belonging to an exponential family. Our method relies on importance sampling of the characteristic function, with insights brought by the standard saddlepoint approximation scheme with exponential tilting. We show that saddlepoint Monte Carlo makes it possible to perform exact inference on particularly challenging problems and datasets. We focus on the ecological inference problem, where one observes only aggregates at a fine level. We present in particular a study of the carryover of votes between the two rounds of various French elections, using the finest available data (number of votes for each candidate in about 60,000 polling stations over most of the French territory). We show that existing, popular approximate methods for ecological inference can lead to substantial bias, which saddlepoint Monte Carlo is immune from. We also present original results for the 2024 legislative elections on political centre-to-left and left-to-centre conversion rates when the far-right is present in the second round.

## Selected Papers from Bayesian Analysis (Room Main Hall)

### *Bayesian inference for spatial-temporal non-Gaussian data using predictive stacking*

Presenter: Sudipto Banerjee

Analysing non-Gaussian spatial-temporal data requires introducing spatial as well as temporal dependence in generalised linear models through the link function of an exponential family distribution. Unlike in Gaussian likelihoods, inference is considerably encumbered by the inability to analytically integrate out the random effects

and reduce the dimension of the parameter space. Iterative estimation algorithms struggle to converge due to the presence of weakly identified parameters. We devise Bayesian inference using predictive stacking that assimilates inference from analytically tractable conditional posterior distributions. We achieve this by expanding upon the Diaconis-Ylvisaker family of conjugate priors and exploiting generalised conjugate multivariate (GCM) distribution theory for exponential families, which enables exact sampling from analytically available posterior distributions conditional upon some process parameters. Subsequently, we assimilate inference over a range of values of these parameters using Bayesian predictive stacking. We evaluate inferential performance on simulated data, compare with full Bayesian inference using Markov chain Monte Carlo (MCMC) and apply our method to analyse spatially-temporally referenced avian count data from the North American Breeding Bird Survey database.

#### ***Group Spike-and-Slab Variational Bayes***

Presenter: Kolyan Ray

We introduce Group Spike-and-Slab Variational Bayes (GSVB), a scalable method for group sparse regression. A fast co-ordinate ascent variational inference (CAVI) algorithm is developed for several common model families including Gaussian, Binomial and Poisson. Theoretical guarantees for our proposed approach are provided by deriving contraction rates for the variational posterior in grouped linear regression. Through extensive numerical studies, we demonstrate that GSVB provides state-of-the-art performance, offering a computationally inexpensive substitute to Markov Chain Monte Carlo (MCMC), whilst performing comparably or better than existing maximum a posteriori (MAP) methods. Additionally, we analyze three real world datasets wherein we highlight the practical utility of our method, demonstrating that GSVB provides parsimonious models with excellent predictive performance, variable selection and uncertainty quantification.

#### ***Bayesian Semi-supervised Multi-category Classification under Nonparanormality***

Presenter: Subhashis Ghoshal

Semi-supervised learning is a classification method that utilizes both labeled and unlabeled data. This paper proposes a fully Bayesian semi-supervised learning algorithm that can be applied to any multi-category classification problem. We assume the labels are missing at random when using unlabeled data in a semi-supervised setting. Suppose we have  $K$  classes in the data. We assume that the observations follow  $K$  multivariate normal distributions, depending on their true class labels, after some common unknown transformation is applied to each component of the observation vector. The function is expanded in a B-spline series, and a prior is added to the coefficients. We consider a normal prior on the coefficients and constrain the values to satisfy identifiability constraints. The precision matrices of the Gaussian distributions are given a conjugate Wishart prior, while the means are given the improper uniform prior. The resulting posterior remains conditionally conjugate, and the Gibbs sampler, aided by a data augmentation technique, can thus be employed. An extensive simulation study compares the proposed method with several other available methods. The proposed method is also applied to real datasets on the diagnosis of breast cancer and the classification of signals. We conclude that the proposed method has a better prediction accuracy in various cases.

## **Multitrack Sessions 12: 11:00–12:30**

### **Model Choice and Interpretation in Large-Scale Inference (Room 1101)**

#### ***Misspecified Yet Credible: A Generalized Bayes Framework for Uncertainty Quantification in High-Dimensional Bayesian Vector Autoregressive Models***

Presenter: Ray Bai

Vector autoregressive (VAR) models are widely used to capture linear dependencies in multivariate time series, yet Bayesian inferential theory for high-dimensional VAR models remains largely undeveloped. We propose a generalized Bayes framework that automatically adapts to sparsity and is robust to misspecification of both the error distribution and covariance structure. Under mild regularity conditions, we show that this approach yields reliable uncertainty quantification for the VAR transition matrices in very high dimensions, both in terms of component-wise coverage of individual entries and uniform coverage of the full matrices. As a corollary, the same strategy also delivers valid inference for sparse high-dimensional stochastic regressions with serially correlated errors.

#### ***A generalized Bayesian tree ensemble method for nonparametric density estimation***

Presenter: Linxi Liu

Tree-based methods cover a class of effective and computationally efficient nonparametric learning algorithms for

a wide range of inference objectives. Ensembles of trees—such as random forests, boosting, and Bayesian additive regression trees—are among the most successful and widely adopted algorithms. In unsupervised learning, tree-based methods have been proven effective in a range of statistical inference tasks, such as density estimation, clustering, multi-resolution sketching of distributional variations, and data compression. In this work, we study the density estimation problem, a fundamental task in unsupervised learning, and introduce a novel generalized Bayesian tree ensemble method for density estimation, referred to as “Gibbs density forest.” In contrast to the classical likelihood-based Bayesian inference, our method employs the risk-based generalized Bayesian inference, achieving an optimal bias-variance tradeoff through information-risk minimization. An appropriately chosen risk function plays a central role under this framework. We propose a new risk function for density forests, showing that this risk function can effectively measure the goodness of each forest. To reduce correlations among the trees and to better approximate smooth objects, we fit each tree under a small perturbation. Under mild conditions, we establish posterior contraction rates for Gibbs density forests under the generalized Bayesian framework. In particular, for density functions in certain Hölder spaces, we show that Gibbs density forests can achieve faster convergence rates than their single tree counterparts. The generalized Bayesian framework also enables built-in uncertainty quantification. We provide numerical results to demonstrate the performance of Gibbs density forests and empirical coverage of the credible sets in different settings.

### ***Assessing variable activity for Bayesian Additive Regression Trees***

Presenter: Akira Horiguchi

Bayesian Additive Regression Trees (BART) are non-parametric models that can capture complex exogenous variable effects. In any regression problem, it is often of interest to learn which variables are most active. Variable activity in BART is usually measured by counting the number of times a tree splits for each variable. Such one-way counts have the advantage of fast computations. Despite their convenience, one-way counts have several issues. They are statistically unjustified, cannot distinguish between main effects and interaction effects, and become inflated when measuring interaction effects. An alternative method well-established in the literature is Sobol indices, a variance-based global sensitivity analysis technique. However, these indices often require Monte Carlo integration, which can be computationally expensive. This paper provides analytic expressions for Sobol indices for BART posterior samples. These expressions are easy to interpret and computationally feasible. We also provide theoretical guarantees of contraction rates. Furthermore, we will show a fascinating connection between first-order (main-effects) Sobol indices and one-way counts. Finally, we compare these methods using analytic test functions and the En-ROADS climate impacts simulator.

## **Scalable Gaussian Processes (Room 1102)**

### ***Vecchia approximated Bayesian heteroskedastic Gaussian processes***

Presenter: Robert Gramacy

Many computer simulations are stochastic and exhibit input dependent noise. In such situations, heteroskedastic Gaussian processes (hetGPs) make ideal surrogates as they estimate a latent, non-constant variance. However, existing hetGP implementations are unable to deal with large simulation campaigns and use point estimates for all unknown quantities, including latent variances. This limits applicability to small campaigns and undercuts uncertainty. We propose a Bayesian hetGP using elliptical slice sampling (ESS) for posterior variance integration, and the Vecchia approximation to circumvent computational bottlenecks. We show favorable performance for our upgraded hetGP capability, compared to alternatives, on a benchmark example and a motivating corpus of more than 9-million lake temperature simulations.

### ***Scaling Gaussian processes with SMC<sup>2</sup>***

Presenter: Sara Wade

Gaussian processes are a key component of many flexible statistical and machine learning models. However, they exhibit cubic computational complexity and high memory constraints due to the need of inverting and storing a full covariance matrix. To circumvent this, mixtures of Gaussian process experts partition data points to independent experts, reducing the complexity by allowing inference based on smaller, local covariance matrices. Moreover, mixtures of Gaussian process experts substantially enrich the model’s flexibility, allowing for behaviors such as non-stationarity, heteroscedasticity, and discontinuities. In this work, we construct a novel parallelizable inference approach based on nested sequential Monte Carlo samplers to simultaneously infer both the gating network and Gaussian process expert parameters.

### ***Vecchia Gaussian processes: Probabilistic properties, minimax rates and methodological developments***



Presenter: Yichen Zhu

Gaussian Processes (GPs) are widely used to model dependency in spatial statistics and machine learning, yet the exact computation suffers an intractable time complexity of  $O(n^3)$ . Vecchia approximation allows scalable Bayesian inference of GPs in  $O(n)$  time by introducing sparsity in the spatial dependency structure that is characterized by a directed acyclic graph (DAG). Despite the popularity in practice, it is still unclear how to choose the DAG structure and there are still no theoretical guarantees in nonparametric settings. In this paper, we systematically study the Vecchia GPs as standalone stochastic processes and uncover important probabilistic properties and statistical results in methodology and theory. For probabilistic properties, we prove that the conditional distributions of the Matérn GPs, as well as the Vecchia approximations of the Matérn GPs, can be characterized by polynomials. This allows us to prove a series of results regarding the small ball probabilities and RKHSs of Vecchia GPs. For statistical methodology, we provide a principled guideline to choose parent sets as norming sets with fixed cardinality and provide detailed algorithms following such guidelines. For statistical theory, we prove posterior contraction rates for applying Vecchia GPs to regression problems, where minimax optimality is achieved by optimally tuned GPs via either oracle rescaling or hierarchical Bayesian methods. Our theory and methodology are demonstrated with numerical studies, where we also provide efficient implementation of our methods in C++ with R interfaces.

## Advances in Sampling & Computation (Room 1103)

### *Identification-aware Markov Chain Monte Carlo*

Presenter: Kyle Kuang

Leaving posterior sensitivity concerns aside, non-identifiability of the parameters does not raise a difficulty for Bayesian inference as far as the posterior is proper, but multi-modality or flat regions of the posterior induced by the lack of identification leaves a challenge for modern Bayesian computation. Sampling methods often struggle with slow or non-convergence when dealing with multiple modes or flat regions of the target distributions. This paper develops a novel Markov chain Monte Carlo (MCMC) approach for non-identified models, leveraging the knowledge of observationally equivalent sets of parameters, and highlights an important role that identification plays in modern Bayesian analysis. We show that our proposal overcomes the issues of being trapped in a local mode and achieves a faster rate of convergence than the existing MCMC techniques including random walk Metropolis-Hastings and Hamiltonian Monte Carlo. The gain in the speed of convergence is more significant as the dimension or cardinality of the identified sets increases. Simulation studies show its superior performance compared to other popular computational methods including Hamiltonian Monte Carlo and sequential Monte Carlo. We also demonstrate that our method uncovers non-trivial modes in the target distribution in a structural vector moving-average (SVMA) application.

### *Rapid mixing of stereographic MCMC for heavy-tailed sampling*

Presenter: Federica Milinanni

Sampling from high-dimensional, heavy-tailed distributions is a fundamental challenge in computational statistics, as many standard Markov chain Monte Carlo (MCMC) methods mix poorly in such settings. Recently, Stereographic MCMC [Yang et al., 2024] was proposed as a promising family of samplers for such task. However, its non-asymptotic convergence properties remain unexplored. In this work, we fill this gap by establishing non-asymptotic upper bounds on the mixing time of the Stereographic Random Walk Metropolis. Our results demonstrate that, under certain conditions on the initial distribution, the mixing time is polynomial in dimension for a broad class of distributions, including light- and heavy-tailed cases. Our proof is based on a conductance argument and leverages weighted isoperimetric inequalities for heavy-tailed distributions. These proof techniques provide new insights into the geometric properties of heavy tailed distributions that govern rapid mixing in high dimensions.

### *Can Microcanonical Langevin Dynamics Leverage Mini-Batch Gradient Noise?*

Presenter: David Rügamer

Scaling inference methods such as Markov chain Monte Carlo to high-dimensional models remains a central challenge in Bayesian deep learning. A promising recent proposal, microcanonical Langevin Monte Carlo, has shown state-of-the-art performance across a wide range of problems. However, its reliance on full-dataset gradients makes it prohibitively expensive for large-scale problems. This talk addresses a fundamental question: Can microcanonical dynamics effectively leverage mini-batch gradient noise? We present the first systematic study of this problem, revealing two critical failure modes: a limitation due to anisotropic gradient noise and numerical instabilities in complex high-dimensional posteriors. We resolve both issues by proposing a principled gradient

noise preconditioning scheme and developing a novel, energy-variance-based adaptive tuner that automates step size selection and informs dynamical numerical guardrails. The resulting algorithm is a robust and scalable microcanonical Monte Carlo sampler that consistently outperforms strong stochastic gradient MCMC baselines on challenging high-dimensional inference tasks like Bayesian neural networks. Combined with recent ensemble techniques, our work unlocks a new class of stochastic microcanonical Langevin ensemble (SMILE) samplers for large-scale Bayesian inference.

### ***Exact MCMC Inference for Doubly Intractable Posterior Distributions***

Presenter: Bingjing Tang

Posterior inference in Gaussian process models can be challenging when the likelihood involves intractable integrals, resulting in a doubly-intractable posterior that renders standard Markov chain Monte Carlo (MCMC) algorithms inapplicable. Such posteriors arise in applications including density estimation, point process intensity modeling, and evolutionary inference.

We develop exact MCMC methods for doubly-intractable posteriors across three scenarios. First, we address the problem of incorporating practitioners' distributional knowledge about a subset of observation coordinates into nonparametric Bayesian models. We introduce a constrained Bayesian model, where the conditional density is expressed as a perturbation of a parametric distribution with a transformed Gaussian process prior on the perturbation. Exact MCMC inference is performed using a rejection-based data augmentation scheme that avoids discretization.

Second, we propose a nonparametric Bayesian approach for estimating the intensity function of an inhomogeneous Poisson process without relying on large data augmentation or likelihood approximations. The method jointly models the intensity and cumulative intensity using transformed Gaussian processes.

Third, we consider inference under the bounded coalescent model, obtained by conditioning the coalescent tree so that the time to the most recent common ancestor is upper-bounded by a fixed time. We first develop a probabilistic generative model to simulate from the bounded coalescent, surpassing naive rejection sampling. Posterior inference of effective population size involves multiple intractable integrals in the likelihood, leading to a doubly-intractable posterior. We extend the method from the second scenario to jointly model the function and integrals as a transformed Gaussian process. We demonstrate the method using genetic lineage tracing data.

### ***Metropolis-Adjusted Interacting Particle Sampling***

Presenter: Bjorn Sprungk

In recent years, various interacting particle samplers have been developed to sample from complex target distributions, such as those found in Bayesian inverse problems. These samplers are motivated by the mean-field limit perspective and implemented as ensembles of particles that move in the product state space according to coupled stochastic differential equations. The ensemble approximation and numerical time stepping used to simulate these systems can introduce bias and affect the invariance of the particle system with respect to the target distribution. To correct for this, we investigate the use of a Metropolization step, similar to the Metropolis-adjusted Langevin algorithm. We examine Metropolization of either the whole ensemble or smaller subsets of the ensemble, and prove basic convergence of the resulting ensemble Markov chain to the target distribution. Our numerical results demonstrate the benefits of this correction in numerical examples for popular interacting particle samplers such as ALDI, CBS, and stochastic SVGD.

## **Quantum Bayesian Statistics (Room 1104)**

### ***Parameter estimation about quantum states: from Bayesian point of view***

Presenter: Jun Suzuki

The Bayesian approach offers a powerful methodology for estimating unknown probability distributions, with its success evident across diverse fields including data science, machine learning, and artificial intelligence. Currently, quantum estimation has become a pivotal area of research in advancing near-term quantum technologies, particularly quantum sensing, quantum imaging, and quantum metrology, where it promises enhanced accuracy and efficiency. Bayesian statistics is expected to play a crucial role in these quantum applications, offering advantages over frequentist methods. In this talk, we briefly provide the formulation of quantum Bayesian estimation, tracing its origins back to the late 1960s. We will discuss the fundamental difference between classical and quantum Bayesian statistics based on optimal design of experiments. Finally, we will discuss the precision limits for the Bayes risk and highlight several open problems in the field.

### ***Comparison between indefinite causal order strategy nor adaptive strategy does not improve the estimation of group action via Bayesian approach***

Presenter: Masahito Hayashi

We consider the estimation of unknown unitary operation under the Bayesian setting when the set of possible unitary operations is given by a projective unitary representation of a compact group. We show that indefinite causal order strategy nor adaptive strategy does not improve the performance of this estimation when error function satisfies group covariance. That is, the optimal parallel strategy gives the optimal performance even under indefinite causal order strategy and adaptive strategy.

### ***Nonparametric Bayesian estimation of quantum states***

Presenter: Cristina Butucea

We revisit the optimal Bayesian measurement of a Gaussian state. Even though these states can be fully described by bi-variate Wigner functions which are Gaussian, the optimal quantum Bayesian measurement gives a different risk than the classical one. Recent advances have been made in nonparametric estimation of pure states. We give here nonparametric Bayesian estimators of pure quantum states.

## **Optimal Bayesian Methods for Function Estimation Adapted to Intricate Structures (Room Main Hall)**

### ***Bayesian Sparse Gaussian Mixture Model for Clustering in High Dimensions***

Presenter: Yanxun Xu

We study the sparse high-dimensional Gaussian mixture model when the number of clusters is allowed to grow with the sample size. A minimax lower bound for parameter estimation is established, and we show that a constrained maximum likelihood estimator achieves the minimax lower bound. However, this optimization-based estimator is computationally intractable because the objective function is highly nonconvex and the feasible set involves discrete structures. To address the computational challenge, we propose a computationally tractable Bayesian approach to estimate high-dimensional Gaussian mixtures whose cluster centers exhibit sparsity using a continuous spike-and-slab prior. We further prove that the posterior contraction rate of the proposed Bayesian method is minimax optimal. The misclustering rate is obtained as a by-product using tools from matrix perturbation theory. The proposed Bayesian sparse Gaussian mixture model does not require pre-specifying the number of clusters, which can be adaptively estimated. The validity and usefulness of the proposed method is demonstrated through simulation studies and the analysis of a real-world single-cell RNA sequencing data set.

### ***Bayesian Adaptive Triangulation***

Presenter: Seonghyun Jeong

Conventional nonparametric regression on a two-dimensional irregular domain often overlooks the shape of the domain, permitting smoothing across its boundaries. In spatial and geostatistical applications, however, such an assumption may be inappropriate, as domain boundaries frequently restrict interactions among observations. Moreover, accounting for spatially varying function smoothness is considerably more challenging than in the univariate setting, and most existing approaches fail to properly address this local property of the target function. To overcome these issues, this study proposes Bayesian adaptive triangulation, based on locally adaptive triangulation over a polygonal domain. Our method explores the optimal constrained Delaunay triangulation of the domain based on the data, capturing boundary geometry and enabling spatial adaptation to rapidly varying local signals. A carefully designed prior distribution enhances the empirical performance of the proposed method. We show that the method achieves the optimal posterior contraction rate and adapts to the unknown function smoothness. Simulation studies demonstrate that our method outperforms existing models, providing improved estimation accuracy while maintaining lower model complexity.

### ***Bayesian Semiparametric Orthogonal Tucker Factorized Mixed Models for Multi-dimensional Longitudinal Functional Data***

Presenter: Arkaprava Roy

We introduce a novel longitudinal mixed model for analyzing complex multi-dimensional functional data, addressing challenges such as high-resolution, structural complexities, and computational demands. Our approach integrates dimension-reduction techniques, including basis function representation and Tucker tensor decomposition, to model complex functional (e.g., spatial and temporal) variations, group differences, and individual heterogeneity, while drastically reducing model dimensions. The model accommodates multiplicative random effects whose marginalization yields a novel Tucker-decomposed covariance-tensor framework. To ensure scalability, we employ semi-orthogonal mode matrices implemented via a novel graph-Laplacian-based smoothness prior with low-rank approximation, leading to an efficient posterior sampling method. A cumulative shrinkage strategy promotes sparsity and enables semi-automated rank selection. We establish theoretical guarantees for posterior

convergence and demonstrate the method's effectiveness through simulations, showing significant improvements over existing techniques. Applying the method to Alzheimer's Disease Neuroimaging Initiative (ADNI) neuroimaging data reveals novel insights into local brain changes associated with disease progression, highlighting the method's practical utility for studying cognitive decline and neurodegenerative conditions.

Discussant: Sergios Agapiou

## Multitrack Sessions 13: 14:00–15:30

### Causal and Interpretable Representation Learning (Room 1101)

#### *Learning Discrete Causal Representations from Heterogeneous Domains: A Bayesian Approach*

Presenter: Ankur Garg

Causal representation learning (CRL) aims to infer abstract latent causal concepts that give rise to potentially high-dimensional observations. This is particularly relevant when heterogeneous data from multiple related domains, or distributions, is available since domain shifts are often thought to arise from sparse and localized changes to some of the underlying causal mechanisms, whereas most other parts of the causal generative process remain unchanged.

In this project, we explore a Bayesian approach to learning discrete causal concepts from such multi-domain data. We propose a model that encodes common CRL structural assumptions, together with a tailored inference scheme; we demonstrate its effectiveness on political survey datasets, where we seek to recover the causal ordering between partisanship and issue opinions from both real survey data and LLM-generated synthetic data.

#### *Probabilistic discrete representations for causal representation learning*

Presenter: Yixin Wang

Causal representation learning seeks to uncover latent variables that capture the causal structure underlying high-dimensional observations. While most prior work assumes a deterministic mapping from latent to observed variables, often to ensure identifiability, this assumption fails to reflect the noisy nature of real-world data, where observation processes are inherently stochastic. In this work, we probabilistic discrete representations for causal representation learning, explicitly modeling the noisy observation process via exponential family distributions. We introduce a latent variable model with structured binary latent variables and demonstrate that causal relationships among the latents can provably be recovered up to label permutation. This model enables both better predictive performance and theoretical identifiability under relaxed conditions, even when the latent-to-observed mapping is non-deterministic.

#### *Nonlinear multi-study factor analysis*

Presenter: Gemma Moran

High-dimensional data often exhibit variation that can be captured by lower dimensional factors. For high-dimensional data from multiple studies or environments, one goal is to understand which underlying factors are common to all studies, and which factors are study or environment-specific. As a particular example, we consider platelet gene expression data from patients in different disease groups. In this data, factors correspond to clusters of genes which are co-expressed; we may expect some clusters (or biological pathways) to be active for all diseases, while some clusters are only active for a specific disease. To learn these factors, we consider a nonlinear multi-study factor model, which allows for both shared and specific factors. To fit this model, we propose a multi-group sparse variational autoencoder. The underlying model is sparse in that each observed feature (i.e. each dimension of the data) depends on a small subset of the latent factors. In the genomics example, this means each gene is active in only a few biological processes. Further, the model implicitly induces a penalty on the number of latent factors, which helps separate the shared factors from the group-specific factors. We prove that the shared factors are identified, and demonstrate our method recovers meaningful factors in the platelet gene expression data.

### Bayesian Latent Models: Beyond Normality (Room 1102)

#### *Condition matrix completion for polypharmacy medication information*

Presenter: Giovanni Toto

We propose a hierarchical Bayesian approach for matrix completion in presence of non-random missingness. Our method is motivated by a polypharmacy application in which patients' medical conditions are only partially observed, and information on their medications can be used to infer the presence or absence of the remaining conditions. In this application, the entries in the partially unknown matrix specify the presence of conditions,

with patients as rows and conditions as columns. The binary entries are independent Bernoulli realizations. The success probabilities are modeled using a factor model, which provides a parsimonious representation of patient- and medication-specific characteristics. Unknown entries are completed block-wise under the assumption that, for any medication taken, a patient has at least one condition for which the medication is typically prescribed.

***Conditional Random Partitions: A Telescopic Clustering Approach for Longitudinal Data***

Presenter: Beatrice Franzolini

Telescopic clustering models extend dependent nonparametric mixtures to panel data by introducing a stack of locally evolving random partitions. Most alternative methods impose global temporal dependence via time-indexed random probability measures and thus only achieve partial exchangeability. In contrast, our telescopic framework captures subject-specific dynamics, making it ideal when the same observational units are tracked over multiple occasions. At each temporal layer, a latent partition is generated conditional on the previous one. This telescopic propagation of cluster assignments induces a natural conditional dependence perfectly aligned with the sequential essence of longitudinal studies. In effect, the model behaves like a nonparametric hidden Markov model whose transition probabilities are governed by the conditional law of the evolving random partition, allowing for richly structured clustering dynamics over time. In this talk, I will (1) introduce the full class of telescopic clustering models and motivate their construction (2) explore their key probabilistic properties and illustrate how local dependence emerges and (3) demonstrate practical advantages through real-world applications, showcasing how telescopic partitions can uncover time-varying patterns that traditional approaches typically miss.

***Poisson process factorization for mutational signature analysis with genomic covariates***

Presenter: Jeff Miller

Mutational signatures are powerful summaries of the mutational processes altering the DNA of cancer cells. The usual approach to mutational signature analysis consists of decomposing the matrix of mutation counts from a sample of patients using non-negative matrix factorization (NMF). However, this ignores the heterogeneous patterns of mutation rates along the genome. We introduce Poisson process factorization (PPF), which addresses this limitation by employing an inhomogeneous Poisson point process model to infer mutational signatures and their activities as they vary across the genome. PPF generalizes the baseline NMF model by representing a patient's exposure to each signature as a locus-specific function that depends on genomic covariates and patient-specific copy numbers via a log-linear model. We develop tractable algorithms for maximum a posteriori estimation and posterior inference via Markov chain Monte Carlo. We demonstrate the method on simulated data and real data from breast cancer.

Discussant: Hedibert Freitas Lopes

**Advances in Stochastic Gradient MCMC for Complex and Structured Data (Room 1103)**

***Safe, Scalable and Accurate Stochastic Gradient MCMC for Bayesian Hierarchical GLMs***

Presenter: Youngsoo Baek

We propose a novel scalable stochastic gradient MCMC algorithm for fitting hierarchical generalized linear models (GLMs) to large data. Stochastic gradient MCMC methods are an attractive option for scaling Bayesian model fitting to large biomedical data, for which it is crucial to quantify uncertainty in the underlying variation within the longitudinal data of each patient. However, the problem of sampling the variance and correlation parameters in such a model has received relatively little attention in the previous literature. We demonstrate that a combination of common variance parameterizations used by methods like HMC and NUTS with previously proposed methods is highly unstable and prone to divergence issues; worse, in fact, it does not even have an invariant distribution. We provide a detailed study of how to detect possible divergence in the model formulation and resolve the issue by using a stochastic mirror Langevin algorithm, a stochastic subsampling variant of the previously studied mirror Langevin dynamics. The resulting algorithm requires an accurate evaluation of the gradients for each independent unit's marginal likelihood, and we discuss varying state-of-art approaches to compute them efficiently besides (another) MCMC. Finally, the posterior variance is re-calibrated post-sampling to reduce artificial noise from subsampling, using a novel correction formula based on solving a simple Lyapunov equation. Accuracy and computational merits are illustrated through simulated experiments and real-world pain trajectory data for breast cancer survivors.

***Scalable Bayesian Inference for Generalized Linear Mixed Models via Stochastic Gradient MCMC***

Presenter: Samuel Berchuck

The generalized linear mixed model (GLMM) is widely used for analyzing correlated data, particularly in large-



scale biomedical and social science applications. Scalable Bayesian inference for GLMMs is challenging because the marginal likelihood is intractable and conventional Markov chain Monte Carlo (MCMC) methods become computationally prohibitive as the number of subjects grows. We develop a stochastic gradient MCMC (SGMCMC) algorithm tailored to GLMMs that enables accurate posterior inference in the large-sample regime. Our approach uses Fisher's identity to construct an unbiased Monte Carlo estimator of the gradient of the marginal log-likelihood, making SGMCMC feasible when direct gradient computation is impossible. We analyze the additional variability introduced by both minibatching and gradient approximation, and derive a post-hoc covariance correction that yields properly calibrated posterior uncertainty. Through simulations, we show that the proposed method provides accurate posterior means and variances, outperforming existing approaches, including control variate methods, in large-n settings. We further demonstrate the method's practical utility in an analysis of electronic health records data, where accounting for variance inflation materially changes scientific conclusions.

#### ***Targeted Stochastic Gradient MCMC for HMMs with Rare Latent States***

Presenter: Alexander Young

Markov chain Monte Carlo (MCMC) algorithms for hidden Markov models often rely on the forward-backward sampler. This makes them computationally slow as the length of the time series increases, motivating the development of sub-sampling-based approaches. These approximate the full posterior by using small random subsequences of the data at each MCMC iteration within stochastic gradient MCMC. In the presence of imbalanced data resulting from rare latent states, subsequences often exclude rare latent state data, leading to inaccurate inference and prediction/detection of rare events. We propose a targeted sub-sampling (TASS) approach that over-samples observations corresponding to rare latent states when calculating the stochastic gradient of parameters associated with them. TASS uses an initial clustering of the data to construct subsequence weights that reduce the variance in gradient estimation. This leads to improved sampling efficiency, in particular in settings where the rare latent states correspond to extreme observations. We demonstrate substantial gains in predictive and inferential accuracy on real and synthetic examples.

### **Beyond likelihood: Loss-based Bayesian Methods (Room 1104)**

#### ***Loss-Based Variational Bayes Prediction***

Presenter: Ruben Loaiza Maya

We propose a new approach to Bayesian prediction that caters for models with a large number of parameters and is robust to model misspecification. Given a class of high-dimensional (but parametric) predictive models, this new approach constructs a posterior predictive using a variational approximation to a generalized posterior directly focused on predictive accuracy. The theoretical behaviour of the new prediction approach is analyzed, and a form of optimality is demonstrated. Applications to both simulated and empirical data using high-dimensional Bayesian neural networks and autoregressive mixture models demonstrate that the approach provides more accurate results than various alternatives, including misspecified likelihood-based predictions.

This talk is based on the article <https://www.tandfonline.com/doi/full/10.1080/10618600.2024.2341899>

#### ***Prequential posteriors***

Presenter: Shreya Sinha Roy

Data assimilation is a fundamental task in updating forecasting models upon observing new data, with applications ranging from weather prediction to online reinforcement learning. Deep generative forecasting models (DGFMs) have shown excellent performance in these areas, but assimilating data into such models is challenging due to their intractable likelihood functions. This limitation restricts the use of standard Bayesian data assimilation methodologies for DGFMs. To overcome this, we introduce prequential posteriors, based upon a predictive-sequential (prequential) loss function; an approach naturally suited for temporally dependent data which is the focus of forecasting tasks. Since the true data-generating process often lies outside the assumed model class, we adopt an alternative notion of consistency and prove that, under mild conditions, both the prequential loss minimizer and the prequential posterior concentrate around parameters with optimal predictive performance. For scalable inference, we employ easily parallelizable wastefree sequential Monte Carlo (SMC) samplers with preconditioned gradient-based kernels, enabling efficient exploration of high-dimensional parameter spaces such as those in DGFMs. We validate our method on both a synthetic multi-dimensional time series and a real-world meteorological dataset; highlighting its practical utility for data assimilation for complex dynamical systems.

#### ***A generalized Bayesian approach for high-dimensional robust regression with serially correlated errors and predictors***

Presenter: Kshitij Khare



We introduce a loss-based generalized Bayesian methodology for high-dimensional robust regression with serially correlated errors and predictors. The proposed framework employs a novel scaled pseudo-Huber (SPH) loss function, which smooths the well-known Huber loss, effectively balancing quadratic ( $\ell_2$ ) and absolute linear ( $\ell_1$ ) loss behaviors. This flexibility enables the framework to accommodate both thin-tailed and heavy-tailed data efficiently. The generalized Bayesian approach constructs a working likelihood based on the SPH loss, facilitating efficient and stable estimation while providing rigorous uncertainty quantification for all model parameters. Notably, this approach allows formal statistical inference without requiring ad hoc tuning parameter selection while adaptively addressing a wide range of tail behavior in the errors. By specifying appropriate prior distributions for the regression coefficients—such as ridge priors for small or moderate-dimensional settings and spike-and-slab priors for high-dimensional settings—the framework ensures principled inference. We establish rigorous theoretical guarantees for accurate parameter estimation and correct predictor selection under sparsity assumptions for a wide range of data generating setups. Extensive simulation studies demonstrate the superior performance of our approach compared to traditional Bayesian regression methods based on  $\ell_2$  and  $\ell_1$ -loss functions. The results highlight its flexibility and robustness, particularly in challenging high-dimensional settings characterized by data contamination. This talk is based on the article <https://arxiv.org/abs/2412.05673v3>

Discussant: Kate Lee

## Advances in Bayesian Machine Learning (Room Main Hall)

### *A new mutual information bound for statistical inference*

Presenter: Pierre Alquier

Recent advances in statistical learning theory have revealed profound connections between mutual information (MI) bounds, PAC-Bayesian theory, and Bayesian nonparametrics. This work introduces a novel mutual information bound for statistical inference. The derived bound has wide-ranging applications. It yields improved contraction rates for fractional posteriors in Bayesian nonparametrics. It can also be used to study variational inference or Maximum Likelihood Estimation (MLE). By bridging these diverse areas, this work advances our understanding of the fundamental limits of statistical inference and the role of information in learning from data. We hope that these results will not only clarify connections between statistical inference and information theory but also help to develop a new toolbox to study a wide range of estimators.

(joint work with El Mahdi Khrifch)

### *Heavy tails, adaptation and benign overfitting*

Presenter: Ismaël Castillo

In nonparametric random design regression, we consider ReLU-neural networks with weight parameters randomly sampled from a well-chosen heavy-tailed distribution. The architecture of the network is taken deterministic and ‘overfitting’, in the sense that the width of the network can be chosen much larger than the oracle width (the one we would choose if knowing the regularity or structure parameters of the regression function). We show that the corresponding Bayesian posterior distribution automatically achieves optimal adaptive rates (up to logarithmic factors) in terms of ‘smoothness’ and ‘structure’ parameters, in a number of settings including compositional structures, Besov anisotropic spaces and design points sitting on a low-dimensional manifold. We also derive contraction rates for mean-field Variational Bayes counterparts. We discuss connections with the setting of random series expansions on a basis, with the phenomenon of benign overfitting, as well as work in progress in these directions.

(Based on joint works with Paul Egels and Sergios Agapiou)

### *Variational Uncertainty Decomposition for In-context Learning*

Presenter: Yingzhen Li

As large language models (LLMs) gain popularity in conducting prediction tasks in-context, understanding the sources of uncertainty in in-context learning becomes essential to ensuring reliability. The recent hypothesis of in-context learning performing predictive Bayesian inference opens the avenue for Bayesian uncertainty estimation, particularly for decomposing uncertainty into epistemic uncertainty due to lack of in-context data and aleatoric uncertainty inherent in the in-context prediction task. However, the decomposition idea remains under-explored due to the intractability of the latent parameter posterior from the underlying Bayesian model. In this work, we introduce a variational uncertainty decomposition framework for in-context learning without explicitly sampling from the latent parameter posterior, by optimising auxiliary inputs as probes to obtain an upper bound to the aleatoric uncertainty of an LLM’s in-context learning procedure. Through experiments on synthetic and real-world tasks, we show quantitatively and qualitatively that the decomposed uncertainties obtained from our

method exhibit desirable properties of epistemic and aleatoric uncertainty.

Discussant: Randal Douc

## Multitrack Sessions 14: 16:00–17:30

### Bayesian Innovations in High-Dimensional Count Data Analysis from Modern Omics Technologies (Room 1101)

#### *Outlier-Robust Bayesian Multivariate Analysis with Correlation-Intact Sandwich Mixture*

Presenter: Kaoru Irie

Handling outliers is a fundamental challenge in multivariate data analysis because outliers may distort the structures of correlation or conditional independence. Although robust Bayesian inference has been extensively studied in univariate settings, theoretical results ensuring posterior robustness in multivariate models are scarce. We propose a novel scale mixture of multivariate normals called correlation-intact sandwich mixtures, in which the scale parameters are real-valued and follow an unfolded log-Pareto distribution. Our theoretical results on posterior robustness in multivariate settings emphasize that the use of a symmetric, super heavy-tailed distribution for those scale parameters is essential for achieving posterior robustness against element-wise contamination. The proposed method was applied to empirical studies of gene expression data using graphical models, suggesting evidence of cellwise contamination by outliers.

(Joint work with Yasuyuki Hamura and Shonosuke Sugawara)

#### *Bayesian nonparametric clustering with feature selection for spatial count data*

Presenter: Guanyu Hu

The advent of next-generation sequencing-based spatially resolved transcriptomics (SRT) techniques has reshaped genomic studies by enabling high-throughput gene expression profiling while preserving spatial context. Nevertheless, inherent challenges associated with these new high-dimensional spatial data (e.g., zero inflation) pose obstacles to effective clustering, a fundamental problem in SRT data analysis. Current computational approaches often rely on heuristic data preprocessing and arbitrary cluster number specification, leading to information loss and, consequently, sub-optimal downstream analysis. In response, we present BNPSpace, a novel Bayesian nonparametric spatial clustering framework that directly models SRT count data and automatically estimates the optimal number of spatial domains. BNPSpace facilitates the partitioning of the heterogeneous spatial domain while identifying a parsimonious set of discriminating genes among spatial domains, enhancing the interpretability of the findings. Additionally, it incorporates spatial information through a Markov random field prior model, encouraging a biologically meaningful partition pattern. We assess the performance of BNPSpace utilizing both simulated and real SRT data, demonstrating each innovation above is essential for the accurate identification of spatial domains. We also verified its scalability with large-scale SRT data. In the application to the human dorsolateral prefrontal cortex (DLPFC) 10x Visium data, comprising 4788 spots, BNPSpace outperforms existing methods by identifying more coherent spatial domain patterns. Furthermore, the discriminating genes identified by BNPSpace showed significant enrichment with odd ratio= 1.877 (p-value= 0.00162) in DLPFC gene sets validated by real biological experiments, underscoring its effectiveness in revealing biologically relevant insights.

#### *Tree-Embedded Bayesian Factor Models for Multidimensional Categorical Distributions*

Presenter: Naoki Awaya

We propose a Bayesian latent factor model for distribution-valued data, providing a parsimonious representation of multiple observed distributions through a low-dimensional latent structure. The method is motivated by applications involving grouped data, such as age-composition and income distributions observed across locations. To enable factor analysis for distributions, we employ a tree-based transformation that embeds distributions into a Euclidean space and construct a Bayesian hierarchical factor model in the transformed domain. We incorporate an infinite factor model to adaptively determine the effective number of latent factors and further extend the framework by introducing spatial dependence through a simultaneous autoregressive (SAR) prior. The proposed approach can accommodate both univariate and multivariate distributions within a unified framework. Through experiments using real population data, we demonstrate that the proposed model provides improved performance over conventional mixture models.

### Bayesian Clustering in Rank Aggregation Modeling (Room 1102)

#### *The Clustered Mallows Model*

Presenter: Nial Friel

Rankings represent preferences that arise from situations where assessors arrange items, for example, in decreasing order of utility. Orderings of the item set are permutations ( $\pi$ ) that reflect strict preferences. However, strict preference relations can be unrealistic for real data. Common traits among items can justify equal ranks and there can also be different importance attribution to decisions that form  $\pi$ . In large item sets, assessors might prioritise certain items, rank others low, and express indifference towards the remaining. Rank aggregation may involve decisive judgments in some parts and ambiguity in others. In this paper, we extend the famous Mallows (Biometrika 44:114–130, 1957) model (MM) to accommodate item indifference. Grouping similar items motivates the proposed Clustered Mallows Model (CMM), a MM counterpart for tied ranks with ties learned from the data. The CMM provides the flexibility to combine strictness and indifferences, describing rank collections as ordered clusters. CMM Bayesian inference is a doubly-intractable problem since the normalized model is unavailable. We overcome this with a version of the exchange algorithm (Murray et al. in Proceedings of the 22nd annual conference on uncertainty in artificial intelligence (UAI-06), 2006) and provide a pseudo-likelihood approximation as a computationally cheaper alternative. Analysis of two real-world ranking datasets is presented, showcasing the practical application of the CMM and highlighting scenarios where it offers advantages over alternative models.

### ***Bayesian Rank-Clustering with Applications to Survey and Election Data***

Presenter: Michael Pearce

This paper proposes a new statistical model to infer interpretable population-level preferences from ordinal comparison data. Such data is ubiquitous, e.g., ranked choice votes, top-10 movie lists, and pairwise sports outcomes. Traditional statistical inference on ordinal comparison data results in an overall ranking of objects, e.g., from best to worst, with each object having a unique rank. However, the ranks of some objects may not be statistically distinguishable. This could happen due to insufficient data or to the true underlying object qualities being equal. Because uncertainty communication in estimates of overall rankings is notoriously difficult, we take a different approach and allow groups of objects to have equal ranks or be rank-clustered in our model. Existing models related to rank-clustering are limited by their inability to handle a variety of ordinal data types, to quantify uncertainty, or by the need to pre-specify the number and size of potential rank-clusters. We solve these limitations through our proposed Bayesian Rank-Clustered Bradley-Terry-Luce (BTL) model. We accommodate rank-clustering via parameter fusion by imposing a novel spike-and-slab prior on object-specific worth parameters in the BTL family of distributions for ordinal comparisons. We demonstrate rank-clustering on simulated and real datasets in surveys, elections, and sports analytics.

### ***Bayesian genome-wide clustering and variable selection of transcriptomic data via rank-based mixtures***

Presenter: Valeria Vitelli

With the increasing availability of ranking data, there has been a growing demand for appropriate unsupervised rank-based inferential frameworks capable of handling high-dimensional datasets and providing uncertainty quantification for all estimates. Rank-based methods have also seen a growing popularity in -omics pipelines, as ranking continuous measurements provides a robust means of handling non-normally distributed data. The Bayesian Mallows model (BMM) has emerged as a promising choice because of its adaptability to various types of ranking data and its flexible framework, integrating cluster-wise rank aggregation with inference at the individual level. However, the scalability of BMM to ultra-high-dimensional settings, such as -omics analyses, has remained limited. The present paper addresses this issue by introducing the first rank-based model generalizing BMM to jointly handle clustering and variable selection, namely the lower-dimensional Bayesian Mallows Model Mixtures (lowBM3). The proposed method provides a novel Bayesian framework that simultaneously handles heterogeneity in the sample, unsupervised parameter estimation, and model selection in a scalable manner for ultra-high-dimensional data. Additionally, a companion postprocessing framework is introduced to provide posterior summaries of the discrete posterior distributions of both the consensus ranking and the variable selector. Posterior summaries are extremely useful for the interpretation of results, especially in the context of molecular data applications. Simulation studies are performed to assess the performance of the method. The usefulness of the method is also shown in an application to signature discovery for cancer genomics, where RNA-seq bulk gene expression data obtained from breast cancer patients are clustered genome-wide.

Discussant: Brendan Murphy

## Statistical Methods for Diffusions: Inference, Computation, and Generative Modeling (Room 1103)

### *Dimension-free guarantee for guidance scale adaptation of diffusion models through PAC-Bayesian analysis.*

Presenter: Shogo Nakakita

Reward-guided alignment of conditional diffusion models—such as text-to-image diffusion models—is crucial for improving image aesthetics and enhancing text-to-image consistency. In this scheme, we aim to select the hypothesis that maximizes the expected reward, defined as the expectation of a given reward function with respect to the distribution of prompts and diffusion outputs. In practice, however, since the prompt distribution is generally unknown, the true expected reward associated with each hypothesis is unavailable, and models are typically fine-tuned using empirical rewards. This raises concerns about generalization, particularly in the high-dimensional regimes that characterize modern diffusion models, and the conditions under which such fine-tuning avoids collapse remain unclear. In this work, we derive a high-probability generalization bound that depends only on the 2-norms of the fine-tuning hypotheses and the sample size. Remarkably, the bound is independent of the dimensionality of both the latent/state space and the prompt space, as well as the model architectures and any implicit constants. Our result therefore suggests that controlling the 2-norms of the hypotheses is sufficient to guarantee generalization in reward-guided fine-tuning. The proof leverages a novel PAC-Bayesian argument combined with an intricate coupling construction for diffusion models conditioned on different prompts.

### *Inferring diffusivity from killed diffusion*

Presenter: Fanny Seizilles

We consider diffusion of independent molecules in an insulated Euclidean domain with unknown diffusivity parameter. At a random time and position, the molecules may bind and stop diffusing in dependence of a given ‘binding potential’. The binding process can be modeled by an additive random functional corresponding to the canonical construction of a ‘killed’ diffusion Markov process. We study the problem of conducting inference on the infinite-dimensional diffusion parameter from a histogram plot of the ‘killing’ positions of the process. We show first that these positions follow a Poisson point process whose intensity measure is determined by the solution of a certain Schrödinger equation. The inference problem can then be re-cast as a non-linear inverse problem for this PDE, which we show to be consistently solvable in a Bayesian way under natural conditions on the initial state of the diffusion, provided the binding potential is not too ‘aggressive’. In the course of our proofs we obtain novel posterior contraction rate results for high-dimensional Poisson count data that are of independent interest. A numerical illustration of the algorithm by standard MCMC methods is also provided. This is joint work with Richard Nickl, available at <https://arxiv.org/abs/2503.14978>

### *Statistical inference for multi-dimensional reversible diffusions*

Presenter: Sven Wang

NA

## Advances in Bayesian Nonparametrics II (Room 1104)

### *Enhancing scalability in Bayesian nonparametric factor analysis of spatiotemporal data*

Presenter: Cheng Li

We propose novel computational strategies for Bayesian nonparametric latent factor spatiotemporal models that are computationally feasible for moderate to large spatiotemporal data. Although such Bayesian models are flexible and enable spatial clustering, they face a prohibitively high computational cost in posterior sampling when the spatial and temporal dimensions increase to a couple hundred. We address this challenge with several speed-up proposals. We integrate a new slice sampling algorithm that permits varying numbers of spatial mixture components, which are guaranteed to be non-increasing through the posterior sampling iterations and thus effectively reducing the number of mixture parameters. Additionally, we introduce a spatial latent nearest-neighbor Gaussian process prior and new sequential updating algorithms for the spatially varying latent variables in the stick-breaking process prior. Our new proposals lead to significantly enhanced computational scalability and storage efficiency while maintaining capabilities for both spatiotemporal prediction and clustering of locations with similar temporal trajectories. The improvement in computational efficiency and inferential performance is substantiated by extensive numerical experiments and an application to sea surface temperature data.

### *Bayesian Mixture Models for Histograms: With Applications to Large Datasets*

Presenter: Richard Warr

In many real-world scenarios—especially those involving privacy constraints or data summarization—data is available only in aggregated forms such as histograms or frequency tables. This work introduces a novel Bayesian method for inferring the underlying population distribution by fitting a mixture model to binned data. While we focus on mixtures of normal distributions, the framework is flexible, and can be extended to other distributional families. We place a prior on the number of mixture components, accommodating both finite and countably infinite mixtures, and perform inference using reversible jump MCMC. The proposed approach demonstrates strong performance on large-scale data, showcasing the potential of nonparametric Bayesian modeling in practical applications. Furthermore, we extend the method to model multiple histograms simultaneously and clustering them using the Dirichlet process. This enables information sharing across populations and provides a principled posterior probability for assessing homogeneity between groups.

***L2-posterior contraction rates for Gaussian process and random series priors in Bayesian nonparametric regression models***

Presenter: Paul Rosa

The nonparametric regression model with normal errors has been extensively studied, both from the frequentist and Bayesian viewpoint. A central result in Bayesian nonparametrics is that under assumptions on the prior, the data-generating distribution (assuming a true frequentist model) and a semi-metric  $d(\cdot, \cdot)$  on the space of regression functions that satisfy the so called testing condition, the posterior contracts around the true distribution with respect to  $d(\cdot, \cdot)$ , and the rate of contraction can be estimated. In the regression setting, the semi-metric  $d(\cdot, \cdot)$  is often taken to be the Hellinger distance or the empirical L2 norm (i.e., the L2 norm with respect to the empirical distribution of the design) in the present regression context. However, extending contraction rates to the “integrated” L2 norm usually requires more work, and has previously been done for instance under sufficient smoothness or boundedness assumptions, which may not necessarily hold. In this work we show that, for priors based on truncated random basis expansions and in the random design setting, a high probability two sided inequality between the empirical L2 norm and the integrated L2 norm holds in appropriate spaces of functions of low frequencies, under mild assumptions on the underlying basis (which can be for instance a Fourier, wavelet or Laplace eigenfunction basis), allowing us to directly deduce an L2 contraction rate from an empirical L2 one without further assumption on the true regression function. We also present an extension of these arguments to Gaussian process priors with Sobolev type RKHS, allowing to show optimal L2 posterior contraction rates under weaker assumptions (to the best of the author’s knowledge) than the ones already existing in the literature.

***Model-based clustering for network data via a latent shrinkage position cluster model***

Presenter: Claire Gormley

Low-dimensional representation and clustering of network data are tasks of great interest across various fields. Latent position models are routinely used for this purpose by assuming that each node has a location in a low-dimensional latent space and by enabling node clustering. However, these models fall short through their inability to simultaneously determine the latent space dimension and number of clusters. Here we introduce the latent shrinkage position cluster model (LSPCM), which addresses this limitation. The LSPCM posits an infinite-dimensional latent space and assumes a Bayesian nonparametric shrinkage prior on the latent positions’ variance parameters resulting in higher dimensions having increasingly smaller variances, aiding the identification of dimensions with non-negligible variance. Further, the LSPCM assumes the latent positions follow a sparse finite Gaussian mixture model, allowing for automatic inference on the number of clusters related to non-empty mixture components. As a result, the LSPCM simultaneously infers the effective dimension of the latent space and the number of clusters, eliminating the need to fit and compare multiple models. The performance of the LSPCM is assessed via simulation studies and demonstrated through application to two real social network datasets from sporting and political contexts. Open-source software is available to facilitate widespread use of the LSPCM.

***Generalized Bayesian noise contrastive learning with additive tree ensembles***

Presenter: Benedetta Bruni

Bayesian tree-based models for probability distributions estimate the density of an IID sample as piece-wise constant over a partition of the sample space. Fine-grained partitions allow trees to represent local features of a density in detail - which is highly desirable for generative modeling - but the resulting deep trees are computationally burdensome and prone to overfitting. Additive tree ensembles are widely used in the supervised literature to mitigate these concerns but cannot be extended to unconditional density learning without significant implementation effort. For example, BART exploits the conditional conjugacy of the Gaussian leaf coefficient priors to the Gaussian regression model, which cannot be easily replicated in the unsupervised setting. We

propose a framework to extend Bayesian additive tree ensembles from prediction to generative modeling retaining their computational simplicity. Adopting a Noise-Contrastive perspective, we target a new estimand: the ratio between our density of interest and a known density - representing a base measure. This enables inference under the Balancing Loss which allows to estimate of the log density ratio via additive tree ensembles and supports conditionally conjugate Inverse-Gaussian priors for the exponentiated leaf coefficients. Given the limited theoretical analyses of additive tree ensembles for unconditional density learning and the recent introduction of the Balancing Loss, we present a Gibbs Posterior contraction theory for the proposed model. The obtained posterior contraction rate matches the minimax one for Hölder density ratios with smoothness parameter  $\alpha < 1$ .

## Memorial for Harry van Zanten (Room Main Hall)

### *On uncertainty quantification for nonparametric multivariate Hawkes processes*

Presenter: Judith Rousseau

Multivariate Hawkes processes form a class of point processes describing self and inter exciting/inhibiting processes. There is now a renewed interest of such processes in applied domains and in machine learning, but there exists only limited theory about inference in such models apart from parametric models. After reviewing results on convergence rates for Bayesian nonparametric approaches to such models, I will present new results on uncertainty quantification for important functionals.

To be more precise, the intensity function of a linear Hawkes process has the following form: for each dimension  $k \leq K$

$$\lambda^k(t) = \sum_{\ell \leq K} \int_0^{t-} h_{\ell k}(t-s) dN_s^\ell + \nu_k, \quad t \in [0, T]$$

where  $(N^\ell, \ell \leq K)$  is the Hawkes process,  $\nu_k > 0$ , and  $h_{\ell k} \geq 0$ . The parameters are the functions  $h_{\ell k}$  and the constants  $\nu_k$ , with  $\ell, k \in [K]$ .

I will explain the specificities for deriving efficient semiparametric theory in these models, corresponding to the fact that we allow the functions  $h_{\ell k}$  to be equal to 0; I will then provide conditions for verifying the semiparametric Bernstein von Mises property on functionals in the form  $\Psi(\nu, h) = \Psi(\nu)$  or  $\Psi(\nu, h) = \Psi(h)$ , where  $\Psi$  is smooth. I will apply these results to the functionals  $\Psi(\nu, h) = \nu$  and  $\Psi(\nu, h) = \int h_{\ell k}(x) dx$  and deduce from these a sharp result on the estimation of the graph of interaction. The latter is defined as the adjacency matrix  $\Delta$  where  $\delta_{\ell k} = 1$  if and only if  $\rho_{\ell k} > 0$ .

### *A prior construction and BvM theorem for non-smooth linear functionals*

Presenter: Lasse Vuursteen

Bayesian nonparametric methods are typically tuned so that the posterior contracts at the optimal rate for the parameter as a whole, but inference often targets a specific functional of the parameter rather than the parameter itself. Several well-known examples show that tuning a nonparametric prior for optimal global risk contraction can generically give the wrong bias-variance tradeoff for a non-smooth linear functional — for instance, pointwise evaluation under a generic Gaussian series prior tuned for optimal L2 contraction over a Sobolev space. In this talk, I will present a general construction that focuses a prior on a non-smooth linear functional so that it satisfies a Bernstein-von Mises-type theorem for that functional, with credible intervals of optimal width, whilst also attaining the optimal risk for the parameter in a global sense.

### *Generalization bounds of transformers for next token prediction in a Bayesian language model*

Presenter: Johannes Schmidt-Hieber

A refined statistical understanding of LLM pre-training requires the analysis of the transformer architecture for data distributions that encapsulate key characteristics of text data. Building on the NLP literature, we propose a Bayesian language model. For this data generating process, we derive generalization bounds of deep transformer architectures. Joint work with Insung Kong and Niklas Dexheimer

### *Exact and optimal uncertainty quantification with Gaussian posteriors*

Presenter: Dennis Nieman

We construct two elementary (uniform-type and average-type) posterior credible sets in the white noise model and study their frequentist coverage. The Gaussian process posterior and the size of the credible sets are related to the theory of optimal estimation. This naturally leads to a set of smoothness classes over which the probability of coverage is at least as much as the posterior mass. Since the bounds are exact, this generalises and improves upon existing asymptotic theory.





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